I

VOLUME ONE OF THREE

Notes In Orthodontics

A complete revision companion · Revised edition

Crash notebook: forty-two schematics and four number cards
The twenty essay topics, with their skeletons
The sixteen chapters
Model viva answers and the twelve traps
New: ethics and consent · radiation protection · infection control · governance and audit

Referenced throughout to Cobourne MT & DiBiase AT, Handbook of Orthodontics, *3rd edition (Elsevier, 2025), and cross-referenced to Volumes II and III.*

For candidates sitting the examination of 5 November 2026

Dr. Kam O Man, Owen

BDS (HK) · MFDS RCSEd · MFDS RCPS (Glasg)
MClinDent (Fixed and Removable Prosthodontics) (KCL)
MImpDent (RCSEd) · MGDS RCSI

An independent revision aid. Not an official publication of, nor endorsed by, the Hong Kong Institute of Orthodontics, Shanghai Jiao Tong University, or the Royal College of Surgeons of Edinburgh.

The set

Three volumes, in the order you should use them. Each is cross-referenced to the other two, so you can enter from whichever you happen to have open.

I	**Notes In Orthodontics** Revised edition · 275 pp	The book that teaches the material. A crash notebook of forty-two schematics and four number cards; the twenty essay topics with their skeletons; the sixteen chapters; model viva answers; and four new units on ethics and consent, radiation protection, infection control, and governance. **Reach for it** when you are learning something for the first time, planning an essay, or want the detail behind a fact.
II	**A Candidate's Revision Handbook** 161 pp	The book that maps the material onto the examination. Thirty sections against the sixty-five curriculum items, the numbers on six cards, the diagnostic cases, the case presentation, the oral, the literature by theme, and the dated countdown. **Reach for it** when you are revising against the syllabus, preparing the components that are not written, or covering what Volume I does not.
III	**The Question Bank** 300 pp	The book that tells you whether it has stuck. 1,016 single best answers in thirty blocks with explanations and references, a mark sheet, 207 viva questions with model answers, and a viva tracker. **Reach for it** at the end of every study session, and let the score decide what you re-read.

LEARN, REVISE, TEST

Volume I teaches the material. Volume II checks it against the sixty-five curriculum items and covers the four assessed components that are not knowledge — the diagnostic cases, the case presentation, the oral and examination technique. Volume III tells you whether any of it has stuck. Work in that order, and let a poor score in Volume III send you back to the section Volume II names.

The plan

Seven working weeks and a weekend, ending **Thursday 5 November 2026**. The written paper is the engine — revising for it is simultaneously revising for the oral and the diagnostic cases, because all three test the same knowledge. The case submissions depend on other people, so their *requests* go out first.

Before Monday — one session, about three hours

1. Pick your three write-up cases and your two presentation cases. For each, list what is missing: pre-treatment records, radiographs, photographs, models or scans, progress notes, debond records, PAR scoring. **2.** Send every request the same afternoon — patient recalls, radiograph retrieval, trainer sign-off. **3.** Ask the Programme Management Office when the three written-up cases are due. If that date falls before 5 November, it outranks everything below. **4.** Sit Question Bank Blocks **H** and **AD** cold. 55 questions, ethics and governance — the likeliest thin spot and the cheapest marks in the paper.

The spine is the curriculum, not the book. Volume II's thirty sections and Volume III's thirty blocks run one-to-one — §1 is Block A, §30 is Block AD — so the weeks below advance through them in order. The Volume I column then names the units whose *main* treatment sits in that week's sections. **Volume I has two separate series and they are not the same thing**: the twenty essay **topics** of Part II and the sixteen **chapters** of Part III. Chapter 1 is Evidence-Based Orthodontics; Topic 1 is the biological basis of tooth movement. Every chapter and every topic appears exactly once below.

Week	Dates	Curriculum	Read in Volume I	Alongside
1	14–20 Sep	§1–§5 · Blocks A–E	Chapters 2–6 · Topic 1	Viva V1–V20 aloud
2	21–27 Sep	§6–§10 · Blocks F–J	Chapters 1, 7 · Topic 2 · Part V.1	V21–V48. First diagnostic drill
3	28 Sep–4 Oct	§11–§15 · Blocks K–O	Chapters 8, 11 · Topics 3–5, 7	V49–V80. Write up case 1
4	5–11 Oct	§16–§20 · Blocks P–T	Chapters 9, 10, 12–14 · Topics 6, 8, 13–15, 20	V81–V111. Write up case 2
5	12–18 Oct	§21–§25 · Blocks U–Y	Topics 11, 12, 16, 17, 19	V112–V151. Write up case 3
6	19–25 Oct	§26–§30 · Blocks Z–AD	Chapters 15, 16 · Topics 9, 10, 18 · Part V.2–V.4	V152–V207
7	26 Oct–1 Nov	Re-sit every block scored under 70%	Only what the mark sheet sends you back to	Mock oral. Present both cases aloud
Final	2–4 Nov	Two or three blocks a session at examination pace	Crash notebook · number cards	Nothing new

Week 5 lists no chapters because §21–§25 are Chapter 13 territory, read in week 4; that week is topic-led. Part V has no chapter or topic of its own — it is new in this edition.

The daily loop — about two hours

One **Volume I** unit (45 min) → the **Volume II** section its teal strap names (20 min) → that **Volume III** block, timed at 60 seconds a question (30 min) → read the explanation for *every* question, including the ones you got right (25 min).

Log the score on the mark sheet. Anything under 70 per cent goes on the week 7 list. Close each session with one viva section, aloud, to the end, before reading the model.

THREE RULES THAT DECIDE THIS

1. **Speak from week one, not week seven.** The oral tests whether you can produce a structured spoken answer under pressure — a separate skill from knowing the material. Question blocks are measurable and satisfying, which is exactly why they expand to fill the time and crowd out the one thing you cannot cram.
2. **Protect the case deadline.** You cannot revise your way out of a missing pre-treatment photograph. A patient recall is a three-to-four week loop you do not control.
3. **Do not re-read Volume I cover to cover.** Use it as a textbook — when a Volume III explanation does not land, follow the strap back to that one unit.

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How to use this volume

This is **Volume I** of three, and the book the other two were built from. It is where you *learn* the material. **Volume II**, the Revision Handbook, is where you revise it against the curriculum and prepare the components that are not written. **Volume III**, the Question Bank, is where you find out whether it has stuck. The three are cross-referenced in both directions, so you can enter from whichever you happen to have open.

What is in each of the three volumes

	Volume	What it is	When to reach for it
I	*Notes In Orthodontics* (this volume)	The full treatment: crash notebook, twenty essay topics, sixteen chapters, model viva answers, and the four new units of Part V	Learning something for the first time; writing an essay plan; wanting the detail behind a fact
II	A Candidate's Revision Handbook	Thirty sections mapped to the sixty-five curriculum items, plus the numbers, the diagnostic cases, the case presentation, the oral and the dated countdown	Revising against the syllabus; the last six weeks; the four assessed components that are not knowledge
III	The Question Bank	1,016 single best answers with explanations, and 207 vivas with model answers	Testing whether you actually know it; the mark sheet tells you what to re-read

What changed in this edition

Change	Detail
Cross-references throughout	Every topic and every chapter now opens with a teal strap naming the Revision Handbook section, the Question Bank block with its live question range, and the viva range that test it
Part V added	Four new units — ethics and consent, radiation protection, infection control, and governance and audit — covering the curriculum items the first edition had no home for
Crash notebook redrawn	All forty-two schematics rebuilt, each carrying its own Handbook and Question Bank pointers. Three whose captions ran off the page have been restored in full
Running heads and contents	Every unit starts on a fresh page, carries its name along the top of every page, and appears in the table of contents
Numbers checked	The four number cards have been reconciled against the figures used in the Question Bank explanations, so a value learned here matches the value marked there
Typesetting	Reset to match the Revision Handbook and Question Bank, so the three volumes read as one set

Reading the panels

Panel	What it is for
In one minute	The whole unit compressed. If you can say this, you can start the answer
Essay skeleton	The numbered headings. Reproduce them and the essay writes itself
Facts to have at your fingertips	The numbers, with the page they came from
Must-know / Nice-to-know	What a pass requires, and what lifts an answer above it
Evidence to name	The studies worth naming. A mis-attributed reference is worse than none
Examiner traps	Where marks are lost
Viva questions	Answer these aloud, to the end, before reading anything
Self-test	Chapter-level recall checks
Teal strap	Where this unit sits in the other two volumes

Concordance with the Revision Handbook and Question Bank

Use this to move between the three volumes. Question ranges are generated at build time, so they stay correct as questions are added.

The twenty essay topics

	Essay topic	Handbook	Questions	Viva
1	The biological basis of tooth movement	§4; force §14	Q96–140	V21–V30
2	Cephalometry: uses and limitations	§10	Q286–325	V59–V68
3	Factors in formulating a treatment plan	§12; space §11	Q326–350, Q351–390	V69–V80
4	Factors affecting the choice of extraction	§13	Q391–430	V81–V90
5	Anchorage and its clinical significance	§14	Q431–481	V91–V102
6	The three orders and fixed appliances	§17	Q567–606	V112–V121
7	The ideal archwire; steel versus NiTi	§15	Q482–525	V103–V111
8	Functional appliances	§16; §22	Q526–566	V112–V121
9	Benefits and complications of treatment	§29; consent §8	Q962–991	V49–V58
10	Medical conditions relevant to orthodontics	§29.3	Q962–991	V174–V183
11	Overbite and methods of overbite reduction	§25.1	Q842–871	V162–V173
12	Reducing overjet in a growing child	§22; §16	Q747–786	V152–V161
13	Management of a missing maxillary central	§20.6	Q677–721	V140–V151
14	Management of an impacted maxillary canine	§20.4	Q677–721	V140–V151
15	Management of an impacted lower second molar	§20.5	Q677–721	V140–V151
16	Methods of crossbite correction	§24; §15.5	Q817–841	V162–V173
17	Correcting a Class III incisor relationship	§23	Q787–816	V152–V161
18	Malocclusion and temporomandibular disorder	§26	Q872–901	V174–V183
19	Mild facial asymmetry in a growing child	§24.4	Q817–841	V162–V173
20	Late lower incisor crowding	§19.4	Q642–676	V130–V139

The sixteen chapters

	Chapter	Handbook	Questions	Viva
1	Evidence-Based Orthodontics	§7	Q196–225	V39–V48
2	Occlusion and Malocclusion	§5; indices §6	Q141–170, Q171–195	V31–V48
3	Prenatal Development	§2.1; clefts §28	Q26–55	V1–V10
4	Postnatal Growth	§1, §2.2–2.4	Q1–25	V1–V10

5	Development of the Dentition	§3; §19.4	Q56–95	V11–V20
6	Orthodontic Tooth Movement	§4; anchorage §14	Q96–140	V21–V30
7	Examination and Diagnosis	§9; ceph §10	Q256–285	V59–V68
8	Treatment Planning	§12; §11; §13	Q326–350–Q391–430	V69–V90
9	Functional Jaw Orthopaedics	§16; §24.2; §23.3	Q526–566	V112–V121
10	Contemporary Fixed Appliances	§17; §15; §18	Q482–525–Q607–641	V103–V129
11	Skeletal Anchorage	§14.6	Q431–481	V91–V102
12	Management of the Developing Dentition	§20; §3.3	Q677–721	V140–V151
13	Management of the Permanent Dentition	§18, §21–§23, §25	Q607–641–Q842–871	V122–V173
14	Clear Aligners	§17.5	Q567–606	V112–V121
15	Orthodontics and Orthognathic Surgery	§27; §24.4	Q902–931	V184–V193
16	Cleft Lip and Palate, and Syndromes	§28	Q932–961	V184–V193

Part V — and the two blocks with no home here

	New unit	Handbook	Questions	Viva
V.1	Ethics, consent and capacity	§8	Q226–255	V39–V48
V.2	Radiation protection and imaging	§30.2; §9.4	Q992–1016	V39–V48
V.3	Infection control	§30.1	Q992–1016	V39–V48
V.4	Governance, audit and referral	§30.3–§30.4	Q992–1016	V39–V48

HOW TO USE THIS VOLUME INSIDE THE PLAN

The dated plan sits at the front of all three books, and this one is the *read* step of its daily loop. Take one unit — a chapter in weeks 1 to 5, a topic thereafter — read it, then follow its teal strap to the **Volume II** section and the **Volume III** block that test it. Reproduce the essay skeleton from memory before you read it; that is the difference between recognising the material and being able to produce it. Save **Part V** for week 6, when the governance and consent blocks come round. In the last three days this volume reduces to the crash notebook: the forty-two schematics, the four number cards, and the twelve traps on the morning.

Part I — Crash notebook

Forty-two schematics, the numbers on four cards, the twelve traps. Almost no prose. Work the schematics forty-eight hours out; the number cards the night before; the traps on the morning.

WHAT CHANGED IN THIS EDITION

Every schematic has been redrawn, and each one now carries the Revision Handbook section and the Question Bank block that test it, so a card you cannot reproduce from memory tells you where to go. Three figures whose captions ran off the page in the first edition have been restored in full, and the numbers have been checked against the sources named in Part IV.

Orientation

Handbook The whole of Part 1 **Questions** All thirty blocks **Viva** All twenty sections

Two pages. If you read nothing else, read these.

The twenty essay topics — one line each

	Topic	The one line
1	Biology of tooth movement	Force is mechanical; movement is biological. An intact PDL is essential.
2	Cephalometry	Reproducible numbers, but not exact ones. It supplements clinical examination.
3	Treatment planning	A fixed sequence, not a list. Each stage constrains the next.
4	Choice of extraction	Extraction is one of four ways to make space, and the most predictable.
5	Anchorage	Newton's third law. Anchorage cannot be avoided, only planned.
6	Three orders · appliances	The three orders have not changed since Angle. Only their delivery has.
7	Archwires	No wire has every property, and the properties you need change through treatment.
8	Functional appliances	They posture the mandible forwards. Most of the correction is dentoalveolar.
9	Benefits and risks	Elective treatment. The benefits are narrower than commonly claimed.
10	Medical conditions	A careful medical history changes force, appliance, extraction and timing.
11	Overbite reduction	Reducing an overbite is easy. Holding it is the difficulty.
12	Overjet in a growing child	A timing question disguised as a technique question.
13	Missing central incisor	No adjacent tooth substitutes for a central. Usually a prosthetic problem.
14	Impacted maxillary canine	~3% impacted, up to 85% palatal. Position decides everything.
15	Impacted lower second molar	Just over 2%, mostly mandibular. Escalate step by step.
16	Crossbite correction	Classify first: dental, functional or skeletal. That decides the appliance.
17	Class III correction	Monitoring is often the right answer. Growth declares the discrepancy.
18	Malocclusion and TMD	An evidence-appraisal question, not a joint question.
19	Facial asymmetry	Minor asymmetry is normal. The task is to spot the progressive case.
20	Late lower incisor crowding	It happens in untreated people too. It is ageing, not relapse.

The sixteen chapters — one line each

	Chapter	The one line
1	Evidence-Based Orthodontics	Hierarchy, appraisal and the difference between statistical and clinical significance.
2	Occlusion and Malocclusion	Classification, aetiology, prevalence, and what treatment does and does not achieve.
3	Prenatal Development	Five processes, two palates, two separate failures that produce two different clefts.
4	Postnatal Growth	Sites, mechanisms and control; rotations; the maturity indicators that matter.
5	Development of the Dentition	Incisor liability against leeway space; the eruption sequence explains the crowding.
6	Orthodontic Tooth Movement	Three phases, the RANKL/OPG axis, optimal force, and root resorption.
7	Examination and Diagnosis	A reproducible sequence, in three planes, with the records justified.
8	Treatment Planning	Aims reconciled against the skeletal pattern; the mandibular arch is the template.
9	Functional Jaw Orthopaedics	Posture, unlock, dentoalveolar change; expansion and protraction.
10	Contemporary Fixed Appliances	Prescription, materials, mechanics, and the claims that failed testing.
11	Skeletal Anchorage	It removed compliance from the anchorage equation. Site decides survival.
12	Managing the Developing Dentition	Interception is timing. Know the ages and the escalating ladders.
13	Managing the Permanent Dentition	Stage I, II, III; finishing; and retention planned from the outset.
14	Clear Aligners	Better for comfort and periodontal indices, worse for occlusal outcome.
15	Orthognathic Surgery	Decompensation, the hierarchy of stability, and the psychological screen.
16	Cleft Lip and Palate	A timeline from birth to the late teens, delivered by a team.

The numbers

Handbook Part 2, Cards 1–4 **Questions** Every block **Viva** Every section

Four cards. Learn them the night before. A number offered unprompted, correctly, is the cheapest mark in the examination.

NUMBERS 1 — BIOLOGY, FORCE AND ANCHORAGE

Tooth movement	
Phase 1, initial — PDL displacement	~2 days
Phase 2, lag — hyalinisation cleared	20–30 days
Phase 3, post-lag — frontal resorption	sustained
Optimal force, bodily movement	50–100 cN
Canine retraction fastest at	272 cN
Normal rate of movement	~1 mm/month
PDL width	~0.2 mm
RANKL peaks at	3 days
Root resorption	
OIIRR, radiographic	up to 73%
OIIRR, histological	up to 90%
OIIRR, severe (≥1/3 root)	1–5%
Pause force application for	3–6 months
Anchorage and headgear	
Space lost without anchorage support	~two-thirds
Headgear, anchorage support	10–12 h/day · 250–350 g/side
Headgear, distalisation	14 h/day · 450–500 g/side
Miniscrew insertion torque	5–10 Ncm
Miniscrew failure, median palate	1.3%
Miniscrew failure, mandibular buccal shelf	65.0%

Compare Handbook **§4** and **§14**; Question Bank Blocks D and N.

NUMBERS 2 — APPLIANCES, WIRES AND EXPANSION

Wires	
Stainless steel, modulus relative to NiTi	∼5× stiffer
Beta titanium stiffness	≈1/2 steel
Superelastic NiTi plateau	constant force over range
Expansion	
RME activation	1–4 turns/day over 2–4 weeks
RME force	up to 10 kg
RME retention	3–6 months
Proportion of RME expansion that is skeletal	∼50%
Skeletal relapse at 4–8 months	∼10%
Slow expansion, removable plate	0.25 mm × 2/week
Quadhelix versus removable	∼3 months faster
SARPE indicated beyond	∼5 mm in the non-growing
Overbite and incisors	
Incisor proclination per mm of overjet	0.1–0.2 mm per degree
Utility/bypass arch intrusive force	∼60 g total, 15 g per tooth
Soft tissue response to incisor retraction	∼0.7 mm per mm

Compare Handbook **§15**, **§16**, **§24**, **§25**; Blocks O, P, X and Y.

NUMBERS 3 — CLINICAL THRESHOLDS AND EPIDEMIOLOGY

The developing dentition	
Incisor liability, maxilla	≈7 mm
Incisor liability, mandible	≈5 mm
Leeway space, maxilla	≈1.5 mm/quadrant
Leeway space, mandible	≈2.5 mm/quadrant
Impaction and anomalies	
Maxillary canine impaction	∼3%
Of those, palatal	up to 85%
Interceptive extraction of the primary canine	10–11 y (RCS 10–13)
Eruptive path improves if	>20 mm movement, review at 12 mo
Impacted mandibular second molar	just over 2%
Growth	
Peak height velocity, CS3	peak within 1 y
CS1 / CS2 / CS4	peak in 2 y / 1 y / 1–2 y ago
Juvenile growth spurt	6–8 y
Clefts	
Cleft lip ± palate	1:700, male 2:1, 70% non-syndromic
Isolated cleft palate	1:2000, female 4:1, 50% non-syndromic
Lip repair / palate repair	3–6 mo / 6–12 mo
Alveolar bone graft	8–10 y, before canine eruption

Compare Handbook **§3**, **§20**, **§1**, **§28**; Blocks C, T, A and AB.

NUMBERS 4 — RISK, EVIDENCE AND OUTCOME

Risks	
White spot lesions	**up to half of patients**
Appear within	**6 months**
Commonest sites	**6s · upper 2s · lower 3s**
Professional fluoride	**every 6–8 weeks**
Pain peaks at	**48 hours, gone by 7 days**
Nickel allergy	**10–30% F · 1–3% M**
Latex allergy	**4.3% general · 9.7% healthcare**
Body dysmorphic disorder in orthodontic patients	**5–13%**
Evidence and outcome	
Early overjet reduction, trauma	**RR 0.58 — 42% reduction**
Facemask avoiding surgery	**60% versus 36%**
IOTN thresholds	**25 low · 50 moderate · 75 high**
Power / significance	**80–90% / 0.05**
PAR: greatly improved	**≥22 points**
Aligners needing no refinement	**6%**
Aligners converting to fixed	**>17%**
Acceptable lower incisor relapse	**<3.5 mm**
Retainers still worn at 4–5 y	**a minority**
Prevalence	
TMD: at least one sign	**~75%**
TMD: at least one symptom	**~33%**
Obstructive sleep apnoea	**~2%**
Dental trauma at age 12	**up to 18%**
Traumatic loss of a central	**~3% of children**
Infective endocarditis mortality	**~21%**

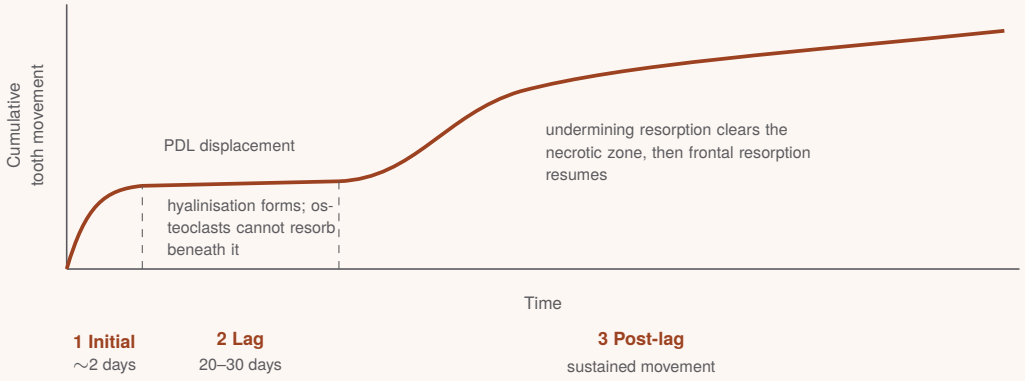
Compare Handbook **§29**, **§7**, **§6**, **§26**; Blocks AC, G, F and Z.

Biology and mechanics

Handbook §4, §14, §15 **Questions** Blocks D, N, O **Viva** V21–V30, V91–V111

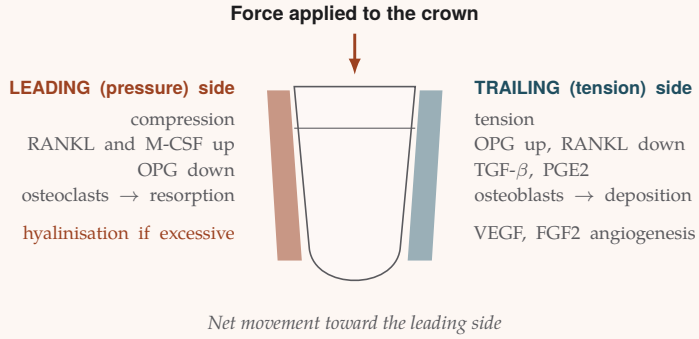
Topics 1, 5, 6, 7 · Chapters 6, 10, 11.

THE THREE PHASES OF TOOTH MOVEMENT



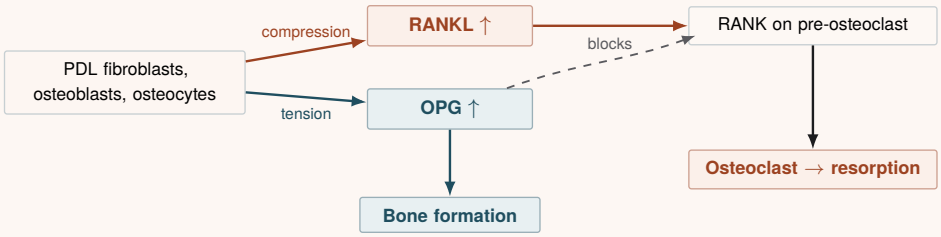
Three phases. The lag is the rate-limiting step, and it is why heavier force does not move teeth faster. Compare Fig. 6.3, p. 295.

PRESSURE AND TENSION



Pressure–tension. Compare Fig. 6.2, p. 294 — the histology is worth seeing.

THE RANKL / OPG AXIS



RANKL is essential for osteoclastic resorption; OPG is a soluble decoy receptor. RANKL peaks at 3 days and returns to baseline from day 6–58 (Danz et al., 2021).

Compare Fig. 6.6, p. 302.

THE MECHANICS IN ONE PAGE

Centre of resistance

1/3–1/2 down the root (single-rooted)

1–2 mm apical to the furcation (multirooted)

Moment

force × perpendicular distance to CRes

greater distance = greater rotation

Couple

two equal, opposite forces

NO net force; acts about CRes wherever applied

Centre of rotation

force nearer CRes → CRot further away

force THROUGH CRes → CRot at infinity = bodily

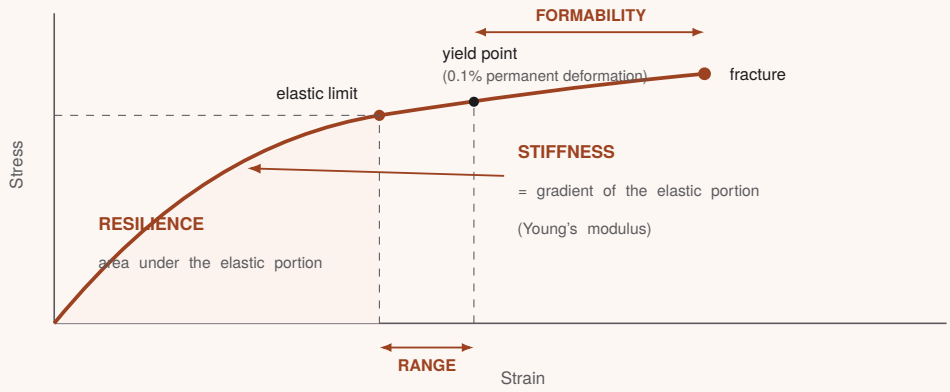
The fundamental problem

CRes is not accessible — force must be applied at the crown, so a moment is **always** produced. **Varying the moment-to-force ratio is how you control the type of movement.**

Periodontal bone loss moves CRes apically = same force, bigger moment, more tipping. Ch 6, pp. 307–312.

Compare Figs 6.11–6.15, pp. 310–314.

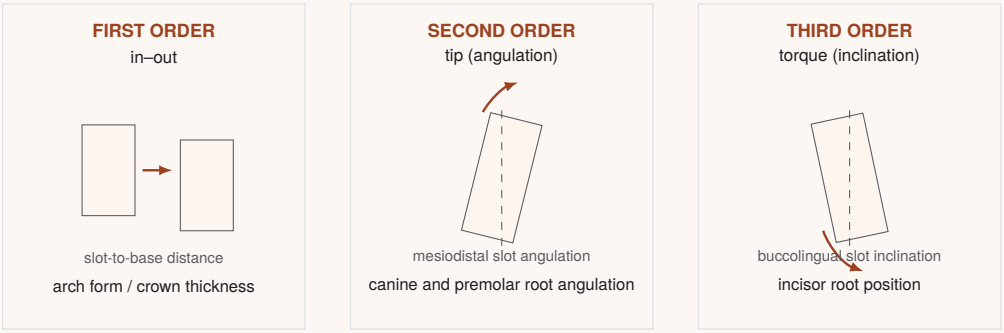
ARCHWIRE PROPERTIES — THE STRESS–STRAIN CURVE



Springback = yield strength ÷ modulus of elasticity — effectively the working range. Not on the curve: biocompatibility · environmental stability · joinability · friction · ductility.

Compare Box 10.2, p. 652; Fig. 10.28, p. 654.

THE THREE ORDERS



Standard edgewise: all three built into the **wire**, tooth by tooth. Pre-adjusted edgewise: all three built into the **bracket** (Andrews, 1972, 1979).

Compare Fig. 10.9, p. 623.

Anchorage

Handbook §14 **Questions** Block N **Viva** V91–V102

Topic 5 · Chapters 6, 11.

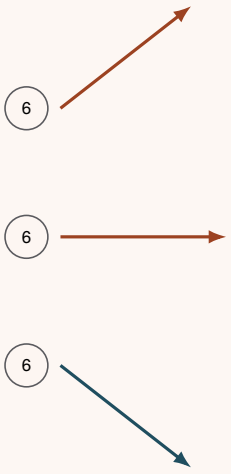
ANCHORAGE DEMAND



Reciprocal anchorage = equal movement of both units is acceptable (expansion, midline diastema). **Unless anchorage is supported, around two-thirds of the space created is lost** (Haque et al., 2017).

Compare Fig. 6.35, p. 339.

HEADGEAR — DIRECTION OF PULL



HIGH / occipital

increased vertical proportions — intrusive

STRAIGHT / combi

normal proportions — molar distalisation

LOW / cervical

reduced vertical proportions — extrusive

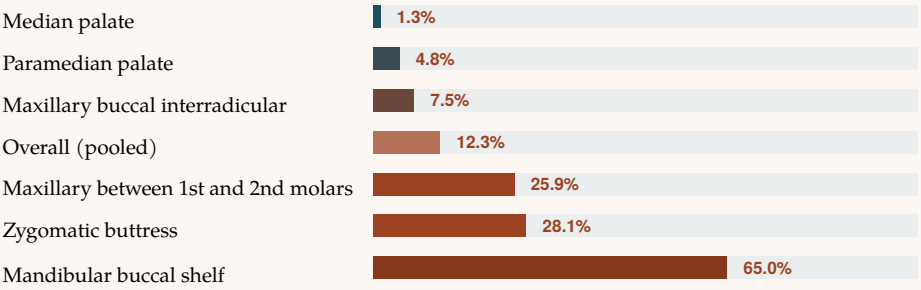
Force and duration

Anchorage support	10–12 h/day	250–350 g per side
Distalisation / orthopaedic	14 h/day	450–500 g per side

Two independent safety devices are mandatory: a self-releasing strap and a locking facebow.

Compare Figs 6.42–6.43, pp. 346–347; injury Fig. 6.45, p. 349.

MINISCREW FAILURE RATE BY SITE



Placement in the palate is not restricted by the dentition — which is why it is by far the most reliable site. **Insertion torque 5–10 Ncm**: too low gives poor primary stability, too high causes microfracture and poor secondary stability. Failure is higher below 8 mm length and 1.2 mm diameter, and root contact is a major cause.

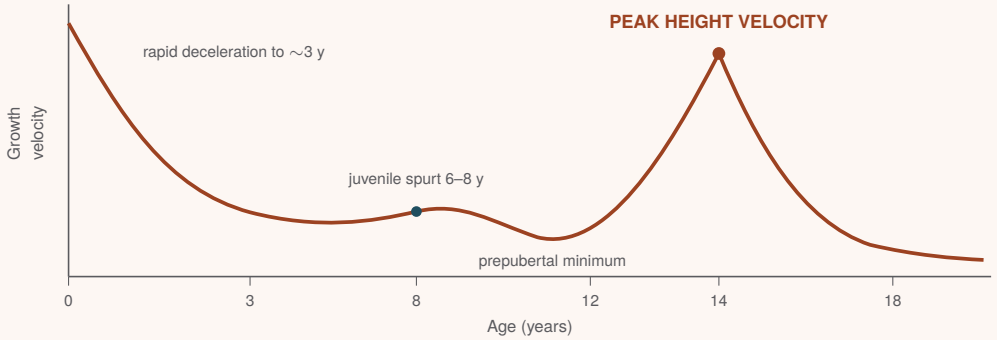
Compare Fig. 11.5, p. 716.

Growth and development

Handbook §1–§3 **Questions** Blocks A, B, C **Viva** V1–V20

Chapters 3, 4, 5.

GROWTH VELOCITY AND CERVICAL VERTEBRAL MATURATION



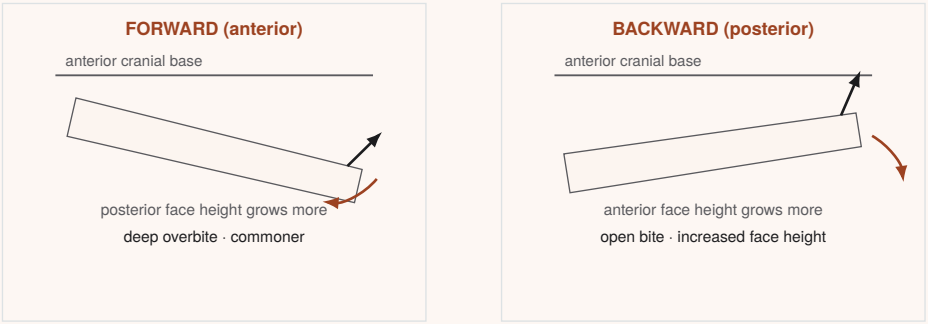
CVM staging (Baccetti, Franchi & McNamara)

CS1 9.7 y → peak in 2 y CS2 10.5 y → peak in 1 y CS3 12.0 y → peak within 1 y (start now) CS4 13.4 y → peak was 1–2 y ago

Caveat: none of the underlying principles is universally accepted.

Compare Figs 4.3, p. 171 and 4.47, p. 240.

MANDIBULAR ROTATION — FORWARD AND BACKWARD



The three rotations (Björk & Skieller; simplified by Solow & Houston)

Total (true) = mandibular body versus cranial base, measured on implants.

Matrix (apparent) = soft tissue matrix versus cranial base — what a cephalogram shows without implants.

Intramatrix (angular remodelling) = the difference; remodelling of the lower border. Total = matrix + intramatrix.

Compare Fig. 4.37, p. 223; extremes Fig. 4.40, p. 229.

THE ARITHMETIC OF THE DEVELOPING ARCH

Incisor liability

Permanent incisors are WIDER than their predecessors.

Maxilla	≈ 7 mm
Mandible	≈ 5 mm

a DEFICIT to be met

Leeway space

Primary canine + molars are WIDER than permanent canine + premolars.

Maxilla	≈ 1.5 mm/quadrant
Mandible	≈ 2.5 mm/quadrant

an EXCESS to be used

Note the asymmetry: the larger liability is MAXILLARY; the larger leeway is MANDIBULAR.

Eruption sequence and its consequence

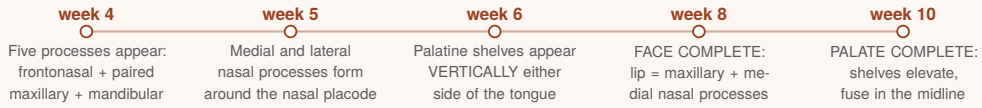
Mandible: canine → 1st premolar → 2nd premolar → **2nd PREMOLAR crowded**

Maxilla: 1st premolar → 2nd premolar → canine → **CANINE crowded**

The last tooth to erupt in each arch is the one that ends up crowded.

Compare Figs 5.23–5.26, pp. 268–280.

DEVELOPMENT OF THE FACE AND PALATE — AND WHERE CLEFTS ARISE



Cleft lip ± primary palate

Maxillary process fails to fuse with the MEDIAL NASAL process, by week 8

1:700 · male 2:1 · 70% non-syndromic · 80% unilateral

Isolated cleft palate

Palatine shelves fail to elevate or fuse, weeks 6–10

1:2000 · female 4:1 · 50% non-syndromic

The sex ratios are opposite because these are *different* developmental failures occurring at *different* times. Ch 3, pp. 121–133; Ch 16, pp. 1113–1118.

Compare Figs 3.8–3.10, pp. 123–133.

Planning, extraction and records

Handbook §9–§13 **Questions** Blocks I, J, K, L, M **Viva** V59–V90

Topics 2, 3, 4 · Chapters 2, 7, 8.

THE TREATMENT PLANNING SEQUENCE — EACH STAGE CONSTRAINS THE NEXT

1. **DIAGNOSIS**
   examination · records · the patient's own concerns
2. **PROBLEM LIST**
   crowding · overjet · overbite · crossbite · skeletal pattern in 3 planes
3. **AIMS**
   FACIAL aims and OCCLUSAL aims — reconciled against the skeletal pattern
4. **PATIENT FACTORS**
   age and growth · dental and periodontal health · medical history · compliance
5. **SPACE ANALYSIS**
   6 areas: crowding · levelling · arch width · incisor A/P · tip · torque
6. **ANCHORAGE**
   follows directly from the space analysis and the extraction pattern
7. **APPLIANCE**
   aims · patient preference · operator expertise
8. **PHASING**
   align & level → overjet/overbite → space closure → detailing
9. **RETENTION**
   planned and AGREED before treatment begins

Planning begins with the position of the dentition — particularly the MAXILLARY INCISORS — within the FACE.

The soft tissues define the limits of orthodontic treatment in each individual (Ch 8, p. 490).

Compare the envelope of discrepancy, Fig. 8.3, p. 494.

EXTRACTION PATTERN BY CLASS OF MALOCCLUSION



Rust = extraction site. **Rationale:** put the space where the movement is needed, and let the differential in space requirement between the arches reflect the molar relationship you intend to finish to.

Compare Figs 8.10–8.12, pp. 517–518.

WHY CEPHALOMETRIC NUMBERS ARE NOT EXACT

The four errors

1 Projection — magnification and the two sides not superimposing. **2 Landmark identification** — the largest error; each landmark has its own envelope, and several (Ba, Or, Go, Gn) are inherently unreliable. **3 Measurement** — reference planes are themselves variable; Frankfort is hard to locate, sella–nasion easier but not anatomically meaningful. **4 Interpretation** — norms are population means with wide standard deviations, derived largely from white populations, and a single value cannot separate which jaw is at fault.

What follows from this

Cephalometry **supplements** clinical examination; it does not replace it. Report trends over serial films rather than absolute values, use the Eastman correction where the cranial base is atypical, and never plan from a single number. Growth prediction from cephalometry has been discredited.

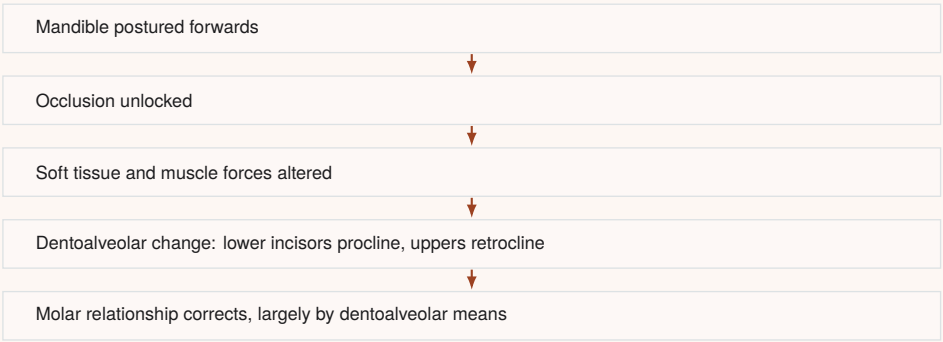
Compare Fig. 7.60, p. 465 — the envelope of error in landmark identification.

Clinical problems

Handbook §16, §20–§25 **Questions** Blocks P, T, U–Y **Viva** V112–V173

Topics 8, 11–17, 19, 20 · Chapters 9, 12, 13.

WHAT A FUNCTIONAL APPLIANCE ACTUALLY DOES



The sentence to write

Any extra mandibular growth is "a matter of only a few millimetres at best, and these effects are essentially transient" (Ch 9, p. 560). The likely mechanism is that the appliance *unlocks the occlusion* and permits normal mandibular growth to express itself, combined with dentoalveolar change (Ch 9, p. 562).

Compare Ch 9, pp. 560–562.

EARLY CLASS II TREATMENT: WHAT THE TRIALS SHOWED

Early treatment DID

· reduce the overjet (dentoalveolar) · give a transient rise in self-esteem · improve profile attractiveness · **REDUCE INCISOR TRAUMA — RR 0.58, a 42% reduction**

Early treatment did NOT

· improve the final skeletal outcome · improve the final occlusal outcome · shorten overall treatment · reduce total appointments · **the LATE group finished slightly better**

The pragmatic policy

Treat early ONLY for children at high risk of trauma — very large overjet, vertical excess, incompetent lips — or significant teasing. Otherwise a single phase in the late mixed or early permanent dentition.

Caveat: much trauma occurs before age 9, and headgear was marginally better. Trials: Florida · North Carolina (USA) · Manchester twin block (UK). O'Brien et al., 2003, 2009; Tulloch et al., 1997, 2004.

Compare Fig. 12.71, p. 847.

OVERBITE REDUCTION — FOUR ROUTES, ONE RULE

A **Growing patient**
Buccal segment EXTRUSION
→ relative incisor intrusion
condylar growth compensates

B **Non-growing patient**
Buccal segment EXTRUSION
→ mandible rotates down/back
worsens Class II; unstable

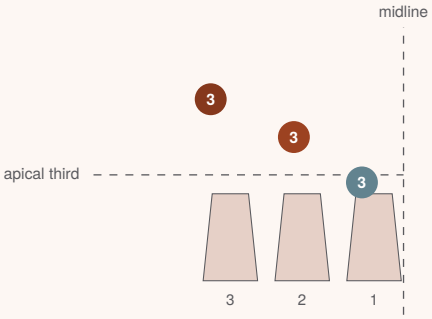
C **High MMPA / adult**
True incisor INTRUSION
→ utility or bypass arch
~60 g total, 15 g per tooth

D **Retroclined incisors**
Incisor PROCLINATION
→ interincisal angle reduces
0.1–0.2 mm per degree; unstable

STABILITY: whichever route, an occlusal stop must be established or the overbite relapses (Houston, 1989).

Compare Fig. 13.40, p. 934.

PROGNOSIS FOR ORTHODONTIC ALIGNMENT OF AN IMPACTED CANINE WORSENS AS...



1 HEIGHT
worse above the apical third of the incisor roots

2 HORIZONTAL POSITION
worse at or across the lateral incisor midline

3 ANGULATION
worse above ~55° to the midsagittal plane

INTERCEPTION CRITERIA
10–11 y (RCS 10–13) · distal to the lateral root midline · 20–30° to midsagittal · no crowding · review at 12 months

~3% impacted; up to 85% palatal; eruptive path improves if the canine moves >20 mm.

Compare Fig. 13.12, p. 878; CBCT Fig. 7.27, p. 407.

CHOOSING AN EXPANSION APPROACH

Unilateral crossbite WITH displacement?

YES ↓ | NO — bilateral, no shift ↓

Dental or mild skeletal

Removable plate: 0.25 mm × 2/week · Quadhelix: ∼3 months faster (Cochrane) · overcorrect · retain nights 3–6 months

High-certainty evidence, ages 7–11

True transverse skeletal deficiency

RME: 1–4 turns/day over 2–4 weeks · forces up to 10 kg · retain 3–6 months · ∼50% of screw expansion is skeletal · ∼10% skeletal relapse at 4–8 months

Growth complete?

SARPE beyond ∼5 mm, or bone-borne expansion. Sutural fusion by the late teens, but correlation between sutural maturation and chronological age is poor.

Slow expansion CAN split the suture in the early mixed dentition — the constraint is sutural interdigitation with age, not the appliance category (Ch 9, p. 543).

Compare Fig. 9.2, p. 544.

CLASS III: THE DECISION PATHWAY

MIXED DENTITION — is the case FAVOURABLE?

Favourable (Table 8.4)

· Class I or mild Class III base · average or reduced lower face height · average or INCREASED overbite · proclined lower / upright upper incisors · anterior displacement on closing

→ **Early intervention** facemask ± RME, 12–14 h/day, before age 10. >60% avoid surgery versus 36% (Mandall, 2016).

Unfavourable

· moderate to severe Class III base · increased lower face height · REDUCED overbite · retroclined lower / proclined upper incisors · no displacement · Class III molars

→ **Monitor into the late teens** Class III worsens with growth. Avoid mandibular extractions if surgery is likely.

Permanent dentition — camouflage or surgery?

Surgery is more likely if: Wits ≤ -11 mm · ANB $\leq -4°$ · overjet ≤ -5 mm · Holdaway $\leq 3.5°$ · gonial $\leq 120°$ · LAFH $\geq 66\%$ · lower incisor inclination $\leq 80°$

Compare Table 8.4, p. 505; decompensation Fig. 13.37, p. 930.

MISSING MAXILLARY CENTRAL INCISOR

Avulsion — ALWAYS attempt re-implantation (maintains space AND alveolar bone)

Maintain the space

Denture or retainer with a pontic

— risks alveolar bone loss

Let it close, re-open later

Preserves alveolar bone

— commits to further orthodontics

Definitive options

UNILATERAL → resin-retained bridge or implant (a lateral rarely substitutes convincingly). **BILATERAL** → move the laterals mesially if size and form allow; premolar autotransplant if premolars are coming out anyway.

Implants: defer until growth is complete

Continued vertical alveolar growth leaves the crown in **infra-occlusion** — the implant behaves like an ankylosed tooth while the adjacent dentition continues to erupt.

Compare Figs 12.58–12.59, pp. 821–822.

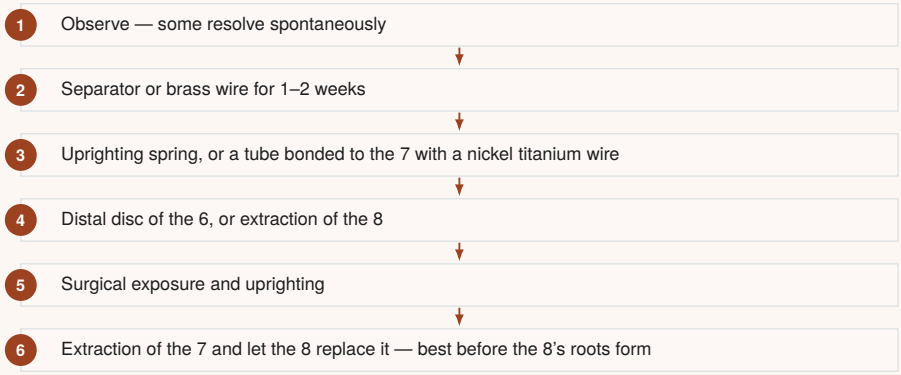
DENTAL TRAUMA: REST PERIOD BEFORE ORTHODONTIC FORCE

Injury	Rest period	Note
Concussion, subluxation	3 months	monitor vitality
Extrusive / lateral luxation	3 months	
Intrusive luxation	3 months	highest resorption risk
Avulsion and re-implantation	3–6 months	expect replacement resorption
Root fracture, healed	3 months	light forces only
Root-filled tooth	no extra delay	no greater resorption risk than a vital tooth

Move root-treated teeth with the same light forces as vital teeth; the evidence does not support the belief that they resorb more.

Compare Ch 12, dental trauma.

IMPACTED LOWER SECOND MOLAR — THE ESCALATING LADDER



Just over 2% prevalence, mostly mandibular. Climb the ladder — do not start at the top.

Compare Ch 12, impacted second molars.

FACIAL ASYMMETRY IN A GROWING CHILD

The clinical test

Bite on a spatula. Is the occlusal plane canted? **Cant present** → likely skeletal, and possibly progressive. **No cant** → likely dental or functional.

Is it progressive?

Serial photographs and measurements at 6-monthly intervals; bone scintigraphy where condylar hyperplasia is suspected. Mild asymmetry is present in almost everyone; the task is to identify the minority that is growing worse.

Obwegeser and Makek: hemimandibular elongation (horizontal, chin deviates) versus hemimandibular hyperplasia (vertical, ramus and body enlarge, occlusal cant).

Do not operate on a growing asymmetry

Defer definitive surgery until growth has ceased and the asymmetry is static on serial records — otherwise it recurs.

Compare Ch 15, asymmetry.

LATE LOWER INCISOR CROWDING

What the evidence says

It occurs in treated *and untreated* dentitions alike. It cannot be prevented by any extraction pattern or archwire sequence. Third molars are not the cause (Ch 5, p. 284). It is a manifestation of continued late facial growth and the forward component of occlusal force — that is, **it is ageing, not relapse**.

What follows

The only reliable answer is **indefinite bonded retention**, and that must be said at the consent conversation before treatment begins — not discovered by the patient at debond.

Compare Ch 5, pp. 283–284; Ch 13, p. 979.

Risk, evidence and the modern debates

Handbook §7, §26, §28, §29 **Questions** Blocks G, Z, AB, AC **Viva** V39–V58, V174–V193

Topics 9, 10, 18 · Chapters 1, 2, 14, 16.

BENEFITS AND RISKS — WITH THE RIGHT LEVEL OF CONFIDENCE

Claimed benefit	Confidence	What can honestly be said
Dental aesthetics	STRONG	consistently improved, and it is the commonest reason patients seek treatment
Trauma reduction (large overjet)	STRONG	RR 0.58 — a 42% reduction
Plaque scores	MODERATE	small and inconsistent effect
Caries reduction	WEAK	not demonstrated; appliances increase risk during treatment
Speech	WEAK	no reliable effect
TMD prevention or cure	NONE	no evidence of causation or prevention
Airway or sleep apnoea	WEAK	extraction does not damage the airway (Ch 2, p. 91)

Orthodontics is **elective**. Stating the benefits at the confidence the evidence supports — rather than overclaiming — is itself a markable act of professionalism.

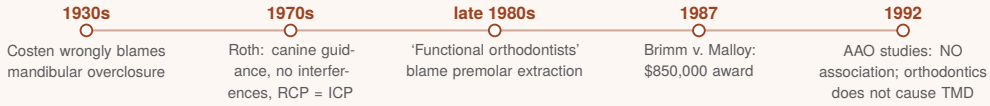
Compare Ch 2, pp. 85–105.

MEDICAL HISTORY: WHAT IT CHANGES

Condition	What it changes
Bisphosphonates	tooth movement slowed or arrested; MRONJ risk on extraction
Diabetes, poorly controlled	periodontal breakdown; defer until controlled
Epilepsy	avoid removable appliances that could be inhaled; gingival hyperplasia with phenytoin
Bleeding disorders, anticoagulants	extraction planning and haemostasis; liaise with the physician
Immunosuppression	infection risk; slower healing
Cardiac lesions	antibiotic prophylaxis is NOT routinely indicated (NICE); maintain oral health
Nickel allergy	titanium or gold-plated brackets; nickel-free wires
Latex allergy	latex-free elastics and gloves
Juvenile idiopathic arthritis	condylar resorption; monitor for progressive retrognathia and open bite
Head and neck radiotherapy	avoid orthodontic movement in irradiated bone
Cleft and craniofacial syndromes	team care; timing dictated by the reconstruction

Compare Ch 7, medical history; Handbook **§29.3**.

MALOCCLUSION AND TMD — A FRACTIOUS HISTORY



The position to state

Weak statistical associations exist. There is no evidence of **causation**. Orthodontics neither causes nor prevents TMD. Symptoms often begin in late adolescence — exactly when we treat.

In practice (Kandasamy et al., 2022)

Screen and document before treatment · counsel on the natural history · pause mechanics if symptoms appear · manage conservatively with reassurance, soft diet, analgesia and a splint if needed · do not offer orthodontics as a treatment for TMD.

Prevalence: at least one *sign* ~75%; at least one *symptom* ~33%.

Compare Where is the evidence 2.3, pp. 104–105.

CLEAR ALIGNERS VERSUS FIXED APPLIANCES

Aligners better

✓ Less pain in the first few days ✓ Better periodontal index scores ✓ Less eating and chewing disturbance ✓ Far less visible ✓ Shorter chairside appointments

Fixed appliances better

✓ Occlusal outcome (ABO-OGS) ✓ Buccolingual inclination ✓ Occlusal contacts and relationship ✓ Overjet and root angulation ✓ Severe crowding and complex cases ✓ Less relapse

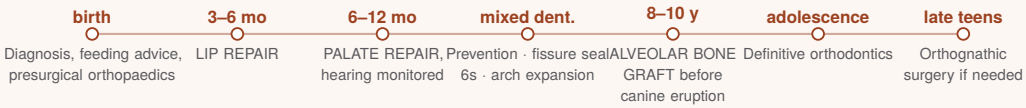
PAR shows NO difference; ABO-OGS shows aligners significantly worse. PAR is the coarser measure — the two do not correlate.

The two numbers to quote

Only **6%** complete treatment without a refinement; over **17%** convert to fixed appliances to finish. Ch 14, pp. 1029–1031.

Compare Fig. 14.21, p. 1031.

THE CLEFT CARE TIMELINE



Epidemiology — learn as a contrast

Cleft lip ± palate	1:700	male 2:1	70% non-syndromic	80% unilateral
Isolated cleft palate	1:2000	female 4:1	50% non-syndromic	

Opposite sex ratios and different syndromic proportions, because these are different developmental failures at different times (lip by 8 weeks; palate weeks 6–10).

The rule of tens for lip repair: 10 weeks, 10 lb, haemoglobin 10 g/dL.

Compare Fig. 16.9, p. 1130.

The twelve traps and ten overturned beliefs

Handbook Part 2, Card 6 **Questions** Every block **Viva** V39–V58

Read these on the morning of the examination.

The twelve traps that cost most marks

	The trap	What to say instead
1	Saying functional appliances "grow the mandible"	They posture it forwards; most of the correction is dentoalveolar, and any skeletal effect is small and largely transient
2	Quoting one optimal force	There is no universal value; 50–100 cN for bodily movement, and lighter is safer
3	Blaming third molars for late lower incisor crowding	It occurs in untreated dentitions too; it is ageing, not relapse
4	Offering orthodontics as a treatment for TMD	Association is weak; causation is unsupported
5	Claiming self-ligation is faster	It is not; nor does it reduce discomfort or expand arches more
6	Planning from a single cephalometric number	Four sources of error; it supplements clinical examination
7	Forgetting to justify the radiographs	Every film needs a clinical justification recorded
8	Omitting retention from the plan	Retention is agreed before active treatment begins
9	Treating every Class II early	Only for trauma risk or significant teasing; the late group finishes slightly better
10	Expanding the lower arch	The mandibular arch is the template; intercanine width is not amenable to change
11	Placing an implant in a growing patient	Continued alveolar growth leaves it in infraocclusion
12	Overstating the benefits of treatment	Aesthetics and trauma reduction are strong; caries, speech and TMD are not

Ten things the evidence overturned

	The belief	The finding, and where
1	Early Class II treatment improves the final result	It does not, except for trauma reduction. O'Brien 2003, 2009; Tulloch 1997, 2004. Ch 9, pp. 588–589
2	Third molars cause late lower incisor crowding	Ch 5, p. 284
3	Orthodontics causes or cures TMD	Ch 2, pp. 104–105
4	Self-ligating brackets are faster	Ch 10, pp. 646–647
5	Customised appliances shorten treatment	Ch 10, p. 631
6	Extractions damage the airway	Ch 2, p. 91
7	Extractions flatten the face	Ch 2, p. 101

	The belief	The finding, and where
8	Root-filled teeth resorb more	They do not; move them with the same light forces
9	Growth can be predicted from a cephalogram	Discredited
10	Aligners match fixed appliances for finish	6% need no refinement; over 17% convert to fixed. Ch 14, pp. 1029–1031

THE LAST HOUR — SAY THESE SIX THINGS AND YOU WILL NOT BE FAR WRONG

1. **Growth status first.** Whether the patient is growing determines what is possible.
2. **The mandibular arch is the template.** Ch 8, p. 525 — quote it.
3. **Anchorage is Newton's third law.** It cannot be avoided, only planned; two-thirds of the space is lost if it is not supported.
4. **Most "skeletal" correction is dentoalveolar.** A few millimetres at best, and largely transient.
5. **Retention is planned at the start.** And for late lower incisor crowding it is indefinite.
6. **Say what the evidence supports, and no more.** Naming the study that overturned the tradition is the cheapest way to show scholarship.

Part II — The twenty essay topics

The twenty topics are not twenty separate subjects. Seven themes run through nearly all of them, and examiners reward candidates who recognise this. If you find yourself short of material, ask which theme applies.

Cross-cutting themes

Seven ideas that recur across the twenty topics. If you are short of material, ask which applies.

Theme 1 — Growth status determines everything

Topics 3, 4, 8, 11, 12, 13, 16, 17, 19. The single most powerful discriminator in orthodontic planning is whether the patient is growing. Growth permits growth modification; it makes overbite reduction by buccal segment extrusion stable, because condylar growth compensates for the increased posterior vertical dimension (Ch 13, p. 932); it allows facemask protraction and sutural expansion; and it permits spontaneous alignment after well-timed extraction (Ch 8, p. 513). Absence of growth converts the same manoeuvres into dentoalveolar compromise or a surgical problem. State the patient's growth status early and return to it in the conclusion.

Theme 2 — Most "skeletal" correction is actually dentoalveolar

Topics 8, 12, 17. Functional appliances, facemasks and chin cups produce far less true skeletal change than their advocates claimed. The textbook's position is that any extra growth is "a matter of only a few millimetres at best, and these effects are essentially transient" (Ch 9, p. 560). The mechanism of functional appliances is probably that they *unlock the occlusion* and permit normal mandibular growth to express itself, combined with dentoalveolar change (Ch 9, p. 562) — a considerably more sophisticated statement than "they grow the mandible", and one that will read well.

Theme 3 — Anchorage is a consequence of Newton's third law, not an optional extra

Topics 3, 4, 5, 6, 11, 12, 14, 17. Anchorage cannot be avoided; it can only be planned. Extraction choice determines the anchorage unit. Appliance philosophy — edgewise versus Begg versus Tip-Edge — is largely a debate about anchorage (Ch 10, p. 615). Skeletal anchorage changed the calculus by removing compliance. The sharpest single fact: unless anchorage is supported, **around two-thirds of the space created is lost** (Haque et al., 2017; Ch 8, p. 528).

Theme 4 — Evidence frequently contradicts tradition

Topics 8, 9, 12, 14, 18, 20. Several long-held beliefs are unsupported: that early Class II treatment improves the final result (it does not, except for trauma reduction — Ch 9, pp. 588–589); that third molars cause late lower incisor crowding (Ch 5, p. 284); that orthodontics causes or cures temporomandibular disorder (Ch 2, pp. 104–105); that self-ligation is faster (Ch 10, pp. 646–647); that customised appliances shorten treatment (Ch 10, p. 631); that extractions damage the airway (Ch 2, p. 91) or the face (Ch 2, p. 101). Naming the study that overturned the tradition is the most efficient way to demonstrate scholarship.

Theme 5 — Retention is planned at the start, not the end

Topics 3, 9, 11, 16, 20.

Relapse is near-universal. Late lower incisor crowding occurs in treated and untreated dentitions alike and cannot be prevented by any extraction pattern or archwire sequence (Ch 5, pp. 283–284). Overbite reduction relapses unless an occlusal stop is established (Houston, 1989; Ch 13, p. 930). Expansion relapses — roughly 10% of skeletal change within 4–8 months (Ch 9, p. 544). Retention "should be discussed prior to the commencement of active treatment" (Ch 13, p. 979).

Theme 6 — Consent, radiation and risk are examinable, not decorative

Topics 2, 3, 9, 10, 18. Justification of radiographs, the risks of demineralisation and root resorption, the medical history's effect on force levels and extraction feasibility, and honest discussion of an uncertain evidence base are all markable content. Candidates routinely omit them.

Theme 7 — The mandibular arch is the template

Topics 3, 4, 11, 16, 17, 20. "As a general principle, the mandibular arch dentition is less amenable to change when planning orthodontic treatment, particularly intercanine width and mandibular incisor inclination. For this reason, the mandibular arch generally forms the template for any proposed orthodontic tooth movement" (Ch 8, p. 525). This single sentence explains why lower arch expansion is avoided, why the lower canine position drives the maxillary plan, why Class III mandibular extractions are planned cautiously, and why late lower incisor crowding is so intractable. Quote it and you have a thread running through half the syllabus.

Topic 1. The biological basis of tooth movement

Revision Handbook: §4; force levels also §14
Question Bank: Block D (Q96–140)
Viva bank: V21–V30

IN ONE MINUTE

Force is mechanical; movement is biological. An intact PDL is essential.

- Pressure–tension: resorption on the leading side, deposition on the trailing side
- Hyalinisation → lag phase of 20–30 days; RANKL drives resorption, OPG opposes it
- No universal optimal force: 50–100 cN for bodily movement, and lighter is safer

Where to read it

Ch 6, *Biological Basis of Tooth Movement*, pp. 287–307. Optimal force: Where is the evidence 6.1, pp. 296– 297. Rate of movement: Where is the evidence 6.2, pp. 304–305. Root resorption: pp. 303–307.

Opening definition

- Orthodontic tooth movement is a biological phenomenon, not a mechanical one.
- The clinician applies force; the movement is produced by remodelling of the periodontal ligament and alveolar bone.
- An intact PDL is an absolute prerequisite — neither implants nor ankylosed teeth will move under orthodontic force (Ch 6, p. 291) — which localises the response to the ligament rather than to bone alone.
- The histological foundation was laid independently by Carl Sandstedt and Albin Oppenheim at the turn of the twentieth century, giving the classic pressure–tension theory (Ch 6, pp. 291–292).

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition; primacy of biology over mechanics; the PDL as prerequisite.
2. Physiology of bone — composition, the three cell types, the BMU.
3. The RANKL/RANK/OPG axis and the Wnt– sclerostin counterbalance.
4. The periodontal ligament as a specialised connective tissue.
5. Pressure–tension theory; leading and trailing sides.
6. Hyalinisation, undermining resorption, the three phases.
7. Alternative hypotheses — bone-bending, piezoelectricity, streaming potentials.
8. Mechanotransduction: the four-step model.
9. The inflammatory cascade and the molecular sequence.
10. Optimal force — the search, the evidence, why no universal value exists.
11. The biological cost: orthodontically induced inflammatory root resorption.
12. Conclusion — clinical implications for force selection.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

ITEM	DETAIL	PAGE
Bone composition	ECM ~30% (type I collagen, proteoglycans), mineral ~60% (hydroxyapatite), cells ~10%	Ch 6, p. 287
Osteocytes	85–90% of bone cells; osteoblasts enveloped during mineralisation; linked by canaliculi	Ch 6, pp. 287–288
Osteoclasts	Multinucleate, from haematopoietic stem cells, circulate as monocytes	Ch 6, p. 287
Remodelling unit	Basic Multicellular Unit (BMU) — apposition and resorption tightly coupled	Ch 6, p. 288
PDL width	~0.2 mm on average; **thinnest around the middle third of the root**	Ch 6, p. 290
PDL contents	Type I collagen bundles (Sharpey's fibres), transeptal fibres, elastic oxytalan fibres, fibroblasts, cementoblasts, immune cells, epithelial rests of Malassez, pluripotent stem cells, GAG-rich ground substance	Ch 6, pp. 290–291
Hyalinisation	Cell-free glass-like necrotic zone; occurs even at **30 g**; **more with tipping than bodily movement**	Ch 6, p. 293
Phase 1 — initial	PDL displacement, first **2 days**	Ch 6, p. 293
Phase 2 — lag	**20–30 days** while hyalinised tissue is cleared by undermining resorption	Ch 6, p. 293
Phase 3 — postlag	Sustained movement by direct frontal resorption	Ch 6, p. 293
Historical optimal force	150–200 g (Storey & Smith, 1952)	Ch 6, p. 296
Modelled maximum	Canine retraction fastest at **272 cN** (Ren et al., 2004); 1 cN ≈ 1 g	Ch 6, p. 297
Systematic review	Bodily movement at **50–250 cN** gives similar rates of **0.1–0.47 mm/week**; >250 cN marginally faster but with greater adverse effects; **50–100 cN considered optimal**	Ch 6, p. 297
Normal rate of movement	Approximately **1 mm per month**; faster in the maxilla and in younger patients	Ch 6, p. 304
OIIRR incidence	Radiographic up to **73%**; histological up to **90%**; **severe (≥1/3 root length) 1– 5%**	Ch 6, p. 303
OIIRR distribution	Maxillary incisors > canines > mandibular incisors > first permanent molars	Ch 6, p. 303
OIIRR management	Pause force application **3–6 months** where significant resorption is identified (Levander et al., 1994)	Ch 6, p. 306

The molecular sequence, in order

- Matrix strain and altered fluid flow → distortion of nerve endings releases substance P and CGRP
- reduced blood flow causes hypoxia, apoptosis and ECM denaturation
- necrotic tissue triggers sterile inflammation with ILs and prostaglandins
- **M-CSF** drives monocyte differentiation, **RANKL** (from osteoblasts, osteocytes and PDL cells) binds RANK on pre-osteoclasts and is *essential* for osteoclastic resorption
- **OPG**, a decoy receptor, opposes this and is increased on the tension side, reduced under compression
- osteoclasts resorb via cathepsins, MMPs and hydrogen ions, forming Howship's lacunae.
- **RANKL peaks at 3 days** and returns to normal from day 6 to 58, implying other pathways sustain the post-lag phase (Danz et al., 2021;

- Ch 6, p. 299). On the tension side, IL-1β, IL-10 and PGE2 raise OPG and lower RANKL;
- TGF-β drives fibroblast and osteoblast proliferation; VEGF and FGF2 support angiogenesis (Ch 6, p. 302). **Two refinements most candidates miss.** First, **osteocytes secrete sclerostin**, which inhibits osteoblast differentiation and canonical Wnt signalling — a safety mechanism against excess bone formation (Ch 6, p. 290). Second, **bone-bending and bioelectric theories** were proposed to explain how differential forces arise if the PDL behaves as a continuous hydrostatic system under Pascal's law; piezoelectric currents are too small and short-lived to matter, streaming potentials are of unclear significance, and "there is little evidence that orthodontic forces bend bone to any clinically meaningful degree" (Ch 6, pp. 294–295).

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCE	WHAT TO BE ABLE TO QUOTE
Sandstedt; Oppenheim	pressure–tension theory (Ch 6, pp. 291–292).
Reitan (1967)	hyalinisation at 30 g, greater with tipping (Ch 6, p. 293).
Storey & Smith (1952)	150–200 g (Ch 6, p. 296).
Quinn & Yoshikawa (1985)	dose–response plateau (Ch 6, p. 297).
Ren et al. (2003, 2004)	no universal optimum; 272 cN model (Ch 6, p. 297).
Theodorou et al. (2019)	50–100 cN optimal for bodily movement (Ch 6, p. 297).
Kanzaki et al. (2004)	OPG gene transfer reduces tooth movement (Ch 6, p. 290).
Yang et al. (2018)	RANKL deletion in transgenic mice blocks movement (Ch 6, p. 301).
Henneman et al. (2008)	four-step mechanotransduction model (Ch 6, p. 296).
El-Angbawi et al. (2023)	photobiomodulation does not shorten overall treatment (Ch 6, p. 305).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not present this as mechanics.

Mention the ankylosis and implant observation early; it is the cleanest available proof that the ligament, not the bone, is the site of response.

2Hyalinisation is not a complication of heavy force alone.

It occurs at 30 g and throughout treatment, not just initially, and **more with tipping than with bodily movement**.

3Do not state an optimal force as settled.

The examinable position is that no universal optimum exists because of large inter-individual variation; what matters more than absolute force is the **differential stress** generated, which is why different movements have different recommended forces — and why **intrusive forces must be very light**, being dissipated through the small surface area of the apices (Ch 6, p. 297).

4RANKL up on compression, OPG up on tension.

Reversing this is a common and costly slip.

5Do not claim acceleration techniques work.

Corticotomy, micro-osteoperforation and piezocision can increase the rate locally through the regional acceleratory phenomenon, but the effects appear short-lived and evidence that they reduce *overall* treatment time is sparse (Ch 6, p. 305).

6Finish in the clinic.

Light forces minimise the lag phase, anchorage loss and root resorption.

undermining resorption clears

Cumulative PDL displacement

the necrotic zone, then frontal

tooth

resorption resumes

movement

hyalinisation forms; osteoclasts

cannot resorb beneath it

Time

1 Initial 2 Lag 3 Post-lag

~2 days 20–30 days sustained movement

The three phases of orthodontic tooth movement. Compare Fig. 6.3, Ch 6, p. 295.

Force applied to the crown

LEADING (pressure) side TRAILING (tension) side

• compression • tension

• RANKL and M-CSF up • OPG up, RANKL down

• OPG down • TGF-β, PGE2

• osteoclasts → resorption • osteoblasts → deposition

• hyalinisation if excessive • VEGF, FGF2 angiogenesis

Net movement toward the leading side

The pressure–tension theory. Compare Fig. 6.2, Ch 6, p. 294 (histology).

compression

PDL fibroblasts, RANKL ↑ RANK on pre-osteoclast osteoblasts, osteocytes blocks

tension

OPG ↑

Osteoclast → resorption

Bone formation

RANKL is essential for osteoclastic resorption; OPG is a soluble decoy receptor.

RANKL peaks at 3 days and returns to baseline from day 6–58 (Danz et al., 2021).

The RANKL/RANK/OPG axis. Compare Fig. 6.6, Ch 6, p. 302.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 1. 1Why does an ankylosed tooth not move, and what does that tell you? 2What is undermining resorption and why does it delay tooth movement? 3If lighter forces are biologically preferable, why not use the lightest possible force? 4How would you explain the risk of root resorption to a patient at consent? 5What happens if you apply a force greater than the anchorage unit's threshold?

Topic 2. Cephalometry: uses and limitations

Revision Handbook: §10
Question Bank: Block J (Q286–325)
Viva bank: V59–V68

IN ONE MINUTE

Reproducible numbers, but not exact ones. It supplements clinical examination.

- Uses: diagnosis, prognostic tracing, monitoring, research, pre-surgical planning
- Limits: landmark error, invalid landmarks, projection error, unproven superimposition
- Justify every exposure — never for growth prediction alone

Where to read it

Ch 7, *Cephalometric Radiography*, pp. 411–482. Uses: pp. 413–416. Errors: pp. 459–462. Superimposition: pp. 462–474.

Opening definition

- Cephalometric radiography is a specialised technique for imaging the craniofacial region in a standardised, reproducible manner
- cephalometric *analysis* is the identification of defined anatomical landmarks on the film and measurement of the angular and linear relationships between them (Ch 7, p. 412).
- Standardisation is achieved with a cephalostat holding the mid-sagittal plane at a fixed distance from source and film, keeping magnification constant.
- Because films are reproducible, longitudinal views of one subject or views of different subjects can be compared.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition; the cephalostat; lateral and postero- anterior views; NHP and RCP.
2. Uses — diagnosis and treatment planning; prognostic tracing.
3. Uses — unerupted teeth, pathology, airway, pre- orthognathic assessment.
4. Uses — monitoring progress, particularly lower incisor position.
5. Uses — research, normative data, growth studies.
6. Growth prediction — historical use, and why it is no longer justified.
7. Limitation — landmark identification error, non- circular envelope, cumulative error.
8. Limitation — validity of what is measured.
9. Limitation — projection error and magnification.
10. Limitation — superimposition reliability.
11. Limitation — population-specific norms.
12. Limitation — ionising radiation and justification.
13. Reducing error; computer and AI-assisted landmarking.
14. Conclusion: supplements, never replaces, clinical examination.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

ITEM	DETAIL	PAGE
Positioning	Lateral: mid-sagittal plane perpendicular to beam, parallel to film. PA: midsagittal parallel to beam, perpendicular to film. Subject in **natural head position or Frankfort horizontal, teeth in RCP**	Ch 7, p. 412
Exposure	Modern solid-state cephalostats use horizontal scanning over a **10-second exposure**	Ch 7, p. 415
Landmark error	Large differences in reliability between landmarks; each has a **characteristic non-circular envelope of error** in x and y (Baumrind & Frantz, 1971a)	Ch 7, p. 460
Cumulative error	The same landmarks are reused across an analysis, so error accumulates and can produce **artificial correlation between measurements** (Baumrind & Frantz, 1971b)	Ch 7, p. 460
Validity	Points A and B represent the anterior limit of the apical bases but **no specific anatomical region exists**, and both **remodel as the incisors move**	Ch 7, p. 461
Projection	Every film has an enlargement factor which must be known to compare linear values between machines; landmarks not parallel to the film distort, and correction is "often ignored because of the difficulty in calculating correction values"	Ch 7, pp. 461– 462
Superimposition	Requires stable reference structures; the evidence base is limited and **no method has been proved to deliver completely accurate results** (Graf et al., 2022a)	Ch 7, p. 463
Frankfort plane	Porion to orbitale; both difficult to locate, bilateral and must be averaged, and it does not lie in the mid-sagittal plane so is sensitive to head positioning	Ch 7, p. 418
Justification	Not justified in all cases, "particularly if only minor tooth movements are planned"; **essential before planning surgical movement of the jaws**	Ch 7, pp. 413, 415
Growth prediction	Ricketts' visualised treatment objective (1957, 1972) and Björk's implant studies (1969) have been shown "at best unreliable"; radiographs **cannot be justified on the basis of growth prediction alone**. Serial films **about 1 year apart** give a more accurate assessment, especially in Class III	Ch 7, p. 416
Error reduction	High-quality film, correct positioning, optimum viewing conditions, hand-trace then digitise, repeat measurements, know the magnification, calibrate examiners for large studies	Ch 7, p. 462

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCE	WHAT TO BE ABLE TO QUOTE
Houston (1983)	"cephalometrics is definitely not an exact science" (Ch 7, p. 460).
Baumrind & Frantz (1971a, 1971b)	landmark identification, then angular and linear measures. Distinguishing the two correctly marks a well-prepared candidate (Ch 7, p. 460).

Ricketts (1957, 1972); Björk (1969)	growth prediction, since discredited (Ch 7, p. 416).
Graf et al. (2022a)	superimposition reliability (Ch 7, p. 463).
Baumrind et al. (1976); Ghafari et al. (1987)	all superimposition methods carry appreciable error and none is significantly more reliable (Ch 7, p. 469).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Weight the two halves equally.

Candidates habitually write a long ”uses” section and a thin ”limitations” section.

2Do not confuse the two Baumrind & Frantz papers.

If unsure, cite them jointly rather than mis-attribute.

3Present growth prediction as a use the evidence has undermined

— a clean Theme 4 opportunity — and add the constructive alternative: serial films a year apart, particularly valuable in Class III where treatment decisions are deliberately delayed.

4Radiation justification is easy marks and frequently omitted.

No threshold dose; justify per patient; not routine; not for minor tooth movement.

5Do not claim AI has solved measurement error.

It reduces it; the main advantage of software is versatility — multiple analyses, superimposition, orthognathic prediction planning with soft tissue profile overlay.

6Close with the correct stance:

cephalometric analysis supplements a thorough clinical examination and does not attempt to replace it.

WHY CEPHALOMETRIC NUMBERS ARE NOT EXACT

REPRODUCIBILITY Landmark identification error; each landmark has its own non-circular envelope. Error is cumulative and can create artificial correlation between measurements.

VALIDITY Points A and B are convenient, not anatomical — and both remodel as the incisors move.

PROJECTION A 3D object rendered in 2D. Every machine has its own magnification; off-parallel landmarks distort.

SUPERIMPOSITION Assumes stable reference structures. No method has been proved completely accurate.

Houston (1983): ”Cephalometrics is definitely not an exact science.” Ch 7, pp. 459–463.

The four sources of cephalometric error. Compare Fig. 7.60, Ch 7, p. 465.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 2. 1Why is point A a poor reference point in a case involving incisor retraction? 2A colleague wants a lateral cephalogram before aligning a

mildly crowded lower arch. Is that justified? 3What is prognostic tracing and what are its limitations? 4How would you minimise measurement error in a research study using cephalometrics? 5Why is superimposition problematic, and what assumption does it rest on?

Topic 3. Factors in formulating an orthodontic treatment plan

Revision Handbook: §12; space analysis §11
Question Bank: Block K (Q326–350), Block L (Q351–390)
Viva bank: V69–V80

IN ONE MINUTE

A fixed sequence, not a list. Each stage constrains the next.

- Diagnosis → problem list → aims → patient factors → space analysis → anchorage → appliance → phasing → retention
- Facial aims and occlusal aims must be reconciled against the skeletal pattern
- Planning starts with the maxillary incisors within the face, not the occlusion

Where to read it

Ch 8, *Treatment Planning: Basic Principles*, pp. 487–489; *Aims of Treatment*, pp. 489–505; space analysis Box 8.6, pp. 507–509.

Opening definition

Orthodontic treatment planning should be a careful, systematic process aimed at creating an individualised plan that realistically addresses the patient's specific problems and expectations (Ch 8, p. 487). Every case is unique, but the *sequence* is not: diagnosis, problem list, aims, patient factors, individualised plan, appliance selection, phasing, retention.

Essay skeleton

1. Diagnosis — clinical examination, records, and the patient's own concerns and expectations.
2. Problem list — crowding or spacing, overjet, overbite, crossbites, skeletal discrepancy in all three planes.
3. Aims — facial and occlusal, and the difficulty of reconciling them.
4. The skeletal pattern as arbiter: no discrepancy → occlusal correction; growing with discrepancy → growth modification; non-growing → camouflage or surgery.
5. Patient factors — age and growth, dental and periodontal health, medical history, compliance, expectations.
6. Space analysis — the six areas assessed and the two-part chart.
7. Anchorage — arising directly from the space analysis.
8. Appliance selection.
9. Phasing — alignment and levelling, overjet and overbite correction, space closure, detailing and finishing.
10. Retention — planned and agreed before treatment starts.
11. Conclusion — the elements are connected, not independent.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

ITEM	DETAIL	PAGE
The two aims	Positioning the dentition within the skeletal and soft tissue environment for optimal facial and dental aesthetics; **and** achieving a stable and ideal static and functional occlusion	Ch 8, p. 489
Where planning starts	With the position of the dentition, particularly the **maxillary incisors**, within the face — a deliberate shift from the classical approach of planning around mandibular incisor position and isolated static occlusal goals	Ch 8, p. 490 (Box 8.2)
Growth modification	Relies on compliance and favourable growth, results are variable, and it is **much more effective for Class II than Class III**	Ch 8, p. 490
Space required for	Relief of crowding; arch levelling and overbite reduction; correction of tooth angulation; overjet reduction; correction of inclination or torque	Ch 8, p. 506
Space analysis part 1	Six areas: crowding/spacing; levelling; arch width change; incisor A/P change; angulation (tip); inclination (torque). Space required is negative, space available positive	Ch 8, p. 507
Space analysis part 2	How space will be created and used: tooth reduction/enlargement, extraction, space opening for prosthesis, molar distal movement, molar mesial movement, differential growth, residue	Ch 8, p. 508
Interpretation	The **difference in space requirement between arches should reflect the molar relationship** — equal for Class I, greater in the maxilla for Class II	Ch 8, p. 508
Why bother	Formal analysis gives "a more disciplined and less varied approach", realistic aims, **greater anticipation of anchorage requirements** and better patient information (Al-Abdallah et al., 2008)	Ch 8, p. 506
Method to name	Royal London Hospital space analysis (Kirschen et al., 2000a, 2000b)	Ch 8, p. 506
Class II camouflage limits	Surgery indicated if overjet >10 mm, pogonion to nasion perpendicular ≥18 mm, mandibular body length <70 mm, or anterior facial height >125 mm (Proffit et al., 1992a)	Ch 8, p. 491
Soft tissue response	For every 1 mm of maxillary incisor retraction the lips retract **0.7 mm** and the nasolabial angle increases **1.6°** (Konstantonis et al., 2018)	Ch 8, p. 492
Retention	"Retention should be discussed prior to the commencement of active treatment"; patient must be told about relapse *and* normal age-related change such as late lower incisor crowding, and must agree the plan	Ch 8, p. 489; Ch 13, p. 979

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCEWHAT TO BE ABLE TO QUOTE

Kirschen, O'Higgins & Lee (2000a,Royal London space planning (Ch 8, p. 506).

2000b)

Al-Abdallah et al. (2008); Ahmed & Sharma (2023)	the value of formal space analysis, particularly for trainees (Ch 8, p. 506).
Proffit et al. (1992a)	cephalometric thresholds for Class II surgery versus camouflage (Ch 8, p. 491).
Konstantonis et al. (2018)	lip response to incisor retraction (Ch 8, p. 492).
Ackerman & Proffit (1997);	the soft tissues define the limits of treatment (Ch 8, p. 490).

Holdaway (1983)

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not write a list.

The question invites one, and a list will pass. A structured argument organised around the planning *sequence* scores higher: show that each stage constrains the next — problem list sets aims, aims set the space requirement, space requirement sets the extraction pattern, extraction pattern sets the anchorage demand.

2Facial aims are routinely forgotten.

Say explicitly that a Class I incisor and canine relationship "must not be achieved at the expense of the soft tissue profile and facial aesthetics", and that reconciling the two depends on the skeletal pattern.

3Quote the shift in planning philosophy

— from mandibular incisor position and static occlusal goals to the maxillary incisor within the face.

4Compliance is a planning factor, not an afterthought.

A technically ideal plan the patient will not wear is not a plan.

5Do not neglect teeth of poor prognosis

— a compromised tooth should always be removed in preference to a healthy one, even if it sits in a less convenient place (Ch 8, p. 519).

6Give retention a paragraph in the plan, not a sentence in the conclusion. THE TREATMENT PLANNING SEQUENCE — EACH STAGE CONSTRAINS THE NEXT

DIAGNOSIS

1

examination · records · the patient's own concerns

PROBLEM LIST

2

crowding · overjet · overbite · crossbite · skeletal pattern in 3 planes

AIMS

3

FACIAL aims and OCCLUSAL aims — reconciled against the skeletal pattern

PATIENT FACTORS

4

age and growth · dental and periodontal health · medical history · compliance

SPACE ANALYSIS

5

6 areas: crowding · levelling · arch width · incisor A/P · tip · torque

ANCHORAGE

6

follows directly from the space analysis and the extraction pattern

APPLIANCE

7

aims · patient preference · operator expertise

PHASING

8

align & level → overjet/overbite → space closure → detailing

RETENTION

9

planned and AGREED before treatment begins

Planning begins with the position of the dentition — particularly the MAXILLARY INCISORS — within the FACE.

The soft tissues define the limits of orthodontic treatment in each individual (Ch 8, p. 490).

The planning sequence. Compare Box 8.6 space analysis, Ch 8, pp. 507–509.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 3. 1Your space analysis shows 6 mm of crowding in the lower arch. Take me through how that figure changes your plan.
2How do you decide between growth modification, camouflage and surgery in a Class II patient? 3What would make you decline to treat, or defer treatment? 4How do you assess motivation and compliance at the first visit? 5Which parts of your plan would change if the patient were 32 rather than 12?

Topic 4. Factors affecting the choice of extraction

Revision Handbook: §13
Question Bank: Block M (Q391–430)
Viva bank: V81–V90

IN ONE MINUTE

Extraction is one of four ways to make space, and the most predictable.

- Crowding sets the starting point: mild 1–4 mm, moderate 5–8 mm, severe ≥9 mm
- Class II → further back in the mandible; Class III → further forward
- Prognosis beats symmetry: remove the compromised tooth, not the convenient one

Where to read it

Ch 8, *Space Creation*, pp. 506–528. The extraction debate: Box 8.7, pp. 514–515. Mandibular guidelines: Table 8.5, p. 516.

Opening definition

Extraction is one of four methods available for creating space in the dental arches, alongside transverse expansion, anteroposterior arch lengthening and interproximal reduction, and it is one of the most effective (Ch 8, pp. 506–507). Which teeth to remove is a multifactorial clinical judgement, not a rule derived from the space deficit alone.

Essay skeleton

1. The four methods of space creation; extraction in context; the historical debate.
2. Degree and site of crowding.
3. Type of malocclusion — Class I, II and III patterns.
4. Anchorage requirements.
5. Presence and position of teeth — agenesis, ectopic canines, trauma.
6. Dental health and prognosis.
7. Facial and soft tissue considerations; patient factors.
8. Conclusion — an interaction of factors, not a formula.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Guidelines for extraction in the mandib	**ular arch (Table 8.5, Ch 8, p. 516):**
AMOUNT OF CROWDING	EXTRACTION CHOICE
Mild (1–4 mm)	Non-extraction or second premolars
Moderate (5–8 mm)	First or second premolars
Severe (≥9 mm)	First premolars

Extraction pattern by class (Ch 8, 13–516): pp. 5

CLASS PATTERN RATIONALE

Class ICorresponding premolar in both archesMaintains a Class I buccal segment relationship. Note that **anchorage loss tends to be greater in the maxilla**

CLASS PATTERN RATIONALE

ClassMandibular second premolars with IImaxillary first premolars

ClassMandibular first premolars with IIImaxillary second premolars; or mandibular premolars alone finishing Class III molar; or a **single mandibular incisor** in mild cases

Other quantities worth knowing:

- **Correctly timed first premolar extractio** growing child, particularly where the mand maxillary canines buccally displaced (Ch 8,
- **Inappropriate first premolar extraction** retraction of the labial segments (Ch 8, p. 5
- **Second permanent molar extraction:** in segment space but little for the labial segm
- in the maxilla it provides little mesial space more reliable third molar eruption (Ch 8, p.
- **Expansion yields only about half its wid** 522).
- **Interproximal reduction:** up to **1 mm pe** anteriorly where enamel is thinner
- avoid in parallel-sided teeth with subgingiv
- the newly exposed enamel is no more caries
- **Canine movement thresholds:** if the max **tooth unit**, consider distalising the whole b
- **more than half a unit**, premolar extractio
- **more than a full unit**, both (Ch 8, p. 528).
- **Anchorage reality check:** first versus sec difference to anchorage loss, and **around t anchorage is supported** (Haque et al., 20 Extract **at least as far forwards in the maxilla**. If mandibular space demand is high, first premolars may be needed in both arches. If demand is mild, extract in the maxilla only and finish to a Class II molar relationship, which is "perfectly acceptable from an occlusal and functional point of view" Extract **further forwards in the mandibular arch**. Any mandibular extraction in a Class III case should be planned with care **n** allows much spontaneous relief of crowding in a ibular canines are mesially angulated and the p. 513). **in mild crowding** leads to excessive space and over-13). the mandible it provides a small amount of buccal ent, and third molar eruption is unpredictable but facilitates buccal segment distalisation and gives 513). **th as usable space** (O'Higgins & Lee, 2000; Ch 8, p. **r contact point** in the buccal segments, less al contacts → air-cool -prone, especially with topical fluoride (Ch 8, p. 525). illary canines need to move distally **less than half a** uccal segment n is almost certainly required ond premolar extraction makes little practical **wo-thirds of the space created is lost unless** 17; Ch 6, p. 338; Ch 8, p. 528).

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCE	WHAT TO BE ABLE TO QUOTE
Angle; Tweed; Begg	the historical arc, from an intact dentition to premolar extraction for stability (Ch 8, p. 514).
Rushing et al. (1995); Paquette et al. (1992); Iared et al. (2017)	soft tissue differences between extraction and non-extraction are minimal and difficult to judge; both groups are perceived to benefit (Ch 8, p. 515).
Bowman & Johnston (2000)	lip retraction must be considerable before professionals or the public regard the profile as unattractive (Ch 8, p. 515).

SOURCEWHAT TO BE ABLE TO QUOTE

Meyer et al. (2014a, 2014b);extractions do not produce larger buccal corridors or a less attractive

Christou et al. (2019); Cheng et al.smile (Ch 8, p. 515). (2016)

Haque et al. (2017)	anchorage loss is the same for first and second premolar extraction (Ch 8, p. 528).
O'Higgins & Lee (2000)	expansion yields half its width as space (Ch 8, p. 522).

EXAMINER TRAPS

The errors that cost marks on this topic.

1The extraction debate is not open.

The examinable position is that properly planned and executed treatment including mid-arch extractions does not damage facial appearance or narrow the smile.

2Crowding is the starting point, not the answer.

Weaker answers discuss only crowding. Class, anchorage, individual tooth prognosis and facial factors carry equal weight.

3Get the direction right:

further *back* in the mandible for Class II, further *forward* in the mandible for Class III.

4Say that prognosis overrides symmetry.

A compromised tooth should always be removed in preference to a healthy one, even where the space created is large and distant from the crowding — first permanent molars being the classic case.

5Mention finishing to a Class II molar relationship

where the lower arch is well aligned. Candidates often assume Class I molars are compulsory; the textbook says a Class II finish is perfectly acceptable occlusally and functionally.

6Do not overstate the anchorage advantage of first premolar extraction.

State the theory, then state Haque's finding that in practice it makes little difference.

EXTRACTION PATTERN BY CLASS OF MALOCCLUSION CLASS I Mx

Corresponding premolar in BOTH arches Md

maintains the Class I buccal segment

CLASS II Mx

Extract further BACK in the mandible Md

Mx 1st premolars + Md 2nd premolars

CLASS III Mx

Extract further FORWARD in the mandible Md

Md 1st premolars + Mx 2nd premolars

Red = extraction site. Rationale: put the space where the movement is needed, and let the differential in space requirement between the arches reflect the molar relationship you intend to finish to.

Extraction pattern by class. Compare Figs 8.10–8.12, Ch 8, pp. 517–518.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 4. 1A 13-year-old has 9 mm of lower crowding and a grossly carious lower first permanent molar. What do you extract, and why? 2Why do we extract further forward in the mandibular arch in Class III cases? 3How does the extraction pattern change your anchorage plan? 4What are the alternatives to extraction, and when would you use each? 5Does extraction flatten the profile? What does the evidence say?

Topic 5. Anchorage and its clinical significance

Revision Handbook: §14
Question Bank: Block N (Q431–481)
Viva bank: V91–V102

IN ONE MINUTE

Newton's third law. Anchorage cannot be avoided, only planned.

- Reciprocal · minimum / moderate / maximum · simple · compound · stationary
- Headgear: 10–12 h at 250–350 g for anchorage; 14 h at 450–500 g to distalise
- Two-thirds of the space created is lost unless anchorage is actively supported

Where to read it

Ch 6, *Anchorage*, pp. 336–352 — sources of anchorage pp. 337–343, headgear pp. 344–350, anchorage loss pp. 350–352. Skeletal anchorage: Ch 11, pp. 706–732.

Opening definition

- Anchorage is the resistance to unwanted tooth movement, or those sites that provide resistance to the reactive forces generated on activation of any orthodontic appliance (Ch 6, p. 336).
- It follows from Newton's third law: any force moving a tooth generates an equal and opposite force on the structures resisting it.
- **Anchorage cannot be avoided; it must be planned at the start of treatment, in all three planes of space.**

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition and the Newtonian basis; planning in three dimensions.
2. Classification by relative movement — reciprocal; minimum, moderate, maximum.
3. Root surface area — simple and compound anchorage; the critical threshold of stress.
4. Type of movement — stationary anchorage and differential force.
5. Palatal and lingual arches — and their real, limited value.
6. Intermaxillary elastics and their side effects.
7. Skeletal anchorage.
8. Extraoral anchorage — headgear: direction, duration, level, safety.
9. Anchorage loss — causes, and deliberate exploitation.
10. Clinical significance.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Classification (Ch 6, pp.	336–341):
TYPE	MEANING
Reciprocal	Equal movement of both units is acceptable — expansion, closure of a midline diastema
Minimum / moderate / maximum	Graded by how far the anchorage unit may move; **maximum = no movement permissible**
Simple	A single-rooted tooth pitted against a multirooted tooth
Compound	Multiple teeth combined to increase total root surface area

TYPE MEANING

StationaryBegg and Tip-Edge: anchor teeth moved bodily (or not at all) while active teeth are permitted to tip

The mechanism of compound anchorage

The force on the active unit should just exceed the **"critical threshold of stress"** for remodelling, but stay below that of the anchorage unit. Incorporating more teeth spreads the force over a greater area and reduces the stress on each ligament (Ch 6, p. 338).

A refinement worth quoting

- Because the force needed to initiate movement is in fact very low, in maximum-anchorage cases — for example patients with periodontal bone loss — there is an argument that anchor teeth should **not** be subject to initial aligning forces at all, since even those stimulate remodelling and disrupt the occlusion.
- They should instead be consolidated with a passive appliance or extra-dental anchorage used (Melsen & Verna, 1999;
- Ch 6, p. 341). **Palatal and lingual arches — the reality (Ch 6, pp. 341–342):**
- **Trans-palatal arch:** bands on the first molars joined by 0.9 mm stainless steel. Molars can still tip, so it **provides very little additional anchorage support in the sagittal plane** (Diar-Bakirly et al., 2017). Its value is in the **transverse and vertical planes** — for example when erupting an ectopic canine.
- **Nance button:** acrylic against the anterior palatal mucosa, fixing arch length. **Must be removed if arch length is to be reduced**, or the acrylic becomes embedded in the palate.
- **Lingual arch:** 0.9 mm steel on the lower first molars. Provides **little anchorage support** and can procline the lower incisors if the wire contacts them (Rebellato et al., 1997). More effective as a space maintainer. **Headgear (Ch 6, pp. 344–350):**

DURATIONFORCE PER SIDE

Anchorage10–12 hours/day250–350 g Distalisation / orthopaedic effect14 hours/day450–500 g

- Direction: **high (occipital)** for increased vertical proportions — intrusive effect on the posterior dentition
- **low (cervical)** for reduced vertical proportions — extrudes posteriorly and can help reduce a deep overbite in a growing patient
- **straight (combi)** for normal proportions and molar distalisation.
- The **outer bow position relative to the centre of resistance** (approximately at the root furcation) determines whether the molar tips mesially or distally.

- Two injuries are described: **catapult injury** (facebow disengages from the tubes while still attached to the strap) and **nocturnal disengagement**. The most serious consequence is ocular damage, which can be relatively asymptomatic initially but is rapidly infected by oral organisms, with the risk of sympathetic ophthalmitis in the *other* eye (Samuels, 1996).
- **Two independent safety devices are mandatory: a self-releasing head-cap or neck-strap, and a locking facebow** (Ch 6, p. 350). **Anchorage loss — the four causes (Ch 6, pp. 351–352):**

1. **Heavy forces** — the differential force theory; above the plateau there is no additional desired movement, but the anchorage unit is pushed above its threshold.
2. **Maxillary arch** — susceptible because of bone density, alveolar width and the greater extent of movement usually required, particularly in Class II (Giannopoulou et al., 2016). Fortunately it is also easier to support there.
3. **Occlusal interferences** — commonly when correcting Class II canines or reducing an overjet before the overbite is fully corrected. A well-intercuspated occlusion helps prevent anchorage loss.
4. **Increased vertical growth pattern** — lower bite forces, reduced intercuspation, downward and backward growth direction, and a more mesial eruptive path. Patients with reduced vertical proportions lose less anchorage but close space more slowly. **Breaking anchorage** — deliberate anchorage loss to close posterior space in Class III cases or in tooth agenesis, particularly absent second premolars and maxillary lateral incisors. Anterior anchorage must then be reinforced with protraction headgear, Class III elastics, increased palatal or lingual root torque, or skeletal anchorage (Ch 6, p. 350).

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCE	WHAT TO BE ABLE TO QUOTE
Newton's third law	state it explicitly; it is the conceptual point being tested.
Begg (1956)	differential force (Ch 10, p. 615).
Jepsen (1963)	root surface area of the permanent dentition, the basis of relative anchorage values (Ch 6, p. 340).
Diar-Bakirly et al. (2017)	the trans-palatal arch gives little sagittal anchorage (Ch 6, p. 341).
Rebellato et al. (1997)	the lingual arch gives little anchorage and may procline incisors (Ch 6, p. 342).
Haque et al. (2017)	no difference in anchorage loss between first and second premolar extraction (Ch 6, p. 338); around **two-thirds of the space created is lost unless anchorage is supported** (Ch 8, p. 528).
Samuels (1996)	headgear injury types (Ch 6, p. 348).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Open with Newton.

Defining anchorage without explaining why it exists loses the conceptual mark.

2Learn the headgear figures and the safety devices.

They are among the most reliably examined numbers in orthodontics, and the safety point is a patient- safety mark you cannot afford to drop.

3Do not overstate palatal and lingual arches.

Saying the trans-palatal arch provides ”very little sagittal anchorage” and that the lingual arch is really a space maintainer shows you have read past the received wisdom.

4Say that anchorage must be planned in all three planes

, not just anteroposteriorly. That single clause elevates the answer.

5Skeletal anchorage is transformative but not universal.

Describe it as providing absolute anchorage independent of compliance, not as having abolished anchorage planning.

6Finish on clinical significance

, since the question asks for it: anchorage must be quantified before treatment, informs extraction pattern and appliance choice, and poor management is a common cause of unsatisfactory outcomes — typically a residual overjet from forward drift of the buccal segments.

molars incisors

MINIMUM

molars may mesialise freely

molars incisors

MODERATE

both units move similar amounts

molars incisors

MAXIMUM

no movement of the anchorage unit

Reciprocal anchorage = equal movement of both units is acceptable (expansion, midline diastema).

Unless anchorage is supported, around two-thirds of the space created is lost (Haque et al., 2017).

Anchorage classification. Compare Fig. 6.35, Ch 6, p. 339.

DIRECTION OF PULL (force applied at the upper first molar) HIGH / occipital

6

increased vertical proportions — intrusive

STRAIGHT / combi

6

normal proportions — molar distalisation

LOW / cervical

6

reduced vertical proportions — extrusive

FORCE AND DURATION

Anchorage support 10–12 h/day 250–350 g per side

Distalisation / orthopaedic 14 h/day 450–500 g per side

Two independent safety devices are mandatory: self-releasing strap + locking facebow

Headgear: direction, force and duration. Compare Figs 6.42–6.43, Ch 6, pp. 346–347.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 5. 1Why is anchorage loss more of a problem in the maxilla? 2How would you assess anchorage demand before starting a case? 3A patient with an increased maxillary–mandibular planes angle needs maximum anchorage. What are your options and which would you choose? 4What safety features must be present on a headgear, and what would you tell the patient and parent? 5What is differential force theory, and which appliance philosophy is built on it?

Topic 6. First, second and third order movements, and the evolution of fixed appliances

Revision Handbook: §17
Question Bank: Block Q (Q567–606)
Viva bank: V112–V121

IN ONE MINUTE

The three orders have not changed since Angle. Only their delivery has.

- First = in-out · second = tip · third = torque
- Standard edgewise puts them in the WIRE; pre-adjusted edgewise puts them in the BRACKET
- Self-ligation and customisation do not shorten treatment — the trials are clear

Where to read it

Ch 10, *The Evolution of Fixed Appliances*, pp. 614–629. Components: pp. 629–666. Self-ligation: Where is the evidence 10.3, pp. 646–647. Customised appliances: Where is the evidence 10.1, pp. 630–631. Andrews' six keys: Ch 2, Box 2.2, p. 58.

Opening definition

Angle developed the edgewise slot after experimenting with several systems, finding that placing the slot horizontally rather than vertically gave greater control: the interaction of a rectangular wire in a rectangular slot provides precise three-dimensional control of tooth position (Angle, 1928; Ch 10, p. 614). **First-order** bends achieve in–out position, **second-order** bends achieve tip or angulation, and **third-order** bends achieve torque or inclination (Ch 10, p. 615).

Essay skeleton

1. Define the three orders precisely with a clinical example of each.
2. Standard edgewise — all information in the wire; strengths and three specific disadvantages.
3. Begg's light wire appliance — differential force; the three stages.
4. Tip-Edge — controlled tipping, then full prescription at finishing.
5. Andrews' pre-adjusted edgewise — the six keys and the four bracket variables.
6. Ligation — conventional versus self-ligating, and the evidence.
7. Lingual appliances and digital customisation, and the evidence.
8. Conclusion — the concepts are unchanged; the delivery has evolved.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

ORDER	CONTROLS	BRACKET VARIABLE IN THE PRE-ADJUSTED APPLIANCE
First	In–out (facial prominence)	Variable **distance from slot to bracket base**
Second	Tip (mesiodistal angulation)	Variable **mesiodistal angulation of the slot** to the long axis
Third	Torque (buccolingual inclination)	Variable **buccolingual inclination of the slot** relative to the base
Andrews centre of	additionally built in **vertical c** the clinical crown — and **mesi**	**ontact point alignment** — slot centre coincident with the **odistal offset** (Ch 10, p. 623).
The evo	**lutionary sequence:**	

- **Standard edgewise (Angle, 1928 → dominant until the late 1970s).** Passive identical brackets, so all detailing depended on bends for each tooth. Three named disadvantages: bending was **time-consuming and required considerable skill**
- space closure required **complicated closing loops**
- and because teeth are moved **bodily** it is **more demanding upon anchorage** (Ch 10, p. 615).
- **Begg light wire appliance (Begg, 1956).** Australian, ex-pupil of Angle. Crowns tipped with intermaxillary elastics, roots uprighted separately with auxiliary springs. Popular through the 1960s–70s. Treatment is compartmentalised into **three stages** with overcorrection emphasised throughout: stage 1 corrects crowding, rotations, overbite and overjet to edge-to-edge, molar relationship and arch coordination
- stage 2 closes all extraction spaces → stage 3 corrects axial inclinations (Cadman, 1975
- Ch 10, pp. 615, 621–622). Limitation: round wires and a simple slot make precise finishing difficult, and the uprighting auxiliaries are complex and time-consuming.
- **Tip-Edge (Kesling, 1988; Kesling et al., 1991).** Tips during the initial phase but **limits how far**, then closes the slot down to express a specific prescription with full-size rectangular archwires (Ch 10, pp. 615–616).
- **Pre-adjusted edgewise (Andrews, 1972, 1979).** Derived from measurements of a large sample of **untreated ideal occlusions**, codified as the **six keys of occlusion: molar relationship**
- **crown angulation → crown inclination → no rotations → no spacing**
- **flat occlusal plane** (Ch 2, p. 58). Remains the most popular system in use.
- **Molar detail worth knowing.** Andrews updated the ideal molar relationship: the distal marginal ridge of the upper first molar should occlude with the mesial marginal ridge of the lower second molar, with the mesiobuccal cusp in the groove between the mesial and middle cusps of the lower first molar — requiring **distal tip and a slight buccal offset**, which he engineered into the appliance (Ch 2, pp. 63–64). **The evidence on the modern refinements — a Theme 4 goldmine:**
- **Self-ligation.** Claims were faster movement, shorter treatment, stable expansion and less pain (Damon, 1998). Randomised trials found **no significant difference in rate of tooth movement, overall treatment time or number of visits** (Scott et al., 2008a
- Fleming et al., 2010 → DiBiase et al., 2011 → Johansson &

- Lundström, 2012) → pain benefit is **marginal and not supported by all studies**
- expansion and incisor proclination findings are **equivocal**
- and there is **no evidence of a better occlusal outcome**. "Apart from saving some chairside time, self-ligating brackets do not perform clinically any differently to those with conventional ligation" (Ch 10, pp. 646–647).
- **Customised appliances.** Retrospective data suggested SureSmile reduces treatment time by 7–15 months, but with high risk of selection bias. A **large high-quality randomised trial found comparable outcomes and no difference in overall treatment time** between Insignia and conventional self-ligating appliances, with **more breakages, more patient complaints and increased treatment planning time** for Insignia. Overall treatment time was dictated by the difficulty of the malocclusion and the identity of the treating orthodontist (Penning et al., 2017
- Ch 10, pp. 630–631).

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCEWHAT TO BE ABLE TO QUOTE

Angle (1928)the edgewise appliance (Ch 10, p. 614).

SOURCE	WHAT TO BE ABLE TO QUOTE
Begg (1956); Cadman (1975)	differential force and the three stages (Ch 10, pp. 615, 621).
Kesling (1988); Kesling et al. (1991)	Tip-Edge (Ch 10, p. 615).
Andrews (1972, 1979)	six keys, then the pre-adjusted prescription (Ch 10, pp. 617–618).
Scott et al. (2008a); Fleming et al. (2010); DiBiase et al. (2011); Johansson & Lundström (2012)	self-ligation trials (Ch 10, pp. 646–647).
Penning et al. (2017)	customised appliance randomised trial (Ch 10, p. 631).

EXAMINER TRAPS

The errors that cost marks on this topic.

1The question has two halves.

Define the orders *and* trace the evolution. Roughly one-third to the definitions, two-thirds to the evolution.

2Explain why each step happened.

Begg exists because standard edgewise was anchorage-hungry; Andrews exists because wire bending was slow and operator-dependent; digital systems exist because bracket positioning is the residual error.

3Andrews' legitimacy rests on his sample.

The prescription was derived from *untreated ideal* occlusions — say so, because it explains why the values mean anything.

4Do not oversell self-ligation or customisation.

The randomised evidence is unambiguous and stating it is one of the easiest ways to demonstrate critical reading.

5Do not forget lingual appliances.

They are the other branch of the same evolution and are **particularly effective at overbite reduction** (Ch 13, p. 933) — a useful cross-reference to Topic 11.

FIRST ORDER SECOND ORDER THIRD ORDER

in–out tip (angulation) torque (inclination)

slot-to-base distance mesiodistal slot angulation buccolingual slot inclination

arch form / crown thickness canine and premolar root angulation incisor root position

Standard edgewise: all three built into the WIRE, tooth by tooth.

Pre-adjusted edgewise: all three built into the BRACKET (Andrews, 1972, 1979).

First, second and third order movements. Compare Fig. 10.9, Ch 10, p. 623.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 6. 1What is torque, and why does it matter during overjet reduction? 2Why was the standard edgewise appliance so anchorage-demanding? 3What is differential force, and how did Begg exploit it? 4What are the practical differences between Roth and MBT prescriptions? 5Does self-ligation reduce treatment time? What does the evidence show?

Topic 7. The ideal archwire; stainless steel versus nickel titanium

Revision Handbook: §15
Question Bank: Block O (Q482–525)
Viva bank: V103–V111

IN ONE MINUTE

No wire has every property, and the properties you need change through treatment.

- Alignment needs springback, low stiffness, stored energy; space closure needs stiffness
- Stainless steel is the working wire; NiTi is the aligning wire — complementary, not rival
- Hysteresis is why NiTi delivers a lighter force than it took to place it

Where to read it

Ch 10, *Archwires*, pp. 647–658. Physical properties: Box 10.2, pp. 651–653. Archwire evidence: Where is the evidence 10.4, pp. 655–656.

Opening definition

The ideal properties required of an orthodontic archwire depend on the stage of treatment and the type of movement being carried out, and **no archwire material offers all of them together** (Kapila & Sachdeva, 1989; Ch 10, p. 647). That single sentence is the answer to the first half of the question; the sequence of wires used through treatment follows directly from it.

Essay skeleton

1. The role of the archwire; the stress–strain curve as the organising framework.
2. Properties derived from the curve — stiffness, range, springback, formability, resilience.
3. Properties not on the curve — biocompatibility, environmental stability, joinability, friction, annealing, ductility, cold-working, flexibility.
4. The requirements change: alignment versus levelling and space closure.
5. Stainless steel — composition, advantages, disadvantages, modifications.
6. Nickel titanium — metallurgy, the three types, advantages, disadvantages.
7. Cobalt chromium and beta titanium.
8. What the trial evidence actually shows.
9. Conclusion — complementary, not competing.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Properties, de	**fined as the textbook defines them (Box 10.2, Ch 10, pp. 652–653):**
PROPERTY	DEFINITION

Stiffness	Force delivered by a wire; the gradient of the straight line on the stress–strain curve, i.e. Young's modulus. Steeper line = stiffer wire
Range	Deflection achievable within the elastic limit; in practice measured to the **yield point**, where 0.1% permanent deformation occurs
Springback	Recovery on unloading when deformed to or beyond the elastic limit; the **ratio of yield strength to modulus of elasticity**; effectively the working range
Formability	Distance on the graph between the **yield point and the fracture point**
Resilience	Energy stored to move teeth; the **area under the elastic portion** of the curve

PROPERTY DEFINITION

Not on theBiocompatib graph-working, flex	ility, environmental stability, joinability, friction, annealing, ductility, coldibility
The two sets of requirem	**ents (Ch 10, pp. 649–650) — learn as a contrasting pair:**
INITIAL ALIGNMENT	LEVELLING, OVERBITE REDUCTION AND SPACE CLOSURE
Large springback	High stiffness
Low stiffness	Low stored energy
High stored energy	Good joinability
Biocompatibility	Biocompatibility
Low surface friction	Low surface friction
Wire diameters: alignmen followed by a short period o **0.016–0.020 inch** for leve	t **0.012–0.018 inch** round NiTi, multistrand or coaxial steel, often f rectangular NiTi to begin torque expression; then round stainless steel lling; then rectangular stainless steel for the later stages (Ch 10, pp.

649–650).

Stainless steel (Ch 10, p. 651)

- Iron **70%**, chromium **17–20%**, nickel **8–12%**, carbon maximum **0.08%** — the "**18–8**" steels.
- **Chromium confers corrosion resistance; nickel increases ductility.** Advantages: high stiffness, excellent formability, biocompatibility, low cost, good joinability — ideal as a working archwire.
- Disadvantage: too stiff for initial alignment; hard to tie into a crowded arch without permanent deformation.
- Modifications: softened by **annealing**, hardened by **cold-working**, springback improved by **incorporating loops** to lengthen the interbracket span, or flexibility increased by **multistrand/twistflex or coaxial** construction — because the force delivered relates to the radius of the individual wires.
- Notably, these multistrand wires "seem to perform as well as nickel titanium archwires during initial alignment", but their tendency to permanent deformation makes NiTi easier to use in practice. **Nickel titanium (Ch 10, pp. 654–655).**
- Developed by the **Naval Ordinance Laboratory** in the early 1960s as *nitinol*, introduced to orthodontics a decade later.
- High flexibility, low force over a long range, **shape memory** and **superelasticity**, arising from a phase transition between the low-stiffness **martensitic** and higher-stiffness **austenitic** crystal structures.

- The **deactivation plateau is lower than the activation plateau — hysteresis** — so the force delivered to the tooth is lower than that needed to activate the wire.
- Disadvantages: low stiffness, **poor formability** and **high friction**, making it unsuitable for space closure or overbite control. **The three generations of NiTi — a precise distinction that scores well:**

1. **Martensitic-stabilised (conventional):** crystal structure fixed in the martensitic form by cold- working; greater flexibility but **no shape memory or superelasticity**.
2. **Austenitic-active (superelastic):** phase shift induced by **stress** — pseudoelasticity. Austenitic at room temperature and in the mouth until deformed.
3. **Martensitic-active copper nickel titanium (thermoelastic, heat-activated):** phase shift induced by **temperature**, the transition temperature tuned by partially replacing nickel with copper and set at around body temperature. Martensitic and very flexible outside the mouth
4. becomes austenitic on insertion and delivers a gentle aligning force.

Other alloys (Ch 10, p. 653)

Cobalt chromium (Elgiloy) — greater formability than steel, similar stiffness, greater friction; heat- treatable; used for auxiliaries such as an intrusion arch or quadhelix. **Beta titanium** — introduced in the 1980s; good formability, stiffness about **one-third** that of stainless steel, higher friction; used for finishing bends and settling.

What the trials show (Ch 10, pp. 655–656)

- A Cochrane review of 12 randomised trials in almost 800 patients found **moderate-quality evidence that coaxial superelastic NiTi produces greater tooth movement over 12 weeks than single-strand superelastic NiTi**, and no real difference in day-1 pain between multi-strand stainless steel and superelastic NiTi.
- Otherwise **insufficient evidence** that any material is superior for alignment rate, time to alignment, pain or root resorption (Wang et al., 2018).
- An earlier review found slightly better irregularity correction with austenitic-active than martensitic-stabilised NiTi, and that martensitic-active copper NiTi sequences took slightly **longer** to reach a working archwire with **more pain** at 4 hours and 1 day (Papageorgiou et al., 2014).
- All differences are marginal: **the choice of initial archwire and sequence is up to the individual orthodontist.**

EXAMINER TRAPS

The errors that cost marks on this topic.

1Define springback and resilience distinctly.

They are routinely conflated. Springback is the ratio of yield strength to modulus of elasticity — the working range.

2The examinable insight is that the requirements change.

An essay listing ideal properties as one fixed set has missed the question.

3Explain hysteresis mechanistically.

It is the reason NiTi delivers a light continuous force.

4Distinguish the three NiTi types by name.

Very few candidates do, and it takes one sentence each.

5Quote the steel composition and the ”18–8” nickname.

Cheap, precise marks.

6Do not say NiTi is ”better”.

They occupy complementary roles. And note the counterintuitive finding that multistrand steel performs comparably during alignment.

FORMABILITY

yield point

(0.1% permanent deformation)

fracture

elastic limit

Stress

STIFFNESS

= gradient of the elastic portion

(Young’s modulus)

RESILIENCE

area under the elastic portion

RANGE

Strain

SPRINGBACK = yield strength ÷ modulus of elasticity — effectively the working range.

Not on the curve: biocompatibility · environmental stability · joinability · friction · ductility.

The stress–strain curve and the properties derived from it. Compare Box 10.2, Ch 10, p. 652.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 7. 1Why can you not place a bend in a superelastic NiTi wire? 2What is superelasticity, and how does it differ from shape memory? 3Why is friction a problem in sliding mechanics, and how would you reduce it? 4When would you choose a multistrand stainless steel wire over NiTi? 5What are beta-titanium wires used for, and why?

Topic 8. Functional appliances

Revision Handbook: §16; Class II application §22
Question Bank: Block P (Q526–566)
Viva bank: V112–V121

IN ONE MINUTE

They posture the mandible forwards. Most of the correction is dentoalveolar.

- Skeletal gain is a few millimetres at best and essentially transient
- Twin block is the UK standard; Herbst removes compliance from the equation
- High-angle patients with a reduced overbite respond poorly — choose another route

Where to read it

Ch 9, *Functional Appliances*, pp. 559–583; clinical use pp. 583–601; fixed Class II correctors pp. 601–611. Long-term effects: Where is the evidence 9.4, pp. 561–562. Predictors of success: 9.5, p. 587. Herbst versus twin block: pp. 606.

Opening definition

- Functional appliances effect their changes by **posturing the mandible forwards and changing the functional environment of the jaws, dentition and surrounding soft tissues** (Ch 9, p. 559).
- Postural adaptation is fundamental to their mode of action and influences four regions: the facial skeleton, the dentition and occlusion, the orofacial soft tissues, and the muscles of mastication.
- Most are designed to correct Class II malocclusion in a growing child; the Herbst is the principal fixed example.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition and the four target regions.
2. Facial skeleton — sutural restraint, condyle as secondary cartilage, glenoid fossa; and what the evidence actually shows.
3. Dentition and occlusion — where most of the correction comes from.
4. Orofacial soft tissues — the lip trap and lip competence.
5. Muscles of mastication — and the contradiction in the electromyographic evidence.

- *Skeletal:* the Class II force component can ap **maxillary growth**, maximised if combined w
- The condyle is a **secondary, adaptive cartil** the local functional environment.
- **"Any extra growth obtained during functi few millimetres at best, and these effects**

1. Removable designs: activators, bionator, Fränkel, twin block.
2. Activators combined with headgear.
3. Fixed functional appliances: the Herbst; fixed Class II correctors.
4. Clinical use — timing, predictors of success, wear, monitoring, retention.

5. Conclusion. ply **some limited restraint on forward** ith headgear. **age** — it does not drive growth but responds to **onal appliance therapy is a matter of only a are essentially transient."***The sophisticated*

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Mechanism — the critical distinction (Ch 9, pp. 559–562)

explanation: untreated Class II patients do not grow into Class I because the occlusion is **locked by intercuspation and dentoalveolar adaptation**.

- Functional appliances probably work by **unlocking the occlusion**, allowing normal mandibular forward growth to express itself, combined with dentoalveolar change (Ch 9, p. 562).
- *Dentoalveolar:* retroclination of the upper incisors, proclination of the lower, plus **guidance of eruption** — acrylic capping or faceting encourages mesial eruption of the mandibular buccal teeth and distal eruption of the maxillary, correcting the buccal segment relationship and contributing to overbite reduction.
- *Soft tissues:* eliminating a lower lip trap and creating lip competence; functional appliance therapy "can often have a very positive effect on the soft tissue profile" (Koretsi et al., 2015).
- *Muscles:* early electromyography suggested lateral pterygoid hyperactivity on protrusion (McNamara, 1973), but later work with **surgically implanted electrodes showed a reduction** in activity (Voudouris et al., 2003) — so the lateral pterygoid may not be the primary factor.
- An alternative hypothesis is viscoelastic stretch of the joint capsule with referred force to the glenoid fossa.

The designs (Ch 9, pp. 568–583, 602–611):

APPLIANCE	DISTINGUISHING FEATURE
Andresen–Häupl activator	Original monobloc; **minimal** vertical opening; worn at night; relies on stimulating the musculature
Harvold / Woodside activator	**Increased vertical bite opening beyond the freeway space**; works by stretching the orofacial connective tissues. Harder to wear
Balters bionator	Acrylic bulk considerably reduced for increased wear; **palatal coffin spring** to encourage a forward tongue position; buccal shields for passive expansion
Fränkel Functional Regulator	Based on the functional matrix; buccal shields and labial pads act on the soft tissue environment. Bulky, prone to breakage, difficult to wear
Clark twin block	Described by William Clark in the early 1980s. Separate upper and lower plates with interlocking blocks whose inclined planes are at approximately **70°** to the occlusal plane. Robust, relatively simple to fabricate, and much easier to wear than conventional activators
Teuscher	Activator–headgear combination with **anterior spurs to torque the upper incisors**; Teuscher suggested 12 hours/day
van Beek	Modified activator with the **headgear bow incorporated directly into the acrylic**
Dynamax	Neville Bass; maxillary splint attached to first molars via bands. A prospective comparison showed the **twin block gave more rapid overjet reduction and fewer adverse effects**

The designs (Ch	9, pp. 568–583, 602–611):
Herbst	Emil Herbst, early twentieth century. Cemented or bonded, with **bilateral rigid telescopic arms** from the maxillary first molars to the mandibular first premolar region. **Overjet reduction in 6–8 months** commonly achieved (Pancherz, 1982). Expensive to fabricate, often difficult to tolerate, prone to catastrophic breakage
Forsus, Powerscope, Jasper	Fixed Class II correctors attached to a fixed appliance; most are prone to breakage or unwanted intrusion and rotation. Forsus is the most successful

Jumper, Eureka

The randomised trial you should quote (Pacha et al., 2023, 2024; Ch 9, p. 606)

- Hanks Herbst versus twin block in significant Class II division 1: **37% of the twin block group were deemed failures versus 17% of the Herbst group**
- twin block patients took **1.5 months longer** and achieved a **smaller mean overjet reduction (5.8 mm versus 7.1 mm)**.
- No major differences in overall skeletal or dental outcome, though the Herbst produced greater mandibular incisor protraction.
- However, the Herbst required **more chairside time, more routine visits and more emergency appointments**.
- Conclusion: the Herbst gives more consistent and predictable overjet reduction and is seemingly preferable to patients, but costs more visits and fails more dramatically. **Predictors of success (Where is the evidence 9.5, Ch 9, p. 587).**
- Favourable: increased overbite; a smaller mandible at the outset; **reduced mandibular plane angle**;
- **smaller gonial angle** — associated with greater increase in mandibular length, more forward chin positioning, greater stability, higher bite force and more skeletal change.
- Unfavourable: **high-angle patients with increased lower anterior face height and mandibular plane angle and a reduced overbite respond poorly**, and alternative modalities should be considered.

Long-term stability (Pancherz et al., 2014; Ch 9, p. 562)

At 32 years post-treatment, **64% of the molar correction and 86% of the overjet correction remained stable**. Early relapse in the first 6 months is mainly dentoalveolar. Recommendation: **at least 2 years of retention using an activator** to maintain the postured bite.

Clinical monitoring (Ch 9, pp. 595–597)

- Record the overjet **and the overjet in maximum protrusion** at each visit — the difference should stay roughly constant, confirming that reduction is physical rather than postural.
- **If no change in overjet or buccal segments within 6 months, review the plan.** Signs of poor wear: continued speech problems, a pristine appliance, and repeated breakages.
- Maximal versus incremental advancement makes little clinical difference.

> **EXAMINER TRAPS**
>
> The errors that cost marks on this topic.

1The mechanism is the question.

Describing functional appliances as producing substantial mandibular growth is describing a position the evidence does not support.

2"Describe some examples of its use" means give indications, not a catalogue.

For each appliance say who it suits: twin block for the compliant growing Class II division 1; Herbst where compliance is doubtful or speed matters; combination headgear designs where maxillary restraint is also wanted; FR III for mild Class III.

3Learn the Fränkel numbering

— FR I for Class II division 1, FR II for Class II division 2, FR III for Class III, FR IV for anterior open bite.

4Quote the Pacha trial.

It is recent, randomised, directly comparative, and few candidates will know it.

5Say who responds badly.

The high-angle, reduced-overbite patient is the one to avoid, and naming that is clinically useful and examinable.

6Do not omit retention.

Two years of activator retention after Herbst treatment is a concrete, quotable recommendation.

WHAT A FUNCTIONAL APPLIANCE ACTUALLY DOES Mandible postured forwards

SKELETAL DENTOALVEOLAR SOFT TISSUE

few mm at best upper incisors retrocline lip trap eliminated

ESSENTIALLY TRANSIENT lower incisors procline lip competence gained

guided buccal eruption

THE SOPHISTICATED EXPLANATION

Class II persists because the occlusion is LOCKED by intercuspation. The appliance unlocks it and lets normal mandibular growth express itself, alongside the dentoalveolar change. Ch 9, p. 562.

What a functional appliance actually does. Compare Fig. 9.12, Ch 9, p. 566.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 8. 1How much true skeletal change does a twin block produce, and how do you know? 2When would you start functional appliance therapy, and why? 3Your patient will not wear their twin block. What are your options? 4What is the functional matrix theory, and which appliance is based on it? 5How does a functional appliance reduce the overbite?

Topic 9. Benefits and complications of orthodontic treatment

Revision Handbook: §29; consent §8
Question Bank: Block AC (Q962–991)
Viva bank: V49–V58, V174–V183

IN ONE MINUTE

Elective treatment. The benefits are narrower than commonly claimed.

- Strong: dental aesthetics and trauma prevention (RR 0.58). Weak: caries, periodontal, speech
- Quantify the risks: white spots in up to half; severe resorption in 1–5% of teeth
- Frame the whole answer around consent

Where to read it

Ch 2, Benefits of Orthodontic Treatment*, pp. 86–92;* Risks of Orthodontic Treatment*, pp. 92–105. Airway* controversy: Where is the evidence 2.1, pp. 90–91. Facial aesthetics: 2.2, p. 101. TMD history: 2.3, pp. 104– 105.

Opening definition

- For any elective medical intervention there should be a clear list of benefits which outweigh the risks.
- Orthodontic treatment provides significant cosmetic advantages, "however, it has proved difficult to provide strong evidence in support of the widely assumed belief that treatment can also improve the oral health and psychological well-being of an individual" (Ch 2, p. 86).
- That sentence, more or less verbatim, is the correct opening posture for this essay.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Framing: elective treatment, benefit–risk balance, consent.
2. Benefits — caries and periodontal health, with the appropriate qualifiers.
3. Benefits — masticatory efficiency and speech.
4. Benefits — breathing and obstructive sleep apnoea; the "airway orthodontics" claim.
5. Benefits — prevention of trauma (the strongest health benefit).
6. Benefits — psychological and quality-of-life.
7. Risks from appliances: demineralisation, enamel fracture, root resorption, pain and pulpal damage, gingivitis, alveolar bone loss and recession, ulceration, allergy, headgear injury.
8. Generalised risks: facial aesthetics, relapse, failure of treatment, TMD.
9. Conclusion — individual weighing, clear explanation, consent.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Benefits, wit	**h the correct confidence attached (Ch 2, pp. 86–92):**
CLAIM	EVIDENCE POSITION
Cleaning	Straight teeth are easier to keep clean and treated patients have **lower plaque scores** — though whether from behaviour change or the treatment is unclear
Caries	**Weak** evidence that crowding is a risk factor, and weak evidence that treatment in adolescence reduces later caries incidence (Macey et al., 2020)

CLAIM	EVIDENCE POSITION
Periodontal	**Lack of evidence** that treatment improves periodontal health; it can contribute to higher recession, alveolar bone loss and pocketing, though often clinically insignificant. **Exception:** correction of an anterior crossbite with recession on a lower incisor, and of a deep traumatic overbite
Mastication	**Weak** evidence, but reduced occlusal contacts do affect chewing efficiency; posterior crossbite in a child is associated with asymmetrical muscle activity; orthognathic correction improves masticatory performance long-term
Speech	Most individuals adapt; a placement error does not always cause auditory distortion; **speech patterns are established long before the permanent dentition erupts**, so treatment is not guaranteed to improve speech
Breathing	RME increases total airway volume short-term, but **RME should not be considered a primary intervention to improve nasal breathing**
OSA	CPAP is the benchmark for severe OSA; **mandibular advancement appliances are effective in mild-to-moderate OSA** (Koretsi et al., 2018); maxillary expansion is **not currently recommended** as a primary treatment
Trauma	An increased overjet is a **significant risk factor**, and randomised trial evidence shows early correction in pre-adolescence **reduces later trauma incidence** (Batista et al., 2018)
Psychological	Increased overjet makes a child more susceptible to **teasing and bullying** (Seehra et al., 2011); but longitudinal studies show **little objective evidence** of improved long-term psychological health (Kenealy et al., 2007). Malocclusion severity correlates with poorer OHRQoL, and treatment **does** improve the **emotional and social well-being** domains (Javidi et al., 2017)
The "airway paragraph. T	**orthodontics" claim (Where is the evidence 2.1, Ch 2, p. 91) — worth a** he theory is that extractions constrict the arches, reduce tongue space, drop the

tongue back and reduce the oropharyngeal airway, increasing OSA. "To date, there is no evidence that orthodontic treatment with extractions results in a reduction in the oropharyngeal airway in either adolescents or adults, or in an increased incidence of sleep apnoea… This phenomenon therefore appears to be based more on scaremongering and marketing than on clinical evidence." Note the textbook explicitly links this to the earlier extraction-and-TMD movement. **Risks from appliances (Ch 2, pp. 92–98):**

RISK	DETAIL
Enamel demineralisation	White spot lesions in **up to half** of fixed appliance patients (Gorelick et al., 1982); **greater prevalence in males** (Tufekci et al., 2011); usually visible **within 6 months**; commonest on **first permanent molars, upper lateral incisors and mandibular canines** (Øgaard, 2008). Fade with time but can remain evident years later. Prevention: excellent hygiene, non-cariogenic diet, **daily high-fluoride toothpaste and professional high-concentration fluoride foam every 6–8 weeks** (Benson et al., 2019)
Enamel fracture	Small risk at the enamel–dentine junction if bond strengths are too high; the historical problem was **early chemically enhanced ceramic brackets**
Root resorption	Almost universal; severe (>1/3 root length) in **1–5%** of treated teeth (Sameshima & Sinclair, 2004); greatest amount and severity in the **anterior maxilla, especially the maxillary lateral incisors**
Pain	Peaks at about **48 hours**, returns to baseline after **7 days**; paracetamol or ibuprofen, which may be taken **pre-emptively** (Monk et al., 2017)
Pulpal damage	Loss of vitality is **rare**; more likely with excessive force, apices pushed through the cortical plate, or a history of trauma
Gingivitis and hyperplasia	Inevitable with fixed appliances, especially bands; improves markedly on removal; **antiepileptics and immunosuppressants** with poor hygiene can require gingival surgery
Alveolar bone loss	Small and **clinically insignificant** with no long-term periodontal effect — **except** in active periodontal disease, which must be treated and stable first
Recession	From movement **out of the osseous envelope**; lower incisors particularly susceptible, and — counterintuitively — related **more to retroclination in high-angle Class III cases with a narrow symphysis and thin biotype than to proclination**
Ulceration	Common early; worse if archwires are not cut or bent back; also acid burns from phosphoric acid etch
Nickel allergy	Prevalence about **10% in the USA and Europe**, more common in females; usually type IV; **oral reactions are rare**
Headgear injury	Predominantly from the inner bow detaching from the molar bands at night; risk of ocular penetration

Generalised risks (Ch 2, pp. 99–101)

- *Facial aesthetics:* extraction produces slightly more retrusive profiles, but in the majority of cases facial change is judged **beneficial by lay and professional judges irrespective of extraction** (Bowman & Johnston, 2000).
- *Relapse:* "almost a universal finding necessitating retention in almost all cases".
- High relapse potential for **rotated teeth, lower incisor crowding, changes in lower incisor labiolingual position, expansion of lower intercanine width, and spacing**.
- Increased crowding later in life occurs in untreated individuals too and **should be regarded as an age-related change rather than relapse**.
- *Failure:* high discontinuation rates in children and adolescents, especially with removable appliances (Klaus et al., 2017)
- a patient who does not complete treatment "may end up with an occlusal result that is unsatisfactory or even worse than the presenting malocclusion, particularly if permanent teeth have been extracted and space closure has not been completed."

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not overclaim the benefits.

The commonest failure is asserting that orthodontics improves periodontal health, caries risk, chewing and speech.

2Quantify the risks.

"Root resorption can occur" is worth little; "severe resorption affecting a third or more of root length occurs in 1–5% of treated teeth, most often the maxillary lateral incisors" is worth a great deal.

3Demineralisation deserves its own paragraph with the prevention regimen

, including the 6–8 weekly professional fluoride.

4The recession point is counterintuitive and worth making:

retroclination in a high-angle Class III with a thin biotype is a greater risk than proclination.

5Relapse and age-related change are different things.

Say so and cross-reference Topic 20.

6Include the airway controversy.

It is current, it is a clean Theme 4 example, and the textbook's language is unusually direct.

BENEFITS AND RISKS — WITH THE RIGHT LEVEL OF CONFIDENCE

Dental aesthetics STRONG
Trauma prevention (RR 0.58) STRONG
OHRQoL: emotional and social MODERATE
Plaque scores MODERATE
Caries reduction WEAK
Periodontal health WEAK / NONE
Masticatory efficiency WEAK
Speech WEAK
Long-term psychological health LITTLE EVIDENCE
Nasal breathing (RME) NOT RECOMMENDED

THE THREE RISKS TO QUANTIFY

White spot lesions up to HALF of patients · within 6 months · commoner in males
Severe root resorption 1–5% of teeth · greatest in the anterior maxilla
Relapse near-universal · distinct from age-related change

Benefits and risks, weighted by the strength of the evidence. Ch 2, pp. 86–105.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 9. 1How do you explain the risk of white spot lesions to a 13-year-old and their parent? 2Which patients are at highest risk of severe root resorption, and what would you do about it? 3Does orthodontic treatment improve periodontal health? 4A patient asks whether treatment will help their child's speech. How do you

answer? 5What would make you stop treatment part-way through?

Topic 10. Medical conditions relevant to orthodontics

Revision Handbook: §29.3
Question Bank: Block AC (Q962–991)
Viva bank: V174–V183

IN ONE MINUTE

A careful medical history changes force, appliance, extraction and timing.

- NICE: routine antibiotic prophylaxis is NOT advised for dental procedures
- Light forces: sickle cell, thalassaemia, malignancy, diabetes
- Bisphosphonates slow tooth movement — simplify the aims

Where to read it

Ch 7, *Medical History*, pp. 356–366.

OPENING DEFINITION

A number of medical conditions may impact upon the provision of orthodontic treatment, and the orthodontist should have a working knowledge of their associated features and essential management (Ch 7, p. 356). The textbook lists them in a defined order, and reproducing that order gives your essay its structure.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

Organise by system. On this topic structure alone earns marks, because the content becomes a jumble otherwise.

1. Heart defects with risk of infective endocarditis.
2. Bleeding and haematological disorders.
3. Childhood malignancies.
4. Diabetes mellitus.
5. Organ transplantation and immunosuppression.
6. Epilepsy.
7. Asthma.
8. Allergies — latex and nickel.
9. Bisphosphonates.
10. Autism spectrum disorder; ADHD; eating disorders; body dysmorphic disorder; anxiety.
11. Conclusion — history-taking and liaison.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

CONDITIONKEY FIGURES AND ORTHODONTIC IMPLICATIONSPAGE

InfectiveIE incidence 3–10 per 1,100,000; **mortality close to 21%**; around 50% need **endocarditis**cardiac surgery. Higher risk: previous IE; any prosthetic valve or prosthetic repair material; any cyanotic heart disease; congenital defects repaired with prosthetic material — **for 6 months, or lifelong if a residual shunt or regurgitation remains**. **NICE advises antibiotic prophylaxis should not be routinely provided** — because there is no consistent association between procedures and IE, **regular toothbrushing almost certainly presents a greater risk than a single dental procedure**, effectiveness is unproven, and prophylaxis may cause more deaths through anaphylaxis. This conflicts with the AHA and ESC, which still recommend it for the highest-risk patients. Most orthodontics is low risk; possible exceptions are **separators and bands, miniscrews, debonding and interproximal reduction** — and even for these the bacteraemia incidence is very low. Judge each case and Ch 7, pp. 356– 358

	liaise with cardiology	
Haemophilia	A: X-linked recessive, **20 per 100,000 males**, factor VIII. B (Christmas disease): factor IX. Von Willebrand: vWF. Not contraindicated; liaise with haematology; **check appliances meticulously for sharp wires** because minor intraoral bleeds are a serious problem in this group; avoid extractions — if unavoidable, **desmopressin** in mild disease, **tranexamic acid**, or factor replacement for extractions and inferior dental blocks. Be aware of increased carriage risk for hepatitis and HIV	Ch 7, p. 358
Sickle cell disorder	HBB mutation; **21.7 per 100,000** in the UK; crisis lasting **3–10 days** precipitated by infection, dehydration or stress. Avoid extractions (osteomyelitis risk); if needed use **local rather than general anaesthesia, preferably without a vasoconstrictor**; **light forces with breaks in force application** to protect the microcirculation and reduce pulpal necrosis risk	Ch 7, pp. 358– 359
Thalassaemia	Alpha and beta, the latter more severe; transfusion-dependent with iron overload risk. **Bone marrow hyperplasia** enlarges the maxilla and skull, producing maxillary protrusion and a "chipmunk" appearance. **Light forces** because of thin cortical plates; **orthognathic surgery generally contraindicated** — bleeding from spongy cancellous bone and failure of rigid fixation	Ch 7, p. 359
Childhood malignancy	Affects ~12 per 100,000 children; **ALL accounts for around 80%** of leukaemias. Treatment causes **arrested root development with V-shaped roots, premature apical closure, microdontia, tooth aplasia**, reduced stature and craniofacial growth. **Delay until remission**; suspend and remove appliances if diagnosed during treatment; **avoid extractions** (osteonecrosis); light forces; short treatment. **Severe root resorption is a contraindication**	Ch 7, pp. 359– 360
Diabetes	Type I **187.7 per 100,000**; type II **5,260 per 100,000** and rising with	Ch 7,

	rigid fixation	
Childhood malignancy	Affects ~12 per 100,000 children; **ALL accounts for around 80%** of leukaemias. Treatment causes **arrested root development with V-shaped roots, premature apical closure, microdontia, tooth aplasia**, reduced stature and craniofacial growth. **Delay until remission**; suspend and remove appliances if diagnosed during treatment; **avoid extractions** (osteonecrosis); light forces; short treatment. **Severe root resorption is a contraindication**	Ch 7, pp. 359– 360
Diabetes	Type I **187.7 per 100,000**; type II **5,260 per 100,000** and rising with obesity. Greater periodontal susceptibility → excellent hygiene and regular periodontal maintenance. Diabetes and obesity may **accelerate tooth movement** via a pro-inflammatory periodontal state → **light forces**. Type I: **morning appointments, having eaten**	Ch 7, pp. 360– 361
Transplant / immunosuppression	**Delay at least 6 months** — the period of highest immunosuppression. Immunosuppressants cause marrow suppression, leucopenia, anaemia, oral mucositis. **Ciclosporin** causes gingival hyperplasia → **bonded tubes rather than bands**, chlorhexidine, possible gingivectomy	Ch 7, p. 361
Epilepsy	Prevalence **760 per 100,000**. **Avoid removable appliances if poorly controlled** — airway risk during seizures. Fixed appliances not contraindicated, but many need regular **cranial MRI**: stainless steel brackets degrade image quality or may dislodge, so **ceramic brackets or clear aligners**. **Phenytoin** causes gingival hyperplasia	Ch 7, pp. 361– 362

CONDITIONKEY FIGURES AND ORTHODONTIC IMPLICATIONSPAGE

AsthmaPrevalence **520 per 100,000** in the UK. Steroid inhalers → **palatal candidiasis, worsened by palate-covering removable appliances**; greater gingivitis risk; **some evidence of greater susceptibility to root resorption Latex allergy**Natural rubber latex contains **15 proven allergenic proteins**. Prevalence **~4.3% general population, 9.7% healthcare workers**; increased after surgery or catheterisation. Usually **type IV** contact dermatitis from manufacturing accelerators; **type I IgE-mediated** is less common but can cause anaphylaxis. Diagnose by skin prick or immunoassay; treat confirmed type I in a **latex-screened environment**; vinyl or nitrile gloves; avoid latex elastomerics — latex-free silicone elastics show **greater force decay** and need more frequent replacement **Nickel allergy**The **commonest contact allergy in industrialised countries**: approximately **10–30% of females and 1–3% of males**. Type IV, from jewellery and piercings. Oral reactions rare; signs non-specific — erythema, metallic taste, burning, angular cheilitis, soreness at the side of the tongue, severe gingivitis despite good hygiene. Diagnosis by **patch testing**. **Stainless steel is considered safe even in diagnosed nickel allergy** because its nickel content is relatively low; **nickel titanium archwires have a much higher content and should be avoided**; titanium or cobalt chromium nickel-free brackets are available **Bisphosphonates**Inhibit osteoclasts by inducing apoptosis; **very long half-life, active for many years**. Used for post-menopausal osteoporosis, malignant hypercalcaemia, and in children for post-transplant osteoporosis prophylaxis, osteogenesis imperfecta and fibrous dysplasia. **Slow or inhibit tooth movement, delay eruption**, compromise results, prolong treatment, impair bone healing, risk osteonecrosis after extraction. **Keep aims simple: avoid extractions, excessive space closure and surgical exposure** Ch 7, p. 362

Ch 7, pp. 362– 363

Ch 7, p. 363

Ch 7, p. 363

Autism spectrum disorder	Impaired social interaction and communication; stimming as emotional selfregulation; hypersensitivity to light, pain and noise; associated dyspraxia, depression, anxiety, epilepsy, eating disorders and a **strong gag reflex**. Higher incidence of malocclusion, possibly from persistent parafunction. Management: **same time, same clinician, same surgery**, minimal distracting stimuli, **gradual desensitisation**, **explicit instructions** because abstract concepts are difficult — and repetitive behavioural traits can be **used to optimise appliance wear**	Ch 7, pp. 363– 364
ADHD	Higher risk of caries and dentoalveolar trauma, poorer hygiene. Keep aims simple, avoid compliance-dependent modalities, **morning appointments**, keep them short, clear simple instructions	Ch 7, p. 364
Eating disorders	Worldwide prevalence now **just below 8%**; commonest in females aged 15–	Ch 7,

1. Oral signs: caries, **palatal erosion from self-induced vomiting**, hypersensitivity, reduced salivary flow, parotid and submandibular swelling, angular cheilitis, candidiasis, glossitis, aphthous ulceration. Anorexia causes **reduced bone turnover with osteopenia → slower tooth movement and poor healing**. Refer; delay while active; **advise against brushing after vomiting**; avoid surgery **Body dysmorphic**An obsessive-compulsive disorder of preoccupation with a minor or non- **disorder**existent appearance concern; associated depression, anxiety and suicidal tendency. Prevalence **0.7–2.5% general population, 5–13% of orthodontic patients, up to 17% seeking orthognathic surgery** p. 365 Ch 7, p. 365

EXAMINER TRAPS

The errors that cost marks on this topic.

1Organise by system.

An unstructured list reads as revision notes, not an essay.

2The NICE position is frequently answered wrongly.

Routine prophylaxis is **not** advised; give the four reasons; name the higher-bacteraemia procedures; note the disagreement with the AHA and ESC; and say that the absence of randomised trials means guidance rests on epidemiology and expert opinion.

3Do not say nickel allergy contraindicates orthodontics.

Stainless steel is safe; it is nickel titanium and nickel-containing brackets that are avoided.

4Bisphosphonates are the classic forgotten answer

— and their effect on the *rate of tooth movement* is the distinctively orthodontic consequence.

5For each condition, say what you would do

, not just what the disease is: force level, appliance type, extraction feasibility, timing, appointment scheduling.

6The neurodevelopmental group is not decoration.

Concrete adaptations — same clinician, same surgery, desensitisation, explicit instructions — are markable and show clinical maturity.

MEDICAL HISTORY: WHAT IT CHANGES

FORCE LEVEL — use light forces Sickle cell · thalassaemia · childhood malignancy · diabetes
EXTRACTIONS — avoid if possible Haemophilia · sickle cell · malignancy · bisphosphonates
APPLIANCE CHOICE Epilepsy → no removable; ceramic/aligner if MRI
Transplant/phenytoin → bonded tubes, not bands
Asthma → avoid palate-covering removable
Nickel allergy → no NiTi; steel is safe
TIMING — delay ≥6 months post-transplant · until malignancy in remission
· while an eating disorder is active
EXPECTATIONS Body dysmorphic disorder in 5–13% of orthodontic patients
NICE: routine antibiotic prophylaxis for dental procedures is NOT advised. Ch 7, pp. 356–366.

What the medical history changes. Ch 7, pp. 356–366.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 10. 1A patient on long-term bisphosphonates needs 6 mm of space closure. What do you do? 2A child with poorly controlled epilepsy needs an upper removable appliance. Would you provide one? 3Do you need antibiotic prophylaxis before placing separators in a patient with a prosthetic valve?
4Which medications commonly cause gingival hyperplasia, and how does that alter your appliance choice? 5How would you adapt a first appointment for a child with autism spectrum disorder?

Topic 11. Overbite and methods of overbite reduction

Revision Handbook: §25.1
Question Bank: Block Y (Q842–871)
Viva bank: V162–V173

IN ONE MINUTE

Reducing an overbite is easy. Holding it is the difficulty.

- Four movements: buccal extrusion (growing / non-growing), true intrusion, proclination
- Growth is the pivot — condylar growth makes relative intrusion possible
- Stability requires an occlusal stop (Houston, 1989)

Where to read it

Ch 13, *Vertical Problems* — increased overbite pp. 927–940. Mechanotherapy pp. 931–937. Dahl concept p. 940.

Opening definition

- An increased overbite occurs when the incisors have erupted past each other, and is often — though not always — associated with reduced vertical facial proportions (Ch 13, p. 927).
- Its reduction is necessary to establish a Class I incisor relationship.
- In Class II division 2 there is typically a hypodivergent growth pattern, reduced lower face height, a retroclined lower labial segment and an **increased curve of Spee arising from lower incisor overeruption in the absence of occlusal contact** with the upper incisors
- the arches tend to be short, wide and mildly crowded.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition and the typical Class II division 2 presentation.
2. Why growth status matters, and the stability requirement.
3. The four fundamental movements.
4. Buccal segment extrusion in the growing patient — relative incisor intrusion.
5. Buccal segment extrusion in the non-growing patient — and why it is undesirable.
6. True incisor intrusion — utility and bypass arches, force levels, miniscrews.
7. Incisor proclination — the arithmetic and the instability.
8. Appliance-specific points: removable bite planes, functional appliances, reverse curve of Spee, lingual appliances, Class II elastics, Begg/Tip-Edge.
9. The Dahl concept in adults.
10. Conclusion — growth status determines the method; stability determines the endpoint.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

The stability rule — learn this as a sentence

- Correction of the **interincisal angle** is essential, because an **occlusal stop** must be established to prevent the incisors erupting past each other again.
- This applies to Class II division 1 as well as division 2: unless maxillary incisor inclination is corrected, no occlusal stop is established and the overbite will relapse (Houston, 1989;
- Ch 13, p. 930). **A practical corollary most candidates miss.** Management of a deep overbite with a low maxillary–mandibular planes angle "is also made easier with a non-extraction approach", because as extraction space closes towards the end of treatment the incisors tend to upright and the overbite can deepen again (Ch 13, p. 930). **The four movements (Ch 13, pp. 931–940):**

METHOD	DETAIL
Buccal segment extrusion in the growing patient	Extruding the lower buccal segments while maintaining lower incisor position produces **relative incisor intrusion**, because vertical condylar growth compensates for the increased vertical dimension. Achieved with an **upper removable appliance with a flat anterior bite plane** (Adams cribs on first molars and first premolars, optional Southend clasp), by many **functional appliances**, or with a **reverse curve of Spee** in the lower archwire — which inevitably also proclines, so **lingual crown torque** is applied to counter it
Buccal segment extrusion in the non-growing patient	Produces **downwards and backwards mandibular rotation**, worsening a Class II skeletal pattern, and is prone to relapse
True incisor intrusion	Indicated with a **high MMPA and Class II**, where molar extrusion would rotate the mandible unfavourably. Mechanically difficult; requires fixed appliances and a buccal segment anchorage unit. **Ricketts-type utility arch**: engaged directly into the brackets, rectangular Elgiloy, a **two-couple** system. **Burstone-type arch**: lies above the incisor brackets which are tied together with a sectional wire, the point of ligation determining the moment — a **one-couple** system; originally rectangular stainless steel with a **double helix**, also made in TMA or pre-formed NiTi (the **Connecticut intrusion arch**), neither of which needs the helix

Incisor proclination **0.1–0.2 mm of overbite change per degree** of incisal angular change (Eberhart et al., 1990); lower incisor proclination reduces the overbite, retroclination increases it

Force and outcome figures worth quoting:

- Intrusion force **about 60 g total, or 15 g per tooth**, to reduce root resorption risk (Ch 13, p. 936).
- Segmental technique in non-growing patients achieves about **1.5 mm of maxillary and 1.9 mm of mandibular incisor intrusion** (Ng et al., 2005).
- **Miniscrews: 2–3 mm** of intrusion reported, with low-quality evidence of **0.67–0.95 mm greater intrusion** than segmental mechanics and less molar extrusion (Sosly et al., 2020; Shakti et al., 2022).
- Segmental techniques give **greater intrusion than continuous reverse curve of Spee mechanics** in both adolescents and adults, but with **more lower incisor proclination when a Ricketts utility arch is used** (Dake & Sinclair, 1989; Weiland et al., 1996).

- **Lingual appliances are particularly effective at overbite reduction**, with less proclination and greater incisor intrusion (Ch 13, p. 933).
- **Dahl concept (Dahl et al., 1975):** localised supra-occlusion producing intrusion of contacting teeth and eruption of unopposed teeth
- **re-establishment of occlusal contacts over an average of 6 months**
- safe and well tolerated provided the increase is only a few millimetres
- adverse events rare, minor and transient — but the evidence for the mechanism and long-term stability is poor (Ch 13, p. 940).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Give the definition and the numbers first

— the question asks "what is overbite?".

2The growing versus non-growing distinction is the pivot.

The same manoeuvre has opposite consequences in each, and an essay that does not separate them cannot score well.

3Distinguish relative from true intrusion explicitly. 4Quote the intrusion force and say why it is light

— intrusive force concentrates stress at the small surface area of the apices, raising resorption risk (cross-reference Topic 1).

5Say whether each arch is one-couple or two-couple.

Very few candidates do, and it demonstrates that you understand the mechanics rather than the names.

6End on stability, not on a list of techniques.

Reducing an overbite is straightforward; holding it is the difficulty, and that turns on the interincisal angle and the occlusal stop.

A Growing patient B Non-growing patient

Buccal segment EXTRUSION Buccal segment EXTRUSION

→ relative incisor intrusion → mandible rotates down/back

condylar growth compensates worsens Class II; unstable

C High MMPA / adult D Retroclined incisors

True incisor INTRUSION Incisor PROCLINATION

→ utility or bypass arch → interincisal angle reduces

~60 g total, 15 g per tooth 0.1–0.2 mm per degree; unstable

STABILITY: whichever route, an occlusal stop must be established or the overbite relapses (Houston, 1989).

The four ways to reduce an increased overbite. Compare Fig. 13.40, Ch 13, p. 934.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 11. 1Why is overbite reduction more difficult in an adult? 2What is relative intrusion, and what makes it possible? 3How would you reduce an

overbite in a patient with a high maxillary–mandibular planes angle? 4What determines whether a reduced overbite will be stable?
5What force would you use for incisor intrusion, and why so light?

Topic 12. Reducing overjet in a growing child

Revision Handbook: §22; the appliances §16
Question Bank: Block V (Q747–786)
Viva bank: V152–V161

IN ONE MINUTE

A timing question disguised as a technique question.

- The trials: no long-term advantage to early treatment, and more appointments
- The one consistent benefit is trauma reduction — RR 0.58, a 42% reduction
- Reserve early treatment for large overjets with incompetent lips, or serious teasing

Where to read it

Ch 9 (functional appliances, as Topic 8); Ch 12, *Skeletal Problems in the Mixed Dentition*, pp. 841–848; early overjet reduction and trauma: Where is the evidence 12.7, pp. 845–847; early Class II trials: Where is the evidence 9.6, pp. 588–589.

Opening definition

In a growing child an increased overjet can be reduced by methods that exploit residual growth alongside conventional dentoalveolar mechanics. The approach depends on the underlying skeletal pattern, the severity of the discrepancy, and the **timing of intervention** — which is the aspect of this topic that the examiners are really testing.

Essay skeleton

1. Framing: skeletal pattern, severity, timing.
2. Mandibular retrognathia → functional appliances; mechanism and options.
3. Headgear alone and in combination.
4. Dentoalveolar problems → tipping, Class II elastics, distalisation, skeletal anchorage.
5. Coexisting crowding → extraction-based space creation. Florida and North Carolina in the USA — more concern **headgear with an anterior bite plane**. Manchester in **modified Clark twin blocks**, in children aged **8–10 ye** *Early treatment did:* reduce the overjet throug **increase in self-esteem**; and increase profile att The American trials also found some skeletal chan *After all treatment was complete:***no real long-te in any of the studies**. Early-treated children w adolescence and had a shorter second phase, b **treatment time**, and the **late-treated children** greater reduction in PAR score and a slightly sm **class II malocclusion is not normally justified**
6. The case for early treatment.
7. The randomised trial evidence against it.
8. Trauma prevention — the one consistent benefit, and its caveats.
9. Timing relative to peak height velocity.
10. Conclusion — a pragmatic policy. ed with **efficacy**, using **bionators, headgear, or** the UK — more concerned with **effectiveness**, using **ars with a mean overjet of 10 mm**. h dentoalveolar change; produce a **transient** ractiveness (O'Brien et al., 2003a, 2003b, 2009b). ge but with wide individual variation. **rm differences between early and late groups** ere less

likely to need a functional appliance in ut had **more attendances and longer overall had a slightly better final occlusal result** — a aller overjet. Conclusion: **"early treatment for "** (O'Brien et al., 2009a).

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

The three landmark trials (Ch 9, pp. 588–589) Trauma — the quantified exception (Ch 12, pp. 845–847)

- Global dental trauma prevalence up to **18% in 12-year-olds**. An increased overjet is a significant risk factor.
- Pooled across the trials, **relative risk of new incisor trauma with early overjet reduction is 0.58 — a 42% reduction**, meaning trauma is almost twice as likely without early treatment.
- But four caveats matter:

1. **Wide confidence intervals** (RR range 0.38 to 0.88).
2. **Many children experienced trauma before the age of 9**, so overjet reduction would have to begin almost as soon as the incisors erupt.
3. Some children experienced **repeated trauma**, potentially inflating incidence in the trial populations.
4. **Early headgear was marginally more effective than a functional appliance** at reducing trauma risk. **The recommended policy (Ch 12, p. 846).** "A reasonable strategy would be to make decisions regarding early overjet reduction for trauma prevention on an individual basis. Children at particularly high risk of trauma (very large overjet, vertical excess, incompetent lips) might benefit from early intervention, whilst for others, preventive advice and the wearing of a mouthguard might suffice until overjet reduction is carried out in the late mixed or early permanent dentition."

Timing relative to growth (Ch 9, pp. 589–590)

Mandibular growth velocity peaks at or just after peak height velocity, so treating in that window is logical. But the onset, duration and intensity of the mandibular spurt vary greatly between individuals and precise correlations with peak height velocity are **far from established**; the supporting studies are heterogeneous and at **high risk of bias**, being largely retrospective with historical control data and positive selection of successful cases.

Where the problem is dentoalveolar

Simple tipping with a removable appliance and spring; fixed appliances with **Class II intermaxillary elastics** — remembering their extrusive and rotational side effects and the risk of leaving the upper labial segment excessively retroclined or the lower excessively proclined; **distalisation** with headgear or skeletal anchorage. Even with good growth, **correcting more than a half-unit Class II molar relationship by distalisation alone is difficult** (Ch 8, pp. 528–529).

Where crowding coexists

Extract at least as far forwards in the maxilla as in the mandible; the common Class II pattern is **maxillary first premolars with mandibular second premolars**; plan anchorage from the space analysis, remembering that two-thirds of the space is lost without support.

EXAMINER TRAPS

The errors that cost marks on this topic.

1This is a timing question disguised as a technique question.

Give the trial evidence a substantial section of its own.

2State the trial conclusion without hedging:

no long-term difference; longer treatment and more appointments if treated early; a slightly better occlusal result in the late-treated group.

3Separate the two timing arguments

— early versus late phase, and treatment relative to peak height velocity.

4Quantify the trauma benefit and then qualify it.

RR 0.58, a 42% reduction — followed by the four caveats, especially that most trauma occurs before age 9 and that headgear was marginally better than a functional appliance.

5Do not assume every increased overjet is a retrognathic mandible.

Proclined or spaced upper incisors are managed quite differently.

6The word "growing" in the question is a prompt.

Say explicitly what growth allows you to do that it would not in an adult.

EARLY CLASS II TREATMENT: WHAT THE TRIALS SHOWED

EARLY TREATMENT DID: EARLY TREATMENT DID NOT:

· reduce the overjet (dentoalveolar) · improve final skeletal outcome

· give a transient rise in self-esteem · improve final occlusal outcome

· improve profile attractiveness · shorten overall treatment

· REDUCE INCISOR TRAUMA · reduce total appointments

· RR 0.58 — a 42% reduction · the LATE group finished slightly better

THE PRAGMATIC POLICY

Treat early ONLY for children at high risk of trauma — very large overjet, vertical excess, incompetent lips — or significant teasing. Otherwise a single phase in the late mixed or early permanent dentition. Caveat: much trauma occurs before age 9, and headgear was marginally better.

Trials: Florida · North Carolina (USA) · Manchester twin block (UK). O'Brien et al., 2003, 2009; Tulloch et al., 1997, 2004.

What the early Class II trials showed. Compare Fig. 12.71, Ch 12, p. 847.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 12. 1A 9-year-old has a 12 mm overjet and incompetent lips. What do you do, and why? 2What did the Manchester twin block trial show, and how has it changed practice? 3How do you identify peak height velocity clinically? 4What are the disadvantages of two-phase treatment? 5If a patient presents at 15 with a 10 mm overjet and no growth left, how does your plan change?

Topic 13. Management of a missing maxillary central incisor

Revision Handbook: §20.6
Question Bank: Block T (Q677–721)
Viva bank: V140–V151

IN ONE MINUTE

No adjacent tooth substitutes for a central. Usually a prosthetic problem.

- Always attempt re-implantation — it maintains space AND alveolar bone
- Interim: denture (risks bone) versus space closure (risks further orthodontics)
- Defer implants until growth is complete or the crown ends up in infra-occlusion

Where to read it

Ch 12, *Early Loss of the Maxillary Central Incisor*, pp. 821–822. Related: absent maxillary lateral incisors, Ch 13, Where is the evidence 13.4, pp. 890–891; implant timing, Ch 13, pp. 896–900; extraction of maxillary central incisors, Ch 8, p. 521.

Opening definition

Traumatic loss of a maxillary central incisor is seen in around **3% of children** and usually occurs **unilaterally, in the mixed dentition, and in boys with an increased overjet** (Järvinen, 1978; Ch 12, p. 821). It is a harder problem than a missing lateral incisor because **a lateral incisor rarely makes a good unilateral substitute for a central**.

Essay skeleton

1. Epidemiology and why a central is harder than a lateral.
2. Immediate management — re-implantation, and why it is attempted regardless of prognosis.
3. Interim space management: the two strategies and their opposite trade-offs.
4. Definitive management of unilateral loss — resin- retained bridge or implant.
5. Timing of implant placement and infra-occlusion.
6. Autotransplantation of a premolar.
7. Bilateral loss and the space closure option.
8. Multidisciplinary planning.
9. Conclusion.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

- **Re-implantation should always be attempted if possible**, because regardless of long-term prognosis these teeth serve as a **useful maintainer of both arch space and alveolar bone** (Ch 12, p. 821).

- **The two interim strategies, with opposite trade-offs (Ch 12, p. 821):**
- *Maintain the space* with a partial denture or retainer with a pontic — simple, but "can be associated with a **loss of alveolar bone height**, which can make subsequent prosthetic replacement more difficult".
- *Allow the space to close and re-open it later* in the permanent dentition before prosthetic replacement — "allows **preservation of alveolar bone**, but it will require fixed appliance treatment and often space creation in the upper arch".
- **Definitive unilateral management:** prosthetic replacement with a **resin-retained bridge or implant**.
- **Implant timing (Ch 13, pp. 890, 896, 900).** "Implants should not be placed until facial growth is complete"
- premature placement leads to **infra-occlusion in the presence of continued vertical growth**, and there is also a risk of **peri-implantitis**. Waiting can mean a significant delay, and a minor degree of infra-occlusion "can be managed by occlusal build-up with composite".
- **Autotransplantation** of a premolar into the central incisor position with subsequent **coronal modification with composite** is an option where premolar extractions are already planned for the underlying malocclusion (Ch 12, p. 821).
- **Bilateral loss:** if space is required to reduce an overjet or relieve crowding **and the lateral incisors are of reasonable size and form**, they can be moved into the position vacated by the centrals — but "it can be difficult to obtain good aesthetics without modifying their coronal morphology restoratively" (Ch 12, p. 821).
- **Elective extraction of a central incisor** (as opposed to traumatic loss): maxillary central incisors are the teeth most commonly affected by trauma, and are not usually electively extracted, but if the prognosis demands it, **whether space closure will give an acceptable result depends on the form and shape of the lateral incisors**
- otherwise leave space for prosthetic replacement or autotransplantation (Ch 8, p. 521).
- **A useful comparator from the lateral incisor literature (Ch 13, pp. 890–891).** In the debate on absent maxillary lateral incisors, space closure with canine substitution and implant replacement both give acceptable outcomes
- in one comparison there were **no issues in the implant group except infra-occlusion of one implant**. The general principle — that the decision turns on the space requirement of the malocclusion as a whole, tooth form, and growth status — transfers directly.

EVIDENCE TO NAME

Cite these accurately, or not at all.

SOURCEWHAT TO BE ABLE TO QUOTE

Järvinen (1978)3% of children, unilateral, mixed dentition, boys with an increased overjet (Ch 12, p. 821). Thiruvenkatachari et al. (2017)smile aesthetics of substituted teeth (in the context of premolar-for-canine substitution, Ch 13, p. 886) — useful when arguing that a substituted tooth can look convincing if properly modified.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Say early why a central is different from a lateral.

This is the conceptual key: no adjacent tooth substitutes convincingly, so unilateral loss is almost always a prosthetic problem.

2Do not skip re-implantation.

Candidates jump to definitive management. The examiner wants the bone- and space-preservation argument articulated: even a tooth destined to fail is valuable in the interim.

3The infra-occlusion point is the highest-yield fact in the topic.

Give the mechanism (continued vertical alveolar growth), the consequence (implant crown left in infra- occlusion), and the management (defer until growth is complete; interim restoration; composite build-up for minor cases).

4Present the interim options as a trade-off, not a preference.

Denture preserves space but risks bone; space closure preserves bone but commits the patient to further orthodontics.

5Bilateral loss changes the answer.

Discussing only unilateral loss answers half the question.

6Mention the restorative colleague and a diagnostic wax-up or digital smile design.

Multidisciplinary planning is markable.

MISSING MAXILLARY CENTRAL INCISOR

Avulsion — ALWAYS attempt re-implantation (maintains space AND alveolar bone)

MAINTAIN THE SPACE LET IT CLOSE, RE-OPEN LATER

Denture or retainer with pontic Preserves alveolar bone

– risks alveolar bone loss – commits to further orthodontics

DEFINITIVE OPTIONS

UNILATERAL → resin-retained bridge or implant (a lateral rarely substitutes convincingly)

BILATERAL → move the laterals mesially if size and form allow. Premolar autotransplant if premolars are coming out anyway.

IMPLANTS: defer until growth is complete — continued vertical alveolar growth leaves the crown in INFRA-OCCLUSION.

Managing a missing maxillary central incisor. Compare Figs 12.58–12.59, Ch 12, pp. 821–822.

DENTAL TRAUMA: REST PERIOD BEFORE ORTHODONTIC FORCE

Ch 8, pp. 536–537. Frequently examined, easily forgotten.

Uncomplicated crown fracture 3 months

Subluxation 3 months

Lateral luxation — minor 3 months

Complicated crown fracture until a hard tissue barrier is seen

Lateral luxation — moderate/severe 12 months

Intrusion 12 months

Extrusion 12 months

RE-IMPLANTATION 12 months

Endodontically treated after trauma usually 12 months

Root fracture 12–24 months if dentine + cementum heal

Root fracture with connective tissue healing → treat as a short root, light forces.

MONITORING A TRAUMATISED TOOTH DURING TREATMENT

Before: vitality testing and periapical radiographs.

During: repeat radiographs 6–9 months after starting movement.

If resorption appears: pause 3 months (cf. 3–6 months for OIIRR generally, Ch 6, p. 306).

THREE COMPLICATIONS OF MOVING A PREVIOUSLY TRAUMATISED TOOTH

Loss of vitality · pulpal calcification · root resorption where resorption has already occurred.

Rest periods after dental trauma before orthodontic force. Ch 8, pp. 535–537.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 13. 1A 9-year-old avulses an upper central incisor. Would you re-implant it, and how does the trauma affect later orthodontics? 2When would you place an implant, and how do you decide growth is complete? 3What are the disadvantages of long-term denture wear in this situation? 4When is autotransplantation a realistic option? 5How would you decide between space opening and space closure?

Topic 14. Management of an impacted maxillary canine

Revision Handbook: §20.4
Question Bank: Block T (Q677–721)
Viva bank: V140–V151

IN ONE MINUTE

~3% impacted, up to 85% palatal. Position decides everything.

- Palpable by 10–11 years; a firm primary canine beyond 14 is a red flag
- Interception: 10–11 y, distal to the lateral root midline, 20–30°, no crowding, review at 12 months
- Poor prognosis above the apical third, at the midline, or beyond 55°

Where to read it

Ch 12, *Unerupted Maxillary Permanent Canine*, pp. 780–795; aetiology Box 12.4, pp. 781–782; interception Where is the evidence 12.4, pp. 793–794; predictors Fig 12.40, p. 796. Ch 13, exposure and alignment pp. 875–890; open versus closed Where is the evidence 13.3, p. 880; autotransplantation pp. 880–882.

Opening definition

- The maxillary permanent canine fails to erupt in around **3% of Caucasian children** (Ferguson, 1990).
- Deviation can be palatal or buccal, but in the majority of cases — **up to 85%** — these teeth are palatal.
- **A palatally displaced canine is ectopic and often fails to erupt, whereas a buccally displaced canine is generally regarded as crowded rather than truly ectopic and often does erupt** (Ch 12, p. 780).

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Epidemiology and the palatal/buccal distinction.
2. Aetiology — the five local factors plus genetic susceptibility.
3. Clinical examination and the radiographic prompts.
4. Localisation and imaging.
5. Interception — the criteria and the evidence.
6. Surgical exposure and orthodontic alignment; open versus closed.

- A developmental position high in the maxilla reliance on the **maxillary lateral incisor ro** diminutive, delayed or absent;
- **retention of the primary canine**; **late chro** premolars, so space is at a premium; and **gen**
- Peck's evidence for the genetic argument: ass reduction and delayed development;
- **bilateral occurrence 17–45%**; **up to three** occurrence; and **five times more common i**

1. Prognostic factors for successful alignment; anchorage.

2. Autotransplantation.
3. Extraction, and space management thereafter.
4. Leaving in situ with review.
5. Conclusion. giving a **long (>20 mm) path of eruption**; **ot for guidance**, which fails if that tooth is **nology** — the canine often erupts after the first **etic susceptibility**. ociation with tooth agenesis, crown-size **times more common in females**; familial **n Europeans than Asians**.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

Aetiology (Ch 12, pp. 780–782)

- The textbook's conclusion is that the palatally displaced canine "almost certainly represents a multifactorial condition". **Clinical examination (Ch 12, p. 781).** By **10–11 years** the canines should be **palpable in the buccal sulcus adjacent to the lateral incisor root**; if not, suspect an abnormal path and investigate radiographically (Ericson & Kurol, 1986). Other prompts: a **palatal bulge**; delayed eruption, mesial angulation, retroclination, microdontia or absence of the lateral incisor; and a **firm primary canine, particularly beyond 14 years**, indicating a lack of resorption.

Imaging

Conventional radiography has an accuracy of **39–85%** in localising maxillary canines (Ch 12, p. 788) — a striking figure and an argument for parallax and, where genuinely needed, CBCT.
Interception — the criteria (Ch 12, p. 792):

1. Patient aged **10–11 years** and in the mixed dentition.
2. Canine positioned **distal to the midline of the lateral incisor root and 20–30° to the mid-sagittal plane** — the closer to the midline, the less successful.
3. **Absence of crowding** in the maxillary arch.
4. If radiographic improvement is **not evident within 12 months**, consider further treatment.

Interception — the evidence (Ch 12, pp. 793–794):

- **Ericson & Kurol (1988)** — normalisation in the majority if carried out between **10 and 13 years** with space available; but no control group.
- **Naoumova et al. (2015)** — a higher-quality randomised trial: **9 of 23 erupted at 1 year after primary canine extraction versus 3 of 22 without**; but **7 of 23 in the extraction group still needed surgical exposure versus 11 of 22** in the control group.
- **Benson et al. (2021), Cochrane** — in children aged 9–14, only **very weak evidence** that primary canine extraction increases the probability of eruption, and **no evidence that it reduces the number of children requiring surgery**. No evidence that double extraction is better than single.
- **RCS England guidelines** support interception in selected cases — **10–13 years of age, space available in the arch, and a maximum of 12 months for radiographic evidence of improvement**.
- **Naoumova's predictive work** found that the **more upright the canine, the lower its height, and the greater its distance from the midline, the more likely it is to erupt regardless of extraction**

- eruption is more successful in younger patients → and **extracting the primary canine is the single variable that affects eruption of a PDC the most**. The **alpha angle** to the mid-sagittal plane and then **sector position** are the most influential positional predictors (Ch 12, p. 796).

Surgical exposure and alignment (Ch 13, pp. 875–880)

- Buccal canines below the mucogingival junction: **apically repositioned flap** gives the best chance of eruption through attached gingiva (Kokich, 2004).
- Palatal canines: open or closed; **Parkin et al. (2017) Cochrane** found no differences in success or operative time, and no differences in aesthetics, periodontal outcomes, pain or treatment time in individual studies, on low to very low quality evidence. **Cassina et al. (2018)** did find open exposure associated with a **reduction of just over 2 months to initial alignment** and **significantly lower odds of ankylosis**. "Overall, it would seem there is little to choose between the two techniques."
- **Sequencing tip worth quoting:** a palatal canine "should be moved distally in the first instance to take the crown away from the maxillary incisor roots before it is moved in the direction of the dental arch", and consideration should be given to **initially avoiding placing a bracket on the lateral incisor** so its root is not moved into the canine (Parkin et al., 2019).
- **Anchorage:** the process is anchorage-demanding; a trans-palatal arch provides **vertical and some anteroposterior** anchorage, and miniscrews are increasingly used either to supplement it or to apply direct force to the canine. **Prognostic factors (Fig 13.12, Ch 13, p. 878).** Prognosis worsens as the **height above the occlusal plane increases**, particularly **above the apical third of the incisor roots**; as the **distance to the midline reduces**, particularly **beyond the midline of the lateral incisor**; and as the **angle of the long axis to the midline vertical plane exceeds 55°**.

Autotransplantation (Ch 13, pp. 880–882)

- Good outcomes of around **88%**. Removal must be as atraumatic as possible to avoid ankylosis — difficult, because the canines that most need it are in the worst positions.
- The canine should be kept **out of occlusion and semi-rigidly splinted for a maximum of 3 weeks**, and **root canal treated once the splint is removed**.
- It can be "parked" under the buccal mucosa while orthodontic space creation is undertaken.
- Susceptible to ankylosis and external root resorption, with a **reduced long-term prognosis compared with canines aligned orthodontically**, and highly operator-dependent. **Leaving in situ (Ch 13, p. 886).** Acceptable if the patient is happy and declines treatment, provided the canine is **not closely associated with the erupted dentition**, there is **no pathological change or root resorption**, and there is **regular radiographic review** in the growing patient for incisor root resorption and, longer term, follicular enlargement or cyst formation.

If the canine is extracted (Ch 13, pp. 882–885)

- Ideally close the space and establish a contact between the lateral incisor and first premolar.
- The **first premolar makes an excellent substitute** if modified: place the root more buccally to create a canine eminence;
- **rotate the crown mesiopalatally** to increase mesiodistal width, hide the palatal cusp and improve the relation with the lower canine; grind the palatal cusp; and intrude the premolar to raise the gingival margin, building up the buccal cusp with composite (Box 13.2, Ch 13, p. 886).

- **Avoid canine extraction where there is severe resorption and poor prognosis of an incisor** — accommodate the canine instead.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Give the diagnostic timings.

Palpable by 10–11 years; a firm primary canine beyond 14 years.

2Give the interception criteria as a numbered list

, with the 12-month review. Note the discrepancy between the textbook's 10–11 years and the RCS's 10– 13 — mentioning both shows you have read the guidance rather than one summary.

3Be honest about the interception evidence.

Cite Naoumova's numbers *and* the Cochrane conclusion *and* the guideline position.

4Do not claim open exposure is superior.

Broadly equivalent, with a possible advantage in time to alignment and ankylosis risk on lower-quality evidence.

5Quote the three prognostic factors with their numbers. 6Do not forget the anchorage burden

— cross-reference Topic 5.

PROGNOSIS FOR ORTHODONTIC ALIGNMENT WORSENS AS…

midline

1 HEIGHT

worse above the apical third of the incisor roots

3

2 HORIZONTAL POSITION

worse at or across the lateral incisor midline

3

3 ANGULATION

apical third

worse above ~55° to the midsagittal plane

3

INTERCEPTION CRITERIA

321

10–11 y (RCS: 10–13) · distal to lateral root midline

20–30° to midsagittal · no crowding · review at 12 mo

~3% impacted; up to 85% palatal; eruptive path >20 mm

Prognostic factors for aligning a palatally displaced canine. Compare Fig. 13.12, Ch 13, p. 878.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 14. 1A 10-year-old has no buccal canine bulge. What do you do next? 2How do you localise an unerupted canine? When would you request CBCT rather than parallax films? 3What is the evidence for extracting the primary canine, and would you still do it? 4Open or closed exposure — how do you choose? 5At what point would you advise a patient that alignment is not worth attempting?

Topic 15. Management of an impacted lower second molar

Revision Handbook: §20.5
Question Bank: Block T (Q677–721)
Viva bank: V140–V151

IN ONE MINUTE

Just over 2%, mostly mandibular. Escalate step by step.

- Observe 6–12 months → disimpact → remove the third molar → surgical uprighting
- Extraction of the second molar gives the LEAST successful outcome
- Exclude primary failure of eruption — traction there causes ankylosis

Where to read it

Ch 12, *Unerupted Second Permanent Molar*, pp. 795–796. The analogous first molar impaction: pp. 793–794. Primary failure of eruption: pp. 800–801.

Opening definition

Unerupted second permanent molars are relatively rare, seen in **just over 2% of the population**, and most commonly affect the **mandibular arch**. The condition "is often not identified until the permanent dentition is established or when orthodontic treatment commences" (Ch 12, p. 795) — which is why routine radiographic vigilance in the late mixed dentition matters.

Essay skeleton

1. Epidemiology and the diagnostic delay.
2. The radiographic warning sign in the mixed dentition.
3. Aetiology.
4. Management principle — stepwise escalation.
5. Observation for mild or potentially reversible impaction.
6. Simple disimpaction.
7. Third molar removal.
8. Surgical exposure with luxation and orthodontic uprighting.
9. Extraction of the second molar — last resort.
10. The differential: primary failure of eruption.
11. Conclusion.

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

- **Prevalence just over 2%**, predominan
- **Aetiology:** most commonly impaction **m against the developing third molar, genetic basis, at least in the mandi** obvious cause of impaction and occasio

- **The warning sign to quote:** in the m **occlusal table of the developing ma** raises suspicion of future impaction (Fi
- Ch 12, p. 795).
- **The management ladder: Observation**Mild, potentially reversible tly mandibular (Ch 12, p. 795). **esially against the first permanent molar, distally or a combination**. There is **some evidence of a bular arch** (Shapira et al., 2011). "There is not always an nally some emergence can take place into the jaw." ixed dentition, **a third molar follicle situated above the ndibular second molar on a panoramic radiograph** g 12.44, redrawn from Kravitz et al., 2016 The **6–12 month rule**, borrowed from first permanent molar impaction: if spontaneous eruption does not occur within that window, the impaction is considered **established or irreversible** and intervention is indicated (Ch 12, p. 793)

Learn the numbers. This is where depth marks are won.

STEP INDICATION DETAIL

STEP	INDICATION	DETAIL
Simple disimpaction	Mild-to-moderate mesial impaction	**Separating elastic, spring or brass wire placed below the contact point** between the two molars — directly analogous to the technique for a first permanent molar impacted against a primary second molar, where **distal grinding of the primary molar** is also an option (Ch 12, p. 793)
Third molar removal	Where the third molar contributes	"A common strategy is extraction of the third molar and surgical exposure of the second molar, with or without post-surgical orthodontic uprighting"

STEP	INDICATION	DETAIL
Surgical uprighting with orthodontic alignment	Established impaction	Following surgical exposure the **third molar is removed and the second molar luxated within its socket to achieve disimpaction; a bracket is bonded at the time of surgery** and alignment completed with a fixed appliance (Pogrel, 1995; Kravitz et al., 2016)
Extraction of the second molar	Severely displaced, ankylosed, poor prognosis, or unwilling patient	**The least successful option.** Longitudinal follow-up suggests that "in many cases, failure of eruption remains even after intervention, although **extraction of the second permanent molar itself often results in the least successful outcome**" (Magnusson & Kjellberg, 2009)

- **The important differential — primary failure of eruption (Ch 12, pp. 800–801).** Diagnosed when there is no clear systemic, pathological or environmental factor preventing eruption. Associated with **loss-of-function mutation in PTH1R**, which encodes a parathyroid hormone receptor with a role in osteoclast function during eruption (Decker et al., 2008). Critically, **active orthodontic extrusion will normally result in ankylosis and failure** — so management is extraction with space closure or prosthetic replacement, localised bony osteotomy with segmental extrusion, or coronal build-up if some eruption has occurred. Recognising this differential is what separates a good answer from an adequate one, because attempting orthodontic traction on a tooth with PFE actively harms the patient.
- **The analogous first permanent molar impaction** (against the primary second molar) is worth citing for the technique: prevalence **0.75–4.3%**, more common in males, **25 times more common in the maxilla than the mandible**, and a good indicator of posterior crowding (Ch 12, p. 793).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Structure as an escalating ladder.

This topic is short on content relative to others, so structure carries proportionally more of the mark.

2Give the observation window and name the decision point

— established or irreversible impaction after 6–12 months.

3Name the disimpaction devices and say they go below the contact point.

Drawing the analogy with first permanent molar impaction shows conceptual understanding rather than rote recall.

4Include primary failure of eruption as a differential.

It changes management entirely and is the sort of point that lifts an answer into a high band.

5Do not present extraction as a neutral alternative.

The evidence is that it produces the least successful outcome, and it makes the patient dependent on unpredictable third molar eruption.

6Mention early detection.

The panoramic warning sign gives you a preventive dimension most candidates omit.

IMPACTED LOWER SECOND MOLAR — THE ESCALATING LADDER

1 OBSERVE

6–12 months. No progress = ESTABLISHED

mild, likely reversible

2 DISIMPACT

separator, brass wire or open coil BELOW the contact point

mild–moderate mesial

3 REMOVE THE 8

relieves the obstruction

third molar contributing

4 SURGICAL UPRIGHTING

expose · remove 8 · luxate · bond bracket at surgery · align

established impaction

5 EXTRACT THE 7

LEAST successful outcome; commits to unpredictable 8 eruption

ankylosed / poor prognosis / unwilling

DIFFERENTIAL: primary failure of eruption (PTH1R). Orthodontic traction causes ankylosis — do not attempt it.

Prevalence just over 2%. Warning sign: a third molar follicle above the occlusal table of the developing 7. Ch 12, p. 795.

The escalating ladder for an impacted lower second molar. Compare Fig. 12.45, Ch 12, p. 801.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 15. 1What would make you observe rather than intervene? 2How does a separator disimpact a molar? 3Would you remove the third molar, and at what stage? 4Why is orthodontic alignment preferred to surgical uprighting alone? 5What happens if you simply extract the impacted second molar?

Topic 16. Methods of crossbite correction

Revision Handbook: §24; appliances §15.5
Question Bank: Block X (Q817–841)
Viva bank: V162–V173

IN ONE MINUTE

Classify first: dental, functional or skeletal. That decides the appliance.

- Unilateral + displacement = dental or mild skeletal; bilateral, no shift = skeletal
- Cochrane: quadhelix or plate beats observation at 7–11 y; quadhelix ~3 months faster
- RME gives ~50% skeletal expansion; ~10% relapses within 4–8 months

Where to read it

Ch 12, *Crossbites in the Mixed Dentition*, pp. 833–841; Cochrane evidence Where is the evidence 12.6, p. 836. Ch 9, maxillary expansion pp. 542–558; RME versus MARPE Where is the evidence 9.2, pp. 546–547. Ch 13, *Transverse Problems*, pp. 949–955; slow versus rapid expansion Where is the evidence 13.7, pp. 952–953.

Opening definition

- A crossbite exists when one or more maxillary teeth occlude palatally or lingually to their normal relationship with the mandibular dentition.
- It may be **anterior or posterior**, **unilateral or bilateral**, and **dental, functional (with a mandibular displacement) or skeletal** in origin.
- Correction is generally indicated early where there is an associated displacement or periodontal damage, since these otherwise perpetuate from the primary into the mixed and permanent dentitions.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition and the three axes of classification.
2. Indications for early correction; eliminating a mandibular displacement.
3. Posterior crossbite — what unilateral versus bilateral tells you.
4. The Cochrane evidence.
5. Removable expansion appliances.
6. Fixed tooth-borne expanders.

- A **unilateral posterior crossbite with a ma** retruded contact to intercuspal position usua
- A **bilateral crossbite without displacemen** skeletal discrepancy.
- Early removal of premature contacts associat posterior crossbite being perpetuated from th dentitions" (Ch 8, p. 487).

1. Slow versus rapid expansion — the mechanisms and the evidence.
2. Miniscrew-assisted expansion.
3. Anterior crossbite — aetiology and correction.

4. The adult: SARPE and orthognathic surgery.
5. Retention and relapse.
6. Conclusion. **ndibular displacement** on closing from lly reflects a dental or mild skeletal discrepancy. **t** suggests a more significant transverse ed with primary teeth ”can reduce the risk of a e primary into the mixed and permanent

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

The diagnostic pearl

The Cochrane evidence (Ugolini et al., 2021; Ch 12, p. 836). 31 randomised trials, approximately 1,400 patients.

- For children aged **7–11 years**, a **quadhelix or expansion plate is beneficial compared with observation** (high-certainty evidence). **HYRAX is probably effective** compared with observation (moderate certainty).
- A **quadhelix is more effective than an expansion plate** (moderate certainty) and takes **around 3 months less time** (moderate certainty).
- **No evidence of any difference** between HYRAX and Haas (moderate certainty); HYRAX and tooth-borne; or HYRAX and bone-borne expansion (both low certainty).

Appliances (C	**h 12, pp. 836–838; Ch 13, pp. 949–955):**
APPLIANCE	DETAIL
Removable expansion plate	Midline screw with **posterior acrylic capping** to disclude the dentition. Activated **twice a week, 0.25 mm per turn**, until **overcorrected** — palatal cusps of the upper buccal teeth occluding on the incline of the buccal cusps of the lower. Then remove the capping and wear at night as a retainer for **3–6 months**
Quadhelix / trihelix	Fixed, tooth-borne. Chaconas & de Alba (1977). Effective and **around 3 months faster** than a plate
HYRAX / Haas	Jackscrew appliances; the **Haas** additionally has acrylic covering the palatal vault to increase skeletal change
MARPE	Hybrid tooth-and-bone-borne or exclusively bone-borne; capable of splitting the suture in the mixed dentition and a **viable alternative to SARPE in adults**
Fixed appliance expansion	Expanded heavy stainless steel archwires — **use rectangular wire to prevent buccal tipping and maintain torque control**. Or a 0.9 mm auxiliary buccal arch in the molar headgear tubes. Or **cross-elastics**, which also produce a vertical force and are **contraindicated where buccal segment extrusion should be avoided**. Bracket type has little effect on expansion — arch form, crowding and archwire matter more
Slow versus ra	**pid expansion (Ch 13, pp. 952–953):**

expansion	headgear tubes. Or **cross-elastics**, which also **contraindicated where buccal segment ex** little effect on expansion — arch form, crowdi	produce a vertical force and are **trusion should be avoided**. Bracket type has ng and archwire matter more
Slow versus r	**apid expansion (Ch 13, pp. 952–953):**	
	SLOW (SME)	RAPID (RME)
Force	Low, continuous	Heavy, intermittent — **up to 10 kg**
Activation	One-quarter turn every 2 days	**One to four times per day, 0.25–1 mm daily**
Duration	—	**2–4 weeks**, then the appliance is left in place **3–6 months**
Outcome	Equally effective for dental expansion; **95% of initial expansion maintained** during retention	**Slightly greater posterior skeletal expansion but greater buccal tipping; 85% maintained**
Patient	Less painful in the first week	More painful in the first week

experience

Both show reasonable stability over 2–14 years on low-level evidence. Both are associated with **bone height loss and reduced buccal bone thickness** — over 1 mm for RME, more for SME.

RME quantities (Ch 9, pp. 543–545)

- Splits the suture and achieves **up to 4 mm of widening**, representing **around 50% of total screw expansion** (Liu et al., 2015).
- Short-term: +2.46 mm maxillary basal bone width; +3.09 mm alveolar palatal crest; **+5.69 mm intermolar width**; +3.75° dental tipping.
- **Around 10% of skeletal change relapses at 4–8 months post-retention.**
- **Sutural fusion generally occurs by the late teenage years** (Melsen, 1975), and the suture becomes increasingly interdigitated with age — but **correlation between sutural maturation and chronological age is poor**. Importantly, **slow expansion with a removable plate, trihelix or quadhelix can readily split the suture in the early mixed dentition**
- it is the late mixed and early permanent dentition that require a higher-force approach.
- **MARPE** gives greater skeletal expansion than conventional RME (about +1.1 to +2 mm more at various landmarks), with **reduced buccal inclination** of premolars and molars (–6.06° and –3.17°), but is more invasive, more expensive, and rests on a **low-level evidence base**. A recent high-quality randomised trial found **limited differences at 5 years** between conventional and hybrid RME (Bazargani et al., 2023). **Anterior crossbite (Ch 12, pp. 838–841).**
- May involve one or several teeth; associated with a local eruption disturbance — often a supernumerary in the anterior maxilla — or crowding, or a Class III incisor relationship.
- **The indication for mixed dentition treatment is gingival recession of a lower incisor from a displacement on closing.** Correct with a removable appliance and palatal **Z- or T- spring**

plus posterior capping where space allows, or with a **2×4 sectional fixed appliance** where bodily movement, space creation or multiple tooth correction is needed.

- Note the stability point: **correction of an anterior crossbite with a positive overbite afterwards is usually stable and requires no retention** (Wiedel & Bondemark, 2015b;
- Ch 13, p. 977).

The adult

Increased sutural interdigitation makes skeletal expansion harder. **SARPE** for more severe cases; bone-borne appliances now allow skeletal expansion in adults without surgery (Ch 13, p. 952).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Classify before you treat.

The marks are in establishing dental versus functional versus skeletal, because that determines everything.

2Quote the Cochrane findings precisely

, including the ”no clear difference between fixed expander types” point, which most candidates omit and which demonstrates you read the review rather than a summary.

3Do not oversimplify the slow/rapid distinction.

Saying that slow expansion can also split the suture in the early mixed dentition, and that the constraint is age-related sutural interdigitation rather than the appliance category, is a genuinely discriminating point.

4Give the activation regimens and the retention periods.

Twice weekly at 0.25 mm; one to four times daily over two to four weeks; three to six months of retention in both cases.

5Always mention overcorrection and retention

— a Theme 5 opportunity, and quantify the relapse (10% of skeletal change at 4–8 months; 85–95% of dental expansion maintained).

6Do not forget the adult.

SARPE, and increasingly bone-borne expansion as an alternative to it.

CHOOSING AN EXPANSION APPROACH Unilateral crossbite WITH displacement?

YES NO — bilateral, no shift

Dental or mild skeletal True transverse skeletal deficiency

Removable plate: 0.25 mm × 2/week RME: 1–4 turns/day over 2–4 weeks

Quadhelix: ~3 months faster (Cochrane) Forces up to 10 kg · retain 3–6 months

Overcorrect · retain nights 3–6 months ~50% of screw expansion is skeletal

High-certainty evidence, ages 7–11 ~10% skeletal relapse at 4–8 months

GROWTH COMPLETE? SARPE beyond ~5 mm, or bone-borne expansion. Sutural fusion by the late teens,

but correlation between sutural maturation and chronological age is poor.

Note: slow expansion CAN split the suture in the early mixed dentition — the constraint is sutural

interdigitation with age, not the appliance category (Ch 9, p. 543).

Choosing an expansion approach. Compare Fig. 9.2, Ch 9, p. 544.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 16. 1How do you distinguish a functional from a true skeletal crossbite? 2Why is a unilateral crossbite with displacement usually not skeletal? 3What proportion of RME expansion is skeletal, and what happens to the rest? 4Why does a midline diastema appear during RME, and what do you tell the patient? 5When would you use a 2×4 appliance rather than a removable appliance for an anterior crossbite?

Topic 17. Correcting a Class III incisor relationship

Revision Handbook: §23
Question Bank: Block W (Q787–816)
Viva bank: V152–V161

IN ONE MINUTE

Monitoring is often the right answer. Growth declares the discrepancy.

- Favourable early cases need a positive overbite and an anterior displacement
- Facemask: >60% avoided surgery versus 36% (Mandall, 2016)
- Decompensation deliberately makes the reverse overjet WORSE before surgery

Where to read it

Ch 8, prognostic indicators Table 8.4, p. 505. Ch 9, maxillary protraction pp. 542–559; Where is the evidence
9.3, p. 557. Ch 12, Skeletal Problems in the Mixed Dentition, *pp. 841–848. Ch 13,* Antero-posterior Problems —
Class III pp. 916–925; Where is the evidence 13.6, pp. 920–921. Ch 15, surgical management.

Opening definition

A Class III incisor relationship exists where the lower incisor edges occlude anterior to the cingulum plateau of the upper incisors, often with a reduced or reverse overjet. It usually reflects an underlying Class III skeletal pattern — maxillary retrusion, mandibular prognathism, or both — but may also arise as a **pseudo- Class III** relationship secondary to a forward mandibular displacement on closing in a skeletal Class I patient.

Essay skeleton

1. Definition; skeletal versus pseudo-Class III.
2. Why monitoring is often correct in the mixed dentition.
3. The prognostic indicators — which cases are favourable.
4. Maxillary protraction with a facemask, with or without RME.
5. Bone-anchored maxillary protraction.
6. Class III functional appliances. **Prognostic indicators (Table 8.4, Ch 8, p. 50 core of the essay:**
7. Chin cup and mandibular restraint.
8. Fixed appliance interception — the 2×4.
9. The permanent dentition: camouflage.
10. The permanent dentition: decompensation and orthognathic surgery.
11. Deciding between them — the discriminant analyses.
12. Conclusion. **5) — reproduce this table and you have the**

The numbered headings. Reproduce these and the essay writes itself.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

INDICATIONS FOR ORTHODONTIC TREATMENT	CONTRAINDICATIONS
Class I or **mild** Class III skeletal base	**Moderate to severe** Class III skeletal base
Average or reduced lower face height	**Increased** lower face height
Average or increased overbite	**Reduced** overbite
Proclined mandibular incisors	**Retroclined** mandibular incisors
Upright or retroclined maxillary incisors	**Proclined** maxillary incisors
Anterior displacement on closing from RCP to ICP	**No** anterior displacement on closing
	A significant **Class III molar** relationship

The underlying logic: the indications describe a case where there is **room to decompensate in the direction you need**, and the contraindications describe one where the dentition has already compensated as far as it can. Stability "not only depends on growth but also on achieving a **positive overbite and overjet** at the end of treatment" (Ch 8, p. 505).

Facemask protraction (Ch 9, pp. 552–558; Ch 12, p. 848)

- Worn **12–14 hours each day**.
- Combining with simultaneous **RME** can improve the advancement achievable by "freeing up the suture system".
- Most effective **before the age of 10 years**, and successful in the short term in the majority of cases.
- **The key long-term outcome (Mandall et al., 2016; Ch 8, p. 487):** while most of the initial skeletal effects are lost with growth, the **success rate in terms of avoiding the need for surgical correction was over 60%, compared with 36% when early treatment is not undertaken**. This is one of the most quotable numbers in the whole syllabus.
- **Who outgrows the benefit:** patients with a **hyperdivergent facial pattern, increased gonial angle and mandibular prognathism**. The difficulty is predicting which cases these will be at the age of 8 or 9.
- **Bone-anchored maxillary protraction (BAMP)** uses Class III intermaxillary traction between miniplates in the anterior mandible and infrazygomatic crests; evidence for efficacy against established techniques is "largely limited to case reports" (Ch 13, p. 920). **Functional appliances for Class III (Ch 12, p. 846).** Their effects are restricted to **upper incisor proclination, lower incisor retroclination, and backwards and downwards rotation of the mandible** — so they can treat only the milder cases. The **FR III** is "bulky, prone to breakage and difficult to wear"; a **reversed twin block** design is also used.

Fixed appliance interception

A **2×4** appliance — attachments on the molars and the four upper incisors — with a **compressed loop or pushcoil** in the buccal segments to procline the upper labial segment.

Camouflage in the permanent dentition (Ch 8, p. 516; Ch 13, pp. 916–925)

Typically **mandibular first premolars with maxillary second premolars**; or mandibular premolars alone

finishing to a Class III molar relationsh **incisor** in milder cases. Any mandibul	ip if the maxillary arch is well aligned; or **a single mandibular** ar extraction in a Class III case should be planned with care.
Deciding between camouflage a 921). Threshold values below which	**nd surgery (Where is the evidence 13.6, Ch 13, pp. 920–** surgery is more likely to be needed:
PARAMETER	THRESHOLD
Wits	≤ –11 mm — **one of the most important parameters**
Holdaway angle	≤ 3.5°
ANB	≤ –4°
Gonial angle	≤ 120°
Lower anterior face height	≥ 66%
Overjet	≤ –5 mm
Mandibular incisor inclination	≤ 80°

Stellzig-Eisenhauer's discriminant analysis achieves **over 90% success** in predicting the need for surgery, combining the Wits value, the maxilla-to-mandible length ratio, the saddle angle (NSAr) and the mandibular midline deviation from the study casts (Kochel et al., 2011).

Surgical management (Ch 13, p. 921)

- Class III malocclusion **tends to worsen with age** as mandibular growth persists, so **surgery should be delayed until all adolescent growth has ceased — the late teens**.
- A crowded maxillary arch may still be treated in the interim if this concerns the patient, even with extractions, provided the upper labial segment is left correctly positioned on its dental base.
- **In the mandibular arch, extractions are best avoided in growing patients who may need surgery**, because any retroclination of the lower labial segment must be undone during pre-surgical decompensation, and that is difficult once the extraction sites have closed.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Decompensation is the concept that separates candidates.

Pre-surgical orthodontics deliberately makes the reverse overjet **worse**, because the incisors must be placed at their correct inclination on their own skeletal bases for the surgeon to achieve full correction.

2Reproduce Table 8.4 as a set of criteria

, including the requirement for a **positive overbite** — which candidates routinely omit and which is essential to stability.

3Quote the Mandall figure

— 60% versus 36% avoidance of surgery — and immediately state the difficulty: predicting which 8-year- old will benefit.

4Be appropriately sceptical about chin cups

, and about BAMP, where the evidence is case-report level.

5The Class III camouflage extraction pattern is the mirror image of Class II

— further forward in the mandible. Cross-reference Topic 4 and get the direction right.

6Do not treat "monitor" as a non-answer.

In the mixed dentition it is often correct, because the discrepancy only declares itself with growth; and the reason to avoid mandibular extractions in a potential surgical case is a strong, specific point.

CLASS III: THE DECISION PATHWAY MIXED DENTITION — is the case FAVOURABLE?

FAVOURABLE (Table 8.4) UNFAVOURABLE

· Class I or mild Class III base · moderate–severe Class III base

· average or reduced lower face height · increased lower face height

· average or INCREASED overbite · REDUCED overbite

· proclined lower / upright upper incisors · retroclined lower / proclined upper incisors

· anterior displacement on closing · no displacement · Class III molars

Early intervention Monitor into the late teens

Facemask ± RME, 12–14 h/day, before age 10 Class III worsens with growth. Avoid mandibular

>60% avoid surgery vs 36% (Mandall, 2016) extractions if surgery is likely.

PERMANENT DENTITION — CAMOUFLAGE OR SURGERY?

Surgery more likely if: Wits $\leq$ –11 mm · ANB $\leq$ –4° · overjet $\leq$ –5 mm · Holdaway $\leq$ 3.5°

gonial $\leq$ 120° · LAFH $\geq$ 66% · lower incisor inclination $\leq$ 80°

The Class III decision pathway. Compare Table 8.4, Ch 8, p. 505.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 17. 1Why do we decompensate before Class III surgery? 2How do you distinguish a true skeletal Class III from a pseudo-Class III? 3Which Class III patients respond best to facemask therapy? 4When is camouflage no longer appropriate? 5Why is a chin cup rarely used now?

Topic 18. Malocclusion and temporomandibular disorder

Revision Handbook: §26
Question Bank: Block Z (Q872–901)
Viva bank: V174–V183

IN ONE MINUTE

An evidence-appraisal question, not a joint question.

- Weak associations exist; evidence of causation does not
- Orthodontics neither causes nor prevents TMD (AAO studies, 1992)
- Screen at baseline, counsel on natural history, manage conservatively and reversibly

Where to read it

Ch 2, *Temporomandibular Joint Dysfunction*, pp. 101–105; the historical account is Where is the evidence 2.3, pp. 104–105. Ideal functional occlusion and mutual protection: Ch 2, pp. 58, 62.

Opening definition

- Temporomandibular joint dysfunction is a common condition associated with a range of signs and symptoms which may include pain or tenderness in or around one or both joints
- clicking, popping or grinding noises → aching facial pain or pain around the ear and temple
- difficulty or pain on chewing → and locking of the joint (Ch 2, p. 102).
- Around **three-quarters of the population have at least one sign and one-third report one symptom** at some stage in life; it has a **higher propensity in females**, tends to be **cyclic**, and the number affected **increases with age**.
- It is often self-limiting but can be debilitating in the long term.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition, presentation, epidemiology and natural history.
2. The occlusal traits with a statistical association.
3. Association is not causation; the biopsychosocial model.
4. The historical controversy — Costen, Roth, the functional orthodontists, Brimm v. Malloy. **Occlusal features associated with TMD** (**Table** overbite; increased overjet; Class II and III displacement; non-working side interferences. **"Wh shown to have a correlation with the signs associations are weak"** (Egermark-Eriksson et al. 2012). **"Associations do not prove a causal rela evidence implicating dental occlusion in the** 2017; Ch 2, pp. 102–103). **The history — a memorable narrative worth de**

- **1930s — James Costen**, an American otolaryng the mandible with TMD signs and symptoms ("C

1. The AAO-funded investigations of 1992 and the subsequent evidence.

2. Whether orthodontics can *prevent* TMD.
3. Contemporary management in orthodontic practice.
4. Conclusion. **2.2, Ch 2, p. 102):** anterior open bite; deep molar relationships; posterior crossbite with **ilst some traits of a malocclusion have been and symptoms of joint dysfunction, these** , 1983; Mohlin et al., 2007; Thilander & Bjerklin, **tionship, and to date there remains a lack of pathophysiology of TMD"** (Manfredini et al., **ploying (Ch 2, pp. 104–105):** ologist, erroneously associated overclosure of osten syndrome").

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

- **1970s — Ronald Roth**, in contrast, argued from within orthodontics that canine guidance, absence of non-working side interferences, and coincidence of intercuspal and retruded contact positions were essential to reduce TMD risk. Mentioning Roth shows you know that the concern about function was not confined to fringe practitioners.
- **Late 1980s — John Witzig and Terrance Spahl** and the "functional orthodontists" claimed that premolar-based retraction mechanics excessively retracted the maxillary incisors, reduced the vertical dimension, deepened the bite, produced anterior interferences and distalised the condyle, causing disc displacement and TMD. They advocated second molar extraction and functional appliance therapy. **"No robust evidence was ever presented to suggest that any of their claims were true."**
- **1987 — Brimm v. Malloy**, Michigan. A teenage girl treated for a mild Class II division 1 with maxillary first premolar extractions, fixed appliances and headgear developed TMD during retention. The plaintiff won **$850,000**.
- **1992 — the American Association of Orthodontists** funded independent studies, published together in a special edition of the *American Journal of Orthodontics and Dentofacial Orthopedics* in January 1992. Collectively: **no evidence of significant associations between dental and skeletal relationships and TMD**
- **TMD is unpredictable and cannot be prevented by any specific intervention**
- **symptoms generally increase with age and often begin in late adolescence**
- **and orthodontic treatment does not cause TMD** (Behrents &
- White, 1992).
- **Subsequently:** conventional treatment with incisor retraction **does not produce distal displacement of the condyles** (Luecke &
- Johnstone, 1992) → orthodontics is not a risk factor with or without extractions (Mohlin et al., 2004
- Rinchuse & → McMinn, 2006) → and there is **also a lack of evidence that orthodontic intervention can reduce TMD signs and symptoms** (Macfarlane et al., 2009
- Luther et al., 2010).

Contemporary management (Kandasamy et al., 2022; Ch 2, pp. 103–104)

- TMD is now understood as a complex group of muscular and neuromuscular conditions, with management "moving away from the more traditional mechanical-occlusal model to a more biopsychosocial one".
- **No treatment modality has been shown to be better than any other**, so management should use **conservative and reversible modalities such as occlusal splints** unless there are specific and justifiable indications otherwise.

- The guidance is to undertake a **thorough TMD examination before starting treatment**;
- **if symptoms appear during treatment, suspend active mechanics** and commence appropriate management; and manage with conservative, reversible therapies. **A useful adjacent fact.**
- Only around **10% of the population have a retruded contact position coincident with intercuspal position**, so a **0–2 mm discrepancy is considered normal** (Posselt, 1952;
- Ch 2, p. 58). And there is a lack of evidence that treating to an ideal functional occlusion eliminates or reduces TMD, or that it produces greater long-term stability (Luther, 2007;
- Ch 2, p. 62) — a point that neatly closes off the counter-argument before it is made.

The Kandasamy guidelines (2022) — managing TMD in orthodontic practice

C	h 2, p. 103. Six points, and they answer most viva questions on this topic directly:
	GUIDELINE
1	Undertake a thorough TMD examination **before** any orthodontic treatment
2	Record any signs and symptoms beforehand, and discuss prognosis with the patient in the context of treatment
3	**TMD pain should be addressed before** starting orthodontics — this may require referral
4	Inform patients that TMD may occur **during and after** treatment
5	If symptoms appear during treatment, **suspend active mechanics** and commence appropriate management
6	Manage with **conservative and reversible** therapies

Two framing points examiners reward

Contemporary management is moving from a mechanical–occlusal model to a **biopsychosocial** one; and because **no treatment modality has been shown to be better than any other**, the argument for conservative reversible treatment is one of proportionality, not superiority (Ch 2, p. 103). **Occlusal features associated with TMD** (Table 2.2, Ch 2, p. 102) — associations are weak and do not prove causation: anterior open bite; deep overbite; increased overjet; Class II and III molar relationships; posterior crossbite with displacement; non-working side interferences.

EXAMINER TRAPS

The errors that cost marks on this topic.

1This is a question about evidence appraisal, not about the joint.

Distinguish correlation from causation explicitly and in those words.

2Use the historical narrative.

It is short, memorable and genuinely differentiating — very few candidates deploy it.

3Do not overcorrect into saying occlusion is irrelevant.

Weak statistical associations do exist for the listed traits; what is absent is evidence of causation.

4Do not claim orthodontics prevents TMD either.

It neither causes nor prevents it, and candidates who correct the first error often commit the second.

5The practical guidance section is where the clinical safety marks are.

Baseline screening, documentation, honest counselling about natural history, pausing mechanics, conservative and reversible management.

6Explain why the belief arose:

symptoms often begin in late adolescence, precisely when orthodontic treatment happens.

MALOCCLUSION AND TMD — A FRACTIOUS HISTORY

1930s 1970s late 1980s 1987 1992

Costen wrongly blames Roth: canine guidance, 'Functional orthodontists' Brimm v. Malloy AAO studies: NO association;

mandibular overclosure no interferences, RCP=ICP blame premolar extraction $850,000 award orthodontics does not cause TMD

THE POSITION TO STATE

Weak statistical ASSOCIATIONS exist. There is no evidence of CAUSATION. Orthodontics neither causes nor prevents TMD. Symptoms often begin in late adolescence — exactly when we treat.

IN PRACTICE (Kandasamy et al., 2022)

Screen and document before treatment · counsel on natural history · pause mechanics if symptoms appear · manage conservatively.

The TMD controversy in five dates. Compare Where is the evidence 2.3, Ch 2, pp. 104–105.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 18. 1A patient develops joint clicking six months into fixed appliance treatment. What do you do? 2Does malocclusion cause TMD? What is your evidence? 3What would you document at the initial assessment regarding the TMJ? 4Would you use orthodontics to treat TMD? 5Why did the belief that orthodontics causes TMD become so widespread?

Topic 19. Mild facial asymmetry in a growing child

Revision Handbook: §24.4
Question Bank: Block X (Q817–841)
Viva bank: V162–V173

IN ONE MINUTE

Minor asymmetry is normal. The task is to spot the progressive case.

- Dental · functional · skeletal — the spatula test and the path of closure separate them
- Cant present = hyperplasia (Type II); chin deviated, plane level = elongation (Type I)
- Do not camouflage a progressive asymmetry — monitor and refer

Where to read it

Ch 15, *Facial asymmetry* and *Condylar hyperplasia*, pp. 1092–1101. Crossbite and displacement: Ch 12, pp. 833–841. Postero-anterior cephalometry: Ch 7. Hemifacial microsomia: Ch 16, pp. 1139–1142.

Opening definition

- **"True facial symmetry is rare and some degree of asymmetry is often seen in even the most attractive of individuals"** (Ch 15, p. 1092).
- The clinical task in a growing child is therefore to distinguish self-limiting or manageable dental and functional causes from progressive skeletal pathology, which may need monitoring into, or surgical correction after, growth completion.
- Asymmetries affecting the mid and lower face fall within the remit of the orthodontist and maxillofacial surgeon; those of the upper face and orbits require more complex craniofacial surgery.

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Framing: normal asymmetry versus pathology; the diagnostic task.
2. Clinical assessment — frontal examination, chin point, occlusal cant, facial thirds.
3. The path of closure and dental centreline discrepancy.
4. Differential diagnosis — dental causes.
5. Differential diagnosis — functional causes.
6. Differential diagnosis — skeletal causes: the Obwegeser and Makek classification. **The classification of condylar hyperplasia asy pp. 1096–1098) — learn all three types: I — Horizontal**Increased width **Hemimandibulardisplacement o elongation**centreline usuall **plane remains**
7. Rarer causes — hemifacial microsomia, trauma and growth arrest, syndromic.
8. Investigation — imaging and the role of scintigraphy.
9. Management of dental and functional asymmetry.
10. Monitoring mild skeletal asymmetry.
11. Definitive management of progressive asymmetry.

12. Conclusion. **mmetry (Obwegeser & Makek, 1986; Ch 15,** of the ascending ramus and body; **marked f the chin point to the unaffected side**; dental y also displaced to the unaffected side; **the occlusal level**

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

TYPE VECTOR FEATURES

TYPE VECTOR FEATURES

II —Vertical Increased height of the ascending ramus with hyperplasia of the lower **Hemimandibular**border; the body is displaced inferiorly and rotated medially. **The hyperplasiadental centreline and chin are often undisturbed**, but a **unilateral open bite and maxillary cant** occur on the affected side **III — Hybrid**BothA combination of unilateral elongation and hyperplasia

Condylar hyperplasia "usually manifests during the **mid to late teenage years** as a **progressive and self-limiting** mandibular pathology", and can be defined histologically into four subtypes (Slootweg & Müller, 1986).

Imaging and its limitations (Ch 15, pp. 1098–1099)

- Diagnosis is clinical and radiological, with plain film and **CBCT**, often supplemented with **technetium-99m bone scintigraphy** to confirm continued active condylar growth, or **SPECT** for three-dimensional visualisation of condylar hyperactivity.
- **Critically: "whilst all these different types of bone scan are sensitive, they are also non- specific, which means that increased activity is seen in association with other forms of pathology, including inflammation, infection and neoplasia.
- Therefore, false-positive results are possible."** **Monitoring without radiation (Ch 15, p. 1099).** Because repeated radioisotope exposure is problematic, longitudinal monitoring is usually by **study models or intraoral scans and facial photographs**. A more recent non-invasive technique is **facial surface superimposition of serial 3D facial photography** with quantitative analysis, with good reproducibility using a small rectangular area on the forehead plus the middle of the nose and the lower infraorbital region as stable reference areas (Häner et al., 2021).

Management (Ch 15, p. 1099)

Correction is by orthodontics and conventional orthognathic surgery **once the excessive condylar growth has ceased** — "however, it can be difficult to ascertain when this has occurred". Where hyperplasia is ongoing and a prolonged wait would produce significant additional deformity, condylar resection may be combined with orthognathic correction.

Dental and functional causes

- Dental: asymmetric crowding, early unilateral loss of primary teeth with drift, asymmetric eruption — managed as part of routine treatment of the malocclusion.
- Functional: a mandibular displacement from a **unilateral posterior crossbite or premature contact**, in which the mandible is guided into an asymmetric intercuspal position despite essentially symmetrical skeletal bases.

- Corrected in the mixed dentition by the interceptive techniques of Topic 16 — with **high-certainty Cochrane evidence** that a quadhelix or expansion plate beats observation in children aged 7–11 (Ugolini et al., 2021).
- This is one of the few situations in orthodontics where an inexpensive early intervention has robust supporting evidence, and it is worth saying so. **Rarer causes.** Hemifacial microsomia and other craniofacial syndromes (Ch 16); **condylar trauma or growth arrest** producing reduced growth on the affected side, with asymmetry *towards* that side; and unilateral condylar hyperplasia in adolescence.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Open by normalising mild asymmetry.

Quoting the textbook's first sentence establishes the correct clinical frame and prevents the answer reading as though every asymmetry is pathology.

2The question asks for differential diagnosis and management.

Weak answers list causes and stop.

3Use the dental / functional / skeletal triad as headings.

They are not mutually exclusive — say so.

4Learn the elongation-versus-hyperplasia distinction precisely, and include the hybrid type.

The discriminator is the occlusal cant: **cant present with the chin and centreline often undisturbed = hyperplasia; chin and centreline deviated to the unaffected side with a level occlusal plane = elongation.**

5Include the spatula test

— asking the patient to bite on a wooden spatula makes a cant immediately visible and is a simple, memorable clinical manoeuvre that few candidates mention.

6Be appropriately cautious about scintigraphy

— sensitive but non-specific, with false positives — and note the newer three-dimensional photographic alternative.

FACIAL ASYMMETRY IN A GROWING CHILD Bite on a spatula. Is there a TRUE OCCLUSAL CANT?

NO YES

Check the PATH OF CLOSURE HEMIMANDIBULAR HYPERPLASIA

(Obwegeser & Makek Type II)

Deviates RCP→ICP → FUNCTIONAL

(crossbite + displacement) Vertical vector · unilateral open bite

Chin and centreline often UNDISTURBED

No deviation, chin to unaffected

side, level plane → ELONGATION (Type I) Type III = hybrid of I and II

PROGRESSIVE? Accelerating chin deviation · new cant · new TMD symptoms in the teens → refer.

Scintigraphy is sensitive but NON-SPECIFIC — false positives. Serial 3D facial photography is the non-invasive alternative.

Sorting facial asymmetry at the chairside. Compare Fig. 15.40, Ch 15, p. 1096.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 19. 1How do you distinguish a functional from a skeletal asymmetry at the chairside? 2What is the significance of an occlusal cant? 3When would you refer, and to whom?
4What is the role of bone scintigraphy, and what are its limitations? 5Would you correct a centreline discrepancy in a 9-year-old with a unilateral crossbite? What would you expect to happen?

Topic 20. Late lower incisor crowding

Revision Handbook: §19.4
Question Bank: Block S (Q642–676)
Viva bank: V130–V139

IN ONE MINUTE

It happens in untreated people too. It is ageing, not relapse.

- Primary (genetic) · secondary (environmental) · tertiary (adolescent onwards)
- Mandibular growth rotation is the major factor; third molars are not
- No extraction pattern prevents it — the answer is permanent bonded retention

Where to read it

Ch 5, Box 5.6, Understanding the Causes of Late Lower Incisor Crowding, *pp. 283–284. Ch 13,* Relapse and
Retention, pp. 974–990; long-term retention Where is the evidence 13.8, pp. 989–990. Ch 2, relapse, p. 100.

Opening definition

- Crowding of the mandibular incisors is one of the most common problems encountered in the permanent dentition, and lower incisor alignment is one of the conditions most likely to relapse after orthodontic treatment.
- Studies of untreated subjects followed from the mixed dentition into adulthood have shown a tendency for **the width and length of the mandibular arch to decrease and for crowding of the anterior teeth to increase** (Sinclair & Little, 1983;
- Ch 5, p. 283). Because it happens in untreated people too, it is an **age-related change rather than relapse** (Ch 2, p. 100).

Essay skeleton

The numbered headings. Reproduce these and the essay writes itself.

1. Definition and the central fact — it occurs in treated and untreated dentitions alike.
2. The longitudinal evidence.
3. Classification: primary, secondary and tertiary crowding.
4. Contributory factors to tertiary crowding.
5. Which of those factors the evidence actually supports.
6. The third molar question and the NICE position.
7. Clinical significance — planning, consent, retention.
8. The retention evidence base.
9. Conclusion.

Facts to have at your fingertips

Learn the numbers. This is where depth marks are won.

The three c	ategories (Ch 5, p. 283):
CATEGORY	BASIS
Primary	Discrepancy of tooth dimension and jaw size, **mainly determined genetically**
Secondary	**Environmental factors** — local space conditions and the position and function of the tongue, lips and buccal musculature
Tertiary	Occurs **during adolescence and post-adolescence**, with a predilection for the lower labial segment. This is "late lower incisor crowding" proper
Contributo occlusal forc	**ry factors (Ch 5, p. 283):** mandibular growth rotations; anterior component of e; physiological mesial drift; soft tissue maturation; degenerative periodontal changes

allowing drift under light pressures; change in diet and lack of interproximal wear; tooth size and shape; tooth loss and drifting altering occlusal function; and mandibular third molars, presence and position. **Which factors the evidence supports — the refinement that distinguishes a strong answer:**

- **Mandibular growth rotation is the dominant factor.** Facial growth continues throughout life, in the same direction as adolescence but to a much smaller degree (Behrents, 1985), and is rotational rather than linear (Solow, 1980). "This can manifest most notably as late lower incisor crowding, **as the lower labial segment uprights to compensate for mandibular growth. This is thought to be a major factor** in late lower incisor crowding seen in both orthodontically-treated and untreated individuals" (Ch 13, p. 978).
- **The anterior component of occlusal force is doubted.** It has been postulated to be causative (Southard et al., 1990); "**however, the short duration of these forces makes this unlikely**" (Ch 13, p. 978).
- **Third molars are not a primary cause.** "The contribution of developing third molars is regarded as being minimal, as crowding can develop even in the absence of their development. **The prophylactic removal of developing third molars is not recommended to prevent late lower incisor crowding**" (Ch 5, pp. 283–284).
- **Typical findings in untreated adults in their 30s and 40s:** shorter dental arch perimeter, decreased mandibular intercanine distance and increased anterior crowding, particularly in the mandible (Thilander, 2009). "It is therefore important to **distinguish between relapse and what are normal changes with ageing**" (Abdulraheem et al., 2020).
- A blunt and quotable observation: "**the changes found in untreated individuals in general are very similar in nature to those found in patients following orthodontic treatment**" (Ch 5, p. 284). **The clinical significance — retention (Ch 13, pp. 989–990):**
- **How long?** "The honest answer is for as long as they want to keep their teeth perfectly straight." Not all patients relapse, "but from the evidence available, **it is not possible to predict who these patients will be**". Little (1975) suggested **under 3.5 mm of lower incisor irregularity** as an acceptable level of relapse, but many patients find even minor deterioration unacceptable.
- **Fixed versus removable.** Both are effective in the upper labial segment. In the lower, fixed retainers are **marginally more effective for the first 12 months**, especially for rotations, but **beyond that both show the same efficacy**. Even with fixed retainers, some reappearance of irregularity occurs. Fixed retainers have a **higher failure rate in the lower arch**, and "the patient will often only notice a breakage when the lower incisors start moving". They accumulate plaque and calculus, though short-term periodontal effects appear minimal
- longer-term evidence is conflicting.

- **Removable retainers.Vacuum-formed retainers appear more effective than Hawley-type at maintaining lower labial segment alignment** (Rowland et al., 2007)
- both can be worn part-time and remain effective → **patients prefer vacuum-formed retainers** — less conspicuous, less effect on speech, and more likely to be worn long term.
- **Compliance decays.** Good in the short term, but "**by 4–5 years post-treatment, the majority of patients will have stopped wearing their retainers altogether**" (Al-Moghrabi et al., 2018
- Krämer et al., 2022) — "ironic because when patients are asked at the start of treatment, almost all of them rate keeping their teeth straight as extremely important".
- **A shared decision.** The choice for long-term retention should be made in partnership with the patient, giving shared responsibility — and there remains "the thorny issue of who should actually manage and be responsible for long-term retention: the orthodontist, dentist or indeed the patient".

EXAMINER TRAPS

The errors that cost marks on this topic.

1Lead with the fact that it occurs in untreated dentitions too.

This reframes the topic from orthodontic failure to normal ageing, and it is the fact the examiner most wants to see.

2Use the primary/secondary/tertiary classification

— it gives the essay a structure and places late crowding in a wider framework.

3Do not present the contributory factors as a flat list of equals.

Say that mandibular growth rotation with uprighting of the lower labial segment is regarded as the major factor, and that the anterior component of occlusal force is doubted because the forces are of such short duration.

4The third molar section is the set-piece.

Traditional theory, then the disconfirming observation that crowding develops even where third molars never form, then the NICE position.

5Do not omit the clinical significance section.

"Discuss the phenomenon" includes what you do about it.

6Give the retention evidence properly

, including the uncomfortable finding that most patients stop wearing retainers within four to five years.

LATE LOWER INCISOR CROWDING

PRIMARY SECONDARY TERTIARY

tooth size vs jaw size tongue, lips, cheeks adolescence onwards

largely GENETIC ENVIRONMENTAL THIS is 'late crowding'

WHAT DRIVES TERTIARY CROWDING

Mandibular growth rotation → lower labial segment uprights MAJOR FACTOR

Soft tissue maturation

Physiological mesial drift

Reduced interproximal wear (diet)

Periodontal attachment loss

Anterior component of occlusal force DOUBTED — forces too brief

Third molars NOT a primary cause — NICE: no prophylactic removal

It happens in UNTREATED people too. It is age-related change, not relapse — and no extraction pattern

or archwire sequence prevents it. The answer is long-term bonded retention.

Late lower incisor crowding: the three categories and what drives them. Compare Box 5.6, Ch 5, pp. 283–284.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Topic 20. 1A patient asks whether removing their wisdom teeth will stop their lower front teeth crowding. What do you say?
2Why do you recommend permanent bonded retention? 3What is the anterior component of occlusal force? 4How is late lower incisor crowding different from relapse? 5Can any extraction pattern prevent it?

Part III — The sixteen chapters

Chapter 1. Evidence-Based Orthodontics

Revision Handbook: §7
Question Bank: Block G (Q196–225)
Viva bank: V39–V48

IN ONE MINUTE

How to read every other chapter. Orthodontics has a weak evidence base.

- PICO · the hierarchy · five types of bias · confounding
- P values address Type I error only; confidence intervals are more informative
- $I^2 \geq 50\% \rightarrow$ random-effects. ITT measures effectiveness; per protocol measures efficacy

Where to read it

pp. 19–56. PICO pp. 20–21 · study design pp. 22–24 · hierarchy of evidence pp. 24–28 · bias and confounding p. 28 · P values and confidence intervals pp. 28–30 · randomised trials pp. 30–36 · systematic review pp. 37–47 · core outcomes pp. 47–51 · guidelines and GRADE pp. 51–56.

The map

Evidence-based medicine and its five steps → PICO → clinical study design → patient-centred research → the hierarchy of evidence → choosing a design → bias and confounding → hypothesis testing and estimation → randomised controlled trials → systematic review and meta-analysis → core outcomes → clinical practice guidelines and GRADE.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Definition of EBM	Integration of the **best research evidence with clinical expertise, patient values and patient circumstances** (Straus et al., 2018)	p. 19
The five steps	1. Formulate the right clinical question · 2. Find the best evidence · 3. Critically appraise it · 4. Integrate appraisal with practice and the individual patient's needs, values and beliefs · 5. Evaluate effectiveness	p. 21
PICO	**P**atients (who, including demographics) · **I**ntervention (treatment or exposure) · **C**omparison (the main alternative) · **O**utcomes (what you are trying to accomplish or measure)	pp. 20–21
Two basic designs	**Observational** — no intervention by the investigator; **hypothesis generating**. **Interventional (experimental)** — the investigator controls the intervention; **hypothesis testing**	p. 22
Observational subtypes	*Descriptive*: case report, case series — no comparison group. *Analytical*: cohort, case-control, cross-sectional, ecological — quantify an association	p. 22

ITEM	DETAIL	PAGE
What each design is for	Prevalence → cross-sectional, ecological. Incidence → cohort, ecological. Causation → cohort, case-control, cross-sectional. Prognosis → cohort. Treatment effect → prospective and randomised clinical trials	p. 22
The hierarchy	Ascending: expert opinion → case report → case series → ecological → crosssectional → case-control → cohort → RCT → systematic review → meta-analysis → **clinical practice guidelines based on high-quality meta-analysis (the gold standard)**. Primary studies are *unfiltered*; systematic review and metaanalysis are *filtered*	pp. 24–26
The five types of bias	**Selection** (differences in baseline characteristics) · **Performance/ intervention** (differences in care provided) · **Detection** (differences in how outcomes are recorded) · **Attrition/follow-up** (differences in withdrawals) · **Reporting/publication** (differences between reported and unreported findings)	pp. 37–40
Confounding	A form of bias where the association between an intervention and the outcome is influenced by **another variable**, so the apparent effect is not the true effect. **Randomisation balances known and unknown confounders**, particularly with large samples	p. 28
What a P value is	A test of the **null hypothesis**. P <0.05 means a positive result would occur by chance fewer than 5 times in 100 — i.e. it quantifies **Type I error**. It does not *rule out* the null hypothesis; it makes rejecting it reasonable	p. 29
The P value trap	"No difference between groups" because P >0.05 does not mean the numbers were identical; there will almost always be numerical differences worth attending to. Conversely a statistically significant result may not be **clinically** significant — "commonly seen in orthodontic studies"	pp. 29–30
Confidence intervals	The **95% CI** is the range of values with only a 5% chance of excluding the true sample mean. **The wider the interval, the less precise the estimate; larger samples narrow it**	p. 30
CONSORT	25-item checklist and flow diagram for reporting parallel-design RCTs (Schulz et al., 2010). Built around four principles: **pre-intervention equivalence, random allocation, blinding, robust outcome reporting**	pp. 30–31
Allocation concealment	A fundamental part of randomisation — performed by a **third party**; the investigator should have **no involvement in participant allocation**	p. 31
Power and significance	**Power** (usually set at **80–90%**) protects against **Type II error** — wrongly accepting the null hypothesis. **Significance** (usually **0.05**) protects against **Type I error**	p. 33
Blinding in orthodontics	Double blinding is often impractical because treatment is prolonged and interventions hard to disguise — but **data analysis should always be carried out blind**	p. 34
Intention-to-treat vs per protocol	**ITT** analyses all randomised patients regardless of whether they completed treatment; it preserves pre-treatment equivalence and measures **effectiveness** ("how well did it work in all those offered it?"). **Per protocol** includes only compliant patients and measures **efficacy** ("how well did it work in those who received it?")	p. 35
PRISMA	27-item checklist and four-phase flow diagram for systematic reviews and metaanalyses (Moher et al., 2009)	p. 37
STROBE	Reporting guidelines for observational studies (von Elm et al., 2007)	p. 28

ITEM	DETAIL	PAGE
Heterogeneity	Variation in the true effect of an intervention between studies. Causes: **different sample sizes and participants, different precise interventions, different outcome measurement**. Assessed by **Chi-squared (Cochran Q)** and the **Higgins I^2** statistic	pp. 46–47
I^2 thresholds	**25% low · 50% moderate · 75% high** (Higgins et al., 2003). **I^2 <50% → fixed- effect analysis; I^2 ≥50% → random-effects analysis**	pp. 46–47
Forest plot clues to heterogeneity	Individual effect sizes should all be on the same side of the line of no effect as the pooled estimate, and **the confidence intervals should all overlap**	p. 46
Relative risk and odds ratio	**RR** = probability of an event in the exposed group relative to the unexposed. **Odds** = ratio of the likelihood of an event happening to it not happening	p. 45
Core Outcome Set	An agreed standardised set of outcomes that **should be measured and reported in all trials** in a field. The **COMET** initiative maintains these; a COS for orthodontic trials has been developed (Tsichlaki et al., 2020)	pp. 47–51

ITEM DETAIL PAGE

GRADEGrades the **certainty of evidence** and the **strength of recommendation**, offering two grades: **strong** (desirable effects clearly outweigh undesirable) and **weak** (desirable effects less certain) **AGREE II**Appraisal of Guidelines Research and Evaluation — the instrument for appraising the quality of clinical practice guidelines (Brouwers et al., 2010) pp. 51–56

p. 51

Nice-to-know

Enriches an answer. Skip on a first pass.

- **PREMs versus PROMs.** Patient-Reported *Experience* Measures capture how the patient felt about the care pathway; Patient-Reported *Outcome* Measures capture their perception of the outcome and their health.
- **What actually predicts patient satisfaction.** Strong associations with final dental appearance *and* with the perceived quality of care — the interactions and relationships with the orthodontist and staff.
- **Qualitative research** uses small **purposive non-random** samples deliberately selected by the investigator; strong for feelings and experiences, vulnerable to researcher bias and unrepresentativeness (p. 24).
- **Split-mouth trials** are popular in orthodontics because they halve the required participant numbers, but their methodological and reporting quality in dentistry is poor (Qin et al., 2020); specific CONSORT guidance exists for within-person trials (pp. 35–36).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not equate ”randomised controlled trial” with ”good”.

The textbook is explicit: ”if a study describes itself as an RCT, this does not automatically mean that it is methodologically sound”.

2Do not confuse a cross-sectional study with a retrospective cohort study.

The book flags this distinction explicitly.

3Do not over-interpret P values.

State that P <0.05 addresses Type I error only, that statistical and clinical significance are different things, and that confidence intervals are more informative.

4Get the I^2 thresholds and their consequence right

— 50% is the switch between fixed-effect and random-effects analysis.

5Expert opinion is the bottom of the hierarchy

— and the authors note pointedly that "it continues to dominate clinical orthodontics".



The hierarchy of evidence. Compare Fig. 1.3, Ch 1, p. 26.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 1. 1What is the difference between intention-to-treat and per protocol analysis, and why does it matter in orthodontics? 2A systematic review reports an I^2 of 78%. What does that tell you and what should the authors have done? 3What are the limitations of relying on $P < 0.05$?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 1.

1. State the PICO framework.
2. Which study designs establish prevalence, and which establish incidence?
3. Name the five types of bias.
4. What does an I^2 of 30% imply for the statistical model used in a meta-analysis?
5. What is allocation concealment and who should perform it?

Chapter 2. Occlusion and Malocclusion

Revision Handbook: §5; indices §6; TMD history §26.1
Question Bank: Block E (Q141–170), Block F (Q171–195)
Viva bank: V31–V48

IN ONE MINUTE

What the specialty is about: ideal occlusion, how we classify it, and whether treating it helps.

- Andrews' six keys; mutual protection; only ~10% have RCP coincident with ICP
- IOTN records the single WORST trait (MOCDO); PAR ≥22 points = greatly improved
- No evidence that an ideal functional occlusion improves stability or prevents TMD

Where to read it

pp. 57–112. Ideal occlusion pp. 57–58 · classification pp. 59–66 · prevalence p. 66 · aetiology pp. 67–86 · benefits pp. 86–92 · risks pp. 92–105 · provision and indices pp. 105–107 · monitoring pp. 107–112. *Benefits and risks are covered in depth as Essay Topic 9, and the TMD material as Essay Topic 18. This unit covers what those topics do not: ideal and functional occlusion, classification, prevalence, aetiology and the occlusal indices.*

The map

Ideal occlusion and mutual protection → classification (Angle, incisor, Ballard, indices) → prevalence → aetiology (skeletal, soft tissue, local, habits) → benefits → risks → provision (IOTN) → monitoring (PAR).

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Andrews' six keys	1. Molar relationship · 2. Crown angulation (mesiodistal tip) · 3. Crown inclination (torque) · 4. No rotations · 5. No spacing · 6. Flat occlusal plane	p. 58
Mutual protection	Anterior teeth protect posterior: **immediate and permanent posterior disocclusion** in lateral and protrusive movement with no non-working-side interferences, via **canine guidance or group function** laterally and incisal guidance in protrusion. Posterior protect anterior: **multiple, simultaneous, bilateral posterior contacts in ICP with the incisors slightly out of contact**	p. 58
RCP–ICP	Only around **10% of the population** have RCP coincident with ICP; a **0–2 mm discrepancy is considered normal** (Posselt, 1952)	p. 58

ITEM	DETAIL	PAGE
Does ideal function matter?	There is a **lack of evidence** that treating to an ideal functional occlusion eliminates or reduces TMD, or that it produces greater long-term stability (Luther, 2007). "If mutual protection is not achieved at the end of orthodontic treatment, there do not appear to be long-term serious consequences"	p. 62
Angle classification	Based on the **first permanent molars**, whose position is assumed fixed within the arch — the classification's principal weakness, since early primary tooth loss or crowding distorts it. Asymmetric relationships are termed **subdivision**	p. 63
Andrews' molar refinement	The **distal marginal ridge of the upper first molar** occludes with the **mesial marginal ridge of the lower second molar**, with the mesiobuccal cusp in the groove between the mesial and middle cusps of the lower first molar — requiring **distal tip and a slight buccal offset**	pp. 63– 64
IOTN structure	**Dental Health Component (Grades 1–5)** and **Aesthetic Component (1–10 photographs)**. The DHC was adapted from the Swedish National Board for Welfare Index; the AC from the **SCAN** index (Evans & Shaw, 1987)	pp. 105– 107
IOTN is hierarchical	Record only the **single worst occlusal trait**, irrespective of other features. **MOCDO** = Missing teeth, Overjet, Crossbite, Displacement of contact points, Overbite	p. 107
Key DHC thresholds	**5a** overjet >9 mm · **4a** overjet >6 but ≤9 mm · **3a** overjet >3.5 but ≤6 mm *with incompetent lips* · **2a** the same *with competent lips* · **5m** reverse overjet >3.5 mm with masticatory/speech difficulty · **4d** displacement >4 mm · **4f** increased and complete overbite **with** gingival or palatal trauma · **3f** the same **without** · **5i** impeded eruption · **5p** cleft lip and palate · **5s** submerged deciduous teeth	pp. 108– 109
PAR	Accumulative score of deviation from normal occlusion, assessed from study casts, with **no maximum cut-off**. **<30% reduction = worse or no different · >30% = improved · ≥22 points = greatly improved**	pp. 107– 108
IOFTN	Index of Orthognathic Functional Treatment Need, for cases requiring combined orthodontic and surgical correction (Ireland et al., 2014)	p. 107; Ch 15, p. 1041

Nice-to-know

Enriches an answer. Skip on a first pass.

- The AC shows a **high level of agreement between professional and patient scores** (p. 107).
- The AC scale is scored on **perceived aesthetic impairment**, not on morphological similarity to the photographs (p. 107).
- The PAR nomogram divides outcomes into three bands — worse/no difference, improved, greatly improved — and is most useful for auditing a clinician's or a service's overall standard rather than an individual case (p. 108).

EXAMINER TRAPS

The errors that cost marks on this topic.

1The Angle classification's weakness is its assumption.

Say that it presumes the first permanent molars hold a fixed position, which crowding and early primary tooth loss falsify.

2Do not claim that finishing to an ideal functional occlusion prevents TMD or improves stability.

The book is explicit that the evidence is lacking. This is the single most efficient way to pre-empt an examiner's follow-up.

3IOTN records only the worst trait.

Candidates commonly describe adding scores.

4Know both PAR thresholds

— the 30% percentage reduction *and* the 22-point absolute reduction.

IOTN AND PAR

IOTN — need for treatment PAR — outcome of treatment

Dental Health Component Grades 1–5 Accumulative score from study casts.

Aesthetic Component 10 photographs No maximum cut-off.

HIERARCHICAL — record only the

SINGLE WORST trait <30% reduction worse or no different

MOCDO Missing · Overjet · Crossbite · >30% reduction improved

Displacement · Overbite

≥22 points GREATLY improved

Key: 5a OJ >9 · 4a >6–9 · 3a >3.5–6 (lips apart)

Best used to audit a clinician or service.

IOTN and PAR. Compare Figs 2.27–2.28, Ch 2, pp. 110–111.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 2. 1A patient has a 7 mm overjet, 3 mm of lower incisor crowding and a unilateral crossbite with a 2 mm displacement. What is their IOTN DHC grade? 2Why is the Angle classification unreliable?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 2.

1. What proportion of the population have RCP coincident with ICP, and what discrepancy is considered normal?
2. What does MOCDO stand for and what is it for?
3. What PAR reduction indicates a greatly improved result?
4. State Andrews' six keys.
5. What is the evidence that finishing to an ideal functional occlusion improves stability?

Chapter 3. Prenatal Development of the Craniofacial Region

Revision Handbook: §2.1; clefts §28
Question Bank: Block B (Q26–55)
Viva bank: V1–V10

IN ONE MINUTE

Where the face comes from — learn it as the basis of clefting.

- Five processes at week 4; face complete by week 8; palate weeks 6–10
- Upper lip = maxillary processes laterally + medial nasal processes in the midline
- Primary palate = premaxilla to the incisive foramen; secondary palate = the shelves

Where to read it

pp. 113–166. Embryonic origins pp. 113–115 · early organisation pp. 115–121 · development of the face pp. 121–130 · development of the palate pp. 130–132 · pharyngeal region pp. 132–137 · development of the skull pp. 137–145 · molecular regulation pp. 145–166.

The map

Germ layers and the neural crest → segmentation of the early head → the pharyngeal arches → development of the face → development of the palate → the pharyngeal pouches → development of the skull → molecular regulation.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
The neural crest	Often called the **fourth germ layer**. Cranial neural crest cells arise during neural tube formation and migrate to provide the building blocks for much of the craniofacial skeleton; those from the **posterior midbrain and hindbrain** migrate into the pharyngeal arch system	p. 114
The five facial processes	Appear at approximately **4 weeks post conception** around the stomodeum: the **frontonasal process**, the **paired maxillary processes** and the **paired mandibular processes** of the first pharyngeal arch	p. 122
The arches	**Six** pharyngeal arches. **First → jaws and dentition · second → suprahyoid apparatus · third → infrahyoid region · fourth and sixth → laryngeal structures.** The **fifth degenerates rapidly** after formation	pp. 116, 119
Arch composition	Each arch is covered externally by **ectoderm**, internally by **endoderm**, with a core of **mesoderm** progressively infiltrated by **neural crest**, and contains a characteristic **nerve, cartilage and artery**	p. 124
Week 4	Frontonasal process enlarges as the forebrain expands into cerebral hemispheres; the **paired mandibular processes unite** to form the lower jaw and lip	p. 122

ITEM	DETAIL	PAGE
Week 5	**Medial and lateral nasal processes** form within the frontonasal process around the **nasal placode**; the placodes sink into the **nasal pits**, demarcating the nostrils	pp. 122– 123
Maxillary process fusions	With the **lateral** nasal processes → **nasolacrimal duct, cheek, alar base of the nose**. With the **medial** nasal processes → **central portion of the nose, upper lip philtrum, primary palate**	p. 123
The upper lip	Formed from the **maxillary processes laterally and the medial nasal processes in the midline**. The essential features of the human face are formed by **8 weeks**	p. 123
Palate timing	Development starts at **week 6** and is essentially complete by **week 10**	p. 130
Primary palate	Derived from the **medial nasal processes**; gives rise to the **four maxillary incisors, their alveolus and a triangular region of hard palate — the premaxilla — up to the incisive foramen**	p. 131
Secondary palate	Palatal processes of the maxilla, horizontal processes of the palatine bone, and the musculature of the soft palate. Dominated by the **paired palatine shelves**, outgrowths of the maxillary processes, which are **initially vertical either side of the tongue**, then **elevate above the tongue** and **fuse in the midline**, also fusing with the **nasal septum superiorly and the primary palate anteriorly**	pp. 131, 133
Ossification	Both primary and secondary palate ossify by **intramembranous** ossification of condensed **neural-crest-derived ectomesenchyme**; the soft palate musculature comes from **head mesoderm**	pp. 131– 132
First pharyngeal	Forms the **tubotympanic recess** → tympanic cavity and pharyngotympanic tube; contacts the first pharyngeal cleft ectoderm at the **tympanic membrane**	p. 132

pouch

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Shh signalling from the prechordal plate** patterns the ventral forebrain and produces bilateral subdivision of the eye field — the molecular basis of holoprosencephaly (p. 121).
- **Palatal rugae** appear sequentially within the epithelium of the palatal shelves (p. 131).
- **Facial variation is highly polygenic.** A study of 3D facial scans and genome-wide association data in over 8,000 individuals across three cohorts confirmed **203 genomic regions** associated with normal-range facial variation, **103 of them new**.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Link every element to clefting.

A cleft lip is a failure of fusion between the maxillary and medial nasal processes; a cleft of the primary palate lies anterior to the incisive foramen and involves the premaxilla; a cleft of the secondary palate is a failure of palatal shelf elevation, contact or fusion.

2Get the timing right and quote it:

face by 8 weeks, palate weeks 6 to 10. The difference in timing is why cleft lip and cleft palate have different epidemiology and different genetic architecture (Chapter 16).

3The tongue is the obstacle.

The palatal shelves start vertical *because the tongue is in the way*; elevation above the tongue is the critical step, and its failure is the mechanism in Pierre Robin sequence.

DEVELOPMENT OF THE FACE AND PALATE — AND WHERE CLEFTS ARISE

week 4 week 5 week 6 week 8 week 10

Five processes appear: Medial and lateral nasal Palatine shelves appear FACE COMPLETE PALATE COMPLETE
frontonasal + paired processes form around VERTICALLY either side lip = maxillary + medial shelves elevate, fuse
maxillary + mandibular the nasal placode of the tongue nasal processes in the midline

CLEFT LIP ± PRIMARY PALATE ISOLATED CLEFT PALATE

Maxillary process fails to fuse with Palatine shelves fail to elevate or
the MEDIAL NASAL process, by week 8 fuse, weeks 6–10

1:700 · male 2:1 · 70% non-syndromic 1:2000 · female 4:1 · 50% non-syndromic

Opposite sex ratios because they are DIFFERENT developmental failures at different times. Ch 3, pp. 121–133; Ch 16, pp. 1113–1115.

Facial and palatal development, and where clefts arise. Compare Figs 3.8–3.10, Ch 3, pp. 123–133.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 3. 1Explain the embryological basis of a unilateral cleft lip and palate.

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 3.

1. Name the five processes surrounding the stomodeum and when they appear.
2. Which structures derive from fusion of the maxillary with the *lateral* nasal processes?
3. What does the primary palate comprise, and what is its posterior boundary?
4. Over what weeks does the palate develop?
5. Which pharyngeal arch degenerates?

Chapter 4. Postnatal Growth of the Craniofacial Region

Revision Handbook: §1 and §2.2–2.4
Question Bank: Block A (Q1–25), Block B (Q26–55)
Viva bank: V1–V10

IN ONE MINUTE

The intellectual foundation of everything done to growing patients.

- Height–velocity curve: juvenile spurt 6–8 y, then the adolescent spurt
- Theories are one debate: genetic (Scott, Sicher) versus environmental (Moss), synthesised by van Limborgh
- Rotations: total (true) = matrix (apparent) + intramatrix (angular remodelling)

Where to read it

pp. 167–244. General somatic growth pp. 167–174 · craniofacial growth pp. 174–181 · the skull at birth pp. 181–185 · mechanisms of bone growth pp. 185–192 · **theories of growth pp. 192–202** · cranial vault pp. 202–205 · cranial base pp. 205–207 · face pp. 207–212 · **mandible and growth rotations pp. 212–226** · dentoalveolar compensation p. 226 · **growth prediction pp. 226–243** · adult growth p. 243.

The map

General body growth and growth studies → craniofacial growth and how it is studied → the skull at birth → mechanisms of bone growth → theories of growth → growth of vault, base, face and mandible → mandibular rotations → dentoalveolar compensation → growth prediction and skeletal maturity → adult growth.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
The three phases of the height– velocity curve	1. Rapid growth at birth **decelerating until around 3 years** · 2. Slowly decelerating phase to the early teens, **interrupted by a juvenile spurt at around 6–8 years** · 3. The **adolescent growth spurt**, then progressive deceleration to adulthood	p. 168
Sexual dimorphism	The adolescent spurt occurs **later in males and is more significant**, giving a longer overall growth period, greater acceleration and greater final height	p. 171
Scammon's curves	**General** growth is sigmoid · **brain and head** are well developed at birth and essentially complete by **~10 years** · **reproductive** tissue increases only at puberty · **lymphoid** tissue peaks around puberty then declines, largely through thymic involution	pp. 171– 172

ITEM	DETAIL	PAGE
Growth study designs	**Longitudinal** — repeated data from the same individuals; gives true growth, individual variation and velocity curves, but costly and vulnerable to dropout. **Cross-sectional** — each individual contributes once; quicker and commoner, but only shows what has been *attained* at each age. **Mixed-longitudinal** — a combination	p. 172
The four mechanisms of skull growth	**Primary displacement** (soft tissue growth physically carries the bone away) · **remodelling** (periosteal deposition and endosteal resorption producing cortical drift) · **relocation** (following the direction of external deposition, opposite to primary displacement) · **secondary displacement** (caused by enlargement of other, sometimes distant, bones)	p. 193
Remodelling theory	**James Couper Brash.** Bone remodelling and interstitial growth as the primary mechanism; sutures and cartilages have **no intrinsic growth capability**. "Hunterian growth" after John Hunter's madder-root alizarin staining experiments in pigs	p. 192
Sutural theory	**Weinmann and Sicher.** Growth is **genetically regulated within the sutures and cartilages**, with the sutures able to generate a **tissue- separating force**	p. 194
Cartilaginous theory	**James Scott.** Sutures are merely modified periosteum — **permissive, not force-generating**. Cartilage drives growth: the **nasal septal cartilage** displaces the maxilla downwards and forwards, **synchondroses** elongate the cranial base, and the **condylar cartilage** directs mandibular growth	p. 194
Functional matrix theory	**Moss and Salentijn**, from van der Klaauw. Growth of skeletal units is **secondary to growth of the functional matrix** — soft tissues and functional spaces. **Genes play no major role in postnatal growth.** Two matrices: **periosteal** (muscles, tendons → microskeletal units, size and shape) and **capsular** (neurocranium, orofacial spaces, pharynx → macroskeletal units, displacement)	pp. 194– 198
van Limborgh	**Five factors:** intrinsic genetic; local and general **epigenetic**; local and general **environmental**. The first unifying theory; **genetic factors predominate, environment is moderated** — in explicit contrast to Moss	p. 198
Part–counterpart	**Enlow.** The skull is composed of parts and counterparts that must grow in balance; the key dividing line is the **Posterior Maxillary (PM) plane**, linking the **maxillary tuberosity with the mandibular lingual tuberosity** through the pterygomaxillary fissure. Provides a model for the basis of malocclusion	pp. 198– 199
Servosystem theory	**Petrovic.** Explains growth in terms of cybernetics — feedback control	p. 201
Mandibular growth rotations	**Total rotation** = change in inclination of the mandibular **body (corpus)** relative to the anterior cranial base, measured on implants. **Matrix rotation** = change in inclination of the **soft tissue matrix**, measured on a tangent to the lower border, centred on the condyles. **Intramatrix rotation** = the difference, reflecting **remodelling along the lower border**. Total = matrix + intramatrix	pp. 220– 222
Solow and Houston's simplification	**True rotation** = total rotation (the fundamental rotation between mandible and cranial base) · **apparent rotation** = matrix rotation (what you see cephalometrically without implants) · **angular remodelling of the lower border** = intramatrix rotation	p. 222

ITEM	DETAIL	PAGE
Direction and centres	Convention: cephalometric view faces right, so **forwards (anterior) = anti- clockwise**, **backwards (posterior) = clockwise**. **Forwards rotations are the most common**, with centres through the **condyles, incisors or premolars**; backwards rotations have centres in the **condyles or the most distal-occluding molars**	p. 222
What drives rotation	An **imbalance between anterior and posterior face height growth**. Excess **anterior** face height growth → **backwards** rotation. Increased **posterior** face height growth → **forwards** rotation (Houston, 1988). Dentoalveolar compensation usually maintains a normal occlusion; if the imbalance is severe, anterior open bite or deep overbite results	p. 222
Björk's seven structural signs	Inclination of the condylar head · curvature of the mandibular canal · shape of the lower border · inclination of the symphysis · interincisal angle · interpremolar and intermolar angles · anterior lower face height	p. 224
Can rotation be predicted?	**Conflicting.** Skieller et al. (1984) found a high prognostic estimate using Björk's more extreme cases; but Baumrind et al. (1984) found five experienced clinicians could distinguish extreme backward from forward rotators **no better than chance**. Extreme rotations are easily identified **once growth is complete**	p. 224
CVM	Originally described by **Don Lamparski** (unpublished MSc, Pittsburgh, 1970s), modified by **Baccetti, Franchi and McNamara**. Uses morphology of **C2 (odontoid process), C3 and C4**; **six stages, CS1–CS6**; derived from **18 males and 12 females** in the Michigan longitudinal study	p. 238
CVM staging and timing	**CS1** (mean 9.7 y) or **CS2** (10.5 y) — peak mandibular growth **2 years or 1 year later** respectively; wait at least a year. **CS3** (12.0 y) — peak within **1 year**; the **ideal time to start functional appliance treatment**. **CS4** (13.4 y) — peak was **1–2 years ago**. Girls mature earlier; there is variation between continents	pp. 238– 240
The health warning on CVM	The method assumes that the timing of the adolescent spurt can be accurately predicted from CVM, that it correlates with maximal mandibular growth, and that treating in that window increases mandibular growth. **"None of these principles are universally accepted."** Taking a lateral cephalogram *specifically* to assess CVM is **difficult to justify**	p. 238

Nice-to-know

Enriches an answer. Skip on a first pass.

- The **Harpenden growth study** (Tanner and Whitehouse, 1948–72): 55 boys and 35 girls from a children's home, measured six-monthly until puberty then three-monthly; still one of the most useful references for pubertal growth (p. 173).
- The mandible descends relative to the cranial base partly because **the cervical column lengthens and takes the head away from the shoulder girdle**, stretching the muscle chain from mandible to skull base and mandible to hyoid to shoulder girdle (Houston, 1988; p. 225).

- **Dentoalveolar compensation** is the mechanism that maintains a normal interarch occlusal relationship despite variation in skeletal pattern (Solow, 1980
- p. 226) — the reason many skeletal discrepancies present with a less severe malocclusion than the cephalometry predicts, and the reason decompensation is necessary before surgery.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not present the theories as a chronological list.

Present them as a debate about one question — is craniofacial growth intrinsically determined or

environmentally driven? — with Scott and Weinmann/Sicher at the genetic end, Moss at the environmental end, and van Limborgh offering the synthesis that most people now accept.

2The functional matrix theory's strongest claim is also its weakest.

Moss held that genes play *no major role* in postnatal craniofacial growth.

3Distinguish true from apparent rotation.

Apparent rotation is what a cephalogram shows *without* implants; true rotation requires Björk's implants.

4Be honest about growth prediction.

Björk's structural signs are descriptive of what has happened, not reliably predictive of what will; the Baumrind finding of "no better than chance" is the one to quote.

5Be equally honest about CVM.

Quote the CS3 recommendation *and* the book's statement that none of the underlying principles are universally accepted, and that a radiograph taken solely for CVM is hard to justify.

rapid deceleration to ~3 y PEAK HEIGHT VELOCITY

Growth velocity

juvenile spurt 6–8 y

prepubertal minimum

038121418

Age (years)

CVM STAGING (Baccetti, Franchi & McNamara)

CS1 9.7 y → peak in 2 y · CS2 10.5 y → peak in 1 y · CS3 12.0 y → peak within 1 y (start now)

CS4 13.4 y → peak was 1–2 y ago. Caveat: none of the underlying principles is universally accepted.

The height–velocity curve and CVM staging. Compare Figs 4.3 and 4.47, Ch 4, pp. 171, 240.

FORWARD (anterior) BACKWARD (posterior)

anterior cranial base anterior cranial base

posterior face height grows more anterior face height grows more

deep overbite · commoner open bite · increased face height

THE THREE ROTATIONS (Björk & Skieller; simplified by Solow & Houston)

TOTAL (true) = mandibular body vs cranial base, measured on implants

MATRIX (apparent) = soft tissue matrix vs cranial base — what a cephalogram shows without implants

INTRAMATRIX (angular remodelling) = the difference; remodelling of the lower border. Total = matrix + intramatrix

Mandibular growth rotations. Compare Fig. 4.37, Ch 4, p. 223.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 4. 1What is the functional matrix theory, and how well does it hold up? 2What is a mandibular growth rotation and why does it matter clinically? 3Would you take a lateral cephalogram to assess cervical vertebral maturation before starting a twin block?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 4.

1. At what age is the juvenile growth spurt, and how does the adolescent spurt differ between the sexes?
2. Name the four mechanisms by which skull bones grow and move.
3. Who proposed the cartilaginous theory, and what three cartilages did he implicate?
4. Define true, apparent and angular remodelling rotations.
5. What does CS3 indicate and what is the caveat attached to CVM as a method?

Chapter 5. Development of the Dentition

Revision Handbook: §3; late crowding §19.4
Question Bank: Block C (Q56–95)
Viva bank: V11–V20

IN ONE MINUTE

The arithmetic of the developing arch. Two numbers do most of the work.

- Incisor liability: 7 mm maxilla, 5 mm mandible — a deficit
- Leeway space: 1.5 mm maxilla, 2.5 mm mandible per quadrant — an excess
- The last tooth to erupt in each arch is the one that gets crowded

Where to read it

pp. 245–286. Prenatal development pp. 245–254 · postnatal development pp. 254–286. Late lower incisor crowding: Box 5.6, pp. 283–284.

The map

Prenatal odontogenesis → the primary dentition and gum pads → features of the ideal primary dentition → the mixed dentition in three phases → incisor liability → the ugly duckling stage → leeway space → occlusal changes in the permanent dentition.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Ideal primary dentition features	**Divergent** primary molar roots (making space for the premolars) · **primate/anthropoid spaces** — mesial to the maxillary primary canines and **distal** to the mandibular canines · **Class I** molar and canine relationship · distal edges of the second primary molars **flush** in the vertical plane. "These features are rarely all seen together, and variation is very much the norm"	p. 264
Predicting from the primary dentition	**Generally unreliable**	p. 264
Changes before the mixed dentition	Occlusal wear and more forward mandibular growth can produce an **edge- to-edge** incisor relationship; interproximal wear or caries alters the molar relationship; **prolonged non-nutritive sucking** can induce an anterior open bite and posterior crossbites	p. 264
The three phases of the mixed dentition	1. First permanent molars and incisors · 2. Premolars, canines and second permanent molars · 3. Third permanent molars	pp. 265– 266
Crown and root timing	Permanent crowns take about **4 years** to form (**first molars ~3 years**). Eruption follows crown completion by **3–5 years**. Roots complete **2–3 years after eruption**. From crown completion, teeth take **2–5 years to reach the alveolar crest** and a **further 1–2 years to reach occlusion**	p. 266

ITEM	DETAIL	PAGE
Eruption ages	First permanent molars **6–7 years**; first then second permanent incisors **8– 9 years**. In the mandible the first permanent incisors may erupt **before or with** the first molars. **Mandibular teeth generally erupt before maxillary**	p. 266
Incisor liability	The permanent incisor crowns are collectively **larger than their predecessors by ~5 mm in the mandible and ~7 mm in the maxilla**	p. 266
How the incisor liability is met	Residual **spacing between the primary incisors** · permanent incisors erupting in a **more labial position** (particularly maxillary), occupying a greater arch perimeter · **transverse increase in intercanine arch width**	pp. 266– 268
Intercanine width	The **only significant change in mandibular intercanine width** occurs during eruption of the permanent incisors, when the primary canines move **distally into the primate spaces**. Change is achieved largely by **alveolar rather than skeletal** change. Very little change occurs thereafter. Maxillary intercanine width increases more, and over a longer period	pp. 268, 278
Ugly duckling stage	Physiological spacing with a midline diastema during eruption of the permanent incisors	p. 276
Eruption of premolars, canines, second molars	Normally between **9 and 12 years**. **Mandible:** canine → first premolar → second premolar. **Maxilla:** first premolar → second premolar → canine	pp. 278– 279
The consequence	The **mandibular second premolar and maxillary canine are the most vulnerable to crowding**, because they erupt last in their respective arches	p. 279
Leeway space	The combined mesiodistal width of the primary canine and molars **exceeds** that of the permanent canine and premolars: approximately **1.5 mm per quadrant in the maxilla and 2.5 mm in the mandible** — the difference reflecting the larger lower second primary molar	p. 279
What determines whether canines and premolars erupt uncrowded	The **size of the leeway space** · whether the permanent incisors have already **encroached on the primary canine region** · the **mechanism of molar relationship correction** (forward movement of the mandibular first molar, e.g. after early loss of a primary second molar, leaves less space)	p. 279
Late lower incisor crowding	**Primary** crowding — tooth-size/jaw-size discrepancy, mainly genetic. **Secondary** — environmental, including tongue, lip and buccal musculature. **Tertiary** — during and after adolescence, with a predilection for the lower labial segment. Third molars contribute **minimally**; **prophylactic removal is not recommended** to prevent it	pp. 283– 284
The key observation	"The changes found in untreated individuals in general are **very similar in nature** to those found in patients following orthodontic treatment"	p. 284

Nice-to-know

Enriches an answer. Skip on a first pass.

- Male dental arches grow **larger and for longer** than female, in both the pre-adolescent and adolescent periods (p. 283).
- There is a gradual and progressive **loss of arch length with age, particularly in the lower arch of females** (p. 284).

- Dental mesenchymal stem cell populations are named by source: **DPSC** (dental pulp), **SHED** (exfoliated deciduous teeth), **SCAP** (apical papilla), **DFPC** (dental follicle), **PDLSC** (periodontal ligament), **GMSC/hOM-SC** (gingiva and oral mucosa) (pp. 284–285).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Learn both numbers and both arches.

Incisor liability 5 mm mandible / 7 mm maxilla; leeway space 2.5 mm mandible / 1.5 mm maxilla per quadrant.

2Explain why the second premolar and the maxillary canine are the crowding victims.

It is a consequence of eruption sequence, not of tooth size.

3Do not treat the ideal primary dentition features as a checklist you would expect to find.

The book says they are rarely all present and variation is the norm — and that prediction from the primary dentition is unreliable.

4Intercanine width is established early and by alveolar change.

This is the anatomical reason lower arch expansion is unstable, and it links Chapter 5 to Chapters 8 and 13.

THE ARITHMETIC OF THE DEVELOPING ARCH INCISOR LIABILITY LEEWAY SPACE

Permanent incisors are WIDER than Primary canine + molars are WIDER
their predecessors than permanent canine + premolars

Maxilla ≈ 7 mm Maxilla ≈ 1.5 mm/quadrant Mandible ≈ 5 mm Mandible ≈ 2.5 mm/quadrant

a DEFICIT to be met an EXCESS to be used

NOTE THE ASYMMETRY: the larger liability is MAXILLARY; the larger leeway is MANDIBULAR

ERUPTION SEQUENCE AND ITS CONSEQUENCE

Mandible: canine → 1st premolar → 2nd premolar → 2nd PREMOLAR crowded

Maxilla: 1st premolar → 2nd premolar → canine → CANINE crowded

The last tooth to erupt in each arch is the one that ends up crowded.

Incisor liability and leeway space. Compare Figs 5.23–5.24, Ch 5, pp. 268, 278.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 5. 1A 9-year-old has 3 mm of lower incisor crowding. Would you intervene? 2What is the leeway space and how do you exploit it?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 5.

1. State the incisor liability in each arch.
2. State the leeway space per quadrant in each arch, and explain the difference.
3. Give the eruption sequence for the premolars and canines in each arch, and name the tooth

most vulnerable to crowding in each.

4. When does the only significant change in mandibular intercanine width occur, and by what mechanism?
5. Distinguish primary, secondary and tertiary crowding.

Chapter 6. Orthodontic Tooth Movement

Revision Handbook: §4; anchorage and headgear §14
Question Bank: Block D (Q96–140), Block N (Q431–481)
Viva bank: V21–V30, V91–V102

IN ONE MINUTE

Biology first, then the force system that provokes it.

- Centre of resistance 1/3–1/2 down the root; 1–2 mm apical to the furcation if multi-rooted
- A couple exerts no net force and acts about CRes wherever it is applied
- Anchorage: see Essay Topic 5 for the classification and the headgear numbers

Where to read it

pp. 287–353. Biological basis pp. 287–307 · mechanical basis pp. 307–336 · **anchorage pp. 336–352**. *The biological basis is Essay Topic 1 and anchorage is Essay Topic 5. This unit covers the section between them, which the essay topics touch only in passing: the mechanics.*

Must-know

Examinable. Each fact carries its page reference.

TERM	DEFINITION	PAGE
Force	A load with **magnitude and direction**, represented as a vector. Multiple forces resolve to a **resultant** by the parallelogram method. Measured in grams; **1 Newton ≈ 100 g** (actually 101.97 g), so **1 cN ≈ 1 g**	p. 307
Centre of resistance	The point at which application of a force produces **bodily movement or translation**. In a free body it coincides with the centre of mass, but teeth are fixed in bone. Presumed to be **one-third to halfway down the root** of a healthy singlerooted tooth; **moves apically if bone support is lost**. For a multirooted tooth it lies **between the roots, 1–2 mm apical to the furcation**. Rigidly ligated arches, and each jaw, have their own centre of resistance	p. 307
Moment	The **rotational effect** created when a force is applied at a distance from the centre of resistance. Moment = **force** × **perpendicular distance** to the centre of resistance; the greater the distance, the greater the rotation	p. 309
Couple	**Two equal and opposite forces** producing rotation. Magnitude = one force × the perpendicular distance between them. Exerts **no net force**, so a couple acting alone produces **pure rotation**, and it always acts **around the centre of resistance regardless of where the forces are applied**. Couples are created within edgewise slots using rectangular or square wires	p. 309

TERM	DEFINITION	PAGE
Centre of rotation	The fixed point about which a body rotates. **The closer the force is applied to the centre of resistance, the further away the centre of rotation moves and the less rotation occurs**; a force through the centre of resistance places the centre of rotation at infinity and produces bodily movement. Determined by intersecting the perpendicular bisectors of lines joining two points before and after movement	pp. 309– 310
The fundamental problem	The centre of resistance is **not directly accessible** — force must be applied to the crown, so a moment and some rotation is **always** produced. **Varying the moment-to-force ratio controls the type of movement**	p. 312
Why removable appliances tip	A spring makes **point contact** with the crown; **no reactionary force or couple is created**, so only tipping is possible. This is a principal reason conventional removable appliances have limited contemporary use	p. 312

FMUST KNOWMust-know forces and the headgear numbers

Examinable. Each fact carries its page reference.

Cross-referenced from Essay Topic 5, but reproduce here for completeness: **anchorage 10–12 h/ day at 250–350 g per side; distalisation or orthopaedic effect 14 h/day at 450–500 g per side** (pp. 346–347); **two independent safety devices are mandatory — a self-releasing headcap or neck-strap and a locking facebow** (p. 350).

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Piezoelectricity and streaming potentials** were once proposed as mediators of remodelling; piezoelectric currents are too small and short-lived, and streaming potentials are of unclear significance (p. 295).
- **Bone-bending** was proposed to reconcile orthodontic pressure–tension with the orthopaedic dogma that compression stimulates formation, but "there is little evidence that orthodontic forces bend bone to any clinically meaningful degree" (p. 294).
- **Rate of tooth movement** is around **1 mm per month**, faster in the maxilla and in younger patients; adolescents with a higher body mass index move marginally faster during alignment and have higher baseline inflammatory markers, including RANKL (pp. 304–305).
- **Bisphosphonates and NSAIDs reduce** the rate of tooth movement; **corticosteroids, PTH and thyroxine increase** it (p. 304).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Define the couple properly.

The examinable subtlety is that a couple produces *no net force* and acts about the centre of resistance *wherever* the forces are applied.

2Get the centre of rotation relationship the right way round.

Force *closer* to the centre of resistance → centre of rotation *further away* → *less* rotation.

3Note that periodontal bone loss moves the centre of resistance apically.

That single fact explains why the same force produces more tipping in a periodontally compromised patient, and why maximum-anchorage cases with bone loss need extra-dental anchorage.

THE MECHANICS IN ONE PAGE

CENTRE OF RESISTANCE 1/3–1/2 down the root (single-rooted);
1–2 mm apical to the furcation (multirooted)
MOMENT force × perpendicular distance to CRes
greater distance = greater rotation
COUPLE two equal, opposite forces
NO net force; acts about CRes wherever applied
CENTRE OF ROTATION force nearer CRes → CRot further away
force THROUGH CRes → CRot at infinity = bodily

THE FUNDAMENTAL PROBLEM

CRes is not accessible — force must be applied at the crown, so a moment is ALWAYS produced.
Varying the moment-to-force ratio is how you control the type of movement.
Periodontal bone loss moves CRes APICALLY → same force, bigger moment, more tipping. Ch 6, pp. 307–312.

The mechanics in one page. Compare Figs 6.11–6.15, Ch 6, pp. 310–314.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 6. 1Why can a removable appliance only tip a tooth? 2Where is the centre of resistance, and why does it matter?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 6.

1. Define a couple and state where it acts.
2. What happens to the centre of rotation as the point of force application approaches the centre of resistance?
3. Where is the centre of resistance of a multirooted tooth?
4. What is the approximate normal rate of orthodontic tooth movement, and in whom is it faster?
5. Which drugs slow, and which accelerate, tooth movement?

Chapter 7. The Orthodontic Patient: Examination and Diagnosis

Revision Handbook: §9; cephalometrics §10; medical history §29.3
Question Bank: Block I (Q256–285), Block J (Q286–325)
Viva bank: V59–V68

IN ONE MINUTE

A systematic examination, in a fixed order, recorded reproducibly.

- Complaint and motivation come FIRST; skeletal pattern in all three planes
- Cephalometry is not justified for minor tooth movement or growth prediction
- CBCT test: will the extra information change the plan?

Where to read it

pp. 354–482. Complaint and motivation p. 354 · dental history p. 355 · **medical history pp. 356–366** · extraoral examination pp. 367–380 · intraoral examination pp. 380–395 · orthodontic records pp. 395–411 · **cephalometric radiography pp. 411–482**. *The medical history is Essay Topic 10 and cephalometrics is Essay Topic 2. This unit covers the clinical examination and records, which the essay topics do not.*

The map

Complaint and motivation → dental history → medical history → extraoral examination (skeletal pattern in three planes, soft tissues, temporomandibular joints) → intraoral examination → records → cephalometric radiography.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
The order matters	Complaint and motivation come **first**, before any clinical finding. The patient's concerns and expectations are part of the diagnosis, not an afterthought	p. 354
Assessment planes	Skeletal pattern is assessed in **all three planes** — antero-posterior, vertical and transverse	pp. 367– 380
Natural head position	Subjects are orientated in **NHP or with the Frankfort plane horizontal, and teeth in RCP** for cephalometry	p. 412
Frankfort plane	**Porion to orbitale.** Advantages: identifiable both clinically and radiographically, and used as the principal reference in several analyses. Disadvantages: both landmarks are **difficult to locate**, both are **bilateral and must be averaged**, and the plane **does not lie in the mid-sagittal plane**, so it is sensitive to head positioning	p. 418
When a cephalogram is justified	**Not in all cases**, "particularly if only minor tooth movements are planned"; **essential before planning surgical movement of the jaws**; and **cannot be justified on the basis of growth prediction alone**	pp. 413, 415– 416

ITEM	DETAIL	PAGE
A better approach to growth	**Serial lateral skull radiographs approximately 1 year apart**, especially useful in Class III cases where treatment decisions are delayed until the direction and extent of the discrepancy are known	p. 416

ITEM DETAIL PAGE

ErrorsLandmark identification error with a **non-circular envelope** in x and y; **cumulative** error producing **artificial correlation** between measurements; **validity** problems (points A and B are convenient rather than anatomical, and remodel as incisors move); **projection** error and magnification; and **superimposition** reliability, where **no method has been proved completely accurate CBCT**The key question is **"whether the additional information provided by CBCT imaging does actually influence the treatment planning decision"**. Research is still needed to define robust clinical selection criteria for large- volume CBCT based on quantified patient benefit pp. 459– 463

p. 411

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Optical laser scanning and stereophotogrammetry** produce three-dimensional soft tissue images without radiation, and are now used to study facial growth and treatment change (pp. 411–414).
- **Computer and AI-assisted landmarking** significantly reduces measurement error, but the main advantage of software is **versatility** — multiple or customised analyses, superimposition, and orthognathic prediction planning with profile photographs superimposed on the tracing (p. 417).
- Modern combined cephalostat-panoramic units use **horizontal scanning with a 10-second exposure** (p. 415).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not begin an examination answer with the occlusion.

Complaint, motivation and expectations first; then history; then extraoral; then intraoral.

2Give the Frankfort plane's disadvantages, not just its definition.

Three specific problems, all quotable.

3Justify every radiograph individually.

No threshold dose; not routine; not for minor tooth movement; not for growth prediction.

4Apply the CBCT test as a question, not a rule:

will the extra information change the plan? If not, do not take it.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 7. 1A referring dentist has sent a patient with a CBCT they took to localise an unerupted canine. Comment.

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 7.

1. What are the three disadvantages of the Frankfort plane as a cephalometric reference?
2. In what circumstances is a lateral cephalogram essential, and in what circumstances is it not justified?
3. What is the more accurate alternative to cephalometric growth prediction?
4. What is the test for whether CBCT is justified?
5. Why are points A and B poor reference points?

Chapter 8. The Orthodontic Patient: Treatment Planning

Revision Handbook: §12; space analysis §11; extraction §13
Question Bank: Block K (Q326–350), Block L (Q351–390), Block M (Q391–430)
Viva bank: V69–V90

IN ONE MINUTE

Turning a diagnosis into a plan — starting with the face.

- Planning begins with the maxillary incisors within the face; soft tissues set the limits
- Expansion yields only about half its width as usable space
- The mandibular arch is the template for all proposed tooth movement

Where to read it

pp. 483–538. Timing of treatment pp. 483–487 · basic principles pp. 487–489 · aims pp. 489–505 · **facial aims pp. 496–505** · occlusal aims and space creation pp. 505–531 · limiting factors pp. 531–538. *Planning is Essay Topic 3 and extraction choice is Essay Topic 4. This unit adds the timing-of- treatment section and the limiting factors, which the essay topics do not cover.*

Must-know

Examinable. Each fact carries its page reference.

ITEM DETAIL PAGE

The philosophical shift	Planning begins with the dentition, **particularly the maxillary incisors, within the face** — "something of a shift in emphasis from the classic approach of planning based around mandibular incisor position and treating to isolated static occlusal goals". Early orthodontics assumed ideal occlusion would give ideal aesthetics; it did not, producing protrusion and relapse. Cephalometrics then produced arbitrary hard tissue standards which "bear little relation to the soft tissues and facial profile". Now **the soft tissues define the limits of treatment**	p. 490
Why the shift happened	Greater emphasis on facial aesthetics and a desire for fuller profiles; resurgence of functional appliance therapy; advances in orthognathic surgery; better understanding of relapse; acceptance of long-term retention; and appreciation of **how the face ages**, so that treatment in the teenage years has consequences decades later	p. 490
Growth modification	Relies on good compliance **and** favourable growth; results are variable; and it is **much more effective at correcting Class II than Class III**	p. 490
Class II camouflage limits	Surgery indicated if **overjet >10 mm**, **pogonion to nasion perpendicular ≥18 mm**, **mandibular body length <70 mm**, or **anterior facial height >125 mm** (Proffit et al., 1992a)	p. 491

Lip response to retraction	**0.7 mm of lip retraction per 1 mm of maxillary incisor retraction**, with a **1.6° increase in nasolabial angle** — the **upper lip being less predictable and less sensitive to retraction than the lower** (Konstantonis et al., 2018)	p. 492
Class III prognostic indicators	*For orthodontics:* Class I or mild Class III skeletal base; average or reduced lower face height; **average or increased overbite**; **proclined** mandibular incisors; **upright or retroclined** maxillary incisors; **anterior displacement on closing**. *Against:* moderate–severe Class III; increased lower face height; **reduced overbite**; retroclined mandibular incisors; proclined maxillary incisors; no displacement; a significant Class III molar relationship	p. 505 (Table 8.4)
Class III stability	Depends not only on growth but on **achieving a positive overbite and overjet at the end of treatment**; patients with reduced overbites at the outset, especially without an anterior displacement, "are often not suitable for orthodontic correction alone"	p. 505

ITEM	DETAIL	PAGE
Occlusal correction to eliminate a displacement	Removing premature contacts on primary teeth **reduces the risk of a posterior crossbite being perpetuated** from the primary into the mixed and permanent dentitions. Options: selective grinding, composite onlays, removable expansion appliance or quadhelix — with **treatment superior to no treatment and the quadhelix possibly more successful than a removable appliance**	p. 487
Early maxillary protraction	Clinically appreciable maxillary advancement is achievable, **particularly before the age of 10 years**, and simultaneous RME can improve it. **Over 60% avoided surgery versus 36% without early treatment** (Mandall et al., 2016) — but those with a **hyperdivergent pattern, increased gonial angle and mandibular prognathism** tend to outgrow the effect, and predicting which cases at 8 or 9 is the problem	p. 487
The mandibular arch is the template	"As a general principle, the mandibular arch dentition is **less amenable to change** when planning orthodontic treatment, particularly **intercanine width and mandibular incisor inclination**. For this reason, the mandibular arch generally forms the template for any proposed orthodontic tooth movement"	p. 525
Canine movement thresholds	Maxillary canines needing **<1/2 tooth unit** of distal movement → consider distalising the whole buccal segment. **>1/2 unit** → premolar extraction almost certainly required. **>1 unit** → first premolar extraction **and** distal movement	p. 528
Expansion yield	Space created within the arch is **only about half the amount of expansion** (O'Higgins & Lee, 2000); **any maxillary expansion is associated with some relapse in the long term**; and **expansion of the mandibular arch to relieve labial segment crowding should be avoided**	pp. 521–522
Interproximal reduction	Up to **1 mm per contact point in the buccal segments**, less anteriorly where enamel is thinner. **Air-cool**; leave adequate contact points and no ledges; **avoid in parallel-sided teeth with subgingival contacts**. Newly exposed enamel is **no more caries-susceptible**, especially with topical fluoride. Most useful in **mild crowding**	p. 525

ITEM	DETAIL	PAGE
Trauma rest periods	Before applying orthodontic force: crown fracture 3 months · subluxation 3 months · minor lateral luxation 3 months · **moderate/severe luxation, intrusion, extrusion and re-implantation all 12 months** · root fracture 12–24 months · endodontically treated after trauma usually 12 months	Ch 8, pp. 536– 537
Monitoring a traumatised tooth	Vitality testing and periapicals before treatment; repeat radiographs **6–9 months** after starting movement; **pause 3 months** if resorption appears. Complications of moving such teeth: **loss of vitality, pulpal calcification, further root resorption**	Ch 8, p. 535
Finishing to Class II molars	Where the mandibular arch is well aligned or only mildly crowded and no extractions are planned there, maxillary premolar extraction gives a Class II molar relationship, which is **"perfectly acceptable from an occlusal and functional point of view"**	p. 529

Nice-to-know

Enriches an answer. Skip on a first pass.

- Second permanent molar extraction: in the mandible it gives a small amount of buccal segment space but **little for the labial segment**, and third molar eruption is unpredictable
- in the maxilla it gives little mesial space but **facilitates buccal segment distalisation and is associated with more reliable third molar eruption** (p. 513).
- Even with very good headgear wear, **a full unit Class II molar relationship is difficult to correct by distal movement alone** in the absence of favourable growth (p. 522).
- The "great extraction debate" box (pp. 514–515) is worth reading once for its history — Angle, Tweed, Begg — and for the conclusion that **expansion of the mandibular arch remains inherently unstable however it is re-branded as "arch development"**.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Quote the shift in planning philosophy.

It is one sentence and it reframes an entire answer.

2Reproduce Table 8.4 for Class III

, and include the positive overbite requirement, which candidates omit.

3Use the mandibular-arch-as-template sentence.

It is the single most portable line in the book and applies to Chapters 5, 8, 12, 13 and 15.

4Know that expansion yields only half its width as space.

It is the fact that makes expansion a weaker space-creating option than it appears.

5Do not assume Class I molars are compulsory.

A Class II finish is explicitly endorsed where the lower arch needs no extractions.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 8. 1Why does the textbook say planning should begin with the face rather than the occlusion?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 8.

1. State the four cephalometric thresholds beyond which Class II surgery is indicated.
2. How much does the upper lip retract per millimetre of maxillary incisor retraction?
3. How much usable space does 6 mm of maxillary expansion create?
4. What are the Class III contraindications to orthodontic correction alone?
5. How much interproximal reduction is permissible per contact point in the buccal segments?

Chapter 9. Functional Jaw Orthopaedics

Revision Handbook: §16; expansion §24.2; facemask §23.3
Question Bank: Block P (Q526–566), Block X (Q817–841)
Viva bank: V112–V121, V162–V173

IN ONE MINUTE

Changing the jaws rather than the teeth. Transverse works; sagittal is modest.

- RME splits the suture: ~50% of screw expansion is skeletal, ~10% relapses
- Slow expansion CAN split the suture in the early mixed dentition
- Facemask: >60% avoided surgery versus 36% — a shift in probability, not a cure

Where to read it

pp. 539–613. Maxilla — expansion and protraction pp. 542–559 · **functional appliances pp. 559–583** · designs pp. 568–583 · clinical use pp. 583–601 · **fixed Class II correctors pp. 601–611** · mandibular restraint p. 611. *Functional appliances are Essay Topic 8 and overjet reduction in the growing child is Essay Topic 12. This unit covers the maxillary orthopaedics that opens the chapter — expansion and protraction — which the essay topics treat only within crossbite and Class III.*

FMUST KNOWMust-know maxillary transverse orthopaedics
Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Sutural maturation	The intermaxillary suture becomes **increasingly interdigitated and complex with age**, with **complete fusion generally by the late teenage years** (Melsen, 1975). But there is a **poor correlation between sutural maturational stage and chronological age**	p. 543
SME can split the suture	In the **early mixed dentition**, slow expansion with a removable plate, trihelix or quadhelix **can readily split the intermaxillary suture**. It is the **late mixed and early permanent dentitions** that require the higher-force RME approach	p. 543
RME quantities	Splits the suture and achieves **up to 4 mm of widening**, representing **about 50% of total screw expansion** (Liu et al., 2015). Short-term: **+2.46 mm** maxillary basal bone width, **+3.09 mm** alveolar palatal crest, **+5.69 mm** intermolar width, with **+3.75° dental tipping**	pp. 543– 544
RME relapse	About **10% of the skeletal change relapses at 4–8 months post-retention**, with some uprighting of the dentition	p. 544
A planning caution	Significant maxillary expansion **in the absence of a posterior crossbite** will require compensatory expansion of the mandibular dentition, **which is likely to be unstable**	p. 543

MARPE versus RME	MARPE gives greater skeletal expansion — approximately **+1.21 mm at ANS, +1.13 mm at first premolars, up to +2 mm at first molars, +1.14 mm at PNS** — with **reduced buccal inclination (–6.06° premolars, –3.17° molars**). It also increases sutural opening and palatal width and reduces adverse periodontal effects, **though it does not eliminate them**. But it is **more invasive, more expensive**, and the evidence base for RME generally is low level and for MARPE particularly so	pp. 546– 547
The randomised trial	A high-quality RCT comparing conventional with hybrid tooth-and-bone-borne RME using CBCT found **limited differences in expansion between groups at 5 years** (Bazargani et al., 2023)	p. 547
Expansion and breathing	Evidence relating to improved breathing after RME is **low quality and equivocal**, whether conventional or bone-anchored. Equating airway volume with nasal airflow and breathing efficiency is **"quite simplistic"**	pp. 547– 548

FMUST KNOWMust-know maxillary protraction

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Facemask	Elastic traction from an intraoral appliance to a forehead- and chin-borne facemask, worn **12–14 hours daily**	p. 552
Timing	Clinically appreciable maxillary advancement, particularly **before the age of 10 years**, successful in the short term in the majority	p. 487
The long- term outcome	Most initial skeletal effects are lost with growth, but the **success rate in avoiding surgical correction was over 60%, versus 36% without early treatment** (Mandall et al., 2016)	p. 487
Who outgrows it	Patients with a **hyperdivergent facial pattern, increased gonial angle and mandibular prognathism**	p. 487
BAMP	Bone-anchored maxillary protraction — Class III intermaxillary traction between **miniplates in the anterior mandible and the infrazygomatic crests**. Evidence for efficacy against established techniques is **largely limited to case reports**	pp. 558– 559; Ch 13, p. 920
MUST KNOW F	Must-know functional appliances (summary — see Essay Topic 8)	

Examinable. Each fact carries its page reference.

Mechanism: **predominantly dentoalveolar**, with skeletal change "a matter of only a few millimetres at best... essentially transient" (p. 560); the appliance probably works by **unlocking an occlusion held by intercuspation and dentoalveolar adaptation** and allowing normal mandibular growth to express itself (p. 562). Twin block blocks at **~70°** (p. 577). Herbst overjet reduction in **6–8 months** (p. 602). Pancherz at 32 years: **64% of molar and 86% of overjet correction stable**; retain **at least 2 years** with an activator (p. 562). Pacha RCT: **twin block 37% failure versus Herbst 17%**, overjet reduction **5.8 versus 7.1 mm** (p. 606). Poor responders: **high-angle patients with increased lower anterior face height and mandibular plane angle and a reduced overbite** (p. 587).

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Fixed Class II correctors** other than the Herbst — Jasper Jumper, Eureka, Forsus, Powerscope — are pre-formed springs or rods attached directly to a fixed appliance.
- **Muscle evidence is contradictory.** Early surface electromyography suggested lateral pterygoid hyperactivity on protrusion (McNamara, 1973), but later work with **surgically implanted electrodes showed a reduction** in activity (Voudouris et al., 2003) — so an alternative hypothesis is **viscoelastic stretch of the joint capsule with referred force to the glenoid fossa** (p. 564).
- **Animal models are of limited applicability:** the appliance regimes would not be tolerated by humans
- animals mature far faster → the animals often fail to thrive
- and **Class II malocclusion does not exist in monkeys or rodents, so the experiment actually creates an abnormal Class III** (pp. 564–565).
- **Maximal versus incremental advancement** makes little clinical difference; what probably matters more is **suitable retention of the overjet reduction** to allow consolidation of remodelling (pp. 595–596).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not oversimplify slow versus rapid expansion.

Slow expansion can split the suture in the early mixed dentition; the constraint is **age-related sutural interdigitation**, not the appliance category.

2Do not claim expansion treats nasal breathing or sleep apnoea.

The evidence is low quality and equivocal, and equating airway volume with airflow is explicitly criticised.

3Quote the Mandall figures.

60% versus 36% is the strongest available argument for any form of early Class III intervention, and immediately identifying the difficulty — predicting which 8-year-old will benefit — shows judgement.

4Be sceptical about MARPE and BAMP.

Both are promising; neither has the evidence to displace established techniques.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 9. 1A 9-year-old has a bilateral posterior crossbite with no displacement and a Class III skeletal pattern. What do you do?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 9.

1. What proportion of total screw expansion in RME is skeletal, and how much relapses?
2. Can slow maxillary expansion split the intermaxillary suture?
3. What is the reported success of early maxillary protraction in avoiding surgery?
4. How long is a facemask worn each day?
5. Why is significant maxillary expansion in the absence of a crossbite unwise?

Chapter 10. Contemporary Fixed Appliances

Revision Handbook: §17; materials §15; mechanics §18
Question Bank: Block O (Q482–525), Block Q (Q567–606), Block R (Q607–641)
Viva bank: V103–V129

IN ONE MINUTE

Everything attached to a tooth, and what you do with it in what order.

- Alignment → levelling → space closure → finishing, each with its own wire properties
- Self-ligation: no difference in rate, treatment time, visits or occlusal outcome
- Customisation: comparable results, more breakages, more planning time

Where to read it

pp. 614–705. Evolution pp. 614–629 · components pp. 629–666 · brackets pp. 630–647 · **archwires pp. 647– 658** · auxiliaries and bands pp. 658–666 · using the pre-adjusted appliance pp. 666–687 · **stages of treatment pp. 687–705**. *Appliance evolution is Essay Topic 6 and archwires are Essay Topic 7. This unit adds the stages of fixed appliance treatment, which the essay topics do not cover.*
FMUST KNOWMust-know the stages of treatment
Examinable. Each fact carries its page reference.

STAGE	AIM AND ARCHWIRE	PAGE
1. Alignment	Correct rotations and displacements. Requires **large springback, low stiffness, high stored energy, biocompatibility, low friction**. Round **NiTi, multistrand or coaxial steel, 0.012–0.018 inch**, often followed by a short period of **rectangular NiTi to begin torque expression**	p. 649
2. Levelling and overbite reduction	Increasing stiffness; **round stainless steel 0.016–0.020 inch**; a **reverse curve of Spee** where overbite reduction is needed, with **lingual crown torque** to counter the proclination it causes	pp. 650, 934, 938
3. Space closure	Requires **high stiffness, low stored energy, biocompatibility, low friction, good joinability**. **Rectangular stainless steel**. Either **sliding mechanics** (frictional properties matter) or **looped closure** (formability matters). **En masse retraction combined with fixed anchorage is the most efficient method**; individual canine retraction is less anchorage-demanding but slower	pp. 650, 698
4. Finishing and detailing	Settling the occlusion; **beta titanium** for finishing bends because of its formability and lower stiffness	p. 653
FMUST KNOW	the evidence on refinements Must-know	

Examinable. Each fact carries its page reference.

Self-ligation is**No significant difference** in rate of tooth movement, treatment time or faster, allows stablenumber of visits (Scott 2008a; Fleming 2010; DiBiase 2011; Johansson & expansion, is lessLundström 2012). Pain benefit **marginal and not supported by all studies**. painfulExpansion and incisor proclination findings **equivocal**. **No better occlusal outcome**. "Apart from saving some chairside time... self-ligating brackets do not perform clinically any differently" pp. 646– 647

CLAIM EVIDENCE PAGE

CLAIM EVIDENCE PAGE

CustomisedRetrospective SureSmile data suggested 7–15 months' saving, but with high **appliances** shortenrisk of selection bias. A **large high-quality RCT** found **comparable** treatment**outcomes and no difference in treatment time**, with **more breakages, more patient complaints and more planning time** for Insignia. Treatment time was dictated by **case difficulty and the identity of the treating orthodontist** (Penning et al., 2017) **One archwire**Cochrane review of 12 RCTs in ~800 patients: **moderate evidence that sequence is bestcoaxial superelastic NiTi produces greater movement over 12 weeks than single-strand**; no real difference in day-1 pain between multistrand steel and superelastic NiTi; otherwise **insufficient evidence** on alignment rate, time, pain or root resorption (Wang et al., 2018). Differences are marginal; **the choice is up to the individual orthodontist** pp. 630– 631
pp. 655– 656
FMUST KNOWMust-know archwire materials (summary — see Essay Topic 7)
Examinable. Each fact carries its page reference.

Stainless steel, the "18–8" alloy: Fe 70%, Cr 17–20%, Ni 8–12%, C max 0.08%; **chromium confers corrosion resistance, nickel confers ductility** (p. 651). **NiTi** — developed by the Naval Ordinance Laboratory in the early 1960s as *nitinol*; three generations: **martensitic-stabilised** (flexible, no shape memory), **austenitic-active** (stress-induced, superelastic), **martensitic-active copper NiTi** (temperature-induced, thermoelastic) (pp. 654–655). **Cobalt chromium (Elgiloy)** — greater formability than steel, similar stiffness, greater friction; used for auxiliaries such as intrusion arches and quadhelices. **Beta titanium** — stiffness about **one-third** that of steel, good formability, higher friction; finishing (p. 653).

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Nickel-free brackets** in titanium or cobalt chromium are available; **3D-printed brackets** have been described (p. 630).
- Bracket bases are **roughened or sandblasted** and **curved in both planes** to aid location and seating (p. 630).
- **Lingual appliances** — Incognito and WIN (named for its inventor Dirk Wiechmann *and* the appliance design coming from an idea INside his head) use CAD technology with a manual set- up, scanned to a virtual model, brackets computer-generated, printed in wax and cast in metal, then indirectly bonded via a transfer tray.

EXAMINER TRAPS

The errors that cost marks on this topic.

1The archwire sequence follows from the changing property requirements

, not from convention. State the two contrasting requirement sets and the sequence writes itself.

2State plainly what the evidence shows about self-ligation and customisation.

It is the easiest place in the book to demonstrate critical reading.

3Quote the Penning finding on what actually determines treatment time

— case difficulty and the operator. It is a memorable and slightly humbling point.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 10. 1Take me through your archwire sequence for a moderately crowded extraction case, and justify each stage.

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 10.

1. State the archwire properties required during alignment and during space closure.
2. What did the Penning trial find about customised appliances?
3. Name the three generations of nickel titanium archwire and their trigger.
4. What is the stiffness of beta titanium relative to stainless steel?
5. What does the Cochrane review conclude about initial alignment archwires?

Chapter 11. Skeletal Anchorage

Revision Handbook: §14.6
Question Bank: Block N (Q431–481)
Viva bank: V91–V102

IN ONE MINUTE

What changed orthodontics most in twenty years: anchorage without compliance.

- Survival 80–90%; cortical bone thickness is the single largest factor
- Insertion torque 5–10 Ncm — too low loses primary, too high loses secondary stability
- Palatal placement is by far the most reliable; the mandibular buccal shelf the least

Where to read it

pp. 706–732. Development pp. 706–707 · design pp. 707–709 · benefits and risks pp. 709–719 · placement and removal pp. 719–724 · force application p. 724 · clinical applications pp. 725–732.

The map

History → design (head, neck, body) → benefits and risks → factors affecting success → placement technique → force application → clinical applications.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
The three components	**Head** — emerges into the mouth; various designs for attaching wires, elastics, appliances or arches, some with an integrated screw drive. **Neck** — traverses the mucosa; length dictated by mucosal depth and type; **most are designed for keratinised mucosa**. **Body** — the threaded part engaging bone; **cylindrical or tapered**; most current systems are **self-drilling** (early ones were self-tapping and required a pilot hole)	pp. 707– 708
Material	Majority **Grade 5 machined titanium alloy** for strength and biocompatibility; **stainless steel is an acceptable alternative** with similar properties and failure rates. Surfaces are **polished** — sandblasting or acid-etching does **not** enhance success, unlike dental implants	p. 709
Dimensions	Diameter typically **1.3–2 mm**; length **5–14 mm**	p. 709
Benefits	Reduced reliance on compliance; better anchorage control; **greater range of tooth movements possible**; minimal anchorage loss and potentially more efficient treatment	p. 709
Overall survival	**80–90%**	p. 713
Primary vs secondary stability	**Primary** — initial mechanical interlocking at placement. **Secondary** — longerterm success, biologically mediated	p. 713

ITEM	DETAIL	PAGE
The single largest factor	**Cortical bone thickness** — greater thickness gives lower failure rates	p. 713
Length and diameter	**Higher failure for screws <8 mm long and <1.2 mm diameter.** Short screws (≤8 mm) fail at **12.7%** versus **8.3%** for longer (>8 mm). Higher success with screws ≥**6–8 mm** long, and with diameters ≥**1.4 mm** in the mandible and ≤**1.4 mm** in the maxilla. But bigger is not simply better — larger diameters risk root contact interradicularly	pp. 714–715
Failure by site	**Median palate 1.3%** · **paramedian palate 4.8%** · **maxillary buccal interradicular 5–10%**, rising to **25.9% between the first and second molars** · **zygomatic buttress 28.1%** · **mandibular buccal shelf >65%** in one series (though 7% and 25% reported elsewhere) · overall **12.3%** for some sites	pp. 716–717
Mucosa	Insertion into **keratinised attached gingiva** has a higher success rate than non-keratinised free gingiva, which can also be uncomfortable	p. 716
Root contact	A **major cause of failure**, and the major risk of interradicular placement	p. 716
Insertion torque	Recommended **5–10 Ncm** in alveolar bone (Motoyoshi et al., 2006). **Lower torque → poor primary stability; higher torque → poor secondary stability**, through microfracture. Greater torque is needed posteriorly and for larger-diameter screws; **success rates fall above 10 Ncm**	pp. 718–719
Loading	**Immediate loading does not affect failure rates in adults**; in **adolescents, delayed loading has been associated with higher** failure	p. 719
Sinus perforation	**10%** incidence with one insertion position, rising to **48%** in another. Usually asymptomatic, but **associated with a higher risk of failure**	pp. 711, 719
Other determinants	**Patient sex does not affect success**; **vertical facial type does**, with higher failure in some patterns; **heavy smoking increases failure**; **self-drilling and self-tapping have similar success**; **tapered versus cylindrical body makes no difference**; **clinician experience affects failure rates**	pp. 714–719
Fracture	Reported at about **5%**, though **10–20% of orthodontists** report having experienced it	p. 713
Safe interradicular distances (to the CEJ)	Maxilla: **8 mm or more** between the central incisors and between lateral incisor and canine; **6 mm or more** between first and second premolars; **4 mm or more** between second premolars and first molars, and palatally between premolars and molars. Mandible: **4 mm or more** between first and second premolars, between second premolars and first molars, and between first and second molars	pp. 717–718
The palate	**Placement in the palate is not restricted by the dentition** — which is why median and paramedian sites have the lowest failure rates	p. 718

Nice-to-know

Enriches an answer. Skip on a first pass.

- **History:** Kanomi described miniscrews to intrude lower incisors in 1997
- Costa and colleagues followed in 1998 with 16 screws and only two failures, showing they could be **immediately loaded**

- Hyo-Sang Park's group in South Korea produced the first commercial system (Kyung et al., 2003) (pp. 706–707).
- **Thread pitch and depth** vary between manufacturers; a **finer pitch requires higher insertion torque**, which can itself influence success (p. 707).
- Placement uses **hand-drivers or reduced-speed torque-limiting handpieces**, with speed reduced to **25–50 rpm** and torque limited (p. 720).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Do not describe miniscrews as risk-free.

The greatest risk is **failure and loss of the screw**, and the anatomy — cortical bone and adjacent roots — genuinely constrains their use.

2The torque window is the sort of specific fact that distinguishes candidates.

5–10 Ncm, and know *why* both extremes fail: too low, no mechanical interlock; too high, microfracture of bone and loss of secondary stability.

3Know the failure gradient by site.

Palatal placement is dramatically more reliable than buccal interradicular placement, and the mandibular buccal shelf is the least reliable of all.

4Immediate loading is fine in adults.

Candidates often assume a healing period is mandatory; the evidence says otherwise in adults, with a caveat in adolescents.

MINISCREW FAILURE RATE BY SITE

Median palate 1.3%
Paramedian palate 4.8%
Maxillary buccal interradicular 7.5%
Overall (pooled) 12.3%
Maxillary between 1st/2nd molars 25.9%
Zygomatic buttress 28.1%
Mandibular buccal shelf 65.0%
Placement in the palate is not restricted by the dentition — which is why it is by far the most reliable site.
Insertion torque 5–10 Ncm too low → poor primary stability · too high → microfracture, poor secondary stability
Higher failure below 8 mm length and 1.2 mm diameter · root contact is a major cause of failure
Miniscrew failure rate by site. Compare Fig. 11.5, Ch 11, p. 716.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 11. 1What factors determine whether a miniscrew succeeds? 2You want to retract an upper labial segment without any molar movement in an adult. How would you plan the anchorage?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 11.

1. What is the recommended insertion torque range, and what happens above and below it?
2. Give the failure rates for median palatal and mandibular buccal shelf placement.
3. What is the single largest factor influencing miniscrew success?
4. What length and diameter thresholds are associated with higher failure?
5. Does immediate loading affect failure rates?

Chapter 12. Management of the Developing Dentition

Revision Handbook: §20; anomalies §3.3
Question Bank: Block T (Q677–721)
Viva bank: V140–V151

IN ONE MINUTE

A catalogue of what can go wrong between 6 and 12, and the interceptive response.

- Ask each time: self-correcting, interceptable, or plan around it in the permanent dentition?
- Agenesis 6.4%; supernumeraries up to 4%; the teeth at the end of each series are vulnerable
- Recognise primary failure of eruption — traction causes ankylosis

Where to read it

pp. 733–858. Early loss of primary teeth pp. 733–737 · prolonged retention pp. 737–748 · **tooth agenesis pp. 748–755** · **supernumeraries pp. 755–758** · abnormalities of size pp. 758–761 and form pp. 761–769 · **abnormalities of eruption pp. 769–806** · early loss of permanent teeth pp. 806–822 · sucking habits pp. 822–825 · tongue-tie p. 825 · crowding pp. 826–833 · crossbites pp. 833–841 · skeletal problems pp. 841–849 · abnormalities of structure pp. 849–858. *The impacted canine is Essay Topic 14, the impacted second molar Topic 15, the missing central incisor Topic 13, crossbites Topic 16, and early Class II and Class III intervention Topics 12 and 17. This unit covers the developmental anomalies those topics do not: agenesis, supernumeraries, abnormalities of size, form and structure.*

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Terminology and prevalence	Selective tooth agenesis (STHAG) has an overall prevalence of **6.4% excluding third molars**, is **more common in females**, and varies by continent: **Africa 13.4% · Europe 7% · North America 5% · Latin America 4.4%** (Khalaf et al., 2014)	p. 748
Severity	**Oligodontia** — six or more teeth missing excluding third molars — is the least common, at around **0.25% of Caucasians**	p. 751
Which teeth	**Third permanent molars most commonly absent, up to 30%**, followed in Caucasians by **mandibular second premolars, then maxillary lateral incisors**. **Teeth at the end of each series are the most vulnerable.** Agenesis of canines and of first and second permanent molars is **rare in isolation**	p. 751

ITEM	DETAIL	PAGE
Syndromic associations	**Down syndrome** is one of the commonest causes of syndromic agenesis. The **ectodermal dysplasias** are a heterogeneous group characterised by defective **teeth, hair, nails and sweat glands**, with generalised reduction in crown and root size, conical crowns, enamel hypoplasia and molar taurodontism	pp. 751–752

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Prevalence	Up to **4%** of the population in the permanent dentition; **less than 1%** in the primary dentition; **twice as common in males**	p. 755

ITEM	DETAIL	PAGE
The four types	**Conical** — small, peg-shaped, normal root formation. **Tuberculate** — multicusped, often barrel-shaped. **Supplemental** — duplication of a normal tooth. **Odontome** — a developmental malformation containing enamel and dentine	
Consequences	Occur in around **50% of patients** with supernumeraries: **impaction, displacement or failure of eruption** of the permanent tooth	pp. 755–757 p. 757

FMUST KNOWMust-know abnormalities of size and form

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Megadontia	Around **1%** of the population	p. 760
Microdontia	Around **2.5%**; may affect the whole dentition or individual teeth	p. 760
Transposition	**~0.33%** of most populations; the **canine is almost always involved**, most often **maxillary canine and first premolar**, or **mandibular canine and lateral incisor**. Associations suggesting a genetic basis: **female predilection, unilateral and left-sided dominance, tooth agenesis, peg-shaped maxillary lateral incisors, retained primary teeth and Down syndrome**	pp. 802–803

Must-know

FirstAgainst the primary second molar: prevalence **0.75–4.3%**, **more common in permanent-males**, and **25 times more common in the maxilla**, where it is a good indicator **molar**of **posterior crowding**. If not self-corrected within **6–12 months**, intervention is **impaction**indicated. Treat with a **separating elastic, spring or brass wire below the contact point**, or **distal grinding of the primary molar Primary**Diagnosed when **no clear systemic, pathological or environmental factor** is **failure of**responsible. Associated with **loss-of-function mutation in**

PTH1R, encoding a **eruption**parathyroid hormone receptor with a role in osteoclast function during eruption (Decker et al., 2008). **Critically: active orthodontic extrusion will normally result in ankylosis and failure.** Management is extraction with space closure or prosthetic replacement, localised osteotomy with segmental extrusion, or coronal build-up p. 793

pp. 800– 801

Examinable. Each fact carries its page reference.

ITEM DETAIL PAGE

Nice-to-know

Enriches an answer. Skip on a first pass.

- Autosomal dominant *PTH1R* mutations causing primary failure of eruption **may also be associated with osteoarthritis** (p. 800).
- Transposition may be intercepted by **extracting an overlying primary tooth** if identified early and the transposition is not fully established (p. 803).
- The interceptive extraction, balancing extraction and ankylosed tooth evidence boxes (12.1, 12.2, 12.3) are all worth reading; **balancing first molar extractions are rarely required to preserve a centreline**, and such decisions should be made as part of the overall plan (p. 821).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Primary failure of eruption is the diagnosis you must not miss.

Applying orthodontic traction to a tooth with PFE causes ankylosis and makes things worse.

2Give the agenesis pattern with its logic:

teeth at the end of each series — third molars, second premolars, lateral incisors — are the vulnerable ones.

3The 6–12 month observation window

recurs: it applies to first permanent molar impaction and to second permanent molar impaction alike.

4Supernumeraries cause a problem in about half of cases

— so the presence of one is not automatically an indication for removal, and the decision depends on whether it is obstructing eruption or displacing a tooth.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 12. 1A 9-year-old has an unerupted upper central incisor and a supernumerary on the radiograph. How do you manage it?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 12.

1. What is the prevalence of tooth agenesis excluding third molars, and which teeth are most commonly affected?
2. Name the four types of supernumerary tooth.
3. What proportion of patients with a supernumerary experience a consequence?
4. What gene is associated with primary failure of eruption, and why must it be recognised?
5. Where is first permanent molar impaction most common, and what does it indicate?

Chapter 13. Management of the Permanent Dentition

Revision Handbook: §18, §21, §22, §23, §25; retention §19
Question Bank: Block R (Q607–641), Block S (Q642–676), Block U (Q722–746), Block V (Q747–786), Block W (Q787–816), Block Y (Q842–871)
Viva bank: V122–V173

IN ONE MINUTE

Definitive treatment, organised by plane of space, ending with the adult and retention.

- Class III surgery threshold: Wits $\leq$ –11 mm is the single most important value
- Mandibular advancement appliances work in mild-to-moderate OSA; expansion does not
- Most patients stop wearing removable retainers within 4–5 years

Where to read it

pp. 859–990. Tooth–arch size problems pp. 859–902 · anteroposterior problems pp. 902–925 · vertical problems pp. 925–949 · transverse problems pp. 949–955 · **adult orthodontics pp. 955–972 · obstructive sleep apnoea pp. 972–974 · relapse and retention pp. 974–990**. *Impacted canines are Essay Topic 14, overbite Topic 11, Class III Topic 17, crossbite Topic 16 and late lower incisor crowding Topic 20. This unit covers adult orthodontics, obstructive sleep apnoea and the retention evidence, which the essay topics touch only partially.*
FMUST KNOWMust-know obstructive sleep apnoea
Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Definition and prevalence	Periodic cessation of breathing (apnoea) or collapse of the upper airway during sleep. Affects around **2%** of the population	p. 972
The management hierarchy	**Eliminate aggravating factors** (weight, alcohol, sleeping position — encouraging lateral rather than supine sleep) → **CPAP**, the benchmark for severe disease but associated with **poor compliance** → **mandibular advancement splints**, effective for **mild-to-moderate** OSA → **surgery** for those who cannot tolerate CPAP or an appliance, or who have an underlying skeletal discrepancy	pp. 972– 974; Ch 2, p. 90
Appliance types	**Monobloc** — a single component rigidly fixing the mandible. **Bibloc** — two interconnected components allowing some movement	p. 973
What orthodontics cannot claim	Maxillary expansion is **not currently recommended as a primary treatment** for OSA, and RME is **not** recommended as a primary intervention to improve nasal breathing. There is **no evidence** that extraction-based orthodontic treatment reduces the oropharyngeal airway or increases the incidence of sleep apnoea — the "airway orthodontics" claim "appears to be based more on scaremongering and marketing than on clinical evidence"	Ch 2, pp. 90– 91
MUST KNOW F	Must-know relapse and retention	

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Traits with high relapse potential	**Rotated teeth · lower incisor crowding · changes in lower incisor labiolingual position · expansion of lower intercanine width · spacing**	Ch 2, p. 100
Age-related change is not relapse	Increased crowding later in life occurs in untreated individuals and **should be regarded as an age-related change**. Typical untreated adults in their 30s and 40s show **shorter arch perimeter, decreased mandibular intercanine distance and increased anterior crowding** (Thilander, 2009)	Ch 2, p. 100; p. 978
The dominant mechanism	Facial growth continues throughout life in the same direction as adolescence but to a much smaller degree (Behrents, 1985) and is **rotational** (Solow, 1980); the lower labial segment **uprights to compensate for mandibular growth**, which is "**thought to be a major factor**" in late lower incisor crowding in treated and untreated individuals alike	p. 978
The occlusal force theory is doubted	The anterior component of occlusal force has been proposed as causative (Southard et al., 1990), "**however, the short duration of these forces makes this unlikely**"	p. 978
What is stable	Anterior crossbite corrected **with a positive overbite** is usually stable and **requires no retention** (Wiedel & Bondemark, 2015b). Posterior crossbite corrected in the mixed dentition shows good stability. A well-interdigitated Class I occlusion aids stability, including after Class II correction. **But occlusal outcome does not influence lower incisor irregularity, which increases post-retention irrespective of the finish**	p. 977
Lower lip position	If lips are competent after treatment and the **lower lip rests labial to the upper incisors**, overjet reduction is more stable. Lip competence increases with growth, **especially in boys**, whose lips grow later and more	p. 977
How long?	"The honest answer is **for as long as they want to keep their teeth perfectly straight**." Not everyone relapses, but **it is not possible to predict who**. Little (1975) suggested **under 3.5 mm** of lower incisor irregularity as an acceptable level of relapse — but many patients find any deterioration unacceptable	p. 989
Fixed versus removable	Both effective in the **upper** labial segment. In the **lower**, fixed retainers are **marginally more effective for the first 12 months**, especially for rotations, but **equivalent thereafter**. Even with fixed retainers some irregularity reappears. Fixed retainers have a **higher failure rate in the lower arch**, and "the patient will often only notice a breakage when the lower incisors start moving"	p. 989
Periodontal cost of fixed retainers	Harder to clean, greater plaque and calculus, though **short-term periodontal effects appear minimal**; **longer-term evidence is conflicting** — one study found no detriment at 6 years, another greater recession and probing depths at nearly 10 years	p. 989
Removable retainers	**Vacuum-formed retainers are more effective than Hawley-type** at maintaining lower labial segment alignment (Rowland et al., 2007); both remain effective **part-time**; **patients prefer vacuum-formed** — less conspicuous, less effect on speech, more likely to be worn long term	pp. 989– 990

Compliance decays	Good in the short term, but **by 4–5 years post-treatment the majority of patients have stopped wearing their retainers altogether** (Al-Moghrabi et al., 2018; Krämer et al., 2022) — "ironic because when patients are asked at the start of treatment, almost all of them rate keeping their teeth straight as extremely important"	p. 990
Whose responsibility?	The choice should be made **in partnership with the patient**, giving shared responsibility. There remains "the thorny issue of who should actually manage and be responsible for long-term retention: the orthodontist, dentist or indeed the patient"	p. 990
MUST KNOW F M	ust-know other examinable specifics from this chapter	

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
Class III surgery thresholds	**Wits ≤ –11 mm** (one of the most important parameters) · Holdaway angle ≤3.5° · **ANB ≤ –4°** · gonial angle ≤120° · lower anterior face height ≥66% · **overjet ≤ –5 mm** · mandibular incisor inclination ≤80°. Stellzig-Eisenhauer's discriminant analysis predicts the need for surgery with **>90% success**, combining Wits, the maxillatomandible length ratio, the saddle angle and mandibular midline deviation from the casts (Kochel et al., 2011)	pp. 920– 921
Class III and growth	Class III tends to **worsen with age** as mandibular growth persists, so **surgery is delayed until adolescent growth has ceased — the late teens**. **Mandibular extractions are best avoided in growing patients who may need surgery**, because retroclination must be undone at decompensation and that is difficult once extraction sites have closed	p. 921
Slow versus rapid expansion	**SME** — low continuous force, screw turned a quarter turn every 2 days. **RME** — heavy intermittent force, **1–4 turns per day, 0.25–1 mm daily, over 2–4 weeks**, appliance retained **3–6 months**, forces **up to 10 kg**. Both equally effective for **dental** expansion; RME achieves **slightly greater posterior skeletal expansion but with greater buccal tipping**. Dental expansion maintained: **SME 95%, RME 85%**. Both cause **bone height loss and reduced buccal bone thickness — over 1 mm for RME but more for SME**. RME is **more painful in the first week**	pp. 951– 953
Overbite	Intrusion force **~60 g total, 15 g per tooth**; segmental mechanics achieve **~1.5**	pp.

	extraction sites have closed	
Slow versus rapid expansion	**SME** — low continuous force, screw turned a quarter turn every 2 days. **RME** — heavy intermittent force, **1–4 turns per day, 0.25–1 mm daily, over 2–4 weeks**, appliance retained **3–6 months**, forces **up to 10 kg**. Both equally effective for **dental** expansion; RME achieves **slightly greater posterior skeletal expansion but with greater buccal tipping**. Dental expansion maintained: **SME 95%, RME 85%**. Both cause **bone height loss and reduced buccal bone thickness — over 1 mm for RME but more for SME**. RME is **more painful in the first week**	pp. 951– 953
Overbite reduction	Intrusion force **~60 g total, 15 g per tooth**; segmental mechanics achieve **~1.5 mm maxillary and ~1.9 mm mandibular** intrusion; miniscrews **2–3 mm**, i.e. **0.67–0.95 mm more** than segmental with less molar extrusion; proclination gives **0.1–0.2 mm per degree**; Dahl re-establishes contacts over **~6 months**	pp. 936– 940
Space closure	**En masse retraction combined with fixed anchorage is the most efficient method**; individual canine retraction is less anchorage-demanding but takes longer	p. 698

Nice-to-know

Enriches an answer. Skip on a first pass.

- **Adult orthodontics** (pp. 955–972) is a growing proportion of practice; the key differences are absence of growth, greater periodontal susceptibility, higher motivation but greater aesthetic demands on the appliance, and the frequency of multidisciplinary restorative involvement.
- The **Connecticut intrusion arch** is a pre-formed nickel titanium version of the Burstone bypass arch requiring no helix (Nanda et al., 1998; p. 935).
- **Cross-elastics** for crossbite correction produce a vertical as well as a transverse force component and are therefore **contraindicated where buccal segment extrusion should be avoided** (p. 952).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Distinguish relapse from age-related change explicitly

, and say that no extraction pattern, archwire sequence or occlusal finish prevents the latter.

2Be honest about retention.

The evidence that most patients stop wearing retainers within four to five years is uncomfortable but it is the evidence, and acknowledging it — and therefore favouring bonded retention with a removable back- up and a shared-responsibility conversation — is the mature answer.

3Do not overstate what orthodontics offers in sleep apnoea.

Mandibular advancement appliances for mild-to-moderate disease, yes; expansion as a primary treatment, no.

4The anterior crossbite exception is worth knowing:

corrected with a positive overbite, it is usually stable and needs no retention.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 13. 1How would you counsel a patient about retention at the start of treatment? 2A 45-year-old with treated periodontal disease wants orthodontic treatment. What are your concerns?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 13.

1. Name the five occlusal traits with the highest relapse potential.
2. What is the evidence on fixed versus removable retainers in the lower labial segment?

3. What proportion of patients are still wearing their retainers at 4–5 years?
4. Which occlusal correction is usually stable and requires no retention?
5. State the Wits threshold below which Class III surgery becomes likely.

Chapter 14. Clear Aligners

Revision Handbook: §17.5
Question Bank: Block Q (Q567–606)
Viva bank: V112–V121

IN ONE MINUTE

Taken seriously as a modality — and judged inferior biomechanically.

- Worse on ABO-OGS; no difference on PAR, because PAR is the coarser measure
- Better: pain, periodontal scores, visibility. Worse: relapse, complex cases
- Only 6% need no refinement; over 17% convert to fixed appliances

Where to read it

pp. 991–1033. Introduction pp. 991–1001 · providing treatment pp. 1001–1019 · manufacture pp. 1019–1025 · mechanical properties pp. 1025–1029 · **treatment outcomes pp. 1029–1032** · alternative systems pp. 1032– 1033.

The map

The renaissance of removable appliances → providing treatment with aligners → manufacture → mechanical properties of aligner materials → treatment outcomes → alternative systems.

Must-know

Examinable. Each fact carries its page reference.

ITEM DETAIL PAGE

The headline "Biomechanically they remain **inferior to conventional fixed appliances**" p. 1029
judgement

Quality of the evidence	Based primarily on **matched non-randomised studies** with a few small randomised trials — **moderate quality at best**, and for the synthesised systematic review findings, **moderate to low**	pp. 1029– 1030
ABO-OGS outcome	Using the American Board of Orthodontics Objective Grading System, aligner treatment is associated with **significantly worse occlusal outcomes and more patients with unacceptable results** than fixed appliances. Specifically worse: **buccolingual inclination, occlusal contacts, occlusal relationship, overjet and root angulation** (Papageorgiou et al., 2020)	pp. 1029, 1031
PAR outcome	**No differences** between aligners and fixed appliances in final PAR score or PAR reduction. The apparent contradiction with ABO-OGS is explained by **PAR being a less rigorous evaluation of tooth position** and a more representative measure of baseline case severity — post-treatment ABO-OGS and PAR scores **do not generally correlate**	p. 1029

Where aligners work	Effective for **mild-to-moderate malocclusion requiring simple tooth movement**; **less effective than fixed appliances for more severe crowding and complex cases**	pp. 1029– 1030
Treatment duration	Reported **shorter** for mild-to-moderate cases — but duration studies "need to be carefully controlled for equivalence, particularly with evidence of poorer occlusal outcomes for aligners"	p. 1030
Relapse	**Higher tendency to relapse** than fixed appliances, possibly related to the **increased proclination** that occurs with aligners	p. 1030
Periodontal	**Lower periodontal index scores and better periodontal outcomes** than fixed appliances — but the evidence is limited	p. 1030
Root resorption	**Little robust evidence of any difference**, though the data are sparse	p. 1030

ITEM	DETAIL	PAGE
Pain	**Lower levels of pain in the first few days** than fixed appliances; **placement of attachments is associated with the most pain and discomfort** during aligner treatment	p. 1030
Quality of life	**No conclusive differences** in overall OHRQoL, though **eating, chewing disturbance and discomfort are less** with aligners; **speech difficulty is more likely**, though patients adapt quickly	p. 1030
Refinements	Most manufacturers include at least one refinement — a further course of treatment — in their packages. **Only 6% of patients complete treatment without needing a refinement**, and **over 17% convert from Invisalign to fixed appliances** to complete treatment	pp. 1030, 1032

Nice-to-know

Enriches an answer. Skip on a first pass.

- Aligner materials are assessed against a set of ideal properties including **transparency, biocompatibility, biostability and recyclability** (p. 1029).
- Alternative removable aligner systems, including the older spring aligner concept and commercial modifications incorporating nickel titanium coil springs, are now less commonly used given the growth of commercial aligner systems (p. 1033).

EXAMINER TRAPS

The errors that cost marks on this topic.

1The ABO-OGS versus PAR contradiction is the interesting part.

A candidate who can explain *why* two indices disagree — that PAR is a coarser measure more reflective of baseline severity, and that the two do not correlate post-treatment — is demonstrating exactly the appraisal skill Chapter 1 asks for.

2Do not dismiss aligners, and do not oversell them.

The balanced position is: effective and well-tolerated for mild-to-moderate cases with simple movements; inferior for complex cases and on detailed occlusal outcome measures; better periodontally; more prone to relapse.

3The refinement statistics are the most quotable numbers in the chapter

— 6% need no refinement, 17% convert to fixed. They quantify the gap between the marketed and the achieved.

4Note the evidence quality caveat every time.

Moderate to low, largely non-randomised.

CLEAR ALIGNERS VERSUS FIXED APPLIANCES

ALIGNERS BETTER FIXED APPLIANCES BETTER

✓Less pain in the first few days ✓Occlusal outcome (ABO-OGS)

✓Better periodontal index scores ✓Buccolingual inclination

✓Less eating and chewing disturbance ✓Occlusal contacts and relationship

✓Far less visible ✓Overjet and root angulation

✓Severe crowding and complex cases

✓Less relapse

PAR shows NO difference; ABO-OGS shows aligners significantly worse. PAR is the coarser measure — the two do not correlate.

THE TWO NUMBERS TO QUOTE

Only 6% complete treatment without a refinement · over 17% convert to fixed appliances to finish. Ch 14, pp. 1029–1032.

Aligners versus fixed appliances. Compare Fig. 14.21, Ch 14, p. 1031.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 14. 1An adult asks for Invisalign rather than fixed appliances for a moderately crowded Class I case. How do you counsel them?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 14.

1. What does the ABO-OGS evidence show about aligner outcomes, and how does it differ from the PAR evidence?
2. Which five ABO-OGS components are significantly worse with aligners?
3. What proportion of patients complete aligner treatment without a refinement?
4. What proportion convert to fixed appliances?
5. In what respects do aligners outperform fixed appliances?

Chapter 15. Orthodontics and Orthognathic Surgery

Revision Handbook: §27; asymmetry §24.4
Question Bank: Block AA (Q902–931)
Viva bank: V184–V193

IN ONE MINUTE

What to do when the discrepancy is beyond tooth movement.

- Jaw movements up to 1 cm; assess in a combined clinic with psychological support
- Decompensation deliberately worsens the bite before surgery corrects it
- Le Fort I · BSSO · genioplasty · distraction at ~1 mm/day with 6 weeks' consolidation

Where to read it

pp. 1034–1107. Common facial deformities pp. 1034–1035 · patients presenting for surgery pp. 1035–1041 · assessment pp. 1041–1067 · the process of combined treatment pp. 1067–1086 · common malocclusions and their surgical treatment pp. 1086–1107. Facial asymmetry pp. 1092–1101. *Facial asymmetry in the growing child is Essay Topic 19, and Class III decompensation is Essay Topic 17. This unit covers the surgical process itself.*

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
What surgery adds	Jaw movements of **up to 1 cm** are achievable, which combined with modern fixed appliances means significant malocclusions can be corrected with great accuracy	p. 1041
The team	Assessment in a **combined multidisciplinary clinic**: maxillofacial surgeon, orthodontist, maxillofacial technician, and **access to psychological support**	p. 1041
Minimum records	Study models, radiographs, **cephalometric analysis** (**lateral and, if required, postero-anterior**), and **both intraoral and facial photographs**	p. 1041
IOFTN	The Index of Orthognathic Functional Treatment Need, designed and validated to **prioritise combined cases** (Ireland et al., 2014), addressing the limitation that IOTN's DHC does not capture maxillary incisor show at rest and its AC contains no Class II division 2 or Class III cases	p. 1041
Le Fort I	Disarticulates the maxilla from the skull by cutting along the lateral walls; allows the maxilla to be **advanced, impacted, moved down or set back** as a single unit, and can be **divided bilaterally** for transverse change. **Maxillary advancement increases maxillary incisor show** — relevant when deciding on impaction	pp. 1075, 1092

ITEM	DETAIL	PAGE
Vertical maxillary excess	Managed by **Le Fort I maxillary impaction** in the presence of normal lip length. Beware **broadening of the alar base, elevation of the nasal tip and cheek bulging** giving a ″cherubic″ appearance. Where excess show is primarily on smiling, **botulinum toxin into the levator muscles** is an alternative, lasting around **3 months** and repeatable	p. 1092
BSSO	The bilateral sagittal-split osteotomy moves the mandible **forwards or backwards**; both movements show **good stability, particularly mandibular advancement**	pp. 1079– 1082
Genioplasty	Osteotomy of the inferior border of the chin; can advance, set back, or **reduce the height** of the anterior mandible. Used alone for asymmetries but **most commonly as an adjunct to BSSO**	p. 1082
SARPE	Surgically assisted rapid palatal expansion — a form of **distraction osteogenesis**; used in the non-growing patient, particularly where more than about 5 mm of expansion is needed	p. 1061
Distraction osteogenesis	Osteotomy → appliance fitted → **progressive activation at around 1 mm/day** → **consolidation of around 6 weeks**	p. 1085
Model surgery and planning	Traditional **model surgery** with wafer construction, now increasingly supplemented or replaced by **3D planning and device design services**, with the wafers tried in the patient before surgery	pp. 1070– 1074
Decompensation	The orthodontic phase deliberately **removes dentoalveolar compensation**, placing the incisors at their correct inclination on their own skeletal bases — which **increases** the reverse overjet in a Class III case before surgery corrects it	Ch 13, p. 921
Timing	Surgery is delayed until **growth has ceased**; in Class III this means the **late teens**, because the malocclusion tends to worsen as mandibular growth persists	Ch 13, p. 921
Facial asymmetry	″True facial symmetry is rare and some degree of asymmetry is often seen in even the most attractive of individuals.″ **Obwegeser & Makek (1986)** classification: **Type I hemimandibular elongation** — horizontal vector, chin point markedly displaced to the unaffected side, centreline usually displaced too, **occlusal plane level**. **Type II hemimandibular hyperplasia** — vertical vector, increased ramus height and lower border hyperplasia, **chin and centreline often undisturbed**, but **unilateral open bite and maxillary cant** on the affected side. **Type III hybrid**. Four histological subtypes (Slootweg & Müller, 1986)	pp. 1092– 1098

CondylarManifests in the **mid to late teenage years** as a **progressive and self- hyperplasialimiting** mandibular pathology. Imaging: plain film and CBCT, supplemented by **technetium-99m scintigraphy or SPECT** to confirm continued condylar growth — but these are **sensitive and non-specific**, so **false positives are possible**. Serial **3D facial surface photography and superimposition** is a non-invasive alternative with good reproducibility using the forehead, mid-nose and infraorbital regions (Häner et al., 2021) pp. 1093– 1099

Nice-to-know

Enriches an answer. Skip on a first pass.

- Le Fort II and III level osteotomies are performed by **specialist supra-regional units** (p. 1067).
- Body dysmorphic disorder affects **up to 17% of patients seeking orthognathic surgery** (Ch 7, p. 365) — which is why access to psychological support is part of the multidisciplinary clinic rather than an optional extra.

EXAMINER TRAPS

The errors that cost marks on this topic.

1Decompensation is the concept.

Explain that it makes the malocclusion look worse before surgery, that the patient must be warned at the outset, and that this is why mandibular extractions are avoided in growing Class III patients who may need surgery.

2Give the asymmetry classification precisely, including the hybrid type

, and use the occlusal cant as the discriminator.

3Be cautious about scintigraphy.

Sensitive but non-specific; false positives; and repeated isotope exposure is problematic, which is the argument for serial photographic superimposition.

4Do not forget the psychological dimension.

Multidisciplinary assessment with psychological support, and realistic goal-setting, are markable.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 15. 1Take me through the stages of combined orthodontic-surgical treatment.

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 15.

1. How much jaw movement can be achieved surgically?
2. What are the aesthetic risks of significant maxillary impaction?
3. What is the rate of activation and the consolidation period in distraction osteogenesis?
4. Distinguish hemimandibular elongation from hemimandibular hyperplasia.
5. Why must a Class III patient be warned that their bite will worsen before surgery?

Chapter 16. Cleft Lip and Palate, and Syndromes Affecting the Craniofacial Region

Revision Handbook: §28
Question Bank: Block AB (Q932–961)
Viva bank: V184–V193

IN ONE MINUTE

A timeline. Learn who does what, when, and why.

- Lip 3–6 months · palate 6–12 months · alveolar bone graft 8–10 years
- Graft before canine eruption, at about two-thirds root formation
- CLP 1:700, male 2:1; isolated CP 1:2000, female 4:1 — different embryology

Where to read it

pp. 1108–1161. Candidate genes pp. 1109–1112 · **cleft lip and palate pp. 1112–1135** · cleidocranial dysplasia pp. 1135–1136 · ectodermal dysplasia pp. 1136–1139 · hemifacial microsomia pp. 1139–1142 · 22q11.2 deletion pp. 1142–1145 · Treacher Collins pp. 1145–1149 · Pierre Robin pp. 1149–1151 · craniosynostosis pp. 1151–1154 · oral–facial–digital syndromes p. 1154 · holoprosencephaly pp. 1155–1159 · foetal alcohol spectrum disorders pp. 1159–1161.

Must-know

Examinable. Each fact carries its page reference.

ITEM	DETAIL	PAGE
CLP incidence	Approximately **1:700 live births** in Caucasians. **Highest in some Asian populations (1:500), lowest in African populations (1:2500)**	p. 1113
CLP sex ratio	**Males more commonly affected, approximately 2:1**	p. 1113
Laterality	**Unilateral CLP is around 80%** of all facial clefts, with the **left side** commoner	p. 1113
Syndromic proportion	**70% of CLP is non-syndromic**, 30% syndromic	p. 1113
Isolated cleft palate	Around **1:2000** live births in White populations — about half the frequency of CLP. **Females more commonly affected, approximately 4:1.** Only **50% is non- syndromic**	p. 1113
The two condition clearest indicatio palate arising fro failure of palatal	s have **opposite sex ratios and different syndromic proportions**, whic n that they are embryologically and genetically distinct — cleft lip and m failure of maxillary-to-medial-nasal fusion by 8 weeks, isolated cleft pala shelf elevation or fusion between weeks 6 and 10.	h is the primary te from

Must-know

Examinable. Each fact carries its page reference.

STAGE	DETAIL	PAGE
Aims of care	**Good facial appearance · good orofacial function in speech, eating and swallowing · an aesthetic, functional and stable occlusion · good hearing**	p. 1113
The team	Cleft surgeon, orthodontist, speech therapist, cleft nurse, ENT surgeon, paediatrician, paediatric dentist, restorative dentist, psychologist, paedodontist, audiologist, geneticist	p. 1117
Presurgical orthopaedics	Facial strapping or orthodontic plates may be used to align the lip segments before repair	p. 1121
Lip repair	Usually **3–6 months** of age, as a single procedure; the child is usually **at least 3 months old**. The **"rule of tens"** — at least **10 weeks old, 10 pounds in weight, haemoglobin 10%**	pp. 1122, 1124
Palate repair	Between **6 and 12 months** of age	p. 1122
Named lip repairs	**Millard rotation-advancement** — one of the commonest, effectively hiding the scar. **Tennison triangular flap** — easier to perform. Current practice advocates **primary repair of the lip with correction of the underlying bony structures**, based on functional lip repair principles. **Bilateral cleft lip is repaired in a single procedure**, with the orbicularis oris freed from each lateral element and repaired in the midline	pp. 1124– 1125
The cost of repair	Surgical repair causes **scarring in the maxillary region**, and maxillary hypoplasia is characteristically seen in patients who have undergone palatal repair (Mars & Houston, 1990)	pp. 1123– 1124
Alveolar (secondary) bone grafting	**Cancellous bone, usually from the iliac crest**, placed into the alveolar defect at around **8–10 years of age, before eruption of the permanent canine, when its root formation is about two-thirds complete**. A period of orthodontics precedes it — **expansion of the collapsed maxillary arch with a trihelix or quadhelix, retained with a palatal arch** — to create surgical access and maximise the bone that can be placed. **Do not move teeth into the cleft site where there is no bone**; align those teeth after the graft	p. 1130

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| **Advantages of grafting** | Allows **eruption of the canine** · allows **alignment and orthodontic space closure**, particularly where the lateral incisor is absent · produces a **stable, well-aligned, intact maxillary arch · stabilises the premaxillary segment**, particularly in bilateral cases · establishes **good alveolar contour** | p. 1131 |
| **Problems the graft addresses** | Adjacent teeth displaced, rotated or tipped · teeth in the defect unable to erupt · **collapse of the maxillary arch** with loss of alveolar contour · compromised bony support at the nasal base with flattening and asymmetry · **instability and mobility of the premaxilla** in bilateral cases · **oronasal fistulae** | pp. 1129– 1130 |

Dental anomalies around the cleft	**Supernumerary teeth or agenesis of the permanent lateral incisor**; anomalies of shape and size; enamel defects. **Avoid the occlusal complications of early loss of primary teeth or first permanent molars** — fissure seal and monitor the first permanent molars	p. 1129
CSAG recommendations	Following the Clinical Standards Advisory Group investigation into UK cleft care: **centralise care with expertise and resources concentrated in a small number of centres**; establish a **common nationwide database**; and provide **specialist training only in cleft centres**	p. 1117
Outcome measures	The **GOSLON Yardstick** (Great Ormond Street, London and Oslo) for rating dental arch relationships in unilateral CLP	p. 1117

Nice-to-know

Enriches an answer. Skip on a first pass.

Treat these as pattern recognition — one distinguishing feature each is usually enough.

- **Cleidocranial dysplasia** — absent or hypoplastic clavicles, delayed cranial suture closure, multiple supernumerary teeth, failure of eruption (pp. 1135–1136).
- **Ectodermal dysplasia** — defective teeth, hair, nails and sweat glands; conical crowns, generalised microdontia, enamel hypoplasia, taurodontism, severe hypodontia or anodontia (pp. 1136–1139; Ch 12, p. 752).
- **Hemifacial microsomia** — asymmetric underdevelopment of structures derived from the first and second pharyngeal arches (pp. 1139–1142).
- **22q11.2 deletion syndrome** — cardiac defects, palatal anomalies including velopharyngeal insufficiency, immune deficiency, hypocalcaemia (pp. 1142–1145).

EXAMINER TRAPS

The errors that cost marks on this topic.

1Learn the epidemiology as a contrast, not as two lists.

CLP is 1:700, male 2:1, 70% non-syndromic; isolated CP is 1:2000, female 4:1, 50% non-syndromic.

2The bone graft timing question is really a question about the canine.

8–10 years, before canine eruption, at about two-thirds root formation.

3Say what the orthodontist does before the graft

— expansion of the collapsed arch to create access — and the caution about not moving teeth into an ungrafted cleft.

4Do not forget hearing.

It is one of the four stated aims of cleft care, and ENT involvement is not incidental.

THE CLEFT CARE TIMELINE

birth 3–6 mo 6–12 mo mixed dentition 8–10 y adolescence late teens

Diagnosis, feeding LIP REPAIR PALATE REPAIR Prevention · fissure ALVEOLAR BONE Definitive Orthognathic advice, presurgicalrule of tens hearing monitored seal 6s · arch GRAFT before canine orthodontics surgery if needed

orthopaedics expansion eruption

EPIDEMIOLOGY — LEARN AS A CONTRAST

Cleft lip ± palate 1:700 male 2:1 70% non-syndromic 80% unilateral

Isolated cleft palate 1:2000 female 4:1 50% non-syndromic

Opposite sex ratios and different syndromic proportions — because they are different

developmental failures at different times (lip by 8 weeks; palate weeks 6–10).

The cleft care timeline. Compare Fig. 16.9, Ch 16, p. 1130.

DEVELOPMENT OF THE FACE AND PALATE — AND WHERE CLEFTS ARISE

week 4 week 5 week 6 week 8 week 10

Five processes appear: Medial and lateral nasal Palatine shelves appear FACE COMPLETE PALATE COMPLETE

frontonasal + paired processes form around VERTICALLY either side lip = maxillary + medial shelves elevate, fuse

maxillary + mandibular the nasal placode of the tongue nasal processes in the midline

CLEFT LIP ± PRIMARY PALATE ISOLATED CLEFT PALATE

Maxillary process fails to fuse with Palatine shelves fail to elevate or

the MEDIAL NASAL process, by week 8 fuse, weeks 6–10

1:700 · male 2:1 · 70% non-syndromic 1:2000 · female 4:1 · 50% non-syndromic

Opposite sex ratios because they are DIFFERENT developmental failures at different times. Ch 3, pp. 121–133; Ch 16, pp. 1113–1115.

Facial and palatal development, and where clefts arise. Compare Figs 3.8–3.10, Ch 3, pp. 123–133.

VIVA QUESTIONS

Model answers in Appendix A.

Model answers: Appendix A, Chapter 16. 1When is alveolar bone grafting performed and why then? 2Why do cleft lip and palate and isolated cleft palate have different epidemiology?

Self-test

Cover the answers. Write, then compare.

Answers: Appendix B, Chapter 16.

1. Give the incidence, sex ratio and non-syndromic proportion for CLP and for isolated CP.
2. State the rule of tens.
3. When are lip and palate repair usually performed?
4. At what stage of canine development should alveolar bone grafting be performed?
5. Name the four aims of cleft care.

Part IV — Appendices

Appendix A — Model viva answers: the twenty essay topics

Roughly sixty to ninety seconds spoken. The italic line is the suggested direction — what the examiner is testing and

how to structure your thinking. Do not memorise verbatim.

Topic 1. The biological basis of tooth movement

1. Why does an ankylosed tooth not move, and what does that tell you? *Direction: the examiner wants the principle, not the phenomenon. Answer in one sentence, then draw the inference.* "Because an ankylosed tooth has lost its periodontal ligament — there is direct bone-to-cementum union, so there is no ligament to strain, no fluid flow, no mechanotransduction and therefore no recruitment of osteoclasts. The same is true of an osseointegrated implant. What that tells us is that the biological response to orthodontic force is mediated by the periodontal ligament rather than by alveolar bone responding directly to load, which is why an intact ligament is an absolute prerequisite for tooth movement. Clinically, it also means that suspected ankylosis should be excluded before you attribute failure of movement to appliance design or compliance." **2. What is undermining resorption and why does it delay tooth movement?** *Direction: define, then explain the sequence, then give the clinical consequence.* "When compression exceeds capillary blood pressure, the ligament becomes ischaemic, cells die and the matrix proteins denature, producing a cell-free necrotic zone described as hyalinised because of its glass-like appearance. Osteoclasts cannot resorb bone directly beneath that zone, so movement stops. Macrophages and other phagocytes must first remove the necrotic tissue, and osteoclasts then migrate in from adjacent marrow spaces and resorb the bone from behind — undermining resorption. That process accounts for the lag phase of twenty to thirty days. Once the hyalinised tissue is cleared, direct frontal resorption resumes. The clinical implication is that heavier forces do not move teeth faster; they simply generate more extensive hyalinisation and a longer lag." **3. If lighter forces are biologically preferable, why not use the lightest possible force?** *Direction: this is a dose–response question. Show the curve.* "Because there is a threshold below which you do not exceed the critical stress needed to initiate remodelling, so nothing happens. The relationship is a dose–response curve: at low forces the rate of movement rises with force until a plateau is reached, and beyond that plateau there is no further gain — extreme forces may even reduce movement. Ren's mathematical model puts maximum canine retraction at around 272 centinewtons, but the systematic review evidence is that anything between 50 and 250 centinewtons produces broadly similar rates for bodily movement, so 50 to 100 is regarded as optimal. The practical aim is a force just above the threshold for the teeth you want to move but below the threshold for the anchorage unit — which is the differential force principle." **4. How would you explain the risk of root resorption to a patient at consent?** *Direction: the examiner is testing communication as much as knowledge. Use plain language, quantify, and say what you would do about it.* "I would say that shortening of the root tips happens to some degree in almost everyone who has fixed braces — it shows up on X-rays in around three- quarters of cases — and that in the great majority it is small enough to have no effect at all on the life of the tooth. I would then say that in a small proportion, between one and five per cent of teeth, it is more marked, meaning the loss of a third or more of the root, and that this most often affects the upper front teeth. I would explain that we reduce that risk by using light forces and by not prolonging treatment unnecessarily, that we take a radiograph partway through to check, and that if we saw significant resorption we would pause the appliance for three to six months to let repair occur. I would also flag any previous trauma to those teeth, because that increases the risk."

5. What happens if you apply a force greater than the anchorage unit's threshold? *Direction:*

bridge deliberately to Topic 5 — examiners like candidates who connect subjects. "You lose anchorage without gaining movement. Above the plateau, increasing force produces no additional movement of the teeth you intend to move, but because the anchorage unit has a higher threshold by virtue of its greater combined root surface area, raising the force can push it above that threshold too. The result is mesial drift of the buccal segments — clinically, a residual overjet at the end of overjet reduction. This is Begg's differential force theory used in reverse: light forces preserve anchorage precisely because they sit between the two thresholds."

Topic 2. Cephalometry: uses and limitations

1. Why is point A a poor reference point in a case involving incisor retraction? *Direction: this is the validity objection. Name it as such.* "Point A is chosen for convenience rather than anatomical accuracy — it represents the anterior limit of the maxillary apical base, but there is no discrete anatomical structure there. More importantly it remodels. If you retract the upper incisors, the labial plate over their roots resorbs and point A moves posteriorly, which reduces SNA and ANB. So a change in your cephalometric numbers may reflect local alveolar remodelling rather than any true skeletal change. That is the classic validity error, as opposed to a reproducibility error: you can identify point A very consistently and still be measuring something that does not mean what you think it means." **2. A colleague wants a lateral cephalogram before aligning a mildly crowded lower arch. Is that justified?** *Direction: apply the justification principle rather than reciting regulations.* "I do not think it is. The test is whether the radiograph will materially change the diagnosis or the treatment plan, and in a case of mild lower crowding treated by alignment alone, it will not — the incisor position is being assessed clinically and the plan does not turn on the skeletal pattern. The textbook is explicit that cephalometric radiographs are not justified in all cases, particularly where only minor tooth movements are planned. Since there is no threshold below which radiation risk disappears, every exposure has to be justified individually for that patient. I would take one if the plan involved significant incisor movement, a skeletal discrepancy I intended to camouflage, or any consideration of orthognathic surgery, where a lateral cephalogram is essential." **3. What is prognostic tracing and what are its limitations?** *Direction: define, give the use, then be honest about the constraint.* "Prognostic tracing is simulating the planned tooth and jaw movements on the cephalometric tracing before treatment, so that both the effect and the feasibility of those movements can be studied in advance — for example, whether the incisors can be retracted to the position you want without moving them out of alveolar bone. Its limitations are that it inherits all the errors of the underlying tracing, it assumes the soft tissues will follow the hard tissues in a predictable ratio when in fact individual variation is wide, and in growing patients it requires an assumption about future growth which we know cannot be predicted reliably. It is best used as a feasibility check rather than a prediction." **4. How would you minimise measurement error in a research study using cephalometrics?** *Direction: give a systematic list, ordered from acquisition to analysis.* "I would start at acquisition — correct positioning in the cephalostat, and the highest definition of skeletal and soft tissue structures, because you cannot recover from a poor film. I would view the films under optimum conditions. For analysis I would hand-trace the landmarks first and then digitise, rather than digitising directly, and I would repeat measurements to quantify intra-examiner error, usually reporting a Dahlberg or intraclass correlation. I would establish the magnification factor for the machine so linear values are comparable. And for a large study I would calibrate all examiners

against a reference set before data collection and re-check calibration during it. I would also select landmarks with known good reliability where the research question allows a choice." **5. Why is superimposition problematic, and what assumption does it rest on?** *Direction: state the assumption first — that is the answer.* "It rests on the assumption that the structures you superimpose on are stable over the period between the two films. That assumption is difficult to satisfy: the anterior cranial base is relatively stable but nasion continues to move

with growth, and point A and point B remodel with tooth movement. There is also the practical problem that films taken on different machines have different magnification, which must be accounted for. The evidence base is limited — the studies that have compared methods have found that all of them carry appreciable error and none is clearly more reliable, and no method has been shown to be completely accurate. So superimposition remains the principal way we evaluate growth and treatment change, but the magnitude of change reported should always be interpreted against the known error of the method."

Topic 3. Factors in formulating an orthodontic treatment plan

1. Your space analysis shows 6 mm of crowding in the lower arch. Take me through how that figure changes your plan. *Direction: walk the chain of consequences. This is the question that most rewards structure.* "Six millimetres is moderate crowding, which by the mandibular arch guidelines points to extraction of either first or second premolars rather than a non-extraction approach — although I would not decide on that figure alone. First I would ask what the other five components of the space analysis contribute: levelling an increased curve of Spee, any planned change in incisor anteroposterior position, angulation and inclination, and arch width. So six millimetres of crowding might become nine or ten millimetres of total requirement. Then, because I want to relieve that crowding without proclining the lower labial segment, the lower canines need to move distally by roughly half the crowding on each side, and that distance tells me how far the upper canines must move to reach Class I — which sets the maxillary space requirement and hence the maxillary extraction choice. Finally, the amount of incisor retraction planned sets the anchorage demand: if it is substantial I would plan first premolars and reinforce anchorage, remembering that unsupported, around two- thirds of the space is lost." **2. How do you decide between growth modification, camouflage and surgery in a Class II patient?** *Direction: three variables — growth, severity, patient. Give a threshold if you can.* "Three things decide it: how much growth is left, how severe the skeletal discrepancy is, and what the patient will accept. In a growing child with a mild to moderate skeletal Class II and good compliance, a functional appliance is the first choice, accepting that most of the correction will be dentoalveolar. In a patient with little or no growth and a mild to moderate discrepancy, camouflage with extractions and fixed appliances can be very successful — it will improve alignment and reduce the overjet but cannot change the facial skeleton. Surgery is indicated where the discrepancy is too severe to camouflage without damaging the profile or the periodontium. Proffit's retrospective data give some thresholds — an overjet greater than ten millimetres, pogonion to nasion perpendicular of eighteen millimetres or more, mandibular body length under seventy millimetres, or anterior facial height over a hundred and twenty-five millimetres — but these are guides, and the practical limiting factor is usually the soft tissue profile, since camouflage retracts the upper lip by roughly 0.7 millimetres for every millimetre of incisor retraction." **3. What would make you decline to treat, or defer treatment?** *Direction: cover dental, medical, behavioural and expectation-related reasons. Show you can say no.* "I would defer if oral hygiene were inadequate or there were active caries or untreated periodontal disease, because fixed appliances in that setting cause demineralisation and accelerate bone loss. I

would defer for medical reasons — within six months of a transplant, during active treatment for a childhood malignancy, or while an eating disorder is active. I would decline where the patient's expectations cannot be met by orthodontics, which is a particular concern in body dysmorphic disorder, where prevalence among orthodontic patients is reported at five to thirteen per cent. I would also decline where the balance of benefit and risk is unfavourable — severe pre-existing root resorption, for instance, is a contraindication. And in Class III cases in a growing child I would often actively monitor rather than treat, because the discrepancy only declares itself with growth." **4. How do you assess motivation and compliance at the first visit?** *Direction: be concrete. Examiners want observable indicators, not adjectives.* "I ask whose

idea the treatment was, and listen for whether the concern is the child's or the parent's — a child who cannot name what they dislike about their teeth is a warning sign. I ask what specifically they want changed, because a vague answer makes a satisfactory endpoint hard to define. I look at objective proxies: attendance history with their general dental practitioner, current oral hygiene and caries experience, and whether previous advice has been acted on. I explain what the treatment will actually involve in terms of wear time and duration, and gauge the reaction. Where compliance is doubtful I would bias the plan towards modalities that do not depend on it — fixed rather than removable, skeletal anchorage rather than headgear, a Herbst rather than a twin block — and I would say so at the planning stage rather than discovering it six months in." **5. Which parts of your plan would change if the patient were 32 rather than 12?** *Direction: growth is the hinge. Then add the adult-specific considerations.* "The biggest change is that growth modification comes off the table, so a skeletal discrepancy must be either camouflaged or treated surgically. Overbite reduction becomes harder and less stable, because extruding the buccal segments in an adult rotates the mandible down and back rather than being absorbed by condylar growth, so I would favour true incisor intrusion. Periodontal status matters far more — I would want disease treated and stable first, and I would be more cautious about moving teeth out of the alveolar envelope because recession is a real risk. Root resorption risk and reduced tissue turnover mean lighter forces and realistic timescales. Restorative considerations often drive the plan in adults, so I would plan jointly with a restorative colleague. And motivation is usually better in adults, which makes compliance-dependent mechanics more viable, though appliance aesthetics matter more to them."

Topic 4. Factors affecting the choice of extraction

1. A 13-year-old has 9 mm of lower crowding and a grossly carious lower first permanent molar. What do you extract, and why? *Direction: prognosis trumps the ideal site. Say so first, then deal with the consequences.* "I would extract the compromised first permanent molar rather than a sound premolar, because the principle is that a tooth of poor long-term prognosis should always be removed in preference to a healthy one. Nine millimetres is severe crowding, so the space is needed. The difficulty is that first molar space is large and distant from the site of crowding, so I would need fixed appliances and careful anchorage management to bring that space forward, and I would expect a longer treatment. I would check the timing and the third molar: if the third molar is present and favourably positioned, extraction around the time of bifurcation formation of the second molar gives the best chance of good spontaneous replacement. I would also consider whether a balancing or compensating extraction is needed, and I would plan the contralateral and opposing arch as part of the overall plan rather than reflexively balancing." **2. Why do we extract further forward in the mandibular arch in Class III cases?** *Direction: link space requirement to the direction of the required tooth movement.* "Because in a Class III case the mandibular incisors usually need to move lingually to correct the incisor relationship, and that increases the space requirement in the lower arch rather than the upper.

Extracting first premolars in the mandible and second premolars in the maxilla puts the space where it is needed and creates a differential that helps correct the molar relationship. It also improves anchorage for lower labial segment retraction, because extracting first premolars brings the second premolars into the posterior anchorage unit. In milder cases you can extract in the mandible only and finish to a Class III molar relationship, or even remove a single mandibular incisor. But mandibular extractions in a Class III case need care: if the patient may eventually need orthognathic surgery, retroclining the lower labial segment has to be undone during decompensation, and that is difficult once the extraction sites have closed." **3. How does the extraction pattern change your anchorage plan?** *Direction: theory, then the evidence that qualifies it.* "In theory, extracting first premolars incorporates the second premolars into the posterior anchorage unit, increasing its combined root surface area and its resistance to mesial movement — so first premolar extraction is the more anchorage-conserving choice when significant

incisor retraction is planned. In practice, the evidence is less flattering: Haque's work shows that without anchorage support, anchorage loss is essentially the same whether you extract first or second premolars, and around two-thirds of the space created is lost. So the practical conclusion is that the extraction pattern alone will not protect anchorage — you have to plan active support, whether that is a palatal arch, headgear, or increasingly miniscrews, and the maxillary arch in particular needs it because it is intrinsically more anchorage-prone." **4. What are the alternatives to extraction, and when would you use each?** *Direction: name all four methods, then give the indication and the limitation of each.* "There are four ways to create space and extraction is one. Transverse expansion is appropriate in the maxilla where there is a buccal crossbite, especially with a displacement, but you only gain about half the expansion as usable space, there is always some relapse, and expansion of the mandibular intercanine width is inherently unstable and should be avoided. Anteroposterior lengthening means distalising the buccal segments — realistic in the maxilla with headgear or skeletal anchorage, particularly before the second molars erupt, but difficult in the mandible, and even with good headgear wear you rarely correct a full unit of molar relationship by distalisation alone. Proclination of the incisors also lengthens the arch but is unstable and can damage the profile. Interproximal reduction gives a precise but limited amount, up to a millimetre per contact in the buccal segments and less anteriorly, so it suits mild crowding. Extraction remains the most effective option where the requirement is moderate or severe." **5. Does extraction flatten the profile? What does the evidence say?** *Direction: acknowledge the concern, then give the evidence with numbers.* "It is the most persistent criticism of extraction-based treatment and the evidence does not support it as a routine outcome. Studies comparing patients with morphologically similar profiles treated with and without premolar extractions show that lip procumbency is reduced after four premolar extractions, but the change is small and generally judged favourable, and lip retraction has to be considerable before dental professionals or the public regard a profile as unattractive. Quantitatively, systematic review data suggest around 0.7 millimetres of lip retraction for every millimetre of incisor retraction, with a 1.6 degree increase in nasolabial angle — and wide individual variation, which is why prediction is unreliable. Nor do mid-arch extractions produce larger buccal corridors or a less attractive smile. What does damage the profile is poorly planned treatment, particularly inappropriate attempts to camouflage a severe skeletal discrepancy, or extracting first premolars where crowding is mild and then over-retracting the labial segments."

Topic 5. Anchorage and its clinical significance

1. Why is anchorage loss more of a problem in the maxilla?

Direction: give the three or four contributing factors, then the reassuring corollary. "Several factors combine. The bone density and the width of the alveolus differ from the mandible, and in most Class II cases the amount and rate of tooth movement required in the maxilla is greater — teeth also move faster in the maxilla generally. So the maxillary anchorage unit is both more likely to move and asked to resist more. The corollary is a useful one: it is also considerably easier to support anchorage in the maxilla, because you have the palate available for a trans-palatal arch or Nance button, headgear attaches to the maxillary molars, and the palatal bone is a reliable site for miniscrews with high success rates. So although the maxilla is the vulnerable arch, it is the one where reinforcement is most straightforward." **2. How would you assess anchorage demand before starting a case?** *Direction: root it in the space analysis. That is the answer the examiner wants.* "Through the space analysis. The first part quantifies how much space is needed in each arch for crowding, levelling, incisor anteroposterior change, angulation and inclination. The second part records how that space will be created and used, including how far the molars are permitted to move mesially — and that figure *is* the anchorage requirement. If no mesial movement is permissible, the demand is maximum; if the molars and labial segment can move similar amounts towards each other, it is moderate; if the molars may mesialise freely, it is min-

imum. I would then look at the aggravating factors: is this the maxillary arch, is the vertical pattern increased, are there occlusal interferences, and how much incisor retraction am I planning? And I would build in the expectation that without support around two-thirds of the space will be lost." **3. A patient with an increased maxillary–mandibular planes angle needs maximum anchorage. What are your options and which would you choose?** *Direction: the vertical pattern constrains the answer — say why before you list options.* "The vertical pattern matters twice over. First, these patients lose anchorage more readily — lower bite forces, poorer intercuspation and a more mesial eruptive path. Second, it rules out several conventional options: low-pull headgear would extrude the molars and rotate the mandible further down and back, and Class II elastics would do the same while also extruding the incisors. So I would either use high-pull headgear, which has an intrusive vector on the posterior dentition and helps restrain vertical maxillary development, or — my preference in a case demanding truly maximum anchorage — skeletal anchorage with miniscrews. Miniscrews give absolute anchorage independent of compliance and allow en masse retraction without any molar mesialisation or extrusion. I would combine that with a trans-palatal arch for vertical and transverse control, recognising it contributes little sagittally." **4. What safety features must be present on a headgear, and what would you tell the patient and parent?** *Direction: name both devices, name both injury mechanisms, then give the counselling.* "Two independent safety devices are mandatory: a self-releasing head-cap or neck-strap, which detaches once force exceeds a set level and prevents catapult injury, and a locking facebow, which physically locks into the molar tubes and prevents disengagement at night. They address the two recognised mechanisms — catapult injury, where the facebow springs back into the face or mouth after being pulled out of the tubes while still attached to the strap, and nocturnal disengagement during sleep. I would tell the patient and parent that the risk is small but that the worst outcome is a penetrating eye injury, which can be relatively painless at first, becomes rapidly infected because the facebow carries oral bacteria, and can threaten sight in both eyes. Practically: never pull the facebow out forwards, always release the strap first, remove the headgear before any rough play or sport, and if it comes out at night stop wearing it and contact us. I would demonstrate insertion and removal and have them do it back to me." **5. What is differential force theory, and which appliance philosophy is built on it?**

Direction: define, attribute, then explain how it is exploited. "Differential force theory is Begg's principle that because the force required to tip a tooth is much lower than that required to move it bodily, you can select a force level that lies above the threshold for the teeth you want to move but below the threshold for the anchorage unit. Begg built his light wire appliance on it: the crowns are tipped into position first using very light forces and intermaxillary elastics, with the roots uprighted separately afterwards using auxiliary springs. That made the appliance far lighter on anchorage than the standard edgewise appliance, which moves teeth bodily through bone. Kesling's Tip-Edge appliance refined the idea by limiting how far teeth may tip and then closing the slot to express a full three-dimensional prescription in rectangular wire at the finishing stage. The same principle explains why heavy forces cost you anchorage without buying you speed."

Topic 6. First, second and third order movements, and the evolution of fixed appliances

1. What is torque, and why does it matter during overjet reduction? *Direction: define precisely, then give the mechanical and biological reasons.* "Torque is third-order movement — the buccolingual inclination of a tooth about its long axis, produced by a couple generated between a rectangular archwire and a rectangular slot. It matters in overjet reduction because if you simply tip the upper incisor crowns back, the roots stay forward, the incisors become excessively retroclined, and the interincisal angle increases so no proper occlusal stop is established — which means the overbite will deepen and relapse. You also risk pushing the apices against the labial cortical plate, which raises the risk of root resorption. So palatal root torque is applied

to move the roots back with the crowns, achieving bodily retraction, an acceptable inclination and a stable incisor relationship." **2. Why was the standard edgewise appliance so anchorage-demanding?** *Direction: one sentence of mechanism, then the consequence.* "Because it moves teeth bodily through alveolar bone. A rectangular wire fully engaged in a rectangular slot creates a couple that resists tipping, so the tooth translates rather than tips — and bodily movement requires substantially more force than tipping, since the stress is distributed across the whole ligament rather than concentrated at the alveolar crest and apex. That higher force is transmitted equally to the anchorage unit, so anchorage loss is greater. It was precisely this problem that led Begg to develop the light wire appliance around differential force, tipping the crowns first with very light forces and uprighting the roots afterwards." **3. What is differential force, and how did Begg exploit it?** (*See Topic 5, question 5, for the model answer — expect this question in either topic.*) **4. What are the practical differences between Roth and MBT prescriptions?** *Direction: you are being asked whether you understand that a prescription is a set of compensations, not a truth.* "Both are pre-adjusted edgewise prescriptions built on Andrews' principles, but they compensate differently for the known shortfall between what a bracket prescribes and what is actually expressed. Roth built in overcorrection — more torque and tip than Andrews' measured values — on the basis that some expression is always lost to slot play, and he emphasised gnathological finishing goals, canine guidance and coincidence of retruded contact and intercuspal position. The MBT prescription reduced anterior tip, increased palatal root torque on the upper incisors and lingual crown torque on the lower incisors, and was designed around sliding mechanics with light continuous forces and lacebacks to reduce anchorage demand and control the overbite increase seen during alignment. In practice the differences are modest, and the important point is that no prescription expresses fully — bracket positioning error and slot play mean that finishing bends and torque adjustments are still needed whichever you use." **5. Does self-ligation reduce treatment time? What does the evidence show?**

Direction: answer no, immediately, then support it and be fair about what it does offer. "No. Several randomised clinical trials in the early 2000s compared self-ligating with conventionally ligated brackets and found no significant difference in the rate of tooth movement, overall treatment time or the number of visits. The claim originated from laboratory studies showing reduced friction, but that has not translated clinically. There is some evidence that self-ligating brackets are marginally less painful, though not all studies agree. The evidence on greater posterior expansion or reduced lower incisor proclination is equivocal, with any difference in intermolar width being minimal — so the claim that they permit stable arch expansion and non-extraction treatment is not supported. And there is no evidence of a better occlusal outcome. The genuine advantage is some saving in chairside time when the mechanism works properly. It is a good example of a marketed claim persisting after it has been disproven."

Topic 7. The ideal archwire; stainless steel versus nickel titanium

1. Why can you not place a bend in a superelastic NiTi wire? *Direction: metallurgy first, consequence second.* "Because its behaviour depends on a reversible phase transition between martensite and austenite, and permanently deforming the wire disturbs that. Nickel titanium has poor formability — the distance on the stress–strain curve between yield and fracture is small — so attempting a sharp bend risks fracture, and where it does deform it alters the transitional behaviour locally, changing the force the wire delivers and losing the plateau you were relying on. Practically, that means if I need a bend or a loop, I use a wire that will hold it: stainless steel, cobalt chromium for an auxiliary, or beta titanium for finishing bends. It is also why NiTi cannot be used for looped space closure, quite apart from its low stiffness." **2. What is superelasticity, and how does it differ from shape memory?** *Direction: two mechanisms, one distinguishing variable — stress versus temperature.* "Both arise from the same martensite-to-austenite phase transition, but they are triggered differently. Superelasticity is a stress-induced transition: engaging the wire into a displaced bracket deforms it enough to generate a local

martensitic transformation, and as that region unloads it delivers an almost constant force over a large range — the plateau on the force-deflection graph. That is pseudoelasticity, and wires showing it are austenitic-active. Shape memory is a temperature- induced transition: the wire is martensitic and very flexible at room temperature, easy to engage, and converts to the stiffer austenitic form at mouth temperature, returning towards its original shape and delivering a gentle aligning force. Those are the martensitic-active copper nickel titanium wires, where the transition temperature has been tuned by substituting copper for some of the nickel. In both cases the deactivation plateau sits below the activation plateau — hysteresis — which is why the force delivered to the tooth is lighter than the force needed to place the wire." **3. Why is friction a problem in sliding mechanics, and how would you reduce it?** *Direction: define the problem, then give a systematic list of variables you can control.* "In sliding mechanics the tooth or block of teeth must slide along the archwire, so any force is partly consumed overcoming resistance at the bracket-wire interface. That means less force is available for tooth movement, movement is slower and less predictable, and to compensate you may be tempted to raise the force — which costs anchorage. I would reduce it by choosing a wire with a lower coefficient of friction, so stainless steel rather than nickel titanium or beta titanium at the space closure stage; by using the smallest wire dimension that still gives adequate control, so there is less binding in the slot; by ensuring the teeth are fully levelled and aligned before starting to slide, since binding at the bracket corners rather than classical friction dominates once teeth are tipped; and by considering ligation method, although the randomised evidence shows self-ligation does not translate into faster treatment despite lower laboratory friction." **4. When would you choose a multistrand stainless steel wire over NiTi?**

Direction: the honest answer includes "rarely, but here is when". "They are genuinely comparable for initial alignment — the textbook notes that multistrand and coaxial steel wires seem to perform as well as nickel titanium during alignment, and the Cochrane data show no real difference in day- one pain between multistrand steel and superelastic NiTi. I would choose multistrand where I want the flexibility of a small effective radius but also the option of joining or adapting the wire, since steel has good formability and joinability; where cost matters; or in a patient with a confirmed nickel allergy, where nickel titanium should be avoided but stainless steel is considered safe because of its low nickel content. The practical reason NiTi predominates is that multistrand steel tends towards permanent deformation, so it is less forgiving in a severely crowded arch." **5. What are beta-titanium wires used for, and why?** *Direction: give the property, then the application it enables.* "Beta titanium, introduced in the 1980s, sits between stainless steel and nickel titanium: its stiffness is around a third that of stainless steel, and crucially it has good formability. That combination makes it well suited to the final stages of treatment, where you need to place individual finishing bends to detail tooth position and allow the occlusion to settle, but you want a gentler force than a full-dimension steel wire would deliver. Its drawback is relatively high friction, so it is a poor choice for sliding mechanics. It is also used for intrusion arches, where a flexible rectangular section that can be bent and will deliver a light continuous intrusive force is exactly what is wanted — the Burstone-type arch can be made in titanium-molybdenum alloy without needing the helix required in stainless steel."

Topic 8. Functional appliances

1. How much true skeletal change does a twin block produce, and how do you know? *Direction: give the number, then explain why the historical claims were inflated.* "Very little — the current position is that any additional growth amounts to a few millimetres at best and is essentially transient. The reason older studies claimed more is methodological: they compared small treated case series against controls taken from historical growth studies, which over-emphasises the treatment effect and is unreliable. Where differences were statistically significant they were small and probably not clinically significant. During treatment there is

enhanced mandibular growth, but in the immediate post-functional period growth decelerates, and in the longer term the early skeletal effects largely disappear apart from a possible restraining effect on the maxilla, particularly if headgear is used. The likely explanation for why the appliance works at all is that Class II patients do not grow into Class I because the occlusion is locked by intercuspation and dentoalveolar adaptation — the appliance unlocks it and lets normal mandibular growth express itself, alongside the dentoalveolar changes." **2. When would you start functional appliance therapy, and why?** *Direction: distinguish the two timing questions — early versus late, and relation to peak height velocity.* "There are two separate timing questions. The first is whether to treat early in the mixed dentition or in one phase in adolescence. Three large randomised trials — Florida, North Carolina and Manchester — showed no meaningful long-term difference, with early-treated children having more attendances and longer overall treatment, and in the UK trial the late-treated children finished with a slightly better occlusal result. So early treatment is not normally justified, with the exception of children at high risk of incisor trauma. The second question is timing relative to the adolescent growth spurt: mandibular growth velocity peaks at or just after peak height velocity, so treating in that window is logical, but individual variation in onset, duration and intensity is wide, the supporting evidence is heterogeneous and at high risk of bias, and precise correlation with peak height velocity is not established. In practice I would treat in the late mixed or early permanent dentition, around the growth spurt, and select patients on facial pattern rather than on a maturational index alone." **3. Your patient will not wear their twin block. What are your options?**

Direction: diagnose first, then escalate. Do not jump straight to the Herbst. "First I would establish that non-compliance really is the problem: a pristine appliance, persisting speech difficulty, repeated breakages from removal, and no change in overjet or buccal segment relationship at six months all point that way. I would also check that the appliance fits and is comfortable, because discomfort masquerades as non-compliance. Then I would talk to the patient and parent about what is actually getting in the way — eating, sport, school, embarrassment — and consider a wear calendar, since recording hours can improve compliance. If that fails, the options are to change modality or to change the aim. Changing modality means a fixed functional such as a Herbst, which removes compliance from the equation: the recent randomised trial found 37 per cent of twin block patients failed compared with 17 per cent with the Herbst, with a larger mean overjet reduction, though at the cost of more visits and more breakages. Changing the aim means abandoning growth modification and planning camouflage with fixed appliances and extractions when the permanent dentition is established. What I would not do is persist with an unworn appliance for another six months." **4. What is the functional matrix theory, and which appliance is based on it?** *Direction: state the theory, name the appliance, then say honestly how well it holds up.* "The functional matrix theory holds that skeletal growth is not intrinsically determined by the bone itself but is a secondary response to the growth and functional demands of the surrounding soft tissues and functional spaces — the bone adapts to its matrix. Fränkel's Functional Regulator is the appliance built explicitly on that principle: rather than acting mainly on the teeth, it uses buccal shields and labial pads to hold the cheeks and lips away from the dentition and alveolus, with the intention of altering the soft tissue environment, encouraging bone apposition and reducing hyperactivity of the perioral musculature. There are four versions, for Class II division 1, Class II division 2, Class III and anterior open bite. In practice the appliance is bulky, fragile and hard to wear, and the evidence for a distinctively skeletal mode of action is no stronger than for other functional appliances — most of the correction is dentoalveolar, as it is with the others." **5. How does a functional appliance reduce the overbite?** *Direction: two mechanisms — guided eruption and incisor inclination change.* "Chiefly by guiding differential eruption. Posturing the mandible forward opens the bite and disoccludes the buccal segments, and the acrylic capping or faceting is then relieved selectively so that the mandibular buccal teeth are free to erupt, mesially, while the maxillary buccal teeth

erupt distally. That increase in posterior vertical dimension produces relative intrusion of the incisors, which in a growing patient is compensated by vertical condylar growth so the face height does not simply increase. The second mechanism is the change in incisor inclination — the upper incisors retrocline and the lowers procline, which reduces the interincisal angle and reduces the overbite directly, at roughly a tenth to a fifth of a millimetre per degree. The important caveat is that unless the maxillary incisor inclination ends up correct, no proper occlusal stop is created and the overbite will relapse."

Topic 9. Benefits and complications of orthodontic treatment

1. How do you explain the risk of white spot lesions to a 13-year-old and their parent? *Direction: plain language, honest frequency, and a shared plan. Avoid jargon.* "I would say that braces make teeth much harder to clean, and that if plaque sits around the brackets the enamel starts to dissolve, leaving chalky white marks. I would be honest that this is common — it affects up to half of patients — and that the marks appear quickly, often within six months, most often on the back molars, the upper side teeth and the lower canines. I would explain that they fade but often do not disappear, so the mark can still be visible years later, which matters because the whole point of treatment is appearance. Then I would make it a shared plan: brushing properly twice a day with a high-fluoride toothpaste, cutting out fizzy drinks and sugary snacks between meals, and coming in for a professional fluoride application every six to eight weeks. And I would say clearly that if

hygiene is not good enough we will take the appliance off early, because leaving it on would do more harm than good." **2. Which patients are at highest risk of severe root resorption, and what would you do about it?** *Direction: list risk factors by category, then give a management protocol.* "Patient factors include unusually shaped roots — blunted, pipette-shaped or short — a history of dentoalveolar trauma with pre-existing resorption, and possibly underlying chronic inflammation such as asthma. There may be a genetic contribution, with IL-1 and P2X implicated, though unproven. Treatment factors matter more and are within our control: heavy forces, continuous rather than intermittent forces, prolonged treatment, large tooth movements, intrusive forces, lingual root torque, reduction of large overjets by distal movement of the anterior teeth, and pushing apices into cortical bone. Root- filled teeth are not at higher risk. Practically, I would identify high-risk patients at consent and document it; use light forces, particularly for intrusion; keep treatment as short as the aims allow; and take a mid-treatment radiograph. If significant resorption were found I would pause force application for three to six months to allow repair, reassess the aims, and consider accepting a compromise. I would also counsel the patient that even significantly resorbed teeth are usually retained, but that preventing periodontal breakdown later in life becomes especially important." **3. Does orthodontic treatment improve periodontal health?** *Direction: answer no, then give the nuance and the exception.* "On the current evidence, no. There is a lack of evidence that orthodontic treatment improves periodontal health, and it can contribute to higher levels of recession, alveolar bone loss and increased pocketing, although these changes are usually clinically insignificant. Treated patients do have lower plaque scores, but it is unclear whether that reflects the treatment or a behaviour change prompted by it. There are two genuine exceptions where orthodontics prevents periodontal breakdown: correcting an anterior crossbite where a displacement is stripping the labial gingiva from a lower incisor, and correcting a deep traumatic overbite causing gingival damage. The important qualification is the reverse case — in a patient with active periodontal disease, orthodontic treatment can rapidly increase bone loss, so disease must be treated, stable and well maintained before starting." **4. A patient asks whether treatment will help their child's speech. How do you answer?** *Direction: honest, specific, and give the one situation where it might.* "I would say it is unlikely to, and I would explain why. Speech patterns are established very early in life, well before the permanent teeth erupt and long before the age at which we normally treat, so by the time we intervene the pattern

is set. Most people adapt to even a severe malocclusion — a child with an open bite may protrude the tongue when making an 's' sound, but that placement difference does not always produce an audible distortion. Where a specific occlusal trait is clearly linked to a specific speech problem, such as an anterior open bite affecting 's' and 't' sounds, correcting it may help, and cleft palate is a separate matter entirely because velopharyngeal insufficiency causes hypernasality. But I would not offer speech improvement as a reason to treat, and I would suggest a speech and language therapy assessment if that is the parent's main concern." **5. What would make you stop treatment part-way through?** *Direction: give clear, defensible thresholds. This is a professionalism question.* "Persistently poor oral hygiene with developing demineralisation would be the commonest reason — continuing would cause permanent damage that outweighs any benefit, so I would remove the appliance, stabilise, and reconsider later. Significant root resorption identified on a progress radiograph would prompt at least a pause of three to six months and possibly a compromised finish. A change in medical status — a new diagnosis of childhood malignancy, or starting bisphosphonates — would prompt suspension. Repeated non-attendance or non-compliance that makes the aims unachievable is another, and I would want that documented and discussed rather than allowed to drift. If temporomandibular symptoms developed I would suspend active mechanics while they were managed. In every case I would try to leave the occlusion in a safe, stable state rather than simply debonding — which matters most where premolars have been extracted and space closure is incomplete, because that is the situation in which the patient can genuinely end up worse off than before treatment."

Topic 10. Medical conditions relevant to orthodontics

1. A patient on long-term bisphosphonates needs 6 mm of space closure. What do you do? *Direction: state the two problems — movement and osteonecrosis — then simplify the plan.* "I would not attempt it as originally planned. Bisphosphonates inhibit osteoclastic activity and have a very long half-life, remaining active for years after administration, so tooth movement will be slowed or may not occur at all, treatment will be prolonged and the result compromised. Separately, extraction carries a real risk of osteonecrosis, so creating that space by extraction is unattractive. My approach would be to establish from the prescribing physician what agent, what route — intravenous carries higher risk than oral — what dose and for how long. Then I would radically simplify the aims: avoid extractions, avoid large space closure, avoid surgical exposure of teeth, and accept a compromised but stable and healthy result rather than an ideal one. If the space genuinely must be managed, I would explore restorative options with a colleague, closing part of it and restoring the remainder. And I would consent the patient explicitly about slow movement, prolonged treatment and a compromised outcome before starting." **2. A child with poorly controlled epilepsy needs an upper removable appliance. Would you provide one?** *Direction: no — say why, then give the alternative.* "No. A removable appliance in a poorly controlled epileptic carries a risk of displacement during a seizure with potential compromise of the airway, so it should be avoided. I would use a fixed appliance instead, which is not contraindicated. But I would check two further things. First, whether the child requires regular cranial MRI, because stainless steel brackets can degrade image quality and may become dislodged — in which case I would use ceramic brackets, or clear aligners if the tooth movements were suitable, although aligners are themselves removable and so would need thinking through. Second, whether they are taking phenytoin, because that causes gingival hyperplasia which fixed appliances make worse, so I would insist on an excellent standard of oral hygiene and be prepared for gingivectomy. I would also liaise with the neurologist about seizure control before starting." **3. Do you need antibiotic prophylaxis before placing separators in a patient with a prosthetic valve?** *Direction: give the NICE position, acknowledge the tension, and land on a practical answer.* "Under NICE guidance, no — antibiotic prophylaxis solely to prevent infective endocarditis should not be routinely provided to patients at risk

undergoing dental procedures, on the grounds that there is no consistent association between procedures and endocarditis, that everyday toothbrushing probably confers greater cumulative bacteraemia risk than a single procedure, that effectiveness is unproven, and that prophylaxis may cause more deaths from anaphylaxis than it prevents. Separator placement is one of the few orthodontic procedures with any measurable bacteraemia, along with banding, miniscrew placement, debonding and interproximal reduction, but the incidence remains very low. That said, a prosthetic valve places this patient in the highest-risk category, and the American Heart Association and European Society of Cardiology would still recommend cover for exactly this group. So my practical answer is that I would judge the case on its merits and discuss it with the patient's cardiologist before proceeding, and document that discussion." **4. Which medications commonly cause gingival hyperplasia, and how does that alter your appliance choice?** *Direction: three drug classes, then the practical modifications.* "This textbook names two: ciclosporin, a commonly used immunosuppressant, and phenytoin among the anticonvulsants.

Calcium channel blockers such as nifedipine complete the conventional triad and are worth naming, but note that they are not listed in this source. In orthodontic patients the first two are the ones we meet most, in poorly controlled epilepsy and after organ transplantation. The effect is greatly exacerbated by plaque and by appliances, particularly bands, because bands sit subgingivally and are difficult to clean around. So I would use bonded molar tubes rather than bands wherever possible, keep the appliance as simple as I can, avoid components that trap plaque, reinforce oral hygiene at every visit and support it with chlorhexidine. I would warn the patient and parent that hyperplasia may occur, that it usually improves markedly once the appliance is removed, and that gingivectomy may occasionally be needed before or during treatment. And I would keep the overall treatment time short." **5. How would you adapt a first appointment for a child with autism spectrum disorder?** *Direction: concrete, sequential, environment-focused.* "I would plan before they arrive. I would ask the parent what the child's specific sensitivities are and what has worked at the dentist previously, and I would offer a first or last appointment so waiting is minimal, because patients with autism spectrum disorder respond well to familiar surroundings and being seen on time. I would keep the same clinician and the same surgery at every visit, reduce distracting stimuli such as bright lights and background noise, and consider sending photographs of the room and team in advance. The first appointment might involve no treatment at all — gradual desensitisation, letting the child sit in the chair, look at the mirror, tolerate an examination. I would use explicit, concrete instructions rather than abstractions, because abstract concepts are difficult. I would be mindful of a strong gag reflex when taking impressions and would prefer an intraoral scan. And I would allow extra chair time. One genuinely helpful feature is that a tendency towards repetitive behaviour can be turned to advantage in establishing a routine of removable appliance wear."

Topic 11. Overbite and methods of overbite reduction

1. Why is overbite reduction more difficult in an adult? *Direction: the mechanism, then the consequences, then what you do instead.* "Because you lose the mechanism that makes it easy in a child. In a growing patient you can extrude the lower buccal segments and vertical condylar growth compensates for the increase in posterior vertical dimension, so the effect is relative intrusion of the incisors without an increase in face height. In an adult there is no such compensation, so the same extrusion rotates the mandible downwards and backwards, steepens the occlusal plane, increases lower face height and worsens any underlying Class II pattern — and it is prone to relapse because the extruded molars will re-intrude under occlusal load. So in an adult I would reduce the overbite by true incisor intrusion instead, using segmental mechanics with a utility or bypass arch pitched against a buccal segment anchorage unit, or miniscrews, which give greater intrusion with less unwanted molar extrusion. Whichever

route, the endpoint is the same: correcting the interincisal angle so that a proper occlusal stop is established." **2. What is relative intrusion, and what makes it possible?** *Direction: define by contrast with true intrusion.* "Relative intrusion is a reduction in overbite achieved without moving the incisors apically at all. You extrude the lower buccal segments while holding the lower incisors at their existing eruptive position; that flattens the curve of Spee and reduces the overbite. What makes it possible is vertical growth at the mandibular condyle, which compensates for the increase in posterior vertical dimension so the mandible does not simply rotate down and back. That is why it works in a growing patient and not in an adult. True intrusion, by contrast, moves the incisor apically relative to the alveolus, which is mechanically much more demanding and requires fixed appliances with a segmental approach and very light forces." **3. How would you reduce an overbite in a patient with a high maxillary–mandibular planes angle?**

Direction: rule out extrusion, then give the options in order of preference. "In a high-angle patient, molar extrusion is exactly what I want to avoid, because it would rotate the mandible further down and back, increase the lower face height and worsen the facial pattern — and it is unstable. So I would reduce the overbite by true incisor intrusion. Conventionally that means establishing an anchorage unit in the buccal segments and using a segmental intrusion arch: a Ricketts-type utility arch engaged in the brackets as a two-couple system, or a Burstone-type bypass arch lying above the incisors, tied to a sectional wire, as a one-couple system where the point of ligation determines the moment. Forces must be light, around sixty grams in total or fifteen grams per tooth, because intrusive force concentrates at the apices and raises the risk of root resorption. Realistically that achieves around one and a half to two millimetres. If more is needed, or if I want to avoid any molar extrusion at all, I would use miniscrews, which give two to three millimetres and slightly more intrusion than segmental mechanics with less reciprocal molar extrusion. I would avoid Class II elastics and low-pull headgear entirely in this patient." **4. What determines whether a reduced overbite will be stable?** *Direction: one answer — the occlusal stop. Explain it fully.* "The interincisal relationship at the end of treatment. Stability depends on establishing an occlusal stop, meaning the upper incisor root is correctly positioned so that the lower incisor edge occludes against the cingulum plateau and cannot continue to erupt past it. If you reduce an overjet simply by tipping the upper incisors back, the interincisal angle increases, the lower incisors have nothing to occlude against, they overerupt, and the overbite returns regardless of how it was reduced in the first place. That applies to Class II division 1 as much as division 2 — Houston's point is that unless maxillary incisor inclination is corrected, the stop is not established. A practical corollary is that in a low-angle deep bite case, extraction can make control harder, because as the space closes at the end of treatment the incisors upright and the overbite deepens again. And where correction relied on lower incisor proclination, that is inherently unstable and needs long-term, usually bonded, retention." **5. What force would you use for incisor intrusion, and why so light?** *Direction: give the number, then the biological reason, then the safeguard.* "Around sixty grams in total for the labial segment, roughly fifteen grams per tooth. The reason is that an intrusive force is dissipated through the very small surface area of the root apices rather than across the whole periodontal ligament, so the stress per unit area is high even when the absolute force is modest. That produces hyalinisation adjacent to the apex, and when the necrotic tissue is cleared the cementoid surface of the root is damaged and exposed, allowing clastic cells to resorb the root. Intrusive forces are one of the recognised risk factors for orthodontically induced inflammatory root resorption. So I would use light forces, monitor with a radiograph during treatment, and if I saw significant resorption I would pause force application for three to six months to allow repair."

Topic 12. Reducing overjet in a growing child

1. A 9-year-old has a 12 mm overjet and incompetent lips. What do you do, and why? *Direc-*

tion: this is the exception case. Identify it as such and justify. "This child falls into the group where early treatment is justified, and the reason is trauma prevention rather than occlusal outcome. A very large overjet with incompetent lips is a potent risk factor for incisor trauma, and the pooled trial data show early overjet reduction reduces the relative risk of new trauma to about 0.58 — a forty-two per cent reduction. He may also be being teased, and early treatment produces at least a transient improvement in self-esteem. So I would offer a phase of early overjet reduction, discussing that this is for trauma and psychosocial reasons and that it will not improve his final occlusal result — the trials are clear that early treatment produces no long-term skeletal or occlusal advantage, requires more appointments and lengthens overall treatment. I would also point out that headgear was marginally more effective than a functional appliance for trauma reduction

specifically, so that is a legitimate choice, though a twin block may suit a nine-year-old better. And I would fit a mouthguard for sport regardless." **2. What did the Manchester twin block trial show, and how has it changed practice?** *Direction: describe the design, then the findings in two stages, then the practice change.* "It randomised children aged eight to ten with a mean overjet of ten millimetres either to early twin block treatment or to an untreated control group, then reviewed both groups in early adolescence and treated them according to the clinician's preference. The first-phase findings were positive: effective overjet reduction, achieved dentoalveolarly, a transient increase in self-esteem, and improved profile attractiveness. But when both groups were reassessed after all treatment was complete, there was no meaningful long-term difference. Early-treated children were less likely to need a functional appliance later and had a shorter second phase, but they had more attendances and a longer overall treatment time — and the late-treated group actually finished with a slightly better occlusal result, a greater PAR reduction and a slightly smaller overjet. The conclusion was that early treatment for Class II malocclusion is not normally justified. In practice that has moved us away from routine two-phase treatment towards a single adolescent phase, reserving early intervention for children at high risk of incisor trauma or significant psychosocial harm." **3. How do you identify peak height velocity clinically?** *Direction: give the methods, then be honest about their limitations.* "Several ways, none of them precise. Serial standing height measurements plotted over time are the most direct — you are looking for the point of maximum increase — but that requires records over at least a year and identifies the peak retrospectively. Secondary sexual characteristics give a rough guide, as does the timing of menarche in girls, which occurs after peak height velocity. Radiographically, hand-wrist films were traditionally used, and more recently cervical vertebral maturation staging from the lateral cephalogram, which has the advantage of using a radiograph we may already have. The honest position is that all these indices have limited reliability, that the onset, duration and intensity of the mandibular growth spurt vary widely between individuals, and that the correlation between peak height velocity and mandibular growth velocity is far from established. So I would use them as a general guide to whether a child is pre-pubertal or in the spurt, rather than as a precise trigger for starting a functional appliance." **4. What are the disadvantages of two-phase treatment?** *Direction: list them concretely — this is a question about cost, not just efficacy.* "The first is that it does not deliver what it promises: the randomised evidence shows no long-term skeletal or occlusal advantage over a single adolescent phase, and in the UK trial the single-phase group finished slightly better. The second is burden — early-treated children had more attendances and a longer total treatment time, which means more time out of school and more cost, and in the case of removable appliances a longer period of wear at an age where compliance is fragile. The third is that a long period of appliance wear in childhood risks exhausting cooperation before the definitive phase begins, and there is a higher discontinuation rate with removable appliances in children. The fourth is the extra exposure to appliance-related risks such as demineralisation and root resorption over a longer total treatment span. Against all that, the one consistent benefit is a reduction in incisor trauma, which is why I would reserve two-phase

treatment for children who are genuinely at high risk." **5. If a patient presents at 15 with a 10 mm overjet and no growth left, how does your plan change?** *Direction: growth modification is gone. Give the two remaining pathways and the threshold between them.* "Growth modification comes off the table entirely, so a functional appliance is no longer an option and I am choosing between camouflage and surgery. For camouflage I would need space in the maxilla to retract the upper labial segment bodily, which almost certainly means maxillary premolar extractions, with fixed appliances and torque control, and I would plan the anchorage

carefully because I would need close to maximum anchorage — probably a palatal arch and headgear, or in practice miniscrews. The limiting factor is the soft tissue profile, since retracting the upper incisors takes the upper lip back by around 0.7 millimetres for every millimetre. Proffit's retrospective thresholds suggest that an overjet greater than ten millimetres begins to point towards surgery, along with pogonion to nasion perpendicular of eighteen millimetres or more, mandibular body length under seventy millimetres or anterior facial height over a hundred and twenty-five. So at exactly ten millimetres this is a borderline case, and I would assess it primarily on the face: if the mandible is genuinely deficient and camouflage would flatten an already retrusive profile, I would plan for orthognathic surgery once growth is complete, with pre-surgical decompensation."

Topic 13. Management of a missing maxillary central incisor

1. A 9-year-old avulses an upper central incisor. Would you re-implant it, and how does the trauma affect later orthodontics? *Direction: the textbook's position on re-implantation is unambiguous — give it, then move quickly to the orthodontic consequences, which is what an orthodontic examiner is testing. Do not improvise emergency trauma protocol.* "Yes. The position is unambiguous: re-implantation should always be attempted if possible, because *regardless of the long-term prognosis* the tooth is a useful maintainer of both arch space and alveolar bone. That matters enormously in a nine-year-old, who will not receive a definitive implant for another decade, and alveolar bone lost now cannot easily be recovered. The emergency management itself — extra-oral time, storage medium, splinting regime — follows the dental trauma guidelines rather than any orthodontic protocol, and I would defer to those. What I would add as an orthodontist is that orthodontic attachments have a legitimate role in the acute phase, to stabilise a tooth following avulsion and to reposition luxated or intruded teeth. The orthodontic consequences are the substance of the answer. Intrusion and avulsion injuries in particular lead to replacement resorption, ankylosis and infra-occlusion, so I would counsel the family that the tooth may still be lost, and that where an ankylosed incisor becomes badly infra- occluded the options are extraction with space closure or restorative replacement, with decoronation used occasionally to preserve bone. If the tooth is retained, **no active orthodontic force for twelve months after re-implantation**. When treatment does begin, a traumatised tooth needs vitality testing and periapical radiographs beforehand, repeat radiographs **six to nine months** after starting movement, and a **three-month pause** if resorption appears. And at consent I would name the three recognised complications of moving a previously traumatised tooth: loss of vitality, pulpal calcification, and further root resorption where resorption has already occurred." **2. When would you place an implant, and how do you decide growth is complete?** *Direction: state the rule, explain the failure mode, then give the practical test.* "Not until facial growth is complete. The reason is that an osseointegrated implant behaves like an ankylosed tooth — it does not move with the alveolus. If the face continues to grow vertically, the adjacent natural teeth continue to erupt with their alveolar bone while the implant does not, so the implant crown ends up in infra-occlusion, which is both an aesthetic and a functional problem and is not correctable without removing and replacing the implant. To judge completion I would use serial records rather than a single measurement: standing height plotted over time showing a plateau, and serial lateral cephalograms about a year apart superimposed to demonstrate that growth has ceased. Chronological age alone is unreliable, and growth continues later in

males and continues at a low level throughout adult life. In the interim I would maintain the space with a resin-retained bridge, a denture or a retainer with a pontic, and I would counsel the patient that this interim period may last several years." **3. What are the disadvantages of long-term denture wear in this situation?**

Direction: the key is bone, not the denture itself. "The principal disadvantage is progressive loss of alveolar bone height in the edentulous site. Without a tooth or an implant loading the bone, the ridge resorbs, and a denture does nothing to prevent that — it may even accelerate it through pressure. That matters because the whole plan depends on there being adequate bone for an implant in ten years' time, and a deficient ridge means grafting, greater cost, greater morbidity and a poorer aesthetic result, particularly at the gingival margin where a central incisor is highly visible. Beyond that, a denture in an adolescent is often poorly tolerated and poorly worn, it can traumatise the soft tissues, it accumulates plaque, and it is easily lost or broken. That is exactly why the alternative strategy exists — allowing the space to close in the mixed dentition and re-opening it with fixed appliances immediately before prosthetic replacement, which preserves bone but commits the patient to a further course of orthodontics." **4. When is autotransplantation a realistic option?** *Direction: define the ideal case, then the requirements for success.* "It is realistic when premolar extractions are already planned as part of the orthodontic treatment for the underlying malocclusion, so a donor tooth is available without any additional loss. The great advantage is that you replace the missing incisor with a living tooth that has a periodontal ligament, so it can be moved orthodontically, it maintains the alveolar bone, and it grows with the patient — which means it can be done before growth is complete, unlike an implant. The crown is then modified with composite to resemble an incisor. For success you need an atraumatic removal — the textbook stresses this, because trauma at removal is what causes ankylosis — adequate space and bone at the recipient site, and an experienced surgeon, since outcome is highly operator-dependent. The transplant is kept out of occlusion and semi-rigidly splinted for a maximum of three weeks, then root-canal treated once the splint is removed. Conventional teaching is that the donor should have around two-thirds to three-quarters root formation, though the textbook does not state a figure for premolar donors. The recognised risks are ankylosis and external root resorption. It is under-used, and it is a strong answer to give because it avoids both a long period of denture wear and a permanent prosthesis in a young patient." **5. How would you decide between space opening and space closure?** *Direction: give the deciding variables in order of importance.* "Four things decide it. First, whether the loss is unilateral or bilateral: unilaterally, a lateral incisor rarely makes a convincing central, so space opening and prosthetic replacement is almost always the answer; bilaterally, moving both laterals into the central position is a genuine option because the result is at least symmetrical. Second, whether space is needed elsewhere — if there is significant crowding or an increased overjet to reduce, closing the space serves two purposes at once, whereas in a well-aligned arch closing it would create problems. Third, the size and form of the lateral incisors and canines: substitution only works if they can be restoratively modified to a convincing shape, colour and gingival contour. Fourth, the patient's age and growth status, because space opening for an implant commits them to a long interim period. And I would make that decision jointly with a restorative colleague, ideally supported by a diagnostic wax-up or digital smile design, rather than deciding it alone."

Topic 14. Management of an impacted maxillary canine

1. A 10-year-old has no buccal canine bulge. What do you do next? *Direction: a structured clinical sequence. Do not jump to imaging.* "By ten to eleven years the canines should be palpable in the buccal sulcus adjacent to the lateral incisor root, so absence of the bulge is a genuine warning sign. I would first complete the clinical examination — palpate palatally for a bulge, check the mobility and firmness of the primary canine, look for delayed eruption, mesial angulation or retroclination of the lateral incisor, and assess whether the lateral is diminutive or

absent, because canine guidance depends on its root. I would compare the two sides, because asymmetry is significant. I would check the arch for crowding, since that bears on both aetiology and interception. Then I would take a radiograph — a panoramic view with parallax, either a second panoramic at a different vertical angulation or an anterior occlusal — to localise the canine, remembering that conventional radiography is only 39 to 85 per cent accurate. I would reserve CBCT for cases where three-dimensional information will actually change the plan, particularly where I suspect resorption of the incisor roots." **2. How do you localise an unerupted canine? When would you request CBCT rather than parallax films?** *Direction: explain parallax properly, then give a justification test for CBCT.* "Parallax works on the principle that when the X-ray tube moves, an object further from the film appears to move in the same direction as the tube, and an object closer to the film appears to move in the opposite direction — the SLOB rule, same lingual opposite buccal. In practice I would use either vertical parallax, comparing a panoramic radiograph with an upper anterior occlusal, or horizontal parallax with two periapicals taken at different horizontal angulations. I would also use the panoramic film to assess the canine's sector position, its angle to the mid-sagittal plane and its vertical height, all of which are prognostic. I would request CBCT where the additional three-dimensional information would change management: specifically where I suspect resorption of the lateral or central incisor roots and need to know its extent, where the canine is in a position that makes the decision between alignment, transplantation and extraction genuinely uncertain, or where there is suspected pathology. The test is always whether the information will alter the treatment decision, because otherwise the additional dose is not justified." **3. What is the evidence for extracting the primary canine, and would you still do it?** *Direction: give the trial numbers, the Cochrane conclusion, and then your practical position.* "The evidence is weaker than practice suggests. The original prospective studies by Ericson and Kurol reported normalisation in most cases if done between ten and thirteen with space available, but they had no control group. The best randomised trial, by Naoumova, found nine of twenty-three canines erupted at one year after primary canine extraction compared with three of twenty-two in the control group — a real effect — but seven of the extraction group still required surgical exposure, against eleven of the controls, so the reduction in surgery was modest. The Cochrane review by Benson concluded there was only very weak evidence that extraction increases the probability of eruption and no evidence that it reduces the number of children needing surgery, and no benefit from double extraction over single. Despite that, the Royal College of Surgeons guidelines still support it in selected cases. So yes, I would still do it, in a carefully selected patient — ten to thirteen years, canine distal to the midline of the lateral incisor root and around twenty to thirty degrees to the mid-sagittal plane, no crowding — because it is a low-morbidity intervention with a plausible benefit. But I would review radiographically at twelve months and move on if there were no improvement, and I would consent the family honestly about the limited evidence." **4. Open or closed exposure — how do you choose?** *Direction: acknowledge equivalence, then give the factors that actually decide it in practice.* "The evidence says there is little to choose between them. The Cochrane review found no difference in success or operative time and no differences in aesthetics, periodontal outcome, pain or treatment time, though on low-quality evidence. A broader systematic review including non-randomised prospective studies did find open exposure associated with just over two months less time to initial alignment and significantly lower odds of ankylosis, with no other differences. So in practice I decide on position and on the soft tissue. For a canine that is superficial and palatal, an open exposure with a pack allows spontaneous or assisted eruption and lets me start traction under direct vision. For a deeply placed canine, or one where the crown lies close to the incisor roots, I would use a closed exposure with a bracket and gold chain bonded at surgery, because an open exposure would require extensive bone removal and might not stay open. For a buccally displaced canine below the mucogingival junction I would use an apically repositioned flap so the tooth erupts

through attached gingiva. And I would discuss the plan with the surgeon in advance rather than leaving the technique to the day." **5. At what point would you advise a patient that alignment is not worth attempting?** *Direction: give the prognostic thresholds, then the patient factors, then the alternatives.* "I would use the three positional factors. Prognosis worsens as the canine's height above the occlusal plane increases, particularly above the apical third of the incisor roots; as its horizontal position approaches or crosses the midline of the lateral incisor; and as its angulation to the mid-sagittal plane exceeds about fifty-five degrees. A canine that is high, close to the midline and steeply angled has a poor prognosis, and I would be candid that alignment may take two to three years, is anchorage-demanding and may still fail. Beyond position, patient factors matter as much: an adult who does not want a prolonged course of fixed appliances, or a patient with poor oral hygiene where a long treatment carries real demineralisation risk. Where alignment is impractical I would offer autotransplantation, which has around eighty-eight per cent success and is far quicker, though it risks ankylosis and resorption and depends heavily on the surgeon; or extraction with space closure and substitution of the first premolar into the canine position, which can look very good if the root is placed buccally and the crown rotated and reshaped; or, if the patient declines treatment, leaving the canine in situ with regular radiographic review for incisor root resorption and cyst formation."

Topic 15. Management of an impacted lower second molar

1. What would make you observe rather than intervene? *Direction: define the reversible case, then bound the observation with a decision point.* "I would observe where the impaction looks mild and potentially reversible — a second molar only slightly mesially inclined against the distal of the first molar, with adequate space in the arch, no ankylosis, and no obvious mechanical obstruction from the third molar. Spontaneous self-correction does occur, and the same is well recognised for first permanent molars impacted against primary second molars. But I would bound that observation: six to twelve months, with a review radiograph. If eruption has not progressed within that window the impaction is considered established or irreversible, and continuing to observe simply wastes time and risks the tooth becoming ankylosed or the first molar being damaged by resorption. I would also observe rather than intervene where the patient is not yet ready for surgery, or where I want to see whether the third molar develops into a favourable position first." **2. How does a separator disimpact a molar?** *Direction: describe the mechanics simply, then the placement detail that makes it work.* "It acts as a wedge. The separating elastic, brass wire or light open-coil spring is placed interdentally **below the contact point** between the impacted second molar and the first molar. Because it is below the point of contact, it does not simply push the teeth apart at the occlusal surface — it creates a slow, continuous separating force at the level where the two teeth are locked together, allowing the second molar to move distally and occlusally past the bulbosity of the first molar's distal surface. It acts over a period of weeks and can be repeated or replaced with a larger separator if needed. It is the same principle used to disimpact a first permanent molar caught on the distal of the primary second molar, where distal grinding of the primary molar achieves the same thing. It is attractive precisely because it is cheap, reversible and avoids surgery, so it is worth trying before escalating." **3. Would you remove the third molar, and at what stage?** *Direction: it depends on what the third molar is doing. Give both cases.* "It depends whether the third molar is causing the problem or merely present. Where it is distally impacting the second molar, or where its follicle occupies the space the second molar needs, removing it relieves the obstruction and creates room for eruption, and I would do that early — it may be all that is needed. Where the second molar is mesially impacted against the first molar and the third molar is not implicated, removing it achieves nothing and I would leave it. If I am proceeding to surgical uprighting, the third molar is generally removed at the same operation, because it obstructs luxation of the second molar and would otherwise complicate access. The one situation to think carefully about is where extraction of the second molar is being considered, because

then the third molar becomes the replacement tooth and its position and development matter a great deal — although its eruption into that position is unpredictable, which is one reason extraction is the least attractive option." **4. Why is orthodontic alignment preferred to surgical uprighting alone?** *Direction: control and predictability. Explain the combined technique.* "Because surgery alone gives you one attempt at positioning the tooth, whereas combining it with a fixed appliance lets you control the final position gradually and precisely. Surgical uprighting alone was advocated by Pogrel with good interim results, but the technique has since been revisited and combined with orthodontic alignment. In the combined approach, the tooth is surgically exposed, the third molar removed if it is contributing, and the second molar luxated within its socket just enough to break the impaction — then a bracket is bonded at the time of surgery, while access is good, and the alignment is completed afterwards with a fixed appliance and a light controlled uprighting force. That gives more predictable and controlled positioning than luxating and hoping, it allows the tooth to be uprighted rather than simply tipped, and it lets you monitor and adjust. The risks are ankylosis if the luxation is too aggressive, and loss of vitality." **5. What happens if you simply extract the impacted second molar?** *Direction: state that it is the worst option and explain why, then say when you would still do it.* "It is the least successful option on the follow-up evidence. Magnusson and Kjellberg's longitudinal work found that failure of eruption often persists even after intervention, but that extraction of the second molar itself produces the poorest outcomes overall. The reason is that you then depend on the third molar to erupt into its place, and that is unpredictable — it may be absent, malpositioned, or erupt into a poor position with an inadequate contact and a periodontal defect on the distal of the first molar. You also lose a functional tooth permanently. I would still extract where the tooth is severely displaced or ankylosed, where its long-term prognosis is poor for other reasons such as gross caries or molar-incisor hypomineralisation, where primary failure of eruption is diagnosed since orthodontic traction there simply causes ankylosis, or where the patient is unwilling to undergo prolonged surgical and orthodontic management. But I would present it as a compromise rather than an equivalent alternative, and I would assess the third molar carefully before committing."

Topic 16. Methods of crossbite correction

1. How do you distinguish a functional from a true skeletal crossbite? *Direction: the path of closure is the answer. Describe how you actually examine for it.* "Chiefly by examining the path of closure. I guide the patient into retruded contact position and then ask them to close into intercuspal position, watching the mandible. In a functional crossbite there is a premature contact in retruded contact position — often on the primary canines — and the mandible then displaces laterally into intercuspal position, so what looks like a unilateral crossbite is really a bilateral narrowing with a shift. The give-aways are that the crossbite is unilateral, that the dental centreline is coincident in retruded contact but deviates on closing, and that the maxillary intercanine and intermolar widths are symmetrically narrow rather than genuinely asymmetric. A true skeletal transverse discrepancy more often produces a bilateral crossbite with no displacement, and is commonly associated with a severe Class III pattern. I would also assess the face frontally for a chin point deviation and any occlusal cant, take study models or an intraoral

scan to measure arch widths, and consider a postero-anterior cephalogram where asymmetry is a real concern." **2. Why is a unilateral crossbite with displacement usually not skeletal?** *Direction: explain the geometry.* "Because the jaws are broadly symmetrical structures, and a genuinely skeletal transverse discrepancy tends to affect both sides equally — so it produces a bilateral crossbite. What creates the appearance of unilaterality is a symmetrically narrow maxilla combined with a premature contact: the patient cannot occlude in a symmetrical position without interference, so the mandible displaces to one side to achieve maximum intercuspation, producing a crossbite on that side and an apparently normal relationship on the

other. Correct the transverse discrepancy and eliminate the interference and the displacement disappears, the centreline recovers and the occlusion becomes symmetrical. That is also why treating it early matters: a persistent displacement risks becoming established, is a recognised cause of asymmetric muscle activity during chewing, and can contribute to a genuine asymmetry over time." **3. What proportion of RME expansion is skeletal, and what happens to the rest?** *Direction: give the figure, then account for the remainder and the relapse.* "About half. Conventional rapid maxillary expansion can split the intermaxillary suture and achieve up to four millimetres of widening at the suture, and that represents roughly fifty per cent of the total screw expansion. The remainder is taken up by bending of the alveolar processes and by buccal tipping of the teeth themselves — measured at around three and three-quarter degrees of dental tipping, with intermolar width increasing by nearly six millimetres against basal bone width increasing by around two and a half. Of the skeletal change achieved, about ten per cent relapses within four to eight months of retention, with some uprighting of the dentition. That is why deliberate overcorrection and a retention period of three to six months with the appliance left in place are standard, and why miniscrew-assisted expanders were developed — by applying the force to bone rather than teeth they increase the skeletal proportion and reduce the buccal tipping, though the evidence for their superiority remains low level." **4. Why does a midline diastema appear during RME, and what do you tell the patient?** *Direction: mechanism, reassurance, and the caveat.* "It appears because the two halves of the maxilla are being separated at the intermaxillary suture, and the anterior part of that suture runs between the central incisors — so as the suture opens, the incisor crowns move apart. It is direct clinical confirmation that you have achieved skeletal rather than purely dental expansion, which is reassuring rather than alarming. I would tell the patient and parent to expect it before it happens, so it does not come as a shock — that the gap will appear within the first week or two of turning, that it is a sign the appliance is working, and that it will close spontaneously over the following weeks to months as the transeptal fibres recoil and bony infill occurs at the suture. I would also say that occasionally a small residual space persists and needs closing with a fixed appliance later, and that the incisors may look slightly flared during the process." **5. When would you use a 2×4 appliance rather than a removable appliance for an anterior crossbite?** *Direction: give the three indications, then the practical advantages.* "Three situations. First, where bodily movement is needed rather than simple tipping — a removable appliance with a Z- or T- spring can only tip the crown labially, so if the root needs to move too, or if the tooth is significantly rotated, a fixed appliance is required. Second, where space must be created before the tooth can be brought out of crossbite, since a 2×4 lets me open space with a pushcoil or compressed loop while simultaneously aligning. Third, where several teeth are involved, because correcting three or four incisors with a removable appliance becomes unwieldy. The practical advantages are that correction is faster, it does not depend on the patient wearing the appliance, and I have three-dimensional control including torque, which matters because in a Class III presentation I may want to procline the upper incisors deliberately. I would still use a removable appliance with posterior capping for a

single tipped incisor with adequate space in a compliant child, because it is simpler and cheaper — and it has the useful side effect of reducing the overbite while the capping is in place."

Topic 17. Correcting a Class III incisor relationship

1. Why do we decompensate before Class III surgery? *Direction: this is the highest-yield question in the topic. Explain the mechanism and the consequence of not doing it.* "Because the dentition compensates for the skeletal discrepancy, and unless that compensation is removed the surgeon cannot move the jaws to their correct positions. In an untreated skeletal Class III the upper incisors procline and the lower incisors retrocline, which partly masks the discrepancy and makes the reverse overjet smaller than the skeletal pattern would predict. If you took that patient to surgery unchanged, the teeth would interfere before the jaws reached their planned

position and the surgeon would have to under-correct. So pre-surgical orthodontics does the opposite of what instinct suggests: it proclines the lower incisors and retroclines the uppers back to their correct inclinations on their own skeletal bases, which deliberately increases the reverse overjet. The patient must be warned about this at the outset, because their bite and often their appearance get worse for a year before surgery. It is also why I would avoid mandibular premolar extractions in a growing Class III patient who may need surgery — undoing lower incisor retroclination after the extraction spaces have closed is very difficult." **2. How do you distinguish a true skeletal Class III from a pseudo-Class III?** *Direction: the path of closure again, plus the facial assessment.* "Mainly by examining the path of closure. In a pseudo-Class III the skeletal bases are essentially Class I, but a premature contact — often on the upper incisors, sometimes on a retained primary tooth or after trauma — forces the mandible to displace forwards into intercuspal position. So I guide the patient into retruded contact and look for an edge-to-edge incisor relationship there, with the reverse overjet appearing only on closing into maximum intercuspation. That anterior displacement is itself one of the favourable prognostic indicators for orthodontic correction. I would support that with the facial examination — a pseudo-Class III patient has a normal profile in retruded contact, average or reduced lower face height, and normal chin projection — and with a cephalometric assessment showing an ANB within the normal range rather than markedly negative, and a Wits value that is not grossly negative. In a true skeletal Class III there is no displacement, the profile is concave in retruded contact, and the mandibular incisors are usually already retroclined as compensation." **3. Which Class III patients respond best to facemask therapy?** *Direction: give the favourable phenotype, the timing, the evidence, and who fails.* "The favourable patient is young — under about ten years, because the circummaxillary sutures are still responsive — with maxillary retrusion as the dominant component rather than mandibular excess, average or reduced lower face height, a positive or at least average overbite, and ideally an anterior displacement on closing. Combining protraction with simultaneous rapid maxillary expansion appears to improve the advancement achievable by freeing up the suture system. The wear requirement is twelve to fourteen hours a day, which is a real demand on a young child. The important evidence is Mandall's prospective work: while most of the initial skeletal change is lost with subsequent growth, over sixty per cent of early-treated children avoided the need for surgery, compared with thirty-six per cent of those not treated early — so the benefit is real but it is a shift in probability rather than a cure. The patients who outgrow the benefit are those with a hyperdivergent pattern, an increased gonial angle and genuine mandibular prognathism, and the practical difficulty is that these are hard to identify at eight or nine." **4. When is camouflage no longer appropriate?** *Direction: give the thresholds, then the three limiting factors.* "There are cephalometric thresholds below which surgery becomes more likely: a Wits value of minus eleven millimetres or worse — which is regarded as the single most important parameter — a Holdaway angle of 3.5 degrees or

less, an ANB of minus four or less, a gonial angle of 120 degrees or less, a lower anterior face height of sixty-six per cent or more, a reverse overjet of five millimetres or more, and mandibular incisor inclination of eighty degrees or less. Stellzig-Eisenhauer's discriminant analysis, which combines the Wits value with the maxilla-to-mandible length ratio, the saddle angle and mandibular midline deviation, predicts the need for surgery with over ninety per cent accuracy. But I would not decide on numbers alone. Camouflage becomes inappropriate when achieving a Class I incisor relationship would require moving teeth beyond the limits of the alveolus — retroclining lower incisors that are already retroclined risks dehiscence and recession, particularly in a high-angle case with a narrow symphysis and thin biotype — when it would produce an unacceptable facial result, or when the overbite is reduced or the face height increased, which the prognostic indicators list as contraindications." **5. Why is a chin cup rarely used now?** *Direction: state the intended mechanism, then the evidence, then the side effect.* "The intention was to restrain mandibular growth by applying a compressive force to the

chin directed towards the condyles, on the analogy of headgear restraining the maxilla. The problem is that the evidence for a clinically significant long-term skeletal effect is weak — any apparent restraint tends to be recovered once the appliance is stopped, because mandibular growth continues into the late teens in Class III patients. And the change that is achieved is largely unhelpful: rather than shortening the mandible, the chin cup rotates it downwards and backwards, which increases the lower face height. In a Class III patient with an already increased lower face height and a reduced overbite — which is the commonest presentation — that is exactly the wrong direction, and it makes any subsequent camouflage harder because it reduces the overbite further. Add a demanding wear regime in a young child and the risk of loading the temporomandibular joints, and the balance of benefit to burden is poor. Contemporary practice favours maxillary protraction where the maxilla is deficient, and planning for surgery where the mandible is genuinely excessive."

Topic 18. Malocclusion and temporomandibular disorder

1. A patient develops joint clicking six months into fixed appliance treatment. What do you do? *Direction: assess, reassure with accurate information, manage conservatively, document.* "First I would take a proper history and examine — when it started, whether there is pain, whether there is limitation or locking, and I would compare with what I documented at the baseline TMD screening before treatment, which is exactly why that baseline matters. Clicking alone, without pain or limitation, is common in the general population; three-quarters of people have at least one sign, symptoms generally increase with age and often begin in late adolescence (Behrents & White, 1992), which is exactly when orthodontic treatment happens. So I would explain that this is very likely coincidental rather than caused by the appliance, that the condition is typically cyclical and often self-limiting, and that there is no evidence that orthodontic treatment causes temporomandibular disorder. If there were pain, I would suspend active mechanics — the Kandasamy guidelines of 2022 are explicit on that point — and manage conservatively, referring for management of the pain itself if needed. The textbook's position on management is deliberately narrow: because no treatment modality has been shown to be better than any other, treatment should be based on conservative and reversible modalities, of which it names occlusal splints. I would not offer a specific regimen beyond that, and I would say so rather than invent one. What I would not do is alter my orthodontic plan in an attempt to cure the symptoms, or reach for irreversible occlusal adjustment. And I would document all of it carefully." **2. Does malocclusion cause TMD? What is your evidence?** *Direction: a clear no, with the distinction drawn explicitly.* "No, on the current evidence. Large epidemiological studies have shown weak correlations between certain occlusal traits and the signs and symptoms of joint dysfunction — anterior open bite, deep overbite, increased overjet, Class II and Class III molar relationships, posterior crossbite with displacement and non-working side interferences. But association does not establish causation, and there remains a lack of evidence implicating dental occlusion in the underlying pathophysiology, as Manfredini concluded. The aetiology is now understood as multifactorial, and thinking has moved from a mechanical-occlusal model to a biopsychosocial one that incorporates psychological, muscular and systemic factors. I would be careful not to overstate the negative either: the associations are real, just weak, and correcting a posterior crossbite with a displacement is good practice for other reasons. But I would not offer orthodontic treatment as a treatment for temporomandibular disorder, because there is equally a lack of evidence that orthodontic intervention reduces the signs and symptoms." **3. What would you document at the initial assessment regarding the TMJ?** *Direction: give a concrete checklist. This is a medico-legal question.* "I would record a screening history and examination and date it. History: any pain in or around the joints, the ear or the temple; any joint noises; any difficulty or pain on chewing; any locking or limitation; any headaches; any history of trauma to the jaw; any parafunctional habits such as clenching or bruxism; and any previous treatment such as a splint. Examination: maximum

opening measured in millimetres; any deviation or deflection on opening; palpation of the joints and the muscles of mastication with any tenderness recorded by site; and joint sounds recorded as clicking, popping or crepitus and whether reciprocal. I would also record the path of closure from retruded contact to intercuspal position and any displacement, and note relevant occlusal features. Then I would discuss any positive findings with the patient before treatment starts, explaining that symptoms may develop or fluctuate during or after treatment irrespective of what we do, and record that discussion as part of consent. If there were existing pain I would address it, referring if needed, before commencing orthodontics." **4. Would you use orthodontics to treat TMD?** *Direction: no. Justify on evidence and on the hierarchy of reversibility.* "No. There is a lack of evidence that orthodontic intervention reduces the signs and symptoms of temporomandibular disorder, just as there is a lack of evidence that it causes it. And since no single treatment modality has been shown to be better than any other, the rational approach is to use the least invasive and most reversible options first — the textbook names conservative and reversible modalities such as occlusal splints, and notes that contemporary management is moving away from a mechanical– occlusal model towards a biopsychosocial one. Orthodontic treatment is irreversible, prolonged and expensive, and undertaking it explicitly to cure joint symptoms exposes the patient to risk for an unproven benefit, and exposes the clinician if the symptoms do not resolve — which they may not, since the condition is cyclical. If a patient has both a malocclusion they want treated and temporomandibular symptoms, I would manage the symptoms first, get them stable, and then treat the malocclusion on its own merits, having explained clearly that the orthodontics is not being done to treat the joint." **5. Why did the belief that orthodontics causes TMD become so widespread?** *Direction: three strands — a plausible mechanism, a coincidence of timing, and a legal event.* "Three things came together. The first was a superficially plausible mechanism: the idea, dating back to Costen in the 1930s, that overclosure or distal displacement of the condyle causes joint symptoms — which was later elaborated by Witzig and Spahl in the late 1980s into the claim that premolar extraction and retraction mechanics deepen the bite, create anterior interferences and push the condyle distally. It sounded mechanical and therefore convincing, even though no robust evidence was ever presented and it was subsequently shown that incisor retraction does not distalise the condyles. The second was a coincidence of timing: temporomandibular symptoms characteristically begin in late adolescence and increase with age, which is exactly when orthodontic treatment occurs — so patients who develop symptoms after treatment naturally attribute them to it. The third was the Brimm versus Malloy case in Michigan in 1987, where a family successfully sued for eight hundred and fifty thousand dollars, which gave the claim enormous publicity and made orthodontists defensive. It took the American Association of

Orthodontists commissioning independent studies, published together in 1992, to establish that there is no significant association and that orthodontic treatment does not cause the condition."

Topic 19. Mild facial asymmetry in a growing child

1. How do you distinguish a functional from a skeletal asymmetry at the chairside? *Direction: path of closure plus the occlusal plane. Describe the manoeuvres.* "Two things. First, the path of closure: I guide the patient into retruded contact position and watch them close into intercuspal position. In a functional asymmetry there is a premature contact — usually associated with a narrow maxilla or a retained primary tooth — and the mandible deviates laterally on closing, so the chin point and dental centreline are coincident in retruded contact but deviate in intercuspal position. In a skeletal asymmetry the deviation is present in both positions and does not change. Second, the occlusal plane: I look at the patient frontally and ask them to bite on a wooden spatula, which makes any cant immediately obvious against the interpupillary line. A functional asymmetry does not produce a true cant. I would also assess the facial thirds and the position of the chin point relative to the facial midline, examine for asymmetry of the

ramus and lower border by palpation, take models or a scan to compare arch widths, and if there were any real concern take a postero- anterior cephalogram. And I would look at old photographs, because progression over time is the single most important piece of information." **2. What is the significance of an occlusal cant?** *Direction: it is the key discriminator. Explain what it points to and what it excludes.* "A true occlusal cant indicates a vertical skeletal discrepancy between the two sides — the maxilla and mandible have grown to different vertical dimensions — and that essentially excludes a purely dental or functional cause, because a displacement does not tilt the occlusal plane. In the classification of condylar hyperplasia asymmetries, a cant points towards hemimandibular hyperplasia, Obwegeser and Makek's Type II, where there is increased height of the ascending ramus and hyperplasia of the lower border on the affected side, producing a unilateral open bite and a maxillary cant — and characteristically the chin point and dental centreline are often *undisturbed*. By contrast, hemimandibular elongation, Type I, is a horizontal problem: the chin point and centreline deviate markedly to the unaffected side but the occlusal plane remains level. So identifying whether there is a cant tells me which pattern I am dealing with, and therefore whether the surgical plan will need to address vertical asymmetry, which usually means maxillary surgery as well as mandibular." **3. When would you refer, and to whom?** *Direction: give red flags first, then the destination.* "I would refer to a maxillofacial team where the asymmetry appears skeletal and, particularly, where it appears progressive. The red flags are a history of accelerating chin deviation, a developing or worsening occlusal cant, a unilateral open bite appearing during the teenage years, new-onset temporomandibular symptoms on one side, and any asymmetry that has visibly changed on serial photographs or models. Condylar hyperplasia characteristically manifests in the mid to late teenage years, so a new or worsening asymmetry in that window is significant. I would also refer where there is a history of condylar trauma or growth arrest, or any suspicion of a syndromic diagnosis such as hemifacial microsomia, in which case a craniofacial team is appropriate. What I would not do is refer every mild asymmetry, because minor asymmetry is normal; nor would I attempt orthodontic camouflage of an asymmetry I suspected was progressive, because the result would be unstable and would compromise later surgery." **4. What is the role of bone scintigraphy, and what are its limitations?** *Direction: state the question it answers, then be clear about the limitation.* "It answers one specific question: is the condyle still actively growing? A technetium-99m bone scan, or SPECT for three- dimensional visualisation, demonstrates increased metabolic activity in the condyle, which supports a diagnosis of active condylar hyperplasia and tells you that definitive surgery should be deferred, or that condylar resection may be needed. The limitation is that although these scans are sensitive,

they are non-specific: increased uptake also occurs with inflammation, infection and neoplasia, so false positives are possible, and a positive scan must be interpreted alongside the clinical and radiographic picture rather than in isolation. There is also the problem that you cannot repeat them freely because of the radioisotope exposure, which matters when the clinical question is about progression over time. That is why longitudinal monitoring is usually done with serial study models or intraoral scans and facial photographs, and increasingly with serial three-dimensional facial surface photography and superimposition, which is non-invasive and has been shown to have good reproducibility." **5. Would you correct a centreline discrepancy in a 9-year-old with a unilateral crossbite? What would you expect to happen?** *Direction: yes — say why, cite the evidence, and predict the outcome honestly.* "Yes. A unilateral posterior crossbite with a displacement in a nine-year-old is one of the clearest indications for interceptive treatment, and the Cochrane evidence is high certainty that active intervention with a quadhelix or expansion plate is better than observation in children aged seven to eleven — with the quadhelix correcting the crossbite around three months faster. The rationale is that eliminating the premature contact removes the mandibular displacement, and once the mandible closes symmetrically the dental centreline discrepancy typically resolves along with it, because the discrepancy was positional rather than structural. I would expect the crossbite to correct in a

few months, I would deliberately overcorrect, and I would retain for three to six months because relapse potential is high. I would also warn the parents that a small residual centreline discrepancy may persist if there is a genuine dental component such as asymmetric crowding or early loss of a primary tooth, and that any underlying skeletal asymmetry will not be corrected by this — which is why I would document the chin point and occlusal plane at the outset so I can tell the difference later."

Topic 20. Late lower incisor crowding

1. A patient asks whether removing their wisdom teeth will stop their lower front teeth crowding. What do you say? *Direction: a clear no, the reasoning, and what you offer instead.* "I would say no, and that removing them for that reason is not justified. The theory was that erupting third molars push forwards through the arch and crowd the incisors, but the observation that undermines it is that late lower incisor crowding develops just as readily in people who never form third molars, and in those who had them removed years earlier. Their contribution is regarded as minimal, and NICE specifically states that prophylactic removal of pathology-free impacted third molars is not clinically justified to prevent late lower incisor crowding. I would explain what does cause it — that the mandible continues to grow and rotate slowly throughout adult life and the lower front teeth upright to compensate, alongside soft tissue maturation, physiological mesial drift and reduced wear between the teeth — and that it happens to people who have never had braces. Then I would move the conversation to what actually helps, which is long-term retention, usually a bonded retainer behind the lower front teeth. I would still remove a third molar if there were a genuine indication such as recurrent pericoronitis or caries." **2. Why do you recommend permanent bonded retention?** *Direction: the forces are low-magnitude and lifelong, so the retention must be too. Be honest about the drawbacks.* "Because the forces causing late lower incisor crowding are low in magnitude but act continuously over decades, and because the change is an age-related phenomenon rather than a treatment failure, it will not stop at any point at which it would be safe to discontinue retention. A bonded canine-to-canine retainer works passively without requiring the patient to do anything, which matters because the evidence is that compliance with removable retainers decays sharply — by four to five years after treatment most patients have stopped wearing them altogether. In the lower labial segment, fixed retainers are marginally more effective than removable ones over the

first twelve months, particularly for rotations, although beyond that the two perform similarly. I would be honest about the drawbacks: bonded retainers fail, and the failure rate is higher in the lower arch, and the patient often only notices when the teeth start moving — so they need regular review. They accumulate plaque and calculus and need good hygiene, and the longer-term periodontal evidence is conflicting. So my usual approach is a bonded retainer with a vacuum- formed retainer worn at night as a back-up, with the decision made in partnership with the patient so that responsibility is shared." **3. What is the anterior component of occlusal force?** *Direction: define it, then — importantly — say the textbook doubts it.* "It is the idea that during normal function, occlusal forces have a small forward-directed component which is transmitted mesially through the contact points of the dental arch, gradually driving the teeth forwards and, since the arch is bounded anteriorly, producing crowding in the labial segment. The mechanism is plausible, arising from the mesial inclination of the posterior teeth and the cuspal inclines, and it has been proposed as causative in the development of lower incisor crowding, notably by Southard. But the current view is sceptical: the short duration of these forces during chewing makes it unlikely that they produce meaningful tooth movement, since orthodontic tooth movement requires sustained rather than intermittent force. So I would list it as a contributory factor but not as a major one — the factor regarded as dominant is mandibular growth rotation with uprighting of the lower labial segment to compensate." **4. How is late lower incisor crowding different from relapse?** *Direction: define both, then ex-*

plain why the distinction matters practically. "Relapse is the partial or full return of traits of the original malocclusion after treatment — the tooth going back towards where it started, driven by the elastic recoil of stretched periodontal and gingival fibres, by soft tissue forces resisting the position we created, and by occlusal instability. Late lower incisor crowding is different: it occurs in people who have never been treated, so it cannot be a return to a previous state. It is an age-related change, driven principally by continued mandibular growth and rotation with compensatory uprighting of the lower labial segment, along with soft tissue maturation, mesial drift and reduced interproximal wear. The reason the distinction matters is threefold. Clinically, no extraction pattern, archwire sequence or finishing detail will prevent it, so the answer is retention rather than better mechanics. For consent, the patient must understand before treatment that some deterioration is normal ageing rather than a failure of their treatment. And medico-legally, documenting that conversation protects both parties. In practice the two overlap, because the changes seen in untreated individuals are very similar in nature to those seen after treatment." **5. Can any extraction pattern prevent it?** *Direction: no. Explain why the question keeps being asked.* "No. Late lower incisor crowding develops in untreated dentitions, in extraction cases and in non-extraction cases, and no extraction pattern has been shown to prevent it. The question keeps being asked because there is an intuitive logic to it — if crowding reflects a tooth-size to arch-size discrepancy, creating space ought to solve it — and because Tweed and Begg originally argued for premolar extraction partly on stability grounds after retreating relapsed non-extraction cases. But longitudinal studies over several decades showed that extracting teeth provides no guarantee of long-term stability. The reason is that the aetiology is not primarily a static space discrepancy: it is a continuing process driven by lifelong mandibular growth rotation and soft tissue maturation, which acts on the labial segment regardless of how much space was created twenty years earlier. What extraction can do is allow the lower labial segment to be aligned without proclination, and avoiding proclination and avoiding expansion of the lower intercanine width do improve stability — but that reduces relapse rather than preventing age-related change. The only reliable answer is long-term retention."

Appendix B — Model viva answers: the sixteen chapters

Chapter 1. Evidence-Based Orthodontics

1. What is the difference between intention-to-treat and per protocol analysis, and why does it matter in orthodontics? *Direction: define both, then explain why orthodontics is the specialty where it matters most.* "Intention-to-treat analyses every patient in the group they were randomised to, regardless of whether they completed the allocated treatment. Per protocol analyses only those who actually complied. The difference matters because randomisation is what creates pre-treatment equivalence between groups, and if you drop non-completers from the analysis you destroy that equivalence and reintroduce selection bias. Intention-to-treat therefore measures effectiveness — how well the intervention works when offered in real clinical practice — whereas per protocol measures efficacy, how well it works in those who actually receive it. It matters particularly in orthodontics because our treatments are long and depend heavily on compliance, so attrition is common. The Herbst versus twin block trial is a good illustration: thirty-seven per cent of the twin block group were classified as failures, so an intention-to-treat analysis and a per protocol analysis would tell you quite different things — and for a clinician deciding what to prescribe, the intention-to-treat figure is the more useful one." **2. A systematic review reports an I^2 of 78%. What does that tell you and what should the authors have done?** *Direction: define, interpret against the thresholds, state the consequence for the analysis.* "I^2 is the percentage of the total variation across studies that is due to genuine heterogeneity rather than chance. Seventy-eight per cent is high — the conventional thresholds are twenty-five per cent for low, fifty for moderate and seventy-five for high — so there is substantial real variation in the treatment effect between the included studies. Practically, that means the studies are probably not measuring the same thing: they may differ in sample size and participants, in the precise intervention, or in how the outcome was measured. Because I^2 is above fifty per cent, the authors should have used a random-effects rather than a fixed-effect model, which assumes each study estimates a different true effect. But the more important question is whether they should have pooled the data at all — the guidance is not to undertake a meta-analysis if the available studies are felt to be too different from one another. I would also look at the forest plot for the visual clues: are the individual effect sizes all on the same side of the line of no effect, and do the confidence intervals overlap?" **3. What are the limitations of relying on P < 0.05?** *Direction: three distinct criticisms. Be precise about what a P value is first.* "The P value is a test of the null hypothesis — it tells you how often you would see a result at least this extreme by chance if the null hypothesis were true, so it addresses Type I error. There are three problems with relying on it. First, it does not rule the null hypothesis out; it only makes rejecting it reasonable. Second, statistical significance and clinical significance are not the same thing, and this is common in orthodontics — a functional appliance may produce a statistically significant one-millimetre increase in mandibular length that no patient would notice. The reverse also holds: a result reported as showing no difference because P exceeded 0.05 will almost always show numerical differences in the table that may be clinically interesting. Third, a P value tells you nothing about precision. That is why estimation with confidence intervals is preferable — a 95 per cent confidence interval gives you the range of plausible values, and the wider it is, the less confident you should be in the point estimate, which is exactly the information you need when deciding whether to change your practice."

Chapter 2. Occlusion and Malocclusion

1. A patient has a 7 mm overjet, 3 mm of lower incisor crowding and a unilateral crossbite with a 2 mm displacement. What is their IOTN DHC grade? *Direction: apply MOCDO, take the single worst trait, and state it.* "IOTN is hierarchical, so I record only the single worst trait

and I would work through MOCDO to find it. There are no missing teeth. The overjet of seven millimetres falls in grade 4a, increased overjet greater than six but no more than nine millimetres. The crossbite with a two-millimetre displacement is 4c, since that requires a discrepancy greater than two millimetres — at exactly two millimetres it would be 3c, so I would want to measure carefully. Displacement of contact points of three millimetres is 3d, greater than two but not more than four. So the worst trait is the overjet at 4a, and the DHC grade is 4 — treatment required. I would then separately score the aesthetic component against the ten-photograph scale." **2. Why is the Angle classification unreliable?** *Direction: name the assumption, then give the two ways it fails, then the alternative.* "Because it rests on the premise that the first permanent molars hold a fixed position within the dental arch, and that is not necessarily true. Early loss of primary molars allows the first permanent molar to drift mesially, so a patient with a skeletal Class I pattern can present with a Class II molar relationship that reflects space loss rather than any anteroposterior discrepancy. Crowding in the permanent dentition does the same. The classification is also difficult to apply where the molar relationship is asymmetric, which is why the term subdivision exists. And it describes only the anteroposterior plane, saying nothing about the vertical or transverse. For those reasons the incisor classification, based on the relationship of the lower incisor edge to the cingulum plateau of the upper incisors, is generally preferred for describing malocclusion, with the skeletal pattern assessed separately."

Chapter 3. Prenatal Development of the Craniofacial Region

1. Explain the embryological basis of a unilateral cleft lip and palate

Direction: two separate events, in time order. Name the processes. "There are two distinct failures. The lip and primary palate form when the maxillary process on each side grows medially and fuses with the medial nasal process, which happens by around eight weeks. Failure of that fusion on one side produces a unilateral cleft lip, and because the medial nasal processes also give rise to the premaxilla — the four maxillary incisors, their alveolus and the hard palate up to the incisive foramen — the cleft extends through the alveolus into the primary palate. The secondary palate, behind the incisive foramen, forms separately between weeks six and ten from the paired palatine shelves, which begin vertically either side of the tongue, elevate above it and fuse in the midline and with the nasal septum and the primary palate anteriorly. In a complete unilateral cleft lip and palate, the shelf on the cleft side fails to fuse with its counterpart and with the nasal septum, so the cleft runs back through the secondary palate. That embryology explains the clinical picture: the deficiency is at the alveolus, the segments collapse, and the permanent canine and lateral incisor on the cleft side have no bone to erupt into — which is precisely why alveolar bone grafting is timed before canine eruption."

Chapter 4. Postnatal Growth of the Craniofacial Region

1. What is the functional matrix theory, and how well does it hold up? *Direction: define, give the two matrices, then appraise it fairly.* "The functional matrix theory, developed by Moss and Salentijn from van der Klaauw's concept of functional morphology, holds that growth of the skeletal units of the skull is secondary to growth of the functional matrix — the soft tissues, organs and functional spaces they serve. Bone does not grow of its own accord; it

adapts. Two matrices are described: periosteal matrices, meaning the muscles and tendons intimately related to a bone, which determine the size and shape of microskeletal units; and capsular matrices, meaning organs and tissue spaces such as the neurocranium, orbits and oropharynx, which displace whole macroskeletal units. The classic illustration is the cranial vault, where expansion of the brain drives sutural growth. Its strength is that it explains a great deal — why the vault follows brain growth, why condylar cartilage is adaptive rather than primary, and why abnormal function can distort form. Its weakness is its strongest claim,

which is that genes play no major role in postnatal growth. That is not sustainable, and van Limborgh's five-factor model — intrinsic genetic, local and general epigenetic, local and general environmental, with genetic factors predominating and environment moderating — is closer to current thinking. Clinically the theory matters because it is the intellectual basis of functional appliance therapy, and the modest and largely transient skeletal effects those appliances actually achieve are a fair measure of how far the theory can be pushed." **2. What is a mandibular growth rotation and why does it matter clinically?** *Direction: define the three rotations, then give the clinical consequences.* "Growth in mandibular ramus length occurs at the condyle, but its direction varies and usually involves rotation. Björk and Skieller described three. Total rotation is the change in inclination of the mandibular body relative to the anterior cranial base, measurable only against implants. Matrix rotation is the change in inclination of the soft tissue matrix, measured on a tangent to the lower border and centred on the condyles. Intramatrix rotation is the difference between them, and reflects remodelling along the lower border. Solow and Houston relabelled these as true, apparent and angular remodelling. Rotations are forward or backward, and forward is commoner. Mechanistically they reflect an imbalance between anterior and posterior face height growth: excess anterior face height produces a backward rotation, increased posterior face height a forward one. Clinically this matters in three ways. It explains vertical facial patterns and their associated malocclusions — backward rotation with anterior open bite and increased face height, forward rotation with a deep overbite. It explains why dentoalveolar compensation usually preserves a workable occlusion, and why the malocclusion appears only when the imbalance is severe. And continued slow rotation into adult life, with compensatory uprighting of the lower labial segment, is regarded as the major factor in late lower incisor crowding." **3. Would you take a lateral cephalogram to assess cervical vertebral maturation before starting a twin block?** *Direction: no — and explain both the justification and the evidence problem.* "Not on its own. If I already have a lateral cephalogram taken for a justified diagnostic reason, then assessing cervical vertebral maturation from it costs nothing extra and is worth doing — CS3 indicates that peak mandibular growth will occur within a year and is regarded as the ideal time to begin. But the textbook is explicit that taking a lateral skull radiograph specifically to undertake or assess CVM is difficult to justify, and I agree: there is no threshold dose below which radiation risk disappears, so the exposure must materially change management. More fundamentally, the method rests on three assumptions — that the timing of the adolescent spurt can be accurately predicted from CVM, that it correlates with maximal mandibular growth, and that treating in that window increases mandibular growth — and none of those is universally accepted. The systematic review evidence on timing is heterogeneous and at high risk of bias, and one review found chronological age nearly as good a predictor. So in practice I would judge readiness clinically — dental development, secondary sexual characteristics, serial height measurements, and the parents' account of recent growth — and treat when the patient is dentally and behaviourally ready."

Chapter 5. Development of the Dentition

1. A 9-year-old has 3 mm of lower incisor crowding. Would you intervene?

Direction: no — explain the physiology that will help, and what you would monitor. "Not immediately. At nine the child is between the two phases of mixed dentition eruption, and some of this crowding may resolve without intervention. The permanent incisors are collectively about five millimetres wider than their predecessors in the mandible, and that liability is normally met by residual spacing, a more labial eruptive position and distal movement of the primary canines into the primate spaces — which is the one significant increase in mandibular intercanine width that occurs. Ahead of him there is also the leeway space, about two and a half millimetres per quadrant in the mandible, released when the primary canine and molars are replaced by the permanent canine and premolars. So three millimetres is potentially

recoverable. What I would do is monitor: record the crowding, check the primary molars are caries-free and not at risk of early loss, and confirm the permanent canines are palpable buccally by ten to eleven years. If a primary second molar were lost early I would consider space maintenance, because forward movement of the first permanent molar consumes the leeway space and it is the second premolar that then becomes crowded. I would review in twelve months and reassess when the permanent dentition is nearly complete." **2. What is the leeway space and how do you exploit it?** *Direction: define, quantify per arch, then give the two clinical uses.* "The leeway space is the excess of the combined mesiodistal width of the primary canine and primary molars over that of the permanent canine and premolars that replace them — approximately one and a half millimetres per quadrant in the maxilla and two and a half in the mandible, the difference being the larger lower second primary molar. So around five millimetres becomes available in the lower arch as the buccal segments exchange. It can be exploited in two ways. Passively, by simply allowing it to be taken up by the erupting canines and premolars, which is why mild crowding at eight or nine may resolve. Actively, by preserving it — a lower lingual arch placed before the primary molars are lost prevents the first permanent molars drifting mesially, holding the space for the alignment of the labial segment. The corollary is the commonest way it is lost: early loss of a primary second molar, or mesial movement of the lower first molar during molar correction, uses up the leeway space and the mandibular second premolar becomes crowded."

Chapter 6. Orthodontic Tooth Movement

1. Why can a removable appliance only tip a tooth? *Direction: point contact, no couple. Then give the consequence.* "Because the active component makes only point contact with the crown. A single force applied at the crown, at a distance from the centre of resistance, necessarily creates a moment, and because there is no second point of contact there is no couple to counteract it — so the tooth rotates about a centre of rotation somewhere within the root and tips. A fixed appliance can control movement in three dimensions precisely because the bracket engages the archwire along a slot, so a couple can be generated within the slot to oppose the moment created by the force, and by varying the moment-to-force ratio you can produce controlled tipping, translation or root movement. That is the mechanical reason removable appliances have limited use in contemporary practice, quite apart from being removable and therefore compliance-dependent. It also explains their legitimate niches: tipping a single incisor out of an anterior crossbite with a Z-spring, simple expansion with a midline screw, or a flat anterior bite plane to reduce an overbite." **2. Where is the centre of resistance, and why does it matter?** *Direction: give the anatomy, then the clinical consequences.* "For a healthy single-rooted tooth it is around one-third to halfway down the root; for a multirooted tooth it lies between the roots, one to two millimetres apical to the furcation. It matters for three reasons. First, it defines bodily movement: a force whose line of action passes through the centre of resistance produces translation, and any force that does not creates a moment and therefore rotation. Second, it moves — if alveolar bone is lost through periodontal disease, the centre of resistance moves apically, the

distance from the point of force application increases, and the same force produces a larger moment and more tipping. That is why periodontally compromised teeth tip so readily and why light forces and extra-dental anchorage matter in those patients. Third, it scales: when teeth are rigidly ligated together each arch has its own centre of resistance, as does each jaw, which is what you must consider when applying orthopaedic or extra-oral force — the reason the position and length of a headgear outer bow relative to the molar furcation determines whether the molar tips mesially or distally."

Chapter 7. The Orthodontic Patient: Examination and Diagnosis

1. A referring dentist has sent a patient with a CBCT they took to localise an unerupted canine. Comment. *Direction: be measured — acknowledge the value, then apply the justification test.*

"CBCT can be genuinely valuable for an unerupted canine, because three-dimensional localisation is more accurate than parallax films and it is the only reliable way to assess resorption of the adjacent incisor roots. But the test is whether the additional information will actually influence the treatment planning decision, and that is a question that should be answered before the exposure, not after. For many canines, a panoramic radiograph with a parallax view gives enough information to plan — sector position, angle to the mid-sagittal plane and vertical height are all obtainable, and those are the prognostic variables. I would want to know what question the dentist was trying to answer. If the canine is close to the incisor roots and I need to know whether there is resorption, or if the position is such that the choice between alignment, transplantation and extraction genuinely turns on three-dimensional detail, then it is justified. If it was taken as a default, it is not — and I would say so tactfully, and use the scan now that it exists rather than repeating imaging."

Chapter 8. The Orthodontic Patient: Treatment Planning

1. Why does the textbook say planning should begin with the face rather than the occlusion? *Direction: give the historical argument, because that is what makes the point persuasive.* "Because the alternative has been tried twice and failed twice. In the early years of the specialty it was assumed that if you aligned all the teeth and achieved the occlusal goals, ideal facial aesthetics would follow — Angle's non-extraction, arch-expanding approach — and the result in some cases was extreme protrusion of the dentition and extensive relapse. Then, with the advent of cephalometrics, it was argued that facial growth was genetically determined and orthodontics could not influence the skeleton, so treatment was planned around arbitrary hard tissue standards, particularly mandibular incisor position. But those standard values bear little relation to the soft tissues and the facial profile, so they did not deliver good faces either. Current thinking is that the soft tissues define the limits of treatment in each individual, so planning starts with where the dentition — particularly the maxillary incisors — should sit within the face, and the occlusal plan follows from that. The practical drivers were a public preference for fuller profiles, the resurgence of functional appliances, better orthognathic surgery, a better understanding of relapse, acceptance of long-term retention, and an appreciation that treatment in the teens has consequences for facial appearance decades later."

Chapter 9. Functional Jaw Orthopaedics

1. A 9-year-old has a bilateral posterior crossbite with no displacement and a Class III skeletal pattern. What do you do? *Direction: identify the pattern as skeletal, then combine expansion and protraction, then be honest about prognosis.* "A bilateral crossbite without a displacement suggests a genuine transverse skeletal discrepancy rather than a functional problem, and in combination with a Class III pattern this is the classic presentation of maxillary hypoplasia in all three planes. At nine, this child is in the

favourable window. I would plan rapid maxillary expansion combined with facemask protraction: the expansion corrects the transverse deficiency and simultaneously disarticulates the circummaxillary sutures, which improves the advancement achievable. Facemask wear is twelve to fourteen hours a day, which is a real demand, and expansion is one to four turns daily over two to four weeks with three to six months of retention. I would be honest with the parents that most of the initial skeletal effect is lost with subsequent growth — but that the prospective evidence shows over sixty per cent of children treated this way avoided surgery, against thirty-six per cent of those not treated early, so it meaningfully shifts the odds. I would also say that the children who tend to outgrow the benefit are those with a hyperdivergent pattern, an increased gonial angle and true mandibular prognathism, and that we will monitor growth into the late teens before any definitive decision."

Chapter 10. Contemporary Fixed Appliances

1. Take me through your archwire sequence for a moderately crowded extraction case, and

justify each stage. *Direction: four stages, each justified by the property required.* "I would sequence by the property I need rather than by habit. For alignment I want large springback, low stiffness and high stored energy so the wire can be engaged into displaced teeth without permanent deformation and can deliver a light continuous force over a long range — so a round superelastic nickel titanium, starting around 0.014 inch and progressing to 0.018, or a multistrand steel, which performs comparably. I would then place a rectangular nickel titanium briefly to begin torque expression. Once aligned, I need increasing stiffness to level and to begin overbite reduction, so round stainless steel from 0.016 to 0.020 inch, with a reverse curve of Spee if the overbite requires it and lingual crown torque to counteract the proclination that produces. For space closure I need high stiffness, low stored energy and low friction so that teeth slide along the wire without it distorting, so a rectangular stainless steel working archwire — and I would close en masse against fixed anchorage, which is the most efficient method. Finally, for detailing and settling I want formability with a lower stiffness than full-dimension steel, so beta titanium for finishing bends. I would add that the trial evidence shows the differences between specific alignment archwires and sequences are marginal, so the sequence is legitimately a matter of operator preference within those principles."

Chapter 11. Skeletal Anchorage

1. What factors determine whether a miniscrew succeeds? *Direction: group them — bone, screw, site, technique, patient. That structure is worth more than the list.* "I would group them. Bone factors first: cortical bone thickness is probably the single largest determinant, which is why the palate performs so well. Screw factors: length and diameter, with higher failure below eight millimetres and below 1.2 millimetres — short screws fail at around twelve and a half per cent versus eight per cent for longer ones — although larger diameters increase the risk of root contact interradicularly. Body shape, tapered or cylindrical, and self-drilling versus self-tapping, make no difference. Site: failure rates range from about one per cent in the median palate and five per cent paramedian, through five to ten per cent for maxillary buccal interradicular sites and twenty-six per cent between the first and second molars, up to twenty-eight per cent in the zygomatic buttress and reports above sixty-five per cent for the mandibular buccal shelf. Insertion into keratinised attached gingiva does better than free gingiva. Technique: insertion torque should be five to ten newton-centimetres — too low and you have no primary stability, too high and you microfracture the bone and lose secondary stability — and root contact is a major cause of failure. Operator experience matters. Patient factors: sex does not affect success, but vertical facial type does, and heavy smoking increases failure. And in adults immediate loading is fine; in adolescents delayed loading may be safer."

2. You want to retract an upper labial segment without any molar movement in an adult. How would you plan the anchorage?

Direction: the case for skeletal anchorage, then the specifics of placement. "This is a maximum anchorage case, and conventional means are unlikely to deliver it — without support around two- thirds of the space created is lost, a trans-palatal arch contributes very little sagittally, and headgear in an adult is neither realistic nor acceptable. So I would use miniscrews. I would place them buccally between the second premolar and first molar bilaterally, which requires at least four millimetres of interradicular space measured to the cement-enamel junction, and I would confirm that on a periapical radiograph with the roots aligned first, since root contact is the major cause of failure. I would place into keratinised attached gingiva, use a self-drilling screw of at least eight millimetres length and around 1.4 millimetres diameter, and aim for an insertion torque of five to ten newton-centimetres. I would then retract the labial segment en masse on a rectangular stainless steel working archwire with elastomeric or coil traction from the miniscrews, which is the most efficient approach and avoids the two-stage canine retraction. In an adult I would load immediately. If interradicular space were inadequate, I would place paramedian palatal screws instead, where failure rates are lowest at under five per cent

and placement is not restricted by the dentition, and use an indirect anchorage arrangement."

Chapter 12. Management of the Developing Dentition

1. A 9-year-old has an unerupted upper central incisor and a supernumerary on the radiograph. How do you manage it? *Direction: assess, then a staged plan, then the caveat about space.* "First I would establish the type and position of the supernumerary and the position and morphology of the unerupted incisor, using a panoramic film with parallax or an anterior occlusal, and CBCT only if three-dimensional information would change the plan. Type matters: a tuberculate supernumerary characteristically obstructs eruption, whereas a conical one may not. I would check whether the contralateral incisor has erupted — if it erupted more than six months ago, the delay is significant — and whether space has been lost in the arch, because that is often the limiting factor. The usual management is surgical removal of the supernumerary with exposure of the incisor and, where space is adequate and the tooth is favourably positioned, allowing spontaneous eruption, which occurs in a good proportion of cases over the following year. If space has been lost, I would regain it first or at the same time with a removable appliance or a 2×4 fixed appliance, because an incisor will not erupt into a space that no longer exists. Where the incisor is severely displaced, dilacerated, or fails to erupt after twelve months, I would plan for a bracket and gold chain at the time of surgery and orthodontic traction. And I would warn the parents that dilaceration, which often follows early trauma or an obstructing supernumerary, may make the tooth unsalvageable."

Chapter 13. Management of the Permanent Dentition

1. How would you counsel a patient about retention at the start of treatment? *Direction: honest, quantified, shared. Cover both relapse and ageing.* "I would raise it before treatment starts, because it is part of consent rather than an afterthought at debond. I would explain three things. First, that teeth move after treatment — relapse is almost universal to some degree, and the traits most likely to relapse are rotations, lower incisor alignment, any change in lower incisor position, expansion of the lower intercanine width and spaces. Second, that separate from relapse, teeth continue to crowd slowly throughout adult life even in people who have never had braces, because the mandible keeps growing and rotating and the lower front teeth upright to compensate — so some change is normal ageing, not a failure of treatment. Third, and this is the honest part, that the only reliable answer is retention for as long as they want their teeth to stay straight. I would explain that I usually recommend a bonded retainer behind the lower front teeth with a removable retainer worn at night as a back-up, that bonded retainers do fail and often the first sign is the teeth moving, so regular checks matter, and that the evidence is that most people stop wearing removable retainers within four or five years. I would ask them to make that decision

with me rather than have it imposed, because it is a long-term commitment and shared responsibility works better than instruction." **2. A 45-year-old with treated periodontal disease wants orthodontic treatment. What are your concerns?** *Direction: periodontal first, then mechanics, then expectations.* "My first concern is periodontal stability. Orthodontic treatment in a patient with active periodontal disease can rapidly increase bone loss, so the disease must be treated, stable and well maintained before I start, with a maintenance programme running throughout. Second, mechanics change with reduced bone support: the centre of resistance moves apically, so the same force produces a larger moment and more tipping, and I would use very light forces and be careful about intrusive and torquing movements. Anchorage planning is affected too — the textbook makes the point that in maximum anchorage cases with periodontal bone loss you may not want to place initial aligning forces on the anchor teeth at all, and should consolidate them passively or use skeletal anchorage instead. Third, I would avoid moving teeth out of the alveolar envelope, because recession is a real risk. Fourth, aims: I would plan a realistic, often compromised, result rather than an ideal one, and I would plan it jointly with the periodontist and, if there are restorative implications, with a restorative col-

league. Finally I would be clear at consent that permanent retention is essential, that a bonded retainer will need meticulous hygiene, and that a modest, stable, healthy result is the objective."

Chapter 14. Clear Aligners

1. An adult asks for Invisalign rather than fixed appliances for a moderately crowded Class I case. How do you counsel them? *Direction: this is a consent question. Be balanced, quantified and honest.* "I would say that for a case like theirs — Class I, moderate crowding, essentially simple tooth movements — aligners are a legitimate option and are likely to give a good result, and I would set out the trade-offs honestly. In their favour: they are far less visible, they are more comfortable in the first few days, eating and chewing are easier, and the periodontal outcomes appear slightly better because they are easier to clean around. Against: the evidence shows that on detailed occlusal grading, aligner results are significantly worse than fixed appliances, particularly for buccolingual inclination, occlusal contacts, the occlusal relationship, overjet and root angulation — even though the coarser PAR index shows no difference. They are more prone to relapse afterwards, possibly because teeth end up more proclined. Speech may be affected initially, though people adapt. And I would be candid about refinements: only about six per cent of patients complete treatment without needing a further course of aligners, and over seventeen per cent end up converting to fixed appliances to finish. I would also stress that success depends entirely on wearing them as prescribed, that attachments will be needed and are the most uncomfortable part, and that if we cannot achieve an acceptable result we will need to discuss changing to a fixed appliance. Then I would let them choose."

Chapter 15. Orthodontics and Orthognathic Surgery

1. Take me through the stages of combined orthodontic-surgical treatment

Direction: five stages in order, with the purpose of each. "Five stages. First, joint assessment in a combined clinic with the surgeon, the technician and access to psychological support, with a full record set — study models, lateral and if needed postero-anterior cephalograms, and intra-oral and facial photographs — leading to a provisional plan and a properly informed consent discussion about risks. Second, pre-surgical orthodontics, which is chiefly decompensation: aligning and levelling the arches, coordinating arch widths, and deliberately removing the dentoalveolar compensation so the incisors sit at their correct inclination on their own skeletal bases. In a Class III case that means proclining the lower incisors and retroclining the uppers, which increases the reverse overjet and makes the bite and often the appearance worse — the patient must be warned about that before we begin. Third, planning immediately before surgery: new records, model surgery or three-dimensional planning, and construction of intermediate and final wafers which are tried in. Fourth, surgery itself — a Le Fort I for maxillary movement, a bilateral sagittal split for the mandible, bimaxillary surgery where both are needed, with a genioplasty as an adjunct where the chin requires separate management. Fifth, post-surgical orthodontics to settle and detail the occlusion, followed by retention. Timing throughout is dictated by growth: surgery waits until growth has ceased, which in Class III means the late teens because the discrepancy worsens with continuing mandibular growth."

Chapter 16. Cleft Lip and Palate, and Syndromes Affecting the Craniofacial Region

1. When is alveolar bone grafting performed and why then? *Direction: the age, then the biological reason, then the orthodontic preparation.* "At around eight to ten years of age, and the timing is dictated entirely by the permanent canine. The graft must be placed before that tooth erupts, when its root formation is approximately two-thirds complete, because the whole purpose is to provide bone for it to erupt through — graft too late and the canine has already been obstructed or has erupted into an inadequate site; graft too early and there is a risk to the developing dentition and to maxillary growth. Before the graft I would undertake a period of orthodontics to

expand the collapsed maxillary arch, typically with a trihelix or quadhelix retained afterwards with a palatal arch, which both creates surgical access and maximises the volume of bone that can be placed. Some alignment of the maxillary incisors can be done at the same time, but I would take care not to move any tooth into the cleft site where there is no bone — those teeth are aligned after grafting. The graft is cancellous bone, usually harvested from the iliac crest. Its benefits are that it allows the canine to erupt, permits alignment and orthodontic space closure — which matters because the lateral incisor is often absent — gives a stable intact arch with good alveolar contour, stabilises the premaxilla in bilateral cases and closes oronasal fistulae." **2. Why do cleft lip and palate and isolated cleft palate have different epidemiology?** *Direction: link back to Chapter 3 embryology.* "Because they are different developmental failures at different times. Cleft lip, with or without cleft of the primary palate, arises from failure of the maxillary processes to fuse with the medial nasal processes, which happens by about eight weeks. Isolated cleft palate arises from failure of the palatine shelves to elevate above the tongue and fuse in the midline, between weeks six and ten. Those are separate processes with separate genetic architecture, and the epidemiology reflects that: cleft lip and palate occurs in around one in seven hundred live births in Caucasians, affects males about twice as often as females, is unilateral in around eighty per cent of cases with a left-sided predominance, and is non-syndromic in seventy per cent. Isolated cleft palate is about half as common, around one in two thousand, affects females about four times as often as males, and is non-syndromic in only fifty per cent — so it carries a considerably higher chance of an underlying syndrome, which is why a genetics assessment matters more in that group. The female predominance in cleft palate is usually attributed to the later timing of palatal shelf elevation in female embryos, giving a longer window in which the process can fail."

Appendix C — Self-test answers

Chapter 1. Evidence-Based Orthodontics — 1. Patients, Intervention, Comparison, Outcomes (pp. 20–21). 2. Prevalence — cross-sectional and ecological; incidence — cohort and ecological (p. 22). 3. Selection, performance, detection, attrition, reporting (pp. 37–40). 4. Below 50%, so heterogeneity is low-to-moderate and a fixed-effect analysis is appropriate (p. 47). 5. Ensuring the investigator cannot know or influence which group a participant will be assigned to; it should be performed by a third party, and the investigator should have no involvement in allocation (p. 31). **Chapter 2. Occlusion and Malocclusion** — 1. About 10%; a 0–2 mm discrepancy is normal (p. 58). 2. Missing teeth, Overjet, Crossbite, Displacement of contact points, Overbite — the order in which to search for the single worst trait when assigning an IOTN DHC grade (p. 107). 3. A reduction of 22 points or more (p. 108). 4. Molar relationship, crown angulation, crown inclination, no rotations, no spacing, flat occlusal plane (p. 58). 5. There is none — the evidence is lacking, both for stability and for reducing TMD (p. 62). **Chapter 3. Prenatal Development of the Craniofacial Region** — 1. Frontonasal, paired maxillary, paired mandibular processes; approximately 4 weeks post conception (p. 122). 2. Nasolacrimal duct, cheek and alar base of the nose (p. 123). 3. The four maxillary incisors, their alveolus and a triangular region of hard palate — the premaxilla — bounded posteriorly by the incisive foramen (p. 131). 4. Weeks 6 to 10 (p. 130). 5. The fifth (p. 119). **Chapter 4. Postnatal Growth of the Craniofacial Region** — 1. Around 6–8 years; the adolescent spurt is later and more significant in males, giving a longer growth period and greater final height (pp. 168, 171). 2. Primary displacement, remodelling (cortical drift), relocation, secondary displacement (p. 193). 3. James Scott — nasal septal cartilage, cranial base synchondroses, condylar cartilage (p. 194). 4. True = total rotation, mandibular body relative to cranial base; apparent = matrix rotation, the soft tissue matrix relative to cranial base and what a cephalogram shows without implants; angular remodelling = intramatrix rotation, the difference between them, reflecting lower border remodelling (pp. 220–222). 5. Peak mandibular growth within one year, so the ideal time to start a functional appliance; the caveat is that none of the underlying principles is universally accepted, and a radiograph taken solely for CVM is difficult to justify (p. 238). **Chapter 5. Development of the Dentition** — 1. ~5 mm mandible, ~7 mm maxilla (p. 266). 2. ~1.5 mm maxilla, ~2.5 mm mandible; the mandibular figure is larger because of the greater size of the lower second primary molar (p. 279). 3. Mandible: canine, first premolar, second premolar — second premolar vulnerable. Maxilla: first premolar, second premolar, canine — canine vulnerable (p. 279). 4. During eruption of the permanent incisors, as the primary canines move distally into the primate spaces; achieved largely by alveolar rather than skeletal change (pp. 268, 278). 5. Primary — genetic tooth-size/jaw-size discrepancy; secondary — environmental, local space conditions and soft tissue function; tertiary — adolescent and post-adolescent, predilection for the lower labial segment (p. 283). **Chapter 6. Orthodontic Tooth Movement** — 1. Two equal and opposite forces producing rotation with no net force; it acts about the centre of resistance irrespective of where the forces are applied (p. 309). 2. It moves further away, and less rotation occurs; through the centre of resistance it is at infinity and movement is bodily (pp. 309–310). 3. Between the roots, 1–2 mm apical to the furcation (p. 307). 4. About 1 mm per month; faster in the maxilla and in younger patients (p. 304). 5. Bisphosphonates and NSAIDs slow it; corticosteroids, parathyroid hormone and thyroxine accelerate it (p. 304). **Chapter 7. The Orthodontic Patient: Examination and Diagnosis** — 1. Porion and orbitale are difficult to locate; both are bilateral and must be averaged; the plane does not lie in the mid-sagittal plane, so head positioning affects it (p. 418). 2. Essential before planning surgical jaw movement; not justified for minor tooth movements or on the basis of growth pre-

diction alone (pp. 413, 415– 416). 3. Serial lateral skull radiographs approximately one year apart (p. 416). 4. Whether the additional information will actually influence the treatment planning decision (p. 411). 5. They are chosen for convenience, no specific anatomical region exists at either, and both remodel as the incisors move (p. 461). **Chapter 8. The Orthodontic Patient: Treatment Planning** — 1. Overjet >10 mm; pogonion to nasion perpendicular ≥18 mm; mandibular body length <70 mm; anterior facial height >125 mm (p. 491). 2. About 0.7 mm, with a 1.6° increase in nasolabial angle (p. 492). 3. About 3 mm — roughly half the expansion (p. 522). 4. Moderate-to-severe Class III skeletal base; increased lower face height; reduced overbite; retroclined mandibular incisors; proclined maxillary incisors; no anterior displacement on closing; a significant Class III molar relationship (p. 505). 5. Up to 1 mm, less in the labial segments (p. 525).

Chapter 9. Functional Jaw Orthopaedics — 1. About 50%, up to 4 mm at the suture; about 10% of the skeletal change relapses at 4–8 months (pp. 543–544). 2. Yes, in the early mixed dentition; the constraint on suture splitting is age-related interdigitation, not the rate of expansion (p. 543). 3. Over 60% avoided surgery, versus 36% without early treatment (p. 487). 4. 12–14 hours (p. 552). 5. It requires compensatory expansion of the mandibular dentition, which is likely to be unstable (p. 543). **Chapter 10. Contemporary Fixed Appliances** — 1. Alignment — large springback, low stiffness, high stored energy, biocompatibility, low friction. Space closure — high stiffness, low stored energy, biocompatibility, low friction, good joinability (pp. 649–650). 2. Comparable occlusal outcomes and no difference in treatment time versus conventional, with more breakages, more complaints and more planning time; treatment time was determined by case difficulty and the treating orthodontist (p. 631). 3. Martensitic-stabilised (cold-worked, no shape memory); austenitic-active (stress- induced, superelastic); martensitic-active copper NiTi (temperature-induced, thermoelastic) (pp. 654–655). 4. About one-third (p. 653). 5. Moderate evidence that coaxial superelastic NiTi outperforms single-strand over 12 weeks; otherwise insufficient evidence that any material is superior (p. 655). **Chapter 11. Skeletal Anchorage** — 1. 5–10 Ncm; below, poor primary stability; above, poor secondary stability through microfracture, with reduced success above 10 Ncm (pp. 718–719). 2. Median palate 1.3%; mandibular buccal shelf reported at over 65% in one series, with 7% and 25% elsewhere (pp. 716–717). 3. Cortical bone thickness (p. 713). 4. Length under 8 mm and diameter under 1.2 mm (p. 714). 5. Not in adults; in adolescents delayed loading has been associated with higher failure (p. 719). **Chapter 12. Management of the Developing Dentition** — 1. 6.4%; third molars (up to 30%), then mandibular second premolars, then maxillary lateral incisors — the teeth at the end of each series (pp. 748, 751). 2. Conical, tuberculate, supplemental, odontome (pp. 755–757). 3. About 50% — impaction, displacement or failure of eruption (p. 757). 4. *PTH1R*; because orthodontic extrusion normally causes ankylosis and failure, so it changes management entirely (pp. 800–801). 5. The maxilla, 25 times more often than the mandible; it is a good indicator of posterior crowding (p. 793). **Chapter 13. Management of the Permanent Dentition** — 1. Rotated teeth, lower incisor crowding, changes in lower incisor labiolingual position, expansion of lower intercanine width, spacing (Ch 2, p. 100). 2. Fixed are marginally more effective for the first 12 months, especially for rotations, but equivalent thereafter; fixed have a higher failure rate in the lower arch (p. 989). 3. A minority — the majority have stopped altogether (p. 990). 4. An anterior crossbite corrected with a positive overbite (p. 977). 5. Wits ≤ –11 mm (p. 920). **Chapter 14. Clear Aligners** — 1. ABO-OGS shows significantly worse occlusal outcomes and more unacceptable results; PAR shows no difference, because it is a less rigorous measure of tooth position and more reflective of baseline severity, and the two do not correlate post-treatment (p. 1029). 2. Buccolingual inclination, occlusal contacts, occlusal relationship, overjet, root angulation (p. 1031). 3. About 6% (p. 1032). 4. Over 17% (p. 1032). 5. Less pain in the first few days, better periodontal index scores, less eating and chewing disturbance — though more speech difficulty and more relapse (p. 1030). **Chapter 15. Orthodontics and Orthognathic Surgery** — 1. Up to 1 cm

(p. 1041). 2. Broadening of the alar base, elevation of the nasal tip and subtle cheek bulging, giving a "cherubic" appearance (p. 1092). 3. About 1 mm per day, with around 6 weeks of consolidation (p. 1085). 4. Elongation — horizontal vector, chin and centreline displaced to the unaffected side, occlusal plane level. Hyperplasia — vertical vector, chin and centreline often undisturbed, unilateral open bite and maxillary cant on the affected side (pp. 1096–1098). 5. Because decompensation deliberately removes the dentoalveolar camouflage, proclining the lower and retroclining the upper incisors, which increases the reverse overjet so the surgeon can achieve full correction (Ch 13, p. 921).

Chapter 16. Cleft Lip and Palate, and Syndromes Affecting the Craniofacial Region — 1. CLP — 1:700 in Caucasians, male 2:1, 70% non-syndromic. CP — 1:2000, female 4:1, 50% non-syndromic (pp. 1113, 1115 — CP figures at p. 1113). 2. At least 10 weeks old, 10 pounds in weight, haemoglobin 10% (p. 1124). 3. Lip at 3–6 months; palate at 6–12 months (pp. 1122, 1124). 4. Before eruption, when root formation is about two-thirds complete, usually at 8–10 years of age (p. 1130).

1. Good facial appearance; good orofacial function in speech, eating and swallowing; an aesthetic, functional and stable occlusion; good hearing (p. 1113).

Appendix D — Landmark evidence

Sources worth naming, with the one line you should be able to quote and the page on which to check it. Learn the finding, not just the name.

Biology and mechanics

SOURCE	TOPIC	FINDING	PAGE
Sandstedt; Oppenheim	1	Independent light-microscopy studies at the turn of the last century founded the pressure–tension theory	Ch 6, pp. 291–292
Reitan (1967)	1	Hyalinisation occurs even at 30 g, and **more with tipping than with bodily movement**	Ch 6, p. 293
Baumrind (1969)	1	Bone-bending hypothesis — but there is little evidence forces bend bone to any clinically meaningful degree	Ch 6, p. 294
Storey & Smith (1952)	1	Historical optimal force range 150–200 g	Ch 6, p. 296
Quinn & Yoshikawa (1985)	1	Positive dose–response at low forces until a plateau is reached	Ch 6, p. 297
Ren et al. (2003, 2004)	1	No universal ideal force can be defined; modelled maximum canine retraction at 272 cN	Ch 6, p. 297
Theodorou et al. (2019)	1, 5	50–250 cN give similar rates of 0.1–0.47 mm/week; 50–100 cN considered optimal for bodily movement	Ch 6, p. 297
Kanzaki et al. (2004)	1	RANKL expression increased in compressed PDL cells; OPG gene transfer reduces tooth movement	Ch 6, p. 290
Yang et al. (2018)	1	Deleting RANKL in the periodontium of transgenic mice reduces osteoclasts and tooth movement	Ch 6, p. 301
Henneman et al. (2008)	1	Four-step mechanotransduction: ECM strain and fluid flow → strain transferred to cells → cell recruitment and differentiation → tissue remodelling	Ch 6, p. 296

SOURCE	TOPIC	FINDING	PAGE
Danz et al. (2021)	1	RANKL peaks at 3 days then returns to normal from day 6 to 58	Ch 6, p. 299
Levander et al. (1994)	1	Pause force application for 3–6 months where significant root resorption is identified	Ch 6, p. 306
El-Angbawi et al. (2023)	1	Photobiomodulation may accelerate discrete phases but does not influence overall treatment time	Ch 6, p. 305
Jepsen (1963)	5	Root surface areas of the permanent dentition — the basis of relative anchorage values	Ch 6, p. 340
Kapila & Sachdeva (1989)	7	No archwire material offers all the ideal properties together	Ch 10, p. 647

SOURCE TOPIC FINDING PAGE

Wang et al. (2018),7Coaxial superelastic NiTi produces greater movement over 12 Cochrane-weeks than single-strand; otherwise insufficient evidence that any archwire is superior Papageorgiou et7Austenitic-active slightly better than martensitic-stabilised; copper al. (2014)NiTi sequences slightly slower to a working wire and more painful Ch 10, p. 655

Ch 10, p. 656

Diagnosis and records

SOURCE	TOPIC	FINDING	PAGE
Houston (1983)	2	"Cephalometrics is definitely not an exact science"	Ch 7, p. 460
Baumrind & Frantz (1971a)	2	Every landmark has a characteristic non-circular envelope of error in x and y	Ch 7, p. 460
Baumrind & Frantz (1971b)	2	Landmark error is cumulative and can produce artificial correlation between measurements	Ch 7, p. 460
Ricketts (1957, 1972); Björk (1969)	2	Growth prediction methods, subsequently shown to be at best unreliable	Ch 7, p. 416
Graf et al. (2022a)	2	No superimposition method has been proved to deliver completely accurate results	Ch 7, p. 463

SOURCE	TOPIC	FINDING	PAGE
Baumrind et al. (1976); Ghafari et al. (1987)	2	All superimposition methods carry appreciable error; none is significantly more reliable	Ch 7, p. 469
Posselt (1952)	18	Only ~10% have RCP coincident with ICP; a 0–2 mm discrepancy is normal	Ch 2, p. 58

Planning, extraction and anchorage

SOURCE	TOPIC	FINDING	PAGE
Kirschen, O'Higgins & Lee (2000a, b)	3	Royal London Hospital space analysis	Ch 8, p. 506
Al-Abdallah et al. (2008); Ahmed & Sharma (2023)	3	Formal space analysis gives a more disciplined approach, realistic aims and greater anticipation of anchorage requirements	Ch 8, p. 506
Proffit et al. (1992a)	3, 12	Class II surgery indicated if overjet >10 mm, Pog– nasion perpendicular ≥18 mm, mandibular body <70 mm, or anterior facial height >125 mm	Ch 8, p. 491
Konstantonis et al. (2018)	3, 4	1 mm of maxillary incisor retraction gives 0.7 mm of lip retraction and a 1.6° increase in nasolabial angle	Ch 8, p. 492
Ackerman & Proffit (1997); Holdaway (1983)	3	The soft tissues define the limits of orthodontic treatment	Ch 8, p. 490
Rushing et al. (1995); Paquette et al. (1992); Iared et al. (2017)	4	Soft tissue differences between extraction and nonextraction are minimal; both groups are perceived to benefit	Ch 8, p. 515
Bowman & Johnston (2000)	4	Lip retraction must be considerable before the profile is regarded as unattractive	Ch 8, p. 515

SOURCE TOPIC FINDING PAGE

Meyer et al. (2014a, b);4Extractions do not produce larger buccal corridors or aCh 8, p. Christou et al. (2019); Chengless attractive smile515 et al. (2016)

O'Higgins & Lee (2000)	4, 16	Expansion creates space equal to only about half the amount of expansion	Ch 8, p. 522
Haque et al. (2017)	4, 5	Anchorage loss is the same for first and second premolar extraction; **two-thirds of the space is lost unless anchorage is supported**	Ch 6, p. 338; Ch 8, p. 528
Diar-Bakirly et al. (2017)	5	A trans-palatal arch provides very little additional anchorage in the sagittal plane	Ch 6, p. 341
Rebellato et al. (1997)	5	A lingual arch provides little anchorage support and can procline the lower incisors	Ch 6, p. 342
Melsen & Verna (1999)	5	In maximum anchorage cases, anchor teeth should not be subject to initial aligning forces	Ch 6, p. 341
Samuels (1996)	5	Headgear injuries are of two types — catapult and nocturnal disengagement	Ch 6, p. 348
Giannopoulou et al. (2016)	5	The maxillary arch is particularly susceptible to anchorage loss	Ch 6, p. 351

Appliances

SOURCE	TOPIC	FINDING	PAGE
Angle (1928)	6	The edgewise slot; horizontal orientation gives greater control of tooth movement	Ch 10, p. 614
Begg (1956); Cadman (1975)	5, 6	Differential force; the three stages of Begg treatment	Ch 10, pp. 615, 621
Kesling (1988); Kesling et al. (1991)	6	Tip-Edge — limited tipping, then full prescription in rectangular wire at finishing	Ch 10, p. 615
Andrews (1972)	6	Six keys of occlusion, derived from a large sample of untreated ideal occlusions	Ch 2, p. 58; Ch 10, p. 617
Andrews (1979)	6	The pre-adjusted edgewise prescription built into each bracket	Ch 10, p. 617

SOURCE	TOPIC	FINDING	PAGE
Scott et al. (2008a, b); Fleming et al. (2009a, b, 2010); DiBiase et al. (2011); Johansson & Lundström (2012); Pringle et al. (2009)	6	No difference in rate of movement, treatment time, number of visits or occlusal outcome with self-ligation; pain benefit marginal; expansion claims unsupported	Ch 10, pp. 646– 647
Penning et al. (2017)	6	Randomised trial: Insignia gives comparable outcomes and no reduction in treatment time, with more breakages, more complaints and more planning time. Treatment time was dictated by case difficulty and the treating orthodontist	Ch 10, p. 631

Growth modification and timing

SOURCE TOPIC FINDING PAGE

Tulloch et al. (1997,12The Florida and North Carolina trials: effective early overjetCh 9, p. 2004); Keeling et al.reduction, wide individual variation in skeletal effect, no long-588 (1998); Dolce et al.term difference (2007)

O'Brien et al. (2003a, b)	12	Early twin block treatment reduces the overjet dentoalveolarly and gives a transient increase in self-esteem	Ch 9, p. 588
O'Brien et al. (2009a, b)	12	No long-term difference; early-treated children had more attendances and longer treatment; **the late-treated group had a slightly better final occlusal result**; early treatment is not normally justified	Ch 9, pp. 588–589
Cobourne et al. (2022); Kallunki et al. (2021)	9, 12	Relative risk of new incisor trauma with early overjet reduction is **0.58 — a 42% reduction** — but with wide confidence intervals; headgear was marginally more effective than a functional appliance	Ch 12, pp. 845–847

Batista et al. (2018); Macey et al. (2020); Arraj et al. (2019)	9	Increased overjet is a significant risk factor for incisor trauma; early correction reduces later incidence	Ch 2, p. 90
Pancherz (1982)	8	Herbst overjet reduction in 6–8 months	Ch 9, p. 602
Pancherz et al. (2014)	8	At 32 years, **64% of molar correction and 86% of overjet correction remained stable**; recommend at least 2 years of activator retention	Ch 9, p. 562
Pacha et al. (2023, 2024)	8, 12	Randomised trial: **37% failure with twin block versus 17% with Herbst**; 1.5 months longer; smaller mean overjet reduction (5.8 vs 7.1 mm); but more visits and emergencies with the Herbst	Ch 9, p. 606
Koretsi et al. (2015)	8	Functional appliance therapy has a positive effect on the soft tissue profile in Class II subjects	Ch 9, p. 561
Voudouris et al. (2003)	8	Implanted electrodes showed a *reduction* in lateral pterygoid activity on protrusion, contradicting earlier surface EMG	Ch 9, p. 564
Santana et al. (2020)	8	Little clinical difference between maximal and incremental mandibular advancement	Ch 9, p. 596
Mandall et al. (2016)	17	Facemask protraction: **over 60% avoided surgery versus 36% without early treatment**, though most initial skeletal effects are lost with growth	Ch 8, p. 487; Ch 9, p. 557

Specific clinical problems

SOURCE	TOPIC	FINDING	PAGE
Houston (1989); Millett et al. (2012)	11	Overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop established	Ch 13, pp. 930, 977

SOURCE	TOPIC	FINDING	PAGE
Eberhart et al. (1990)	11	0.1–0.2 mm of overbite change per degree of incisal angular change	Ch 13, p. 939
Ng et al. (2005)	11	Segmental mechanics achieve ~1.5 mm maxillary and ~1.9 mm mandibular incisor intrusion in non-growing patients	Ch 13, p. 936

SOURCE	TOPIC	FINDING	PAGE
Dake & Sinclair (1989); Weiland et al. (1996)	11	Segmental techniques give greater intrusion than continuous reverse curve of Spee, but more lower incisor proclination with a Ricketts utility arch	Ch 13, p. 936
Sosly et al. (2020); Shakti et al. (2022)	11	Miniscrews give 0.67–0.95 mm more intrusion than segmental mechanics with less molar extrusion (low-quality evidence)	Ch 13, p. 937
Burstone (1977); Nanda et al. (1998)	11	One-couple bypass intrusion arch; the Connecticut intrusion arch in NiTi requires no helix	Ch 13, p. 935
Dahl et al. (1975); Poyser et al. (2005)	11	Localised supraocclusion re-establishes contacts over an average of 6 months; adverse events rare, minor and transient	Ch 13, p. 940
Järvinen (1978)	13	Traumatic loss of a maxillary central affects ~3% of children — unilateral, mixed dentition, boys with an increased overjet	Ch 12, p. 821
Ferguson (1990)	14	The maxillary canine fails to erupt in ~3% of Caucasian children; up to 85% are palatal	Ch 12, p. 780
Peck et al. (1994)	14	Bilateral occurrence 17–45%; up to 3× more common in females; 5× more common in Europeans than Asians	Ch 12, p. 781

SOURCE	TOPIC	FINDING	PAGE
Ericson & Kurol (1986, 1988)	14	Canines palpable by 10–11 years; interceptive extraction between 10 and 13 with space available	Ch 12, pp. 781, 793
Naoumova et al. (2015a, b); Naoumova & Kjellberg (2018)	14	**9 of 23 erupted after primary canine extraction versus 3 of 22**; 7 of 23 versus 11 of 22 still needed surgery. Alpha angle and sector position are the most influential predictors	Ch 12, pp. 794, 796
Benson et al. (2021), Cochrane	14	Only very weak evidence that primary canine extraction increases eruption, and **no evidence it reduces the need for surgery**; no benefit from double extraction	Ch 12, p. 794
Parkin et al. (2017), Cochrane	14	No difference between open and closed exposure in success or other outcomes (low to very low quality)	Ch 13, p. 880
Cassina et al. (2018)	14	Open exposure associated with just over 2 months less time to alignment and lower odds of ankylosis	Ch 13, p. 880
Kokich (2004)	14	Apically repositioned flap for buccal canines below the mucogingival junction	Ch 13, p. 877
Patel et al. (2009)	14	Autotransplantation of canines gives good outcomes of around 88%	Ch 13, p. 881
Shapira et al. (2011)	15	Evidence of a genetic basis for second molar impaction, at least in the mandibular arch	Ch 12, p. 795
Pogrel (1995); Kravitz et al. (2016)	15	Surgical uprighting, latterly combined with orthodontic alignment and a bracket bonded at surgery	Ch 12, p. 795
Magnusson & Kjellberg (2009)	15	Failure of eruption often persists after intervention, but **extraction of the second molar itself gives the least successful outcome**	Ch 12, p. 795
Decker et al. (2008)	15	Loss-of-function *PTH1R* mutation in primary failure of eruption	Ch 12, p. 800

SOURCE	TOPIC	FINDING	PAGE
Ugolini et al. (2021), Cochrane	16, 19	31 trials, ~1,400 patients: quadhelix or expansion plate beats observation at 7–11 years (high certainty); quadhelix beats a plate and is ~3 months faster; no difference between fixed expander types	Ch 12, p. 836
Liu et al. (2015)	16	RME achieves up to 4 mm at the suture, about 50% of total screw expansion	Ch 9, p. 543
Lo Giudice et al. (2020)	16	~10% of skeletal change relapses at 4–8 months postretention	Ch 9, p. 544
Bucci et al. (2016); Rutili et al. (2021, 2022); Srivastava et al. (2020)	16	95% of initial expansion maintained with SME versus 85% with RME; RME gives slightly greater posterior skeletal expansion but more buccal tipping and more pain	Ch 13, pp. 952– 953
Bazargani et al. (2023)	16	High-quality randomised trial: limited differences between conventional and hybrid RME at 5 years	Ch 9, p. 547
Wiedel & Bondemark (2015b)	16	Anterior crossbite corrected with a positive overbite is usually stable and requires no retention	Ch 13, p. 977
Kerr et al. (1992); StellzigEisenhauer et al. (2002); Kochel et al. (2011)	17	Thresholds for Class III surgery, Wits $\leq$ –11 mm being among the most important; discriminant analysis predicts the need for surgery with >90% success	Ch 13, pp. 920– 921
Obwegeser & Makek (1986)	19	Three types of condylar hyperplasia asymmetry: hemimandibular elongation, hemimandibular hyperplasia, and hybrid	Ch 15, pp. 1093– 1098
Slootweg & Müller (1986)	19	Four histological subtypes of condylar hyperplasia	Ch 15, p. 1098
Häner et al. (2021)	19	Serial 3D facial surface superimposition — good reproducibility using the forehead, mid-nose and infraorbital regions	Ch 15, p. 1099

SOURCE	TOPIC	FINDING	PAGE
Sinclair & Little (1983)	20	Mandibular arch width and length decrease with age in untreated subjects, with increasing anterior crowding	Ch 5, p. 283
Behrents (1985); Solow (1980)	20	Facial growth continues throughout life and is rotational, driving compensatory uprighting of the lower labial segment	Ch 13, p. 978
Southard et al. (1990)	20	Anterior component of occlusal force proposed as causative — **but the short duration of these forces makes this unlikely**	Ch 13, p. 978
Thilander (2009); Abdulraheem et al. (2020)	20	Shorter arch perimeter, reduced intercanine distance and increased crowding in untreated adults; distinguish relapse from normal ageing	Ch 13, p. 978
Little (1975)	20	Under 3.5 mm of lower incisor irregularity suggested as an acceptable level of relapse	Ch 13, p. 989
Rowland et al. (2007)	20	Vacuum-formed retainers more effective than Hawley type in the lower labial segment	Ch 13, p. 989
Forde et al. (2017, 2018); Martin et al. (2023)	20	Fixed retainers marginally more effective in the lower arch for the first 12 months, especially for rotations; equivalent thereafter	Ch 13, p. 989
Al-Moghrabi et al. (2018); Krämer et al. (2022)	20	**By 4–5 years post-treatment the majority of patients have stopped wearing their retainers**	Ch 13, p. 990

Benefits, risks and evidence appraisal

SOURCE	TOPIC	FINDING	PAGE
Macey et al. (2020)	9	Weak evidence that crowding is a caries risk factor or that treatment reduces later caries incidence	Ch 2, p. 88
Bollen et al. (2008); Papageorgiou et al. (2018)	9	Lack of evidence that orthodontic treatment improves periodontal health	Ch 2, pp. 88, 96
Gorelick et al. (1982); Tufekci et al. (2011); Øgaard (2008)	9	White spot lesions in up to half of fixed appliance patients, more prevalent in males, visible within 6 months, commonest on first molars, upper laterals and lower canines	Ch 2, p. 93
Benson et al. (2019)	9	Daily high-fluoride toothpaste plus professional highconcentration fluoride foam every 6–8 weeks	Ch 2, p. 93
Sameshima & Sinclair (2004)	9	Severe root resorption in 1–5% of treated teeth, greatest in the anterior maxilla	Ch 2, p. 94
Monk et al. (2017)	9	Pain peaks at 48 hours and returns to baseline after 7 days; analgesics may be taken pre-emptively	Ch 2, p. 94
Joss-Vassalli et al. (2010); Aziz et al. (2011); Tepedino et al. (2018)	9	Recession relates more to retroclination in high-angle Class III cases with a narrow symphysis and thin biotype than to proclination	Ch 2, p. 96
Larsen et al. (2015); Orabi et al. (2021)	9	**No evidence that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea** — "scaremongering and marketing"	Ch 2, p. 91
Koretsi et al. (2018)	9	Mandibular advancement appliances effective in mild-tomoderate OSA	Ch 2, p. 90
Seehra et al. (2011)	9	Increased overjet makes a child more susceptible to teasing and bullying	Ch 2, p. 91
Kenealy et al. (2007)	9	Little objective evidence that treatment improves long-term psychological health	Ch 2, p. 91

SOURCE	TOPIC	FINDING	PAGE
Javidi et al. (2017)	9	Treatment improves the emotional and social well-being domains of OHRQoL	Ch 2, p. 92
Klaus et al. (2017)	9	High discontinuation rates in children and adolescents, especially with removable appliances	Ch 2, p. 100
NICE	10	Antibiotic prophylaxis solely to prevent infective endocarditis should not routinely be provided for dental procedures	Ch 7, p. 357
Thornhill et al. (2016)	10	NICE guidance conflicts with the AHA and ESC, which still recommend cover for the highest-risk patients	Ch 7, p. 357
Chakravarthi et al. (2012)	10	Nickel allergy in approximately 10–30% of females and 1– 3% of males	Ch 7, p. 363
Wu et al. (2016)	10	Latex allergy prevalence ~4.3% in the general population and 9.7% in healthcare workers	Ch 7, p. 362
Zymperdikas et al. (2020)	10	Bisphosphonates slow or inhibit tooth movement and delay eruption; keep aims simple	Ch 7, p. 363
Dons et al. (2022)	10	Body dysmorphic disorder in 5–13% of orthodontic patients and up to 17% seeking orthognathic surgery	Ch 7, p. 365

SOURCE TOPIC FINDING PAGE

Egermark-Eriksson et al.18Certain occlusal traits show **weak** correlations with TMDCh 2, p. (1983); Mohlin et al. (2007);signs and symptoms102 Thilander & Bjerklin (2012)

Manfredini et al. (2017)	18	Lack of evidence implicating dental occlusion in the pathophysiology of TMD	Ch 2, p. 103
Brimm v. Malloy (1987)	18	Michigan lawsuit; $850,000 award after TMD following four-premolar extraction treatment	Ch 2, p. 104

Behrents & White (1992)	18	AAO-funded studies: no significant association; TMD cannot be prevented by any intervention; symptoms increase with age and often begin in late adolescence; orthodontic treatment does not cause TMD	Ch 2, p. 105
Luecke & Johnstone (1992)	18	Conventional incisor retraction does not produce distal displacement of the condyles	Ch 2, p. 105
Macfarlane et al. (2009); Luther et al. (2010)	18	Lack of evidence that orthodontic intervention reduces TMD signs and symptoms	Ch 2, p. 105
Kandasamy et al. (2022)	18	Screen before treatment; suspend active mechanics if symptoms appear; manage with conservative, reversible therapies	Ch 2, pp. 103–104
NICE	20	Prophylactic removal of developing third molars is not recommended to prevent late lower incisor crowding	Ch 5, p. 284

Appendix E — The numbers

A final-hour checklist. If you can reproduce this from memory, the quantitative marks in any of the twenty essays are secure. **Biology (Ch 6)**

- Bone: ECM ~30%, mineral ~60%, cells ~10% · osteocytes 85–90% of bone cells
- PDL width ~0.2 mm, thinnest at the middle third · hyalinisation at 30 g, more with tipping
- Phases: initial 2 days · lag 20–30 days · post-lag
- Optimal force: historical 150–200 g · modelled maximum 272 cN · similar rates 50–250 cN (0.1– 0.47 mm/week) · **50–100 cN optimal for bodily movement**
- Rate of movement ~1 mm/month · RANKL peaks day 3
- Root resorption: radiographic 73% · histological 90% · **severe (≥1/3 root) 1–5%** · pause 3–6 months **Anchorage and headgear (Ch 6; last item Ch 8, p. 528)**
- Anchorage: 10–12 h/day, 250–350 g per side
- Distalisation/orthopaedic: 14 h/day, 450–500 g per side
- Two mandatory safety devices: self-releasing strap, locking facebow
- **Two-thirds of space lost without anchorage support Archwires (Ch 10)**
- Stainless steel "18–8": Fe 70%, Cr 17–20%, Ni 8–12%, C max 0.08%
- Alignment 0.012–0.018 in · levelling round steel 0.016–0.020 in
- Beta titanium stiffness ≈ 1/3 of stainless steel **Overbite (Ch 13)**
- Intrusion force ~60 g total, 15 g per tooth
- Segmental intrusion achieved: ~1.5 mm maxillary, ~1.9 mm mandibular
- Miniscrews: 2–3 mm, i.e. 0.67–0.95 mm more than segmental
- Proclination: 0.1–0.2 mm overbite change per degree
- Dahl: contacts re-established over ~6 months **Extraction (Ch 8)**
- Mandibular guidelines: mild 1–4 mm · moderate 5–8 mm · severe ≥9 mm
- Expansion yields ~half its width as space · IPR up to 1 mm per contact in buccal segments
- Canine distal movement: <1/2 unit → distalise · >1/2 unit → extract · >1 unit → both
- Lip response: 0.7 mm per 1 mm of incisor retraction; nasolabial angle +1.6°
- Class II surgery thresholds: overjet >10 mm · Pog–Na perp ≥18 mm · mandibular body <70 mm · AFH >125 mm **Functional appliances (Ch 9)**
- Twin block blocks at ~70° · Herbst overjet reduction in 6–8 months
- Skeletal gain: a few millimetres at best, essentially transient
- Pacha trial: twin block 37% failure vs Herbst 17% · 5.8 vs 7.1 mm overjet reduction
- Pancherz at 32 years: 64% of molar and 86% of overjet correction stable · retain ≥2 years
- Facemask 12–14 h/day · Mandall: **60% avoided surgery vs 36%**
- Early overjet reduction: **RR 0.58, a 42% reduction in trauma risk Expansion (Ch 9, 12, 13)**
- Plate: 0.25 mm per turn twice weekly · retain nights 3–6 months
- RME: 1–4 turns daily, 0.25–1 mm/day, over 2–4 weeks · forces up to 10 kg · retain 3–6 months
- Up to 4 mm at the suture = ~50% of screw expansion · ~10% skeletal relapse at 4–8 months
- Dental expansion retained: SME 95%, RME 85%
- Cochrane: quadhelix ~3 months faster than a plate, ages 7–11
- SARPE beyond ~5 mm in the non-growing patient **Impaction (Ch 12, 13)**
- Maxillary canine: ~3% impacted · **up to 85% palatal** · eruptive path >20 mm
- Bilateral 17–45% · 3× more common in females · 5× more common in Europeans
- Palpable by 10–11 years · firm primary canine beyond 14 years is a red flag

- Conventional radiography 39–85% accurate in localisation
- Interception: 10–11 (RCS: 10–13) years · distal to lateral incisor root midline · 20–30° to mid-sagittal · no crowding · review at 12 months
- Poor prognosis: above the apical third · at or across the midline · angulation >55°
- Autotransplantation ~88% · splint semi-rigidly ≤3 weeks, then root-treat
- Mandibular second molar impaction: just over 2% · observe 6–12 months
- First permanent molar impaction: 0.75–4.3%, 25× more common in the maxilla **Class III** (**Ch 8, 13**)
- Surgery thresholds: Wits ≤ –11 mm · Holdaway ≤3.5° · ANB ≤ –4° · gonial ≤120° · LAFH ≥66% · overjet ≤ –5 mm · lower incisor inclination ≤80°
- Discriminant analysis >90% accurate **Risks, medical and other** (**Ch 2, 7**)
- White spot lesions: up to half of patients · within 6 months · first molars, upper laterals, lower canines · fluoride foam every 6–8 weeks
- Pain peaks 48 h, resolves by 7 days
- Nickel allergy 10–30% of females, 1–3% of males · latex 4.3% general, 9.7% healthcare
- IE mortality ~21% · defer orthodontics ≥6 months post-transplant
- BDD: 0.7–2.5% general, 5–13% orthodontic, up to 17% orthognathic
- Eating disorders: worldwide prevalence just below 8%
- TMD: ~75% have a sign, ~33% a symptom · Brimm v. Malloy $850,000, 1987 · AAO studies 1992
- Traumatic loss of a maxillary central: ~3% of children · dental trauma up to 18% at age 12
- Retention: <3.5 mm irregularity "acceptable" · most patients stop wearing retainers by 4–5 years

Appendix F — Figure finder

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12.6	Early orthodontic treatment for posterior crossbite	Ch 12, p. 836
12.7	Early overjet reduction and dental trauma	Ch 12, p. 845
13.1	Molar distalisation	Ch 13, p. 862
13.2	Implants and anchorage	Ch 13, p. 874
13.3	Open or closed surgical exposure	Ch 13, p. 880
13.4	Absent maxillary lateral incisors	Ch 13, p. 890
13.5	Two-step versus en masse retraction	Ch 13, p. 914
13.6	Orthodontics or surgery for adult Class III?	Ch 13, p. 920
13.7	Slow versus rapid maxillary expansion	Ch 13, p. 952
13.8	Managing long-term retention	Ch 13, p. 989

Appendix H — Corrections log

This handbook has been rebuilt against the textbook itself rather than a secondary summary. Every

topic ga correcti im-preci	ined page references, an ons matter enough to fl se, and because examine	d several gained ma-terial that was previ-ously absent. T ag explicitly, because the earlier versions were eith rs notice precision on exactly these points.	he following er wrong or
TOPIC	PREVIOUSLY STATED	CORRECTED POSI-TION	SOURCE
1	Hyalinisation occurs even at 30 g	Correct, but add: it occurs **more with tip-ping than with bodily movement**, because bodily movement dissipates force more evenly through the ligament	Ch 6, p. 293
1	Contemporary optimal force 50–100 cN	Correct, but the fuller picture is that 50–250 cN produce **similar** rates (0.1–0.47 mm/week); >250 cN is marginally faster with greater adverse effects; and a mathematical model puts maximum canine retraction at **272 cN**	Ch 6, pp. 296–297
2	″Cephalometrics is not an exact science″ attributed loosely	The textbook attributes this to **Houston (1983)**	Ch 7, p. 460
4	Crowding thresholds presented gener-ally	Table 8.5 is specifically **guidelines for extrac-tion in the mandibular arch** — say so	Ch 8, p. 516
5	Trans-palatal arch gives ″modest″ an-chorage	Stronger: it provides **very little additional anchorage in the sagit-tal plane** (Diar-Bakirly et al., 2017); its value is transverse and verti-cal. The lingual arch provides **little anchor-age support** and may procline lower incisors; it is better regarded as a space maintainer	Ch 6, pp. 341–342

4, 5	First premolar extraction gives better anchorage	Theoretically yes, but **in practice anchorage loss is the same for first and second premolar extraction**, and **around two-thirds of the space created is lost unless anchorage is supported** (Haque et al., 2017)	Ch 6, p. 338; Ch 8, p. 528
9	White spot lesions affect "a substantial proportion"	**Up to half** of fixed appliance patients (Gorelick et al., 1982), and they are **more prevalent in males** (Tufekci et al., 2011)	Ch 2, p. 93

TOPIC	PREVIOUSLY STATED	CORRECTED POSITION	SOURCE
9	Severe root resorption most often maxillary incisors	More precisely: greatest amount and severity in the **anterior maxilla, especially the maxillary lateral incisors**; the descending order of frequency is maxillary incisors, canines, mandibular incisors, first permanent molars	Ch 2, p. 94; Ch 6, p. 303
10	Nickel allergy "increasingly prevalent"	Quantify: **10–30% of females and 1–3% of males**; latex **4.3%** general population, **9.7%** healthcare workers	Ch 7, pp. 362–363
12	Early treatment reduces trauma	Quantify: relative risk **0.58**, a **42% reduction** in trauma risk — but with wide confidence intervals, most trauma occurring before age 9, and **headgear marginally more effective than a functional appliance** for this purpose	Ch 12, pp. 845–847

TOPIC	PREVIOUSLY STATED	CORRECTED POSITION	SOURCE
13	Avulsion answer improvised emergency trauma protocol and omitted the orthodontic consequences	Emergency management follows the **dental trauma guidelines**, not an orthodontic protocol. What the textbook does give — and what is examinable — is the **rest period before force is applied** (re-implantation **12 months**), **repeat radiographs at 6–9 months**, a **3-month pause** if resorption appears, and the three complications of moving a traumatised tooth: **loss of vitality, pulpal calcification, further root resorption**	Ch 8, pp. 535–537; Ch 12, pp. 747–748, 821

13	Avulsion answer improvised emergency trauma protocol and omitted the orthodontic consequences	Emergency management follows the **dental trauma guidelines**, not an orthodontic protocol. What the textbook does give — and what is examinable — is the **rest period before force is applied** (re-implantation **12 months**), **repeat radiographs at 6–9 months**, a **3-month pause** if resorption appears, and the three complications of moving a traumatised tooth: **loss of vitality, pulpal calcification, further root resorption**	Ch 8, pp. 535–537; Ch 12, pp. 747–748, 821

14	Interceptive extraction at 9–14 years, canine <30° to midline	The textbook's criteria are **10–11 years**, canine **distal to the midline of the lateral incisor root and 20–30° to the mid-sagittal plane**, no crowding, review at 12 months. RCS England guidance uses 10–13 years	Ch 12, pp. 790–794

16	Quadhelix produces dental expansion; RME produces skeletal	Too neat. **Slow maxillary expansion can also split the intermaxillary suture in the early mixed dentition**; the difficulty is age-related sutural interdigitation, not the appliance category	Ch 9, p. 543
19	Hemimandibular hyperplasia produces a marked occlusal cant	Add: the **dental centreline and chin are often undisturbed**, and a **unilateral open bite** occurs on the affected side. There is also a **hybrid Type III** form	Ch 15, pp. 1096–1098
20	Anterior component of occlusal force is a contributory factor	The textbook is more sceptical: the **short duration of these forces makes this unlikely** (Southard et al., 1990). The dominant factor is **mandibular growth rotation with uprighting of the lower labial segment**	Ch 13, p. 978
| 18 | manage **modali**	TMD answers listed jaw ment | Not in this te **ties such as occlusal s**	rest, soft diet, analgesia, physiotherapy and jaw xtbook. The source names only **conservative and plints**, on the explicit grounds that **no modality has**	exercises as **reversible been shown**

superior to any other. The examinable content is the **Kandasamy et al. (2022) six-point guideline** and the shift to a **biopsychosocial** model, both of which were previously missing | Ch 2, p. 103 | | 18 | Onset in late adolescence stated without attribution | **Verified and retained** — the textbook does say symptoms "generally increase with age and often begin in late adolescence", and it is the AAO-commissioned **Behrents & White (1992)** studies that say so. Now attributed. (Flagged during audit as unsourced against Ch 2, p. 102; found at Ch 2, p. 105) | Ch 2, p. 105 | | 10 | Gingival hyperplasia — "the classic three" including calcium channel blockers | This textbook names only **ciclosporin** and **phenytoin**. Calcium channel blockers are conventional teaching and worth naming, but are flagged as not appearing in this source | Ch 7, pp. 361–362 | **Added after a reader flagged the avulsion answer, and after a systematic re-audit of all 125 model answers:** the full table of **rest periods before orthodontic force after dental trauma** (Ch 8, pp. 536–537), the **monitoring protocol** for a traumatised tooth, and the distinction between

the **3-month** pause for a previously traumatised tooth (Malmgren & Malmgren, 2007) and the **3–6 month** pause for orthodontically induced resorption generally (Levander et al., 1994). **Material now added that was previously missing entirely:** bone-bending and bioelectric signalling (Topic 1); the Wnt/sclerostin axis (Topic 1); the "airway orthodontics" controversy (Topic 9); the Pacha Herbst-versus-twin-block randomised trial (Topics 8 and 12); the Naoumova canine trial and Benson Cochrane review (Topic 14); the Kochel/Stellzig-Eisenhauer discriminant analysis for Class III surgery (Topic 17); slow-versus-rapid expansion evidence (Topic 16); and the retention evidence base (Topic 20).

Appendix I — A revision schedule

The handbook and the textbook are meant to be used together, in this order.

Six weeks out

Read the relevant textbook sections once, using the "Where to read it" line at the head of each topic. Read for comprehension; do not attempt to memorise.

Four weeks out

Work from this handbook. For each topic, cover the page and reproduce the skeleton from memory. If you can produce the skeleton, you can produce the essay. Check any fact you are unsure of against the page reference — it should take under a minute.

Three weeks out

Write four essays under timed conditions, chosen from the topics you find least comfortable. For most candidates those are 6, 7, 15 and 19. Compare what you wrote against the facts table and note the *omissions* rather than the errors; omissions are what cost most candidates marks.

Two weeks out

Work through the viva questions with a colleague. Have them read the question only and judge your spoken answer against the model. Aim for sixty to ninety seconds. The most common faults are answering too briefly, failing to give a number, and failing to say what you would actually *do*.

One week out

Part C, saying the one-line finding aloud for each source. Then Part D. **The night before.** Part A and Part D only. Nothing new.

A final word

The twenty topics here are not a syllabus. They are a representative sample of the ways examiners ask about a much smaller number of underlying principles: the biology of the periodontal ligament, the arithmetic of space, the inescapability of Newton's third law, the difference between what growth permits and what it does not, the primacy of the mandibular arch as a template, and the discipline of preferring evidence to tradition. A candidate who has internalised those six things can construct a creditable answer to a question they have never seen. That, rather than recall, is what the examination is designed to find.

Prepared as structured examination preparation material. All page references are to the supplied electronic edition of Cobourne MT, DiBiase AT, Handbook of Orthodontics, 3rd edition (Elsevier, 2025); this edition carries no printed page numbers, so references are to the PDF page. Primary studies are cited as they appear in that text and should be verified against the originals before formal submission or publication.

Appendix J — A revision route through the book

Weeks 1–2: the foundations

Chapters 1, 3, 4 and 5. These are the chapters that will not come back to you under pressure unless you have laid them down early, and Chapter 4 in particular underpins half of the clinical material. Work the self-tests.

Weeks 3–4: the mechanisms

Chapters 6, 10 and 11. Definitions matter more than narrative here; if you can define a couple, a centre of resistance, springback and resilience, and state the miniscrew torque window, you have the chapter.

Weeks 5–6: the clinical core

Chapters 2, 7, 8, 12 and 13. These overlap most with the essay handbook, so alternate: read the chapter unit here for structure and numbers, then write the corresponding essay from the essay handbook plan.

Week 7: the specialist chapters

Chapters 9, 14, 15 and 16. Each has a small number of highly quotable figures — Mandall's 60 versus 36, the aligner refinement statistics, the rule of tens, the graft timing — and those figures are what you will actually use.

Week 8: consolidation

Work through every "Where is the evidence?" box in the book, using the index in the front of this companion. Then the self-tests, all of them, in one sitting. Then Part D of the essay handbook.

A closing thought

This book is unusual among textbooks in telling you repeatedly how weak its own subject's evidence base is. That is not a flaw to be revised around; it is the most useful thing it teaches. The examiner is not looking for a candidate who knows what orthodontists do. They are looking for one who knows what orthodontists can and cannot demonstrate — and who can still make a decision.

Prepared as structured examination preparation material. All page references are to the supplied electronic edition of Cobourne MT & DiBiase AT, Handbook of Orthodontics, 3rd edition (Elsevier, 2025), which carries no printed page numbers; references are therefore to the PDF page. Primary studies are cited as they appear in that text and should be verified against the originals before formal submission or publication.

Appendix K — Quick index

T = essay topic (Part II) · Ch = chapter (Part III).

TERM	WHERE
Aligner outcomes	Ch 14
Alveolar bone grafting	Ch 16
Anchorage loss	T5, T4, T12
Andrews' six keys	T6, Ch 2
Angle classification	Ch 2
Ankylosis	T1, T14, T15
Autotransplantation	T13, T14
Bisphosphonates	T10
Bone-anchored protraction	T17, Ch 9
Camouflage	T3, T12, T17
Centre of resistance	Ch 6
Cephalometric error	T2, Ch 7
Chin cup	T17
Cleft epidemiology	Ch 16
Condylar hyperplasia	T19, Ch 15
CONSORT / PRISMA	Ch 1
Consent	T3, T9, T18, T20
Couple and moment	T6, Ch 6
Crowding thresholds	T4
CVM staging	Ch 4
Dahl concept	T11
Decompensation	T17, Ch 15
Demineralisation	T9
Differential force	T5, T6
Distraction osteogenesis	Ch 15
Extraction debate	T4, T9
Facemask	T17, Ch 9
Frankfort plane	T2, Ch 7
Functional matrix theory	Ch 4
GRADE / AGREE II	Ch 1
Growth rotations	Ch 4
Headgear force and duration	T5, Ch 6
Herbst	T8, T12
Heterogeneity / I^2	Ch 1
Hierarchy of evidence	Ch 1
Hyalinisation	T1
Incisor liability	Ch 5
Intention-to-treat	Ch 1
IOFTN	Ch 2, Ch 15
IOTN and PAR	T9, Ch 2

TERM	WHERE
Interceptive canine extraction	T14
Intrusion force	T11

TERM	WHERE
Leeway space	T4, Ch 5
Le Fort I	Ch 15
Miniscrews	T5, T14, Ch 11
Mutual protection	Ch 2
Nance button	T5
Neural crest	Ch 3
Nickel allergy	T10
Obstructive sleep apnoea	Ch 13
Occlusal stop	T11, T20
Optimal force	T1
Overjet and trauma	T9, T12
Palate development	Ch 3
Part–counterpart	Ch 4
Peak height velocity	T12, Ch 4
Pharyngeal arches	Ch 3
PICO	Ch 1
Primary failure of eruption	T15, Ch 12
Quadhelix	T16, T19
RANKL / OPG	T1
Rapid maxillary expansion	T16, Ch 9
Retention	T3, T9, T20, Ch 13
Root resorption	T1, T9, T11
Rule of tens	Ch 16
SARPE	T16, Ch 15
Scammon's curves	Ch 4
Self-ligation	T6, Ch 10
Space analysis	T3, Ch 8
Springback and resilience	T7
Superelasticity	T7, Ch 10
Supernumerary teeth	Ch 12
Sutural theory	Ch 4
Tooth agenesis	Ch 12
Trans-palatal arch	T5, T14
Transposition	Ch 12

TERM	WHERE
Twin block	T8, T12
Vertical growth pattern	T5, T11, T17
Wits appraisal	T17, Ch 13

Appendix L — Contemporary Orthodontics (Proffit): a cross-reference

Cross-reference to Proffit WR, Fields HW, Larson BE, Sarver DM. Contemporary Orthodontics, 6th edition (Elsevier, 2019). **A note on these page numbers — they behave differently from the rest of this book.** Every other reference here takes the form *Ch 6, p. 293*, where the page is the page of the supplied electronic Cobourne & DiBiase file, which carries no printed pagination. The Proffit file **does** carry printed page numbers, so every page below is the **true printed page of the 6th edition** and will match any copy you pick up. Chapter numbers are Proffit's own. **The two books are organised on different principles, and that is the main difficulty in using them together.** Cobourne & DiBiase is arranged by *subject* — a chapter on tooth movement, a chapter on anchorage, a chapter on aligners. Proffit is arranged by *patient stage and problem complexity* — preadolescent, adolescent, adult; non-skeletal, skeletal, surgical. So a subject that occupies one Cobourne chapter is often distributed across three or four Proffit chapters, and vice versa. Where that happens the table gives every location rather than the nearest single one.

Proffit's twenty chapters

CH	TITLE	PAGES
1	Malocclusion and Dentofacial Deformity in Contemporary Society	3–18
2	Concepts of Growth and Development	19–60
3	Early Stages of Development	61–84
4	Later Stages of Development	85–110
5	The Etiology of Orthodontic Problems	111–140
6	Orthodontic Diagnosis: The Problem-Oriented Approach	141–208
7	Orthodontic Treatment Planning: From Problem List to Specific Plan	209–248
8	The Biologic Basis of Orthodontic Therapy	249–276
9	Mechanical Principles in Orthodontic Force Control	277–310
10	Contemporary Orthodontic Appliances	311–356
11	Moderate Nonskeletal Problems in Preadolescent Children	357–402
12	Complex Nonskeletal Problems in Preadolescent Children	403–430
13	Treatment of Skeletal Transverse and Class III Problems	431–456
14	Growth Modification in Class II, Open Bite/Deep Bite	457–502
15	Comprehensive Treatment in Adolescents: Alignment and Vertical Problems	503–528
16	Comprehensive Treatment in Adolescence: Space Closure and Class II/III	529–556
17	Comprehensive Treatment: Finishing	557–580

CH	TITLE	PAGES
18	Retention	581–600
19	Special Considerations in Treatment for Adults	601–658
20	Combined Surgical and Orthodontic Treatment	659 onwards

By essay topic (Part II)

TOPIC	READ IN PROFFIT	PAGES
1. Biological basis of tooth movement	Ch 8 — the whole chapter is the equivalent	**249–276**; hyalinization 254; force magnitude 265; root resorption 269–274
2. Cephalometry	Ch 6	**174–193**; analysis 174–179; superimposition 193
3. Formulating a treatment plan	Ch 7 — Proffit's problem-list method is more systematic than Cobourne's and worth reading in full	**209–248**
4. Choice of extraction	Ch 7 planning; Ch 16 space closure	209–248; **548–556**
5. Anchorage	Ch 9 is the core; Ch 16 for anchorage in space closure	**277–310** (densest at 297); 537–544; 266–267
6. Three orders and the evolution of fixed appliances	Ch 10	**311–356**; straight-wire and Andrews 324, 336–337
7. Archwires — steel versus nickel titanium	Ch 9 materials; Ch 15 in use	**278–279**; 503–506
8. Functional appliances	Ch 14	**457–502** (densest 455, 469–470)
9. Benefits and complications	Ch 1 for need and indices; Ch 8 for iatrogenic damage	11–17; **269–275** (resorption, white spot 273–275)
10. Medical conditions	Ch 8 for drug effects; Ch 19 for the adult patient	258–260; **601–658**
11. Overbite and its reduction	Ch 15 vertical; Ch 17 finishing; Ch 6 assessment	**503–528**; 563; 202
12. Reducing overjet in a growing child	Ch 14 — the growth-modification evidence sits here	**457–502**; trial evidence 498–499
13. Missing maxillary central incisor	Ch 12; Ch 16 for space management	**415–416**; 537
14. Impacted maxillary canine	Ch 12 interceptive; Ch 15 exposure and alignment	424–426; **512–517**
15. Impacted lower second molar	Ch 15	**517** — brief; Cobourne is much fuller here
16. Crossbite correction	Ch 13	**431–456** (densest 434–437, 453)
17. Class III	Ch 13 non-surgical; Ch 16; Ch 20 surgical	**440–456**; 552; 659 onwards
18. Malocclusion and TMD	Ch 19; Ch 1 for context	**616–618**; 10
19. Facial asymmetry	Ch 6 diagnosis; Ch 14; Ch 20 surgical	188; 492–496; **686**

TOPICREAD IN PROFFITPAGES

1. Late lower incisorCh 4 for the growth explanation; Ch 18 for**101–103**; 584, 592 crowdingthe consequence

By chapter (Part III)

CHAPTER (COBOURNE &

DIBIASE)	NEAREST PROFFIT EQUIVALENT	PAGES
1. Evidence-Based Orthodontics	Ch 1 — shorter than Cobourne's treatment, and there is no standalone EBO chapter	11–17
2. Occlusion and Malocclusion	Ch 1; classification in Ch 6	3–18; 149
3. Prenatal Development	Ch 3; aetiology in Ch 5	61–84; 111–140
4. Postnatal Growth	Ch 2 and Ch 4 together	**19–60**; 85–110
5. Development of the Dentition	Ch 3 and Ch 4; leeway space 80–82	61–110
6. Orthodontic Tooth Movement	Ch 8 biology; Ch 9 mechanics	**249–310**
7. Examination and Diagnosis	Ch 6	**141–208**
8. Treatment Planning	Ch 7	**209–248**
9. Functional Jaw Orthopaedics	Ch 14	**457–502**
10. Contemporary Fixed Appliances	Ch 10	**311–356**
11. Skeletal Anchorage	**No standalone chapter.** Distributed: Ch 10 appliances, Ch 12 preadolescent, Ch 19 adults, Ch 20 surgical	345; 422–424; 651; 697
12. Management of the Developing Dentition	Ch 11 and Ch 12	**357–430**
13. Management of the Permanent Dentition	Ch 15, 16, 17 and 18 together	**503–600**
14. Clear Aligners	**No standalone chapter.** Distributed: Ch 10, Ch 19	316, 342; 601, 638
15. Orthognathic Surgery	Ch 20	**659 onwards**

1. Cleft Lip and Palate andCh 7 planning for the cleft patient; aetiology in Ch 4 and Ch**241–246**; 108, Syndromes 5 112

Three places where Proffit will not help you

- **Skeletal anchorage** and **clear aligners** have no chapter of their own. Both are treated as techniques within other chapters, and neither is covered at the depth of Cobourne & DiBiase Chapters 11 and 14. For these two, stay with Cobourne.
- **Impacted lower second molars** get roughly a page. Cobourne's escalating ladder has no Proffit counterpart.
- **Cleft care** is considerably shorter than Cobourne Chapter 16, and is framed as a treatment-planning problem rather than as a developmental and multidisciplinary subject. Conversely, Proffit is **stronger** on the problem-oriented diagnostic method (Ch 6–7), on finishing (Ch 17, which Cobourne has no equivalent of), and on the adult patient (Ch 19). Read those even where Cobourne covers the same ground.

Part V — The units this book did not have

Four units written for this edition. Building the Revision Handbook and the Question Bank against the sixty-five curriculum items exposed four areas that the twenty topics and sixteen chapters do not cover as units of their own, because the source textbook does not treat them as chapters. They are, nonetheless, explicit curriculum items, and they are cheap marks.

WHY THESE FOUR

Items 9 and 10 (ethics, consent and communication) are touched on inside Topic 9 but never developed. Item 63 (practice, infection control and governance) has no home at all. Radiation protection appears only as a sentence about justification. Each of these is examinable in the written paper and, more sharply, in the oral — where a candidate who has never rehearsed the consent conversation is easy to spot. Question Bank Blocks **H** and **AD** test this material; sit them early rather than last.

V.1 Ethics, consent and capacity

Revision Handbook: §8
Question Bank: Block H (Q226–255)
Viva bank: V39–V48

IN ONE MINUTE

Orthodontics is elective. Consent is therefore the whole justification for the treatment, not a form signed at the start.

- *Montgomery* (2015) replaced the doctor's standard with the **patient's**: material risks are those a reasonable person *in this patient's position* would attach significance to
- Consent is a process, revisited — not an event
- Under 16: assess **Gillick competence**; over 16: capacity is presumed
- The four principles — autonomy, beneficence, non-maleficence, justice — are the language examiners expect

The Montgomery standard, and what changed

Before 2015 the adequacy of a warning was judged by what a responsible body of practitioners would have disclosed — the *Bolam* test. *Montgomery v. Lanarkshire Health Board* replaced that with a patient-centred test. The clinician must take reasonable care to ensure the patient is aware of any **material** risk in the proposed treatment, and of any reasonable alternative, where material means a risk to which a reasonable person in the patient's position would attach significance, *or* to which this particular patient would.

The practical consequence in orthodontics is that a risk of low incidence but high significance to this patient must be disclosed even if colleagues would not routinely mention it. A professional singer's concern about a bonded lingual retainer, or a patient whose occupation depends on their appearance, changes what is material.

What must be discussed

Heading	Content
The problem	What is wrong, in the patient's own terms, and its consequences if untreated
The proposed treatment	Appliance, duration, extractions, what will be asked of the patient
Realistic benefit	Aesthetics and, where the overjet is large, trauma reduction. Not caries, not speech, not TMD
Material risks	Demineralisation, root resorption, relapse, pain, loss of vitality, gingival recession, allergy, failure to achieve the aim, headgear injury where relevant
Alternatives	Including **no treatment**, and a limited or compromise plan
Retention	That it is lifelong for stability of alignment, and what wearing it involves
Cost and commitment	Appointments, emergencies, the consequences of poor cooperation
Their questions	Recorded, with the answers given

EXAMINER TRAPS

- Saying "I get them to sign the form." The form records the conversation; it is not the consent.
- Omitting **no treatment** from the alternatives. It is always an option in elective care.
- Consenting the parent and not the child. Both must understand; the child's assent matters even where the parent gives legal consent.
- Presenting relapse as a possibility rather than the norm. Some relapse is expected; retention is how it is managed.

Capacity, Gillick competence and the young patient

Capacity is presumed from 16. Below 16, a child may consent if they have sufficient understanding and intelligence to understand fully what is proposed — the **Gillick** test, assessed for the specific decision at the specific time, not as a general status. A Gillick-competent child can consent to treatment; refusal by such a child can, in law, be overridden by a person with parental responsibility or the court, though in elective orthodontics a refusing child should not be treated.

Where the child is not competent, consent is given by a person with parental responsibility, and the child's **assent** should still be sought and recorded. A child who does not want the appliance will not wear it, which makes assent a clinical as well as an ethical requirement.

The four principles, applied

Principle	In an orthodontic decision
Autonomy	The patient's own assessment of whether the appearance of their teeth troubles them is the starting point. IOTN measures need, not desire
Beneficence	Act to produce a benefit the evidence supports; declining to treat a well-aligned dentition is beneficent
Non-maleficence	Weigh demineralisation, resorption and the burden of two years in appliances against the benefit; a poor oral hygiene record is a reason to defer
Justice	Fair use of finite NHS or institutional resource; IOTN exists to make that allocation defensible

Communication, and the difficult conversations

Three conversations recur in the oral. Rehearse each aloud.

1. **The patient whose expectations cannot be met.** Name the limit, name what *can* be achieved, offer the alternative that would achieve the rest (usually surgery), and give them time.
2. **The patient with poor oral hygiene mid-treatment.** State the observation, show them, explain the specific consequence, agree a measurable standard and a review date, and record that appliances will be removed if it is not met.
3. **Body dysmorphic disorder.** Prevalence in orthodontic and orthognathic populations is 5–13%. The signs are a preoccupation disproportionate to any observable feature, repeated consultations, and dissatisfaction after previous treatment. Do not treat; refer for psychological assessment, and say so kindly and plainly.

VIVA QUESTIONS

- What changed after *Montgomery*, and how does it alter your consent conversation?
- A 14-year-old wants treatment; her mother does not. What do you do?
- How would you consent a patient for a bonded retainer?
- A patient asks you to remove appliances early. Walk me through it.

V.2 Radiation protection and imaging justification

Revision Handbook: §30.2; imaging §9.4
Question Bank: Block AD (Q992–1016)
Viva bank: V39–V48

IN ONE MINUTE

Every radiograph needs a clinical justification recorded before it is taken. There is no such thing as a routine orthodontic radiograph.

- The three principles: **justification**, **optimisation**, **limitation**
- ALARP — as low as reasonably practicable
- CBCT only where the diagnostic yield changes management and cannot be obtained from conventional films
- The report on every image is a legal requirement, not an optional extra

Justification, optimisation, limitation

Justification means that the benefit to the individual patient outweighs the detriment. The justification is specific to the patient and the clinical question, and it is recorded in the notes before exposure. "Pre-treatment records" is not a justification; "to locate an unerupted maxillary canine not palpable buccally at age 11" is.

Optimisation means the smallest dose consistent with answering the question: rectangular collimation, the fastest image receptor, appropriate kV and exposure time, no repeat films through poor technique, and a field of view no larger than the region of interest.

Limitation refers to dose limits, which apply to staff and the public, not to patients — a distinction examiners like to test.

Doses, in context

Examination	Effective dose (order)	Equivalent
Intraoral periapical	a few μSv	under a day of background
Panoramic	tens of μSv	a few days of background
Lateral cephalogram	a few μSv	under a day of background
Small field-of-view CBCT	tens of μSv	days of background
Large field-of-view CBCT	up to several hundred μSv	weeks to months of background

Quote the ranking and the order of magnitude rather than precise figures, which vary with equipment and technique. The examinable point is that a large-volume CBCT can carry many times the dose of the conventional films it might replace.

Which film, for which question

Clinical question	Imaging
General assessment of the developing dentition	Panoramic, where the clinical examination raises a question it can answer
Skeletal pattern, incisor inclination, growth monitoring	Lateral cephalogram
Localising an unerupted canine	Two films at different angles — parallax. Panoramic plus anterior occlusal, or two periapicals
Suspected root resorption of an adjacent incisor	CBCT, small field of view, where it changes whether the canine is aligned or removed
Assessing an alveolar cleft before grafting	CBCT or occlusal, as the team protocol directs
Impacted second molar	Panoramic
Periodontal assessment before treatment	Periapicals of the teeth concerned

Parallax, stated correctly

Two images are taken with the tube shifted between them. An object *further* from the tube than the reference — that is, palatal — moves in the *same* direction as the tube shift. An object closer to the tube — buccal — moves in the opposite direction. The mnemonic is SLOB: same lingual, opposite buccal. The shift may be horizontal (two periapicals) or vertical (panoramic and anterior occlusal); the rule is the same.

EXAMINER TRAPS

- Ordering a CBCT "to see it better". The question is whether the extra information changes management.
- Failing to report the whole image. A panoramic taken for a canine still requires the condyles, sinuses and the rest of the dentition to be examined and the findings recorded.
- Repeating a film taken elsewhere without asking for the original.
- Confusing dose limits (staff and public) with dose constraints and diagnostic reference levels.

V.3 Infection control in orthodontic practice

Revision Handbook: §30.1
Question Bank: Block AD (Q992–1016)
Viva bank: V39–V48

IN ONE MINUTE

Standard precautions apply to every patient, because you cannot know who is infectious. Orthodontics has three particular problems: pliers, bands and impressions.

- Decontamination is a **cycle**: cleaning → inspection → disinfection → sterilisation → storage
- Instruments are classified critical, semi-critical or non-critical, and that determines the process
- Orthodontic pliers are hinged and jointed — the hardest instruments in dentistry to clean
- Sharps injuries, and the immediate response, are examinable

The decontamination cycle

Stage	What it involves, and the orthodontic difficulty
Cleaning	Removal of visible soil — the single most important step, because nothing can be sterilised through debris. Ultrasonic or washer-disinfector preferred to manual scrubbing. **Hinged pliers must be opened** and, where the design allows, disassembled
Inspection	Under illuminated magnification. Residual cement or blood in a plier joint means the cycle is repeated
Disinfection	Thermal where possible; chemical only where an instrument cannot tolerate heat
Sterilisation	Steam under pressure. Vacuum (type B) autoclaves for hollow, jointed or wrapped instruments; non-vacuum (type N) only for solid unwrapped items
Storage	Wrapped and dated; the wrap is what maintains sterility, so a torn pack is reprocessed

Classification, and what follows

Class	Definition	Orthodontic examples and handling
Critical	Penetrates soft tissue or bone, or contacts the bloodstream	Band seaters used subgingivally, surgical instruments for exposures or miniscrews — sterilise, use sterile
Semi-critical	Contacts mucosa but does not penetrate	Pliers, mirrors, bracket placement instruments, impression trays — sterilise
Non-critical	Contacts intact skin only	Chair controls, light handles — clean and disinfect, or use barriers

The three orthodontic problems

1. **Pliers.** Jointed, hinged, often with tungsten carbide inserts that corrode. Open the joints for cleaning, avoid prolonged immersion in chloride-containing solutions, lubricate the joint after cleaning and before sterilisation, and inspect the beaks for retained composite.
2. **Bands and brackets.** Bands tried in the mouth and not cemented have been contaminated and must be reprocessed before reuse — a common lapse. Debonded brackets are single-use.
3. **Impressions and appliances to and from the laboratory.** Rinse, disinfect by the method the material tolerates, and label the work as disinfected. Appliances returning from the laboratory are disinfected again before insertion.

Sharps and the immediate response

Sharps in orthodontics are ligature ends, distal end cutters' fragments, band-removing instruments and local anaesthetic needles. Wire ends should be caught by the cutter or held; safety glasses protect the patient and the operator from flying fragments.

After a percutaneous injury: **encourage bleeding** without sucking, wash under running water without scrubbing, cover, and report immediately. Assess the source, seek occupational health advice the same day, and consider post-exposure prophylaxis where indicated. Record the incident.

EXAMINER TRAPS

- "We autoclave everything" — without the cleaning stage, sterilisation fails.
- Using a type N autoclave for wrapped or hinged instruments.
- Reusing try-in bands without reprocessing.
- Forgetting that the laboratory is part of the chain in both directions.
- Not knowing the immediate steps after a sharps injury. Rehearse them aloud.

V.4 Governance, audit and knowing your limits

Revision Handbook: §30.3–§30.4
Question Bank: Block AD (Q992–1016)
Viva bank: V39–V48

IN ONE MINUTE

Governance is how a service demonstrates that its outcomes are good, and improves them when they are not. In orthodontics that means PAR, and it means audit.

- **Audit** compares practice against a standard, changes something, and re-measures — the loop must close
- **Research** asks what the right thing to do is; **audit** asks whether we are doing it
- **PAR** is the outcome measure: ≥22 points, or 30% reduction, is "greatly improved"
- Referral is a clinical skill: knowing the limit of your competence is examinable

The audit cycle

1. **Choose a topic** that matters — high volume, high risk, or known variation.
2. **Set a standard**, with a target and a source. "100% of radiographs have a written justification, per IRMER."
3. **Measure** current practice against it, on a defined sample.
4. **Compare** and identify the gap.
5. **Change** something specific — a form, a checklist, a briefing.
6. **Re-measure**. Without this step it is a survey, not an audit.

AUDIT TOPICS THAT MAKE GOOD ANSWERS

Radiographic justification and reporting · bond failure rate by operator and by tooth · treatment duration against a local standard · PAR score improvement · proportion of cases finishing with an ABO-OGS above a threshold · retention protocol compliance · number of appointments per case · demineralisation at debond · consent documentation · miniscrew survival by site.

Outcome measures

Index	Measures	Use and limitation
IOTN	Treatment *need* — dental health component and aesthetic component	Allocates resource; says nothing about outcome
PAR	Occlusal improvement, pre versus post	The standard outcome measure. ≥22 points or 30% reduction = greatly improved; a coarse measure that under-detects finishing detail
ABO-OGS	Finishing detail: alignment, marginal ridges, buccolingual inclination, occlusal contacts and relationships, overjet, interproximal contacts, root angulation	Sensitive to exactly what PAR misses — which is why aligners score the same on PAR and worse on OGS

Index	Measures	Use and limitation
ICON	Need, complexity, outcome and improvement in one index	Single index, less widely used

Knowing your limits

The curriculum expects a candidate to state when a case should not be treated by them. Refer, or seek advice, where there is: a skeletal discrepancy beyond the envelope of camouflage; a cleft or craniofacial anomaly; suspected condylar hyperplasia or a progressive asymmetry; significant hypodontia requiring a restorative plan; active periodontal disease; a medical condition altering the treatment plan materially; or any case where the patient's expectation cannot be met by what you can deliver.

Saying this plainly is a strength, not a weakness. The examiner is testing insight.

EXAMINER TRAPS

- Describing an audit that never re-measured.
- Confusing audit with research, or with service evaluation.
- Quoting PAR as though it captured finishing quality.
- Claiming to be able to treat anything. Name two case types you would refer, and why.

VIVA QUESTIONS

- Describe an audit you have completed. What changed as a result?
- How would you demonstrate that your treatment outcomes are acceptable?
- What is the difference between IOTN and PAR?
- Which cases would you not treat yourself?